Building Secure and Competitive Clean Technology Supply Chains: Technology and Policy Options for Electric Vehicle Batteries and Critical Minerals

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Abstract

Electric vehicle (EV) battery material supply chains have become a prominent focus in industrial policy contexts, with concerns about supply chain vulnerability, environmental performance, and economic competitiveness driving billions of dollars worth of investments, incentives, and trade restrictions across key nations like the United States and China. Yet, many existing analyses of these supply chains operate at levels of aggregation that obscure the structural features that govern how raw materials become finished batteries and electric vehicles.

These features determine which chemistries are more vulnerable to disruption, which production pathways are cost- and emissions-competitive, and which policy designs could effectively shift investments. This dissertation develops pathway-aware analysis across three studies of battery supply chains. The first maps the material flows of critical minerals used in battery cathodes, and measures the vulnerability of battery chemistry choices to disruptions. The second analyzes the U.S. Inflation Reduction Act (IRA), highlighting how its incentives affected relative competitiveness across battery chemistries and supply chain steps, and how those incentives compare with previous and current incentives in China. Finally, the third develops a bottom-up integrated technoeconomic and life cycle assessment (TEA-LCA) framework framework to evaluate how lithium production pathways depend on resource mineralogy and process choices.

Three structural findings emerge. Lithium iron phosphate (LFP) batteries face a simpler critical-mineral footprint than nickel-based chemistries, with compounding advantages across vulnerability, policy qualification, and upstream economics. Chinese lepidolite-based lithium production carries cost and emissions disadvantages relative to US brine and clay pathways that follow from resource mineralogy rather than from country-level operational or regulatory factors. Further, the IRA's EV supply chain credits provided support that exceeded total modeled battery production costs, but the credit structure imposed uneven qualification requirements across supply chain stages, creating both incentives and barriers to localizing US battery supply chains. Together, these findings show that pathway-resolved analysis surfaces structural features that aggregate approaches could otherwise obscure.

These frameworks can apply beyond batteries to other supply chains and critical minerals, suggesting pathway-resolved engineering analysis as an extensible foundation for analyzing the material supply chain systems underpinning the energy transition.

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Chapter 1

Introduction

Vehicle electrification is an important pathway for reducing transportation's greenhouse gas emissions. Transportation end use emissions (across all types of transportation) accounted for 27% of emissions in the United States [4] and 24% of worldwide emissions in 2020 [5]. Automaker battery design choices, which very often involve specific choices of battery chemistries and firm boundaries (as seen in Appendix A), imply reliance on particular critical materials and associated processing infrastructures. The rapid scale-up of electric vehicle (EV) battery production implies a massive increase in demand for these critical minerals, with projected increases of 5 to 40 times 2020 demand in 2040, depending on the material [3, 6, 7]. Today, electric vehicle batteries mainly use lithium ion chemistries [3, 7]. The primary lithium-ion cathode chemistries are NCA (lithium nickel cobalt aluminum oxide), NMC (lithium nickel manganese cobalt oxide), and LFP (lithium iron phosphate), which depend on varying amounts of four primary critical minerals: lithium, nickel, cobalt, and manganese, as identified in studies of mineral criticality for battery cathode materials [8–10] (Figure 1.1b). Discussion of other minerals that may be deemed as critical or could become critical, including phosphorus, aluminum, and iron, as well as the criticality of other battery chemistries under development, is provided in Appendix A.1.2.

Understanding the choice among battery chemistries is crucial, as this choice largely determines important vehicle-level characteristics such as travel range, battery cost (Fig. Figure 1.1a), charging speed and battery life [11]. There are three primary battery chemistries used in EVs today, which are all lithium-ion technologies. Two are nickel-based: NCA (lithium nickel cobalt aluminium oxide, typically $LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$) and NMC (lithium nickel manganese cobalt oxide, $LiNi_xMn_yCo_{1-x-y}O_2$.). The primary alternative is LFP (lithium iron phosphate, $LiFePO_4$). Though it is less energy dense than either nickel-based chemistry, it also uses lower-cost materials and is considered to be safer and more durable, with better environmental implications [12] (Figure 1.1d), and thus has re-emerged as a viable option [11, 13]. Due to intellectual property limitations and choices made by market leaders such as Tesla, nickel-based chemistries were the dominant choice in US manufactur-

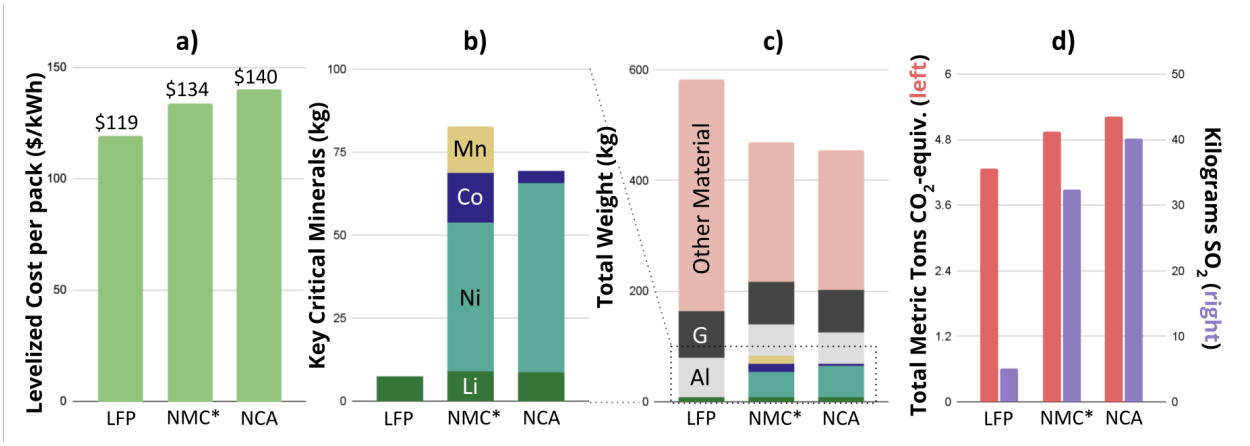


Figure 1.1. A comparison of key characteristics of three lithium-ion battery chemistries key (LFP, NMC, NCA), across four dimensions
a) production cost for a 77 kWh battery, b) weight of key critical minerals, c) total weight of battery materials, and d) environmental impacts across greenhouse gas and sulfur dioxide emissions. *Note NMC is represented by the NMC622 chemistry.

ing whereas LFP-based chemistries were developed by firms in China [14]. As EV adoption is dependent on the aforementioned chemistry-dependent characteristics [15], there is now an open question of which choice will be the dominant technology (if any) across battery chemistries.

1.1 Lithium Ion Battery Supply Chains

The battery supply chain can be separated into three broad segments: upstream (mining and extraction of raw materials), midstream (processing of raw materials into battery-grade components), and downstream (cell and pack manufacturing, as well as end-of-life recycling and reuse). [16] The supply chains for the critical minerals in these batteries differ in terms of the geography of raw material production (Figure 1.2), although a few countries produce the majority of supply for each critical mineral. Arguably the most important choice is the selection of cathode material, as cathodes are over half of the cost of a battery cell and largely determine crucial battery characteristics such as energy density and charging speed. [13, 15]

We build upon the Inflation Reduction Act's definition of critical minerals [18], where aluminium, cobalt, graphite (natural and synthetic), lithium, manganese and nickel (and notably not copper) are critical. Whereas other minerals are defined as critical in the IRA, these six are the only that are relevant to EV batteries [19]. After processing and refining, these materials are combined with other inputs and manufactured into various battery components, such as cathodes, anodes, electrolytes and separators, but also possibly other

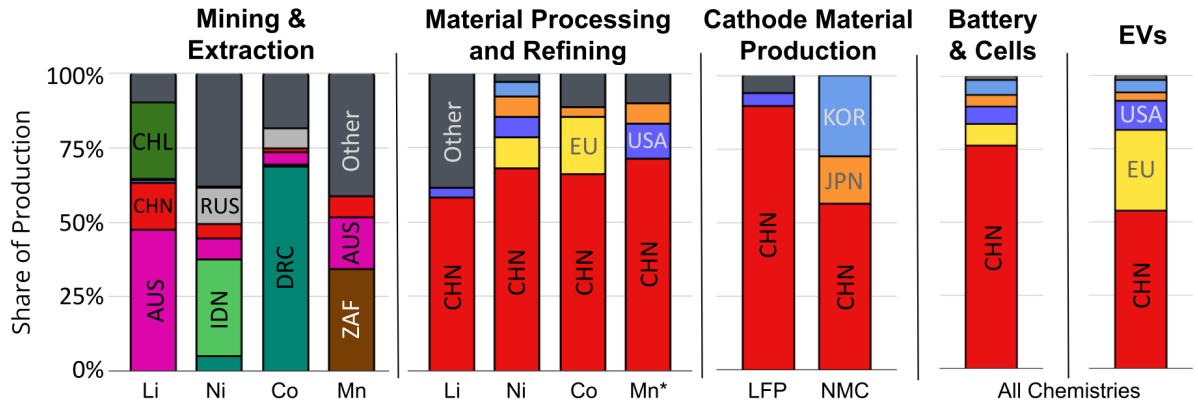


Figure 1.2. Geographical Distribution of EV Battery and Material Supply Chains (2020) Li: lithium; Ni: nickel; Co: cobalt; Mn: manganese (where *electrolytic manganese dioxide is used to represent refined manganese). The following codes are used to represent countries or regions: CHL: Chile, CHN: China, AUS: Australia, IDN: Indonesia, RUS: Russia, DRC: Democratic Republic of the Congo, ZAF: South Africa, KOR: Republic of Korea, JPN: Japan, USA: United States of America, Other: any other country not explicitly listed here. Figure 1 from Cheng et al. [17]; adapted with data from US Geological Survey [1], Sun et al. [2], Agency [7]

non-electroactive components such as aluminium current collectors and structural elements. These components are then manufactured into battery cells, which are combined into modules and packs for electric vehicles.

While upstream mining is geographically distributed by material, China dominates every part of the midstream and downstream supply chains of all materials, from material processing and refining to electric vehicle production. What is unclear from these production data is the degree to which electric vehicle battery material supply chains have vulnerabilities to specific countries when taking into consideration all supply chain steps. In 2020, about 74% of mined lithium [1] and 57% of mined cobalt [20] was used in lithium ion batteries, while only a portion of lithium ion batteries are used in electric vehicles. Furthermore, only about 11% of nickel [21] and 2% of manganese [22] was used to make batteries of any kind.

Despite the scale and policy importance of these supply chains, existing analyses often address vulnerability and competitiveness at relatively aggregate levels, with varying resolutions of underlying structural features like critical mineral combinations in battery chemistries, policy qualification mechanics, and resource-specific processing requirements. These details can determine which chemistries are most vulnerable to disruption, which policy designs effectively shift investment across supply chain stages, and which production pathways are economically and environmentally competitive across geographies (Appendices A.2, B.1,

and C.1 review the relevant literatures, respectively). This dissertation develops pathway-aware analysis to surface these features, through three studies: a material flow analysis of critical minerals in battery cathodes that measures vulnerability to disruptions in Chinese supply (Chapter 2); a comparative analysis of US Inflation Reduction Act incentives and Chinese state support across battery chemistries and supply chain stages (Chapter 3); and a bottom-up integrated technoeconomic-life cycle assessment of three lithium production pathways representing distinct resource-pathway combinations (Chapter 4).

Chapter 2

Electric vehicle battery chemistry affects supply chain disruption vulnerabilities

Modified from a paper published in *Nature Communications* [17].

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2.1 Abstract

We examine the relationship between electric vehicle battery chemistry and supply chain disruption vulnerability for four critical minerals: lithium, cobalt, nickel, and manganese. We compare the nickel manganese cobalt (NMC) and lithium iron phosphate (LFP) cathode chemistries by (1) mapping the supply chains for these four materials, (2) calculating a vulnerability index for each cathode chemistry for various focal countries and (3) using network flow optimization to bound uncertainties. World supply is currently vulnerable to disruptions in China for both chemistries: 80% [71% to 100%] of NMC cathodes and 92% [90% to 93%] of LFP cathodes include minerals that pass through China. NMC has additional risks due to concentrations of nickel, cobalt, and manganese in other countries. The combined vulnerability of multiple supply chain stages is substantially larger than at individual steps alone. Our results suggest that reducing risk requires addressing vulnerabilities across the entire battery supply chain.

2.2 Introduction

To the extent that battery supply chains rely on one or a few countries for specific process steps, many governments have grown concerned about vulnerability to supply disruptions and some have enacted policies to incentivize domestic production. Legislation in the US [18, 23], Europe [24], and elsewhere [25] requires minimum percentages of domestic manufacturing or sourcing from allies (such as free-trade agreement countries in the US context) to qualify for incentives or avoid fines or to meet regulations on carbon footprint and responsible sourcing. As a result, measurements of whole-supply chain vulnerabilities are required to better help understand economic, sustainability, and national security risks. Such measurements may help policymakers evaluate incentives and promulgate regulations on the basis of specific supply chain interventions, such as to reduce such risks, promote domestic production, or internalize economic benefits.

Furthermore, most automotive manufacturers around the world are exercising increasing levels of control over their material supply chains as they design batteries for their vehicles (Appendix Table A.1). As such, this study focuses on the decision-making of firms in the upstream and midstream supply chains, ending at the current typical firm decision boundary, the choice of battery cathode chemistry. While other components such as the anode (graphite) [26] and electrolyte (typically lithium salt solutions) [27] may also suffer from vulnerabilities in their supply chain, the choices among these components are far more limited and thus do not offer the same kind of options to reduce vulnerability by changing technology choices.

2.3 Literature Review

Several streams of literature have developed vulnerability-related concepts and metrics, such as literature on energy security, materials criticality, material flow analysis, input-output analysis, and supply chain disruption propagation. While a detailed literature review is provided (Appendix A.2), we highlight below a few key studies that have framed discussion of battery material supply chains, country-based risks, and measures of vulnerability below.

A number of studies in the material flow analysis literature examine the structure of these supply chains in detail. Using a combination of trade data and production data at each step of the supply chain, these studies describe the relationships between geographies of production, with countries as the unit of geographic resolution. In the context of battery materials, parts of this literature focus on specific stages of the value chain, e.g. raw materials and mining, while others encompass all steps, but the scope is almost always global and limited to one specific battery material – lithium [28], cobalt [29, 30], nickel [31], manganese [32].

We use these methods to reveal the underlying material production and trade networks that undergird the supply chain flows in our analysis. By including the cathode material manufacturing step, we can then compare the relative demand for each material that is required to fulfill material demand for batteries, a novel addition not found in studies of individual critical mineral material flow analyses.

Several materials criticality studies have included measurements of country-based production concentration in assessments of vulnerability or risk [8, 33, 34]. While these studies combine measurements of supply chain structures in ways that aggregate across multiple materials, these studies tend to focus on the relative importance of critical minerals. However, in trying to combine several metrics, these studies can obfuscate the vulnerability of risk caused by specific countries. On the other hand, studies in the recently developing supply chain disruption propagation literature [35–38] identify specific countries, particularly China, as critical nodes in battery material trade networks, by calculating the avalanche size of a disruption – the number of countries affected by the removal of supply or trade from one country. However, these studies do not combine multiple critical minerals and also make the general assumption that disruptions occur as proportional losses of trade from the disrupted country, which then propagate to other countries, which does not encompass the full scope of realities of how disruptions occur.

Furthermore, across all of these studies, the treatment of uncertainty varies substantially. For example, the material flow analysis literature provides insight into methods for accounting for missing trade, but these uncertainties are not always incorporated into materials criticality measurements. A recent study [19] evaluated the feasibility of meeting the recently passed U.S. Inflation Reduction Act's goals for minimum critical mineral requirements across all battery materials for the United States while accounting for uncertainty in imports and maximum material availability. They found that achieving market-value-based targets may be possible with NCA batteries but not necessarily for LFP or NMC batteries. However, this study does not consider additional supply chain steps beyond direct imports to the United States, which can be uncertain.

In this work, we suggest there is an important gap in the existing literature given a lack of analysis of the vulnerabilities in the global flow of multiple battery materials between countries as sources of both supply and demand across stages of the supply chain, particularly in terms of physical quantities of materials. While the criticality literature typically takes the perspective of an importing focal country, our study instead assesses global supply chain dependence on exporting focal countries. Moreover, existing studies do not explore multiple sources of uncertainty. While we do not attempt to determine potential causes of risks and vulnerabilities, nor assess the probability of such risks or vulnerabilities occurring,

we aim to provide a quantitative metric to help assess vulnerabilities and evaluate potential actions to reduce vulnerabilities. In addition to specific insights about the supply chains for electric vehicle batteries, including the explicit inclusion of the cathode material manufacturing step, we provide a novel methodology incorporating material flow analysis methods with optimization methods to account for the impact of uncertainty on measurements of vulnerability.

2.4 Results

We examine the impact of electric vehicle battery design choices on the vulnerability of battery material supply chains. We first describe a static distribution of the supply chains, accounting for trade and production of all materials related to the four primary critical battery minerals used in battery cathodes (lithium, cobalt, nickel, and manganese). We then measure a vulnerability index for a country (or combination of countries) as the total percentage of the end product (in this case, battery cathode material) produced with materials that are either manufactured in or traded from that country, at any step of the supply chain, while accounting for uncertainty due to sourcing of materials at each step and to missing trade data. Our analysis focuses on LFP (lithium iron phosphate) and NMC (lithium nickel manganese cobalt) as they have the minimum (1) and maximum (4) number of critical minerals covered in this assessment.

2.4.1 Battery cathode material flow supply chain mapping

We describe the global supply chains for lithium in Figure 2.1 and for cobalt, nickel and manganese in Figure 2.2, considering the known demand for various lithium ion battery cathode materials. From left to right, each figure shows the portion of global material supply in each country for raw material mining and extraction, raw material trade, refining and processing, refined material trade, and cathode production, which includes trade of scrap or recovered materials that may be involved in the battery material supply chain. For each section labeled as trade (i.e. flows within each rectangular boundary), the flows are between countries, and sections labeled as processing (i.e. flows crossing the rectangular boundaries) are wholly within the country. Boxes that are bisected by the rectangular boundaries could be involved in either trade or production, but are uncertain. We note that not all mined or refined materials are used for battery cathodes. These diagrams map the countries and trade flows that are explicitly involved in these battery material supply chains, as described by the literature. Details about the methodology and data underlying in Figure 2.1 and Figure 2.2 are found in the Methodology section and Appendix A.3, with the simplified version of the

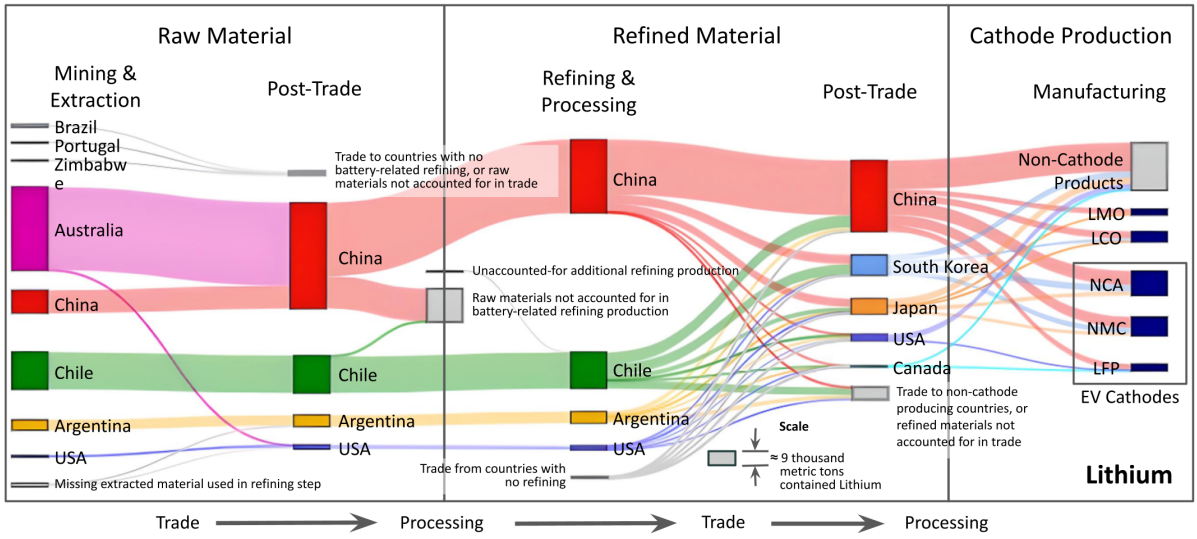


Figure 2.1. A Sankey diagram for global flows of lithium that available data suggest are involved in battery material supply chains.

See Appendix A.3.2 for further details and data sources. Note the scale of production - 82,500 metric tons of lithium were mined and extracted in 2020 (excluding missing production). Note that LMO (lithium manganese oxide) and LCO (lithium cobalt oxide) cathodes are currently rarely used in electric vehicle batteries.

supply chain modeled in this paper presented in Appendix A.1.1.

Figure 1.2 shows that most lithium used in battery production in 2020 was extracted in Australia (49%), Chile (27%), China (16%), Argentina (7%), and the US (1%), where values are rounded to the nearest percentage point. These countries generally processed the lithium they extracted, with the exception of Australia - 99% of which was shipped to China and the remaining 1% of which was shipped to the US. The primary reason for this is because Australia and China are the main producers of spodumene rock, which is refined to lithium products, while the other countries produce brines that must be refined into lithium products in situ [28, 35]. About 31% of the raw Li acquired by China and 3% of Li from Chile appears to have been used or refined in non-battery contexts. The remainder was processed into battery-grade lithium by China (59%), Chile (29%), Argentina (9%), and the US (3%). (Chilean imports and exports exceeded the reported production, which is why Chile had both raw materials not accounted for in battery-related refining and missing additional refining production.) Refined lithium was then traded to China (55%), South Korea (16%), Japan (12%), the US (5%), Canada (1%) and other countries that do not produce cathode materials (TNPC, 12%). Not all of the lithium is used in battery cathode production - 41% (China), 44% (South Korea), 29% (Japan), 95% (United States), and 55% (Canada) of each country's

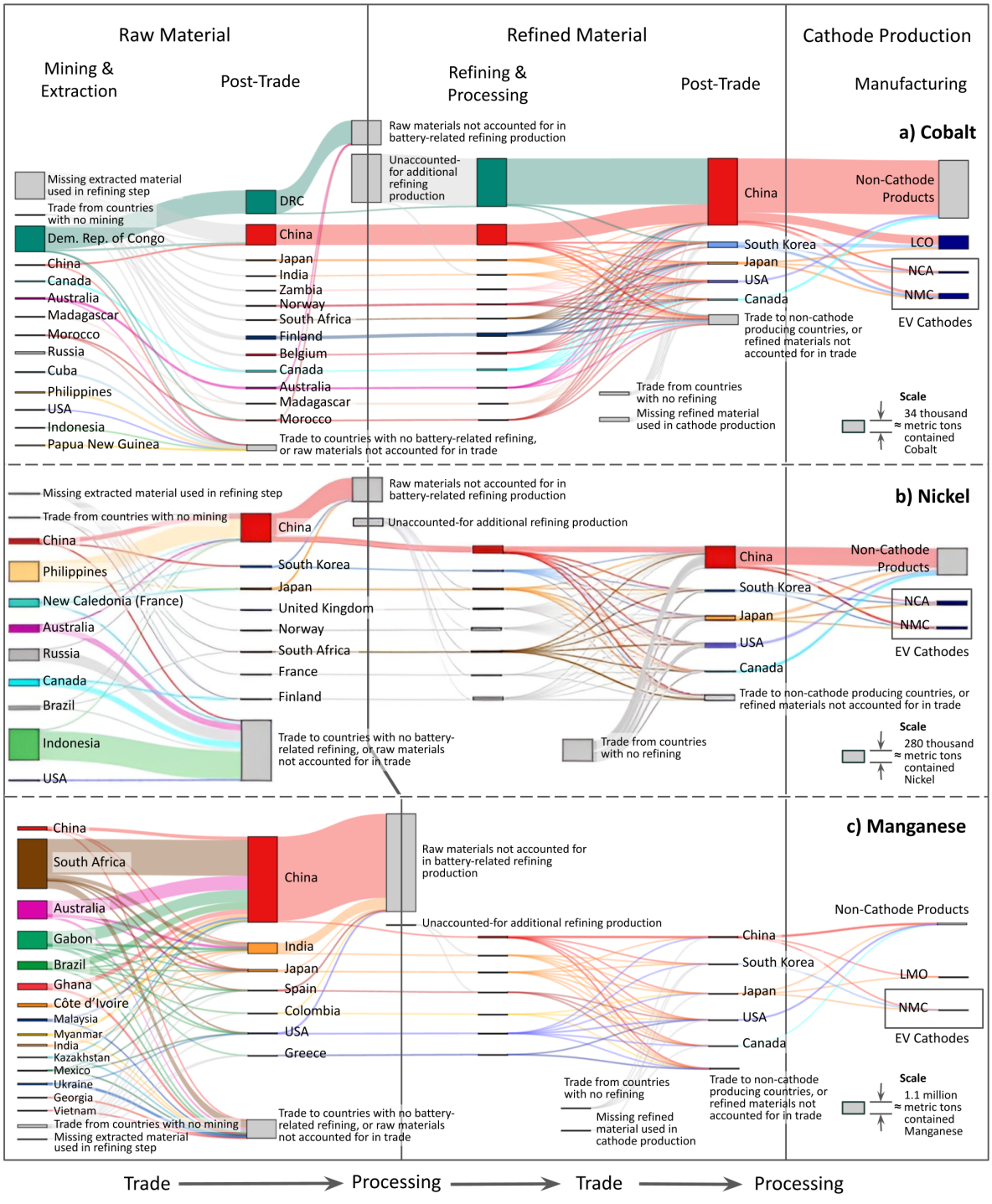


Figure 2.2. Sankey diagrams for global flows of (a) cobalt, (b) nickel, and (c) manganese that available data suggest are involved in battery material supply chains. See Appendix A.3.2 for further details and data sources. Note that the scale of production differs - 142,000 metric tons of cobalt were mined and extracted in 2020 (excluding missing data), compared to 2.5 million metric tons of nickel and 5.67 million metric tons of manganese. Note that LMO (lithium manganese oxide) and LCO (lithium cobalt oxide) cathodes are currently rarely used in electric vehicle batteries.

production were used in non-battery cathode related products, respectively. Finally, LFP cathode material was produced with refined lithium from China (90%), the US (4%) and Canada (6%), while NMC cathode material is produced using refined Li from China (57%), South Korea (27%), and Japan (16%).

The other Sankey diagrams in Figure 2.2 can be interpreted similarly. In general, we see that a large amount of cobalt, nickel, and manganese is used for products other than batteries. It is particularly notable that a large amount of nickel and manganese are transformed into products that are not considered battery related, so even though large producers such as Indonesia, Australia, and Gabon exist, the amount of influence they have on battery material supply chains may be limited. It is also important to note that the amount of uncertainty present in these figures is relatively high, particularly for Cobalt and Nickel, as there is a significant amount of uncertainty in supply in both the raw and refined material steps. While we capture the effect of this uncertainty in our vulnerability index calculation, we note that gathering better data along the supply chain is important to continue to refine and inform our understanding of vulnerability and risk.

2.4.2 Measuring a vulnerability index

Based on these flows, we can quantify measures of supply disruption vulnerability for a focal country or focal set of countries. In this analysis, we define an index of vulnerability to supply disruption from a country i as the total percentage of the end product (in this case, battery cathode material) produced with materials that are either manufactured in or traded from country i at any step of the supply chain. We note that uncertainty in this measurement exists due to:

1. unobserved intranational material flows: the source data report aggregate production within each country and aggregate trade between pairs of countries without observing which materials produced or acquired by a country follow which production and export paths.

2. unobserved international trade: data that should exist to satisfy mass balances when producing battery materials but is not present in the trade data.

3. indirect trade: trade from countries without known processing capability, such as intermediary-based trade used to avoid tariffs [39].

Table 2.1 summarizes how we manage these three sources of uncertainty, defining a base case using proportionality assumptions common in the literature and bounding uncertainty

Table 2.1. Approaches to bounding uncertainty of intranational and international material trade flows, for assessing the portion of materials that are processed by or exported from a particular focal country.

Case	Unobserved Intranational Material Flows	Unobserved International Trade and Indirect Trade
Base Case: Proportional flows	Each country's inputs assigned *proportionally* to its outputs	Also proportionally distributed
Optimistic Case: Minimum portion involving focal country	Each country's inputs are assigned to its outputs using network flow optimization to *minimize* the portion of materials used for the focal battery chemistry that include materials processed or exported by the focal country.	Indirect trade and unobserved trade not assigned to focal country
Pessimistic Case A: Maximum portion involving focal country, observed direct trade only	Each country's inputs are assigned to its outputs using network flow optimization to *maximize* the portion of materials used for the focal battery chemistry that include materials processed or exported by the focal country.	Indirect trade and unobserved trade not assigned to focal country
Pessimistic Case B: Maximum portion involving focal country, all (observed and unobserved) trade	Each country's inputs are assigned to its outputs using network flow optimization to *maximize* the portion of materials used for the focal battery chemistry that include materials processed or exported by the focal country.	Indirect trade and unobserved trade assumed to originate from the focal country

with optimistic and pessimistic cases. These uncertainties also only apply in the context of trade data, as we assume the production data to be accurate.

Our vulnerability index aims to estimate the potential for a disruption in one country (or set of countries) to affect the overall supply of battery cathodes given current supply chain flows. As battery production grows and supply chains shift, vulnerability indices can be updated to reflect changing interdependencies. Our vulnerability index does not capture all possible factors that affect critical material vulnerabilities. For example, concentration of raw material reserves in countries may affect the long term interdependencies in ways that are not reflected by current material flows [40, 41].

As an illustrative example, we present an analysis of the global disruption vulnerability index of LFP and NMC cathode production for China, the country with the most extensive involvement in battery material supply chains (as described in Figure 1.2). This analysis is performed for other countries and regions in Appendix Figure A.7 and Appendix Figure A.8. Summaries for each case are presented in Figure 2.3, with the contribution of each step of the supply chain to the vulnerability index shown as a proportion of the end cathode material affected.

The most simple method to measure a vulnerability index, in line with ideas from the supply disruption propagation and input-output literature, is to calculate a point estimate proportional case. To do this, we only take into account the relative proportion share of imports and domestic production for determining the relationship of each country with the upstream step. Results for the proportional LFP and NMC analyses are summarized in Figure 2.3a,e. For LFP, 90% of cathode production takes place in China, and the US and Canada make up the remaining 10%, suggesting a vulnerability index of at least 90%.

We bound uncertainty in the vulnerability index in Figure 2.3b,c,f,g using network flow optimization to minimize or maximize the portion of the supply chain that could flow through China, given uncertainty about which imports for each country map to which paths. Using only the trade data that is present and accounting for the first type of uncertainty (unobserved intranational flows), we estimate minimum and maximum bounds, as for example we do not necessarily know where the observed production of LFP in China sources lithium from. This corresponds to corresponding to an optimistic case and pessimistic case A. With our base case proportional flow estimates [optimistic estimate to pessimistic case A estimate], we estimate that 17% [0% to 33%] of refined Li imports to the US and Canada are refined in China, increasing the vulnerability index by +2 [+0 to +3] percentage points to a total of 92% [90% to 93%] of cathode material production involving China, depending on which US and Canada imports are used for LFP cathode production. However, it is important to recognize that this scenario, while pessimistic, is not a guaranteed upper bound, as we can have uncertainty due to non-observed and indirect trade that could be attributed to production in China (uncertainties 2 and 3). Subsequently, we include a second, more extreme upper bound (Pessimistic Case B) by simultaneously assuming that all indirect trade and missing materials originate from China, resulting in the vulnerability index for LFP increasing by +7 percentage points between the two pessimistic cases to 100%, as seen in Figure 2.3d. These results suggest that LFP battery material supply chains are highly vulnerable to a disruption in China, as even in the most optimistic case the vulnerability index is over 90%. While not noted on the figure, further analysis indicates that even if all cathode production were moved out of China, an estimated 71% of LFP cathode material would use Li inputs

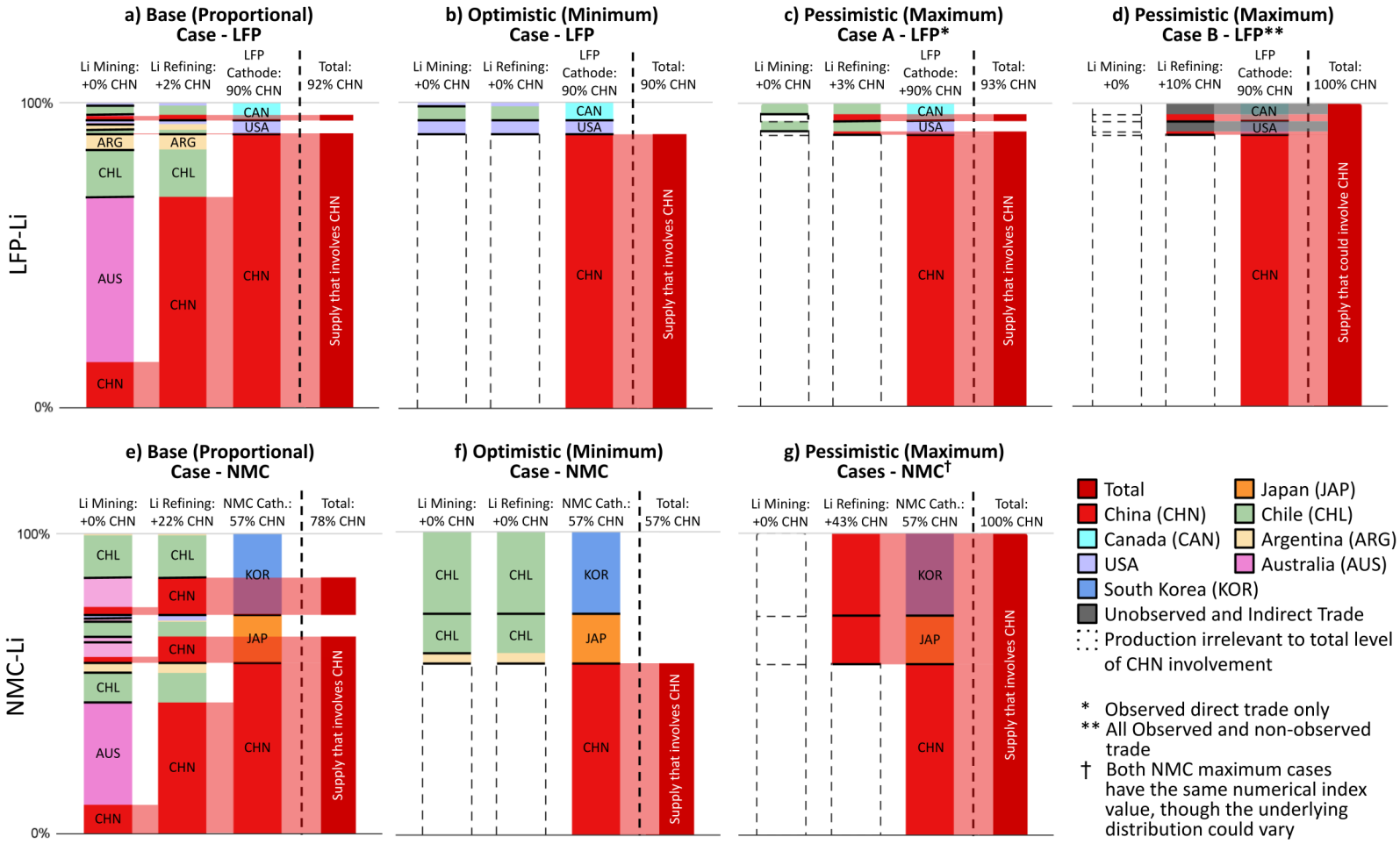


Figure 2.3. Visualization of a vulnerability index for global LFP (Lithium Iron Phosphate [a-d]) and NMC (Lithium Nickel Manganese Cobalt [e-g]) cathode supply, for a lithium supply chain disruption in China.

Table 2.1 defines the four sensitivity cases. The amount of overall vulnerability at the cathode manufacturing step, added at each upstream step, is noted with a plus (+) sign. The solid horizontal black lines are visual aids to indicate separation of countries; semi-transparent red bars represent vulnerability that is propagated downstream. The Li supply chains differ between LFP and NMC cathodes because of differences in the distribution of countries that produce each cathode and the supply chain paths of each. Example solutions for the other NMC materials are in Appendix A.4.1. Indirect trade and unobserved trade do not count towards the measurement of vulnerability in the optimistic case, but are counted towards the pessimistic calculation in the case that includes uncertain data. For the minimum and maximum cases, one possible solution is presented, but other distributions are possible. For all of these cases, we mask the distribution of production and trade upstream of China's supply at each supply chain step, because it does not affect the vulnerability index and may be distributed arbitrarily in the network flow optimization results.

produced or traded from China in the proportional case.

For NMC, 57% of cathode production takes place in China, and South Korea and Japan make up the remaining 43%, suggesting a vulnerability index of at least 57%. We estimate that 50% [0% to 100%] of refined Li imports to South Korea and Japan are refined in China, increasing the vulnerability index by +21 [+0 to +43] percentage points to a total of 78% [57% to 100%] of cathode material production involving lithium and China, depending on which South Korean and Japanese imports are used for NMC cathode production. Assigning indirect trade and missing data to China does not change this range (i.e. in this case, the missing and indirect trade has no effect), because the index cannot increase beyond 100%. Furthermore, though it is similarly not noted on the figure, we note that even if all cathode production were moved out of China, an estimated 68% of NMC cathode material would use Li inputs produced or traded from China in the proportional case. Example solutions for the other NMC materials, similar to Figure 2.3, are in Appendix A.4.1.

Using this method, we also calculate this disruption vulnerability index for the NMC supply chain for nickel, cobalt, and manganese in the case of China. These results, in addition to the LFP-Li and NMC-Li cases, are plotted in the leftmost set of five bars of Figure 2.4, where bars represent the proportional case, error bars capture the range from the Optimistic Case to Pessimistic Case A, and the × represents the extreme Pessimistic Case B. In the proportional and minimum cases, the vulnerability index is highest for manganese [71% and 80%], while in the maximum cases the vulnerability index is highest for lithium [both 100%]. The numerical values presented in this figure are found in Table 2.2 (rounded to the nearest percent).

We aim to determine an overall vulnerability index for NMC as a material, but we suggest that any combination of the vulnerability indices presented here must be done carefully in order to avoid improper comparisons across minerals and chemistry choices. We assume a Leontief production function for multi-material systems like NMC – namely that the mineral that has the lowest amount of supply (and thus highest vulnerability index) defines the overall vulnerability of the entire material system. We then compute an overall vulnerability index for NMC using the maximum vulnerability index across the four critical materials in each of the four cases presented.

Overall, then, LFP has a China vulnerability index of 92% [90% to 93%] due China's dominance in Li refining and LFP cathode production and NMC has a China vulnerability index of 80% [71% to 100%], due most notably to China's dominance in NMC cathode production, Li refining, and Mn refining. In both cases a shift in cathode production away from China could reduce the vulnerability index to a limited degree, but a large reduction would require shifts in both cathode production and material refining. In particular, because NMC

Table 2.2. Summary of vulnerability index results for LFP and NMC battery chemistries for the case of China.

Bold indicates that this material is the limiting factor for calculating vulnerability (i.e., the material with the highest vulnerability) for each chemistry. In Appendix A.1.2, we discuss phosphorus as a potential additional critical material for LFP.

	Supply Chain	**Optimistic Case** (Minimum)	**Base Case** (Proportional)	**Pessimistic Case A** (Known trade only)	**Pessimistic Case B** (Extreme bound)
LFP	Li	**90%**	**92%**	**93%**	**100%**
	Overall	90%	92%	93%	**100%**
NMC	Li	57%	78%	**100%**	**100%**
	Ni	57%	58%	59%	**100%**
	Co	57%	70%	70%	**100%**
	Mn	**71%**	**80%**	76%	94%
	Overall	71%	80%	**100%**	**100%**

has four critical materials with high China vulnerability indices, shifts across the supply chains of all four materials would be needed to achieve large reductions in the vulnerability index. For both cases, the China vulnerability index is higher than what production concentration metrics alone would suggest. In Appendix A.1.2, we discuss how phosphorus might affect this comparison if it were to be considered a critical material.

Figure 2.4 also summarizes vulnerability indices for Russia, the Democratic Republic of the Congo (DRC), and South Africa. With proportionality assumptions, the vulnerability index is high for cobalt in the DRC and high for manganese in South Africa, but the bounding cases show substantial uncertainty – anywhere from 0% to 90% of the NMC supply chain is observed to be vulnerable to disruptions in Russia, the DRC and South Africa. The upper range can approach 100% given uncertainty of indirect trade and unobserved data: for example, the data suggest there is not much observed vulnerability to Russia, but if we were to assume that Russia supplies all of the unobserved nickel and cobalt supply in the battery materials supply chain, the vulnerability index could approach 100%. Of course, when the optimistic and maximally pessimistic vulnerability indices range from 0 to 100, this quantitative measure cannot serve as a precise measure of vulnerability. We suggest that such a calculation indicates that uncertainty is large for this supply chain, given available data, and thus if a policymaker is concerned about this combination of material and focal

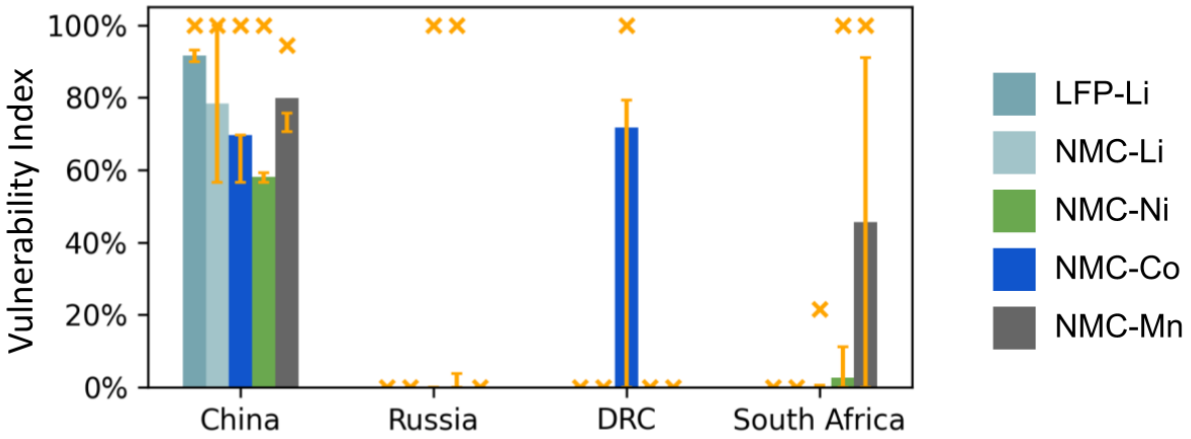


Figure 2.4. Vulnerability index for chemistries LFP and NMC, materials Li, Ni, Co and Mn, and focal countries China, Russia, the Democratic Republic of Congo (DRC), and South Africa.

Bars show results using proportionality assumptions, error bars (in yellow) show range from optimistic (minimum) and pessimistic (maximum) based on uncertainty about intracountry flows, and the yellow × symbols show extreme upper bounds assuming all missing data originate from the focal country. In the case of Russia, a substantial portion of the missing data for nickel and cobalt is known to originate in Russia. [29, 42, 43] We note that the vulnerability index in the proportional case for the NMC-Mn supply chain is greater than the vulnerability index in the pessimistic case that only includes visible trade (Pessimistic Case A). This is because a significant amount of refined manganese used in South Korea and Japan are from non-direct and non-observed trade (Figure 2.4).

country, further investigation and detailed tracking of production and trade data would be warranted.

2.5 Discussion

As vehicle electrification grows worldwide, a variety of decisions will impact firms' and countries' vulnerability to geopolitical and natural disruptions along the supply chain: not only where production happens and the number of firms and sites at each step of the supply chain, but also the chemistry of the battery itself. We mapped the current state of the four most critical battery materials, and then defined a disruption vulnerability index that quantifies, under uncertainty, the portion of LFP and NMC electric vehicle battery cathodes produced by or using materials from a given country. While the index captures disruption potential in current supply chains, not necessarily in future supply chains nor in concentrations of reserves that are not currently used for production, our method could be applied to any current multi-step, multi-material global supply chain. Our results present new perspectives on uncertainty and geopolitical risk in the context of LFP, NMC, and their constituent critical minerals.

Our estimates indicate a high level of potential vulnerability when considering China's influence on the supply chain, with a vulnerability index of 92% [90% to 100%] and 80% [57% to 100%] for LFP and NMC, respectively. Though different countries and administrations may have varying thresholds of concern and risk tolerances, our methodology can help policymakers and researchers identify specific bottlenecks, key relationships, and potential levers for reduction of vulnerability or movement of production location in these interconnected supply chains, such as the key relationship between the Philippines and China regarding nickel or the dependence of South Korea and Japan on refined lithium from China. Using the diagrams, relationships, and analysis generated by this methodology, we hope to spur further research into country-specific vulnerabilities and the effects of specific disruptions on the electric vehicle battery material supply chain.

The uncertainty due to the observability of trade and missing trade data differs in important ways by chemistry, with Figure 2.1, Figure 2.2, and Figure 2.4 shedding light on potential issues of data disclosure and supply chain transparency. The LFP-lithium supply chain has far less uncertainty in terms of unobservable or missing trade data than the supply chains for NMC: much of the production and trade data are observable for the entire lithium supply chain across material extraction, refining, and cathode production. In contrast, the non-lithium inputs into NMC have substantial uncertainty due to unobserved intranational flows, as well as both indirect and unobserved international trade. For example, China ap-

pears to source a substantial amount of refined cobalt from the DRC, but this amount of trade far exceeds the amount of refined cobalt production reported in the DRC. Other studies [29, 43] indicate that unrefined cobalt undergoes initial processing within the DRC and further refining in China, as companies based in China own nearly all of the mines in the DRC [44]. Detailed investigation of our data indicates that nearly all of this traded refined cobalt is categorized as scrap, which also points to potential misclassification of materials as well as high uncertainty in the conversion factors used to calculate material in exports. This finding may help shed light on ongoing transparency and trade tracking issues in the cobalt supply chain, as few material flow analysis studies consider scrap trade in their assessments (see Appendix A.3.3). Furthermore, a substantial portion of battery-grade nickel (about 20% of worldwide supply) is known to be mined and refined in Russia (or neighboring Finland) and used in the European market [45, 46], which is not reflected in the data either. These findings suggest that further emphasis should be placed on understanding how worldwide material production and trade is tracked, particularly around refined and scrap materials. Yet this knowledge of where the data is missing could be considered a further data point on the relative vulnerability when comparing between critical minerals - policymakers may want to choose to promote supply chains where the data is more well-understood and relative uncertainty is minimized.

These results may be unsurprising given the early and consistent support for electric vehicles and novel chemistries in China. Various studies in the literature [47–51] have identified a mix of national science and technology policy directives, government regulations, national and regional subsidies, market incentives and opportunities, and economies of scale (of both labor costs and manufacturing know-how) all contributing to China's currently dominating position in this industry. While lithium ion batteries were initially developed in the United States and commercialized in Japan, battery manufacturers largely started shifting towards China due to state subsidies and directives to invest in new energy technologies [49]. In 2000, the Chinese government initiated the "Going Out Strategy", encouraging companies to expand foreign direct investment for mineral resources in developing countries in Africa and Asia to support a burgeoning domestic materials processing industry underlying the general manufacturing boom in the country, including battery materials like cobalt. In the same year, the national government began massive investment in "New Energy Vehicles" along the entire technology pipeline of research to process commercialization as part of the 10th 5-year plan. Subsidies for electric vehicles started in 2009 that just ended in 2023, totaling in the hundreds of billions of Yuan (tens of billions of USD) [47, 52]. This whole-supply chain approach provides a model for other countries to compare and contrast their domestic policy goals to if they are similarly concerned with building out their electric vehicle manufacturing

and subsequent supply chains.

In the U.S. policy context, the Inflation Reduction Act [18] includes several provisions that encourage firms to change their location of production. A production tax credit equal to 10% of production costs incentivizes firms to domestically manufacture electro-active materials, such as the cathode or anode. In addition, EVs can qualify for two tax credits per vehicle, based on the geography of their supply chains: (1) those that have a minimum amount of critical minerals produced or processed domestically within free trade agreement countries and (2) a minimum amount of battery components manufactured in North America. Our results further suggest that these sourcing requirements may not avoid vulnerabilities due to specific geographies of concern along the entire supply chain (for both requirements). Furthermore, because of the compound nature of vulnerabilities across multiple supply chain stages, reduction of vulnerability at just one across these battery material supply chains is not sufficient.

A possible complementary policy action may be funding research, development, and demonstrations that aim to improve the performance of LFP batteries as well as the lithium extraction and refining processes, and thus reduce reliance on NMC and its compound material risks, as all four critical materials studied are required in order to produce NMC, creating multiple disruption paths. LFP also has an advantage given the relatively small overall physical quantity of materials that is present in the Lithium supply chain (comparing the scales in Figure 2.1 and Figure 2.2) and the fact that uncertainty appears to be the lowest in the Lithium supply chains, as described previously. Immense recoverable deposits of lithium are being rapidly discovered given the recent interest in the material [53–56]. In particular, the U.S. has substantial potential mining capacity (the Thacker Pass mine in Nevada is the third largest individual lithium resource in the world [57]), as well as existing capabilities in both mining and refining (Figure 2.2). While choosing LFP may involve other technological and economic tradeoffs, a shift toward LFP may represent an opportunity to reduce disruption vulnerability, if lithium refining operations and LFP cathode production operations in particular are diversified away from the currently high concentration in China.

While the results presented have primarily focused on China as the single country with the largest influence on battery material supply chains and this vulnerability index, trade blocs or sets of politically aligned countries (e.g. a South American lithium production bloc) appear to also be able to have various levels of dominance on specific supply chains, as seen in Appendix A.4.2. The effect of other countries on multiple stages may not currently be nearly as pronounced, but the methodology generalizes our ability to understand such phenomena and the relative impact of any country or set of countries. Reframing vulnerability in a different context, this methodology could also allow policymakers to understand disruption

vulnerability with respect to any set of countries and assess potential of alternative policy actions to affect overall supply chain vulnerability. Our results thus suggest that because individual countries and trade blocs can dominate multiple stages of the supply chain, reducing battery supply chain risks (or strengthening networks of trade between friendly nations) requires understanding and addressing not just immediate but secondary and tertiary vulnerabilities.

2.6 Methods

A full description of the methodologies and data used to develop this analysis are described in Appendix A.3. We simplify a fully circular supply chain model in our model, which is presented in Appendix A.1.1.

We first build on the methodologies of the Material Flow Analysis literature [27–32, 43, 58–63] to map and characterize the production and flow of materials in battery supply chains in a "data aggregation" step. To do so, we begin with the amount of material known to be produced in each of the mining, refining, and cathode production steps. Each country can produce, import, and export multiple materials at each step of the supply chain, so we convert production and trade statistics into quantities of contained materials. We choose to use reported import data quantities as it tends to be more complete and accurate in comparison to reported export data [29, 30, 64, 65]. For example, we convert trade quantities in nickel ores and concentrates, nickel mattes, nickel sulfates, nickel waste and scrap, etc. into units of contained nickel, as prior literature [2, 34, 63, 66–68] indicates that some or all of these trade flows may contain nickel material that ultimately is used in battery manufacturing. See Appendix A.3 for further discussion on the chosen data sources and trade codes, and corresponding literature. We then balance the amount of each contained material between the mining step and refining step, and between the refining step and the cathode manufacturing step, incorporating trade at each stage.

We then use these flows to calculate the total vulnerability index for cathode material supply for a focal country or set of countries. We estimate our base case assuming the distribution of inputs for each country at each stage flows proportionally to the outputs, in line with ideas of input-output models [69] or the supply chain disruption propagation literature [27]. To bound the vulnerability index, given uncertainty about intracountry flows, we model the supply chain as a network flow problem and optimize the uncertain intracountry flows to minimize or maximize the portion of cathode production using materials that pass through the focal country [70]. To map our network structure to the network flow problem, connections between nodes are assigned capacity values equal to flows calculated through our

data aggregation step, and all mining and mineral extraction is connected to a supersource node. This supersource node is also connected (with infinite capacity) to each country of focus' node(s) along the steps of the supply chain, so that the maximal amount of flow from this supersource to the terminal node can be measured. In the maximal case with uncertain data, the nodes representing uncertain sources of materials are also connected to the supersource. In the minimal case, all connections involving any countries of focus are disconnected from the network, and the maximal flow is calculated. The difference between this quantity and the amount of material demanded is the minimum vulnerability index for the countries of focus. A detailed formulation is provided in Appendix A.3.1.

We note that this methodology is generally flexible and can be augmented with further data as well as applied to any multi-step, multi-material global supply chain given sufficient production and trade data. For example, this study only includes materials scrap trade codes as some measure of circularity. While current trade of end-of-life batteries and battery material supplies is minimal, if trade codes are established for such materials, or trade of such materials could be sufficiently estimated from future trade codes, this methodology could easily incorporate that additional data stream. Furthermore, we believe this methodology would be readily applicable in other supply chains, such as battery chemistries that are not yet on the market, rare earth minerals for motors, turbines, electrolyzers, solar panels, and the like, given sufficient production and trade data.

2.7 Data Availability

Source data are provided with this paper; we only use publicly available production and trade data for this analysis, from the U.S. Geological Service's Mineral Commodity Survey [1], Sun et al. 2021 [2], International Fertilizer Association [71], and IntraCen's TradeMap [65], which is described in detail in Appendix A.3.2 (Data Sources). The source and processed data generated in this study have been deposited in a public GitHub repository with no accession code (https://doi.org/10.5281/zenodo.10607313) [72].

Chapter 3

US industrial policy may reduce electric vehicle battery supply chain vulnerabilities and influence technology choice

Modified from a paper published in *Nature Energy* [73].

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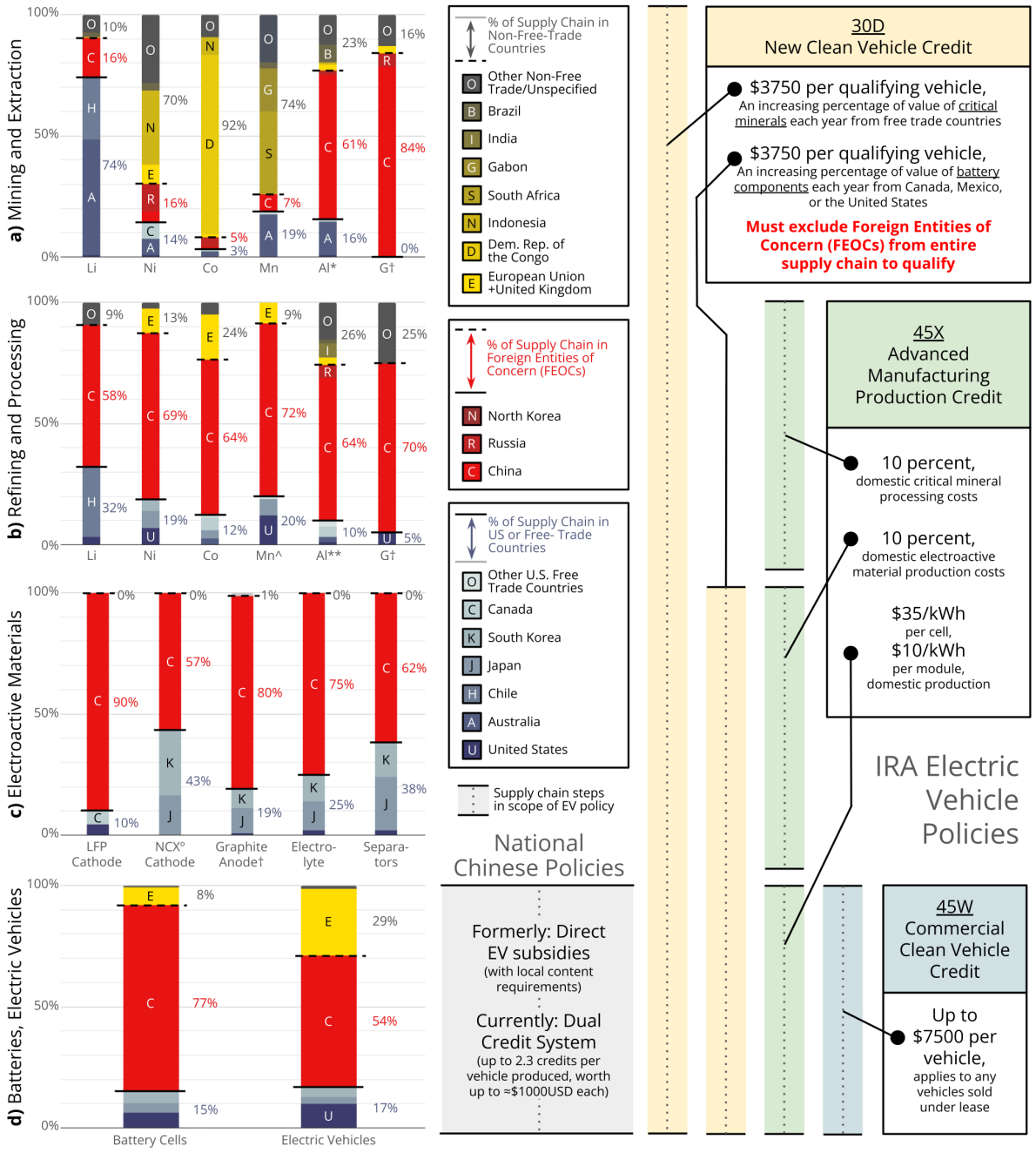
3.1 Abstract

We analyse US Inflation Reduction Act (IRA) incentives for electric vehicle battery technology and supply chain decisions. We find that the total value of available credits exceeds estimated battery production costs, but qualifying for all available credits is difficult. IRA cell and module credits alone bring estimated US battery production costs in line with China. In contrast, IRA material extraction and processing credits are modest. IRA's end-user purchase credits are restricted to electric vehicles whose battery supply chains exclude foreign entities of concern, including China. This incentivizes diversification of the entire supply chain, but leasing avoids these restrictions. Lithium iron phosphate batteries have potential to more easily reduce supply chain vulnerabilities and qualify for incentives, but they have smaller total available incentives than nickel/cobalt-based batteries. Overall, the IRA primarily incentivizes downstream battery manufacturing diversification, whereas upstream supply implications depend on automaker responses to foreign entities of concern and leasing rules.

3.2 Introduction

Industrial policy has garnered considerable attention in both academic and policy circles, particularly with the passage of the Infrastructure Investment and Jobs Act in November 2021 [23] and the Inflation Reduction Act (IRA) [18] and the CHIPS and Science Act in August 2022 [74] in the United States. These acts were designed to explicitly incentivize certain industries in terms of increasing adoption of the products created by these industries; changing the location of production to the United States and its allied countries; and improving the domestic capability to manufacture and innovate [75–77]. In particular, the IRA provided several provisions to promote the adoption of electric vehicles (EVs): vehicles that wholly or partly rely on batteries charged from an external source, namely battery electric vehicles and plug-in hybrid electric vehicles. These policy interventions intend to 'on-shore' (bring back supply chains within domestic borders) or 'friend-shore' (shift supply chains to allied countries), with additional co-benefits around emissions reductions and job creation [18, 77]. Previous studies of industrial policy do not necessarily provide a strong indication of the effect of the IRA policies on the EV supply chain. Analyses of technology-forcing vehicle emissions policies show mixed results in achieving intended objectives, such as emissions reduction and reduced mortality risk [78–81]. Local content requirements, a key restriction in qualifying for IRA EV incentives, have been studied in automotive and renewable energy contexts but have mixed support for their cost efficacy and economic and innovation spillover effects [82, 83]. Analyses of specific IRA policies also have identified mixed outcomes in terms of the size of incentives relative to costs or achieving goals. These studies span many contexts: clean hydrogen and other fuels [84–88], carbon capture and sequestration [89], solar panels [90] and wind power [90, 91], but none encompass the EV context, making it unclear if conclusions from other sectors apply. Studies of the IRA regarding EVs found notable increases in adoption over baseline projections [92, 93], in line with studies of incentives for EVs in general [94, 95], but did not comment on production costs or battery technology choices. Only one peer-reviewed study [90] found that IRA credits would probably make US manufacturing of solar and wind components cheaper than import prices; the impact of IRA policies on the comparative production costs across countries is unclear. Another study [19] has evaluated the choice between battery chemistries, finding that some chemistries will more easily qualify for IRA credits than others, but did not consider impacts on production cost or location. Further discussions of industrial policy writ large and specific analyses of the IRA are found in Appendix B.1.

Other countries, particularly China, have also established incentives for their EV industries and achieve objectives similar to the IRA. Consumer and supplier subsidies, favourable regulations and protectionist policies were implemented at national, regional and even local



Li: Lithium, Ni: Nickel, Co: Cobalt, Mn: Manganese (^Electrolytic Manganese Dioxide only), Al: Aluminum (*Alumina only, **Aluminum from smelting), G: Graphite (†Synthetic and Natural Graphite combined), °NCX represents NCA and NMC combined. Numbers may not sum to 100% due to rounding.

Figure 3.1. The geographical distribution of national production for each EV battery supply chain stage, compared against the value and scope of national-level U.S. (Inflation Reduction Act) and Chinese incentives

Each panel (a) mining and extraction, b) refining and processing, c) electroactive materials, and d) battery and electric vehicle manufacturing) describes the geographical distribution of production of each material or product in each step of the battery supply chain gathered from various sources [2, 7, 96–98], see Supplementary Data 0 for more information. The value of each of the policies is described, and the scope of these policies with regards the steps of the supply chain they target are shown using the colored vertical bars. For more information on the IRA's and China's EV policies, respectively, see Appendix B.3 and Appendix B.2.

levels [48, 99]; China is now the most prominent country within the EV supply chain [7, 17]. Whereas China previously had a direct consumer subsidy for purchasing so-called 'New Energy Vehicles', the supply-side Dual-Credit System is now the national subsidy mechanism. Carmakers gain credits for reducing average fuel consumption and producing EVs and lose credits when failing to do so; EVs currently garner up to 2.3 credits per vehicle produced [100]. As credits can only be traded by firms with a positive surplus to others with negative credits, with emissions and production criteria tightening, credits are valuable: Tesla reportedly sold credits at US$460 each in April 2021, and prices were speculated to be US$900 per credit in 2022 [101]. Other analyses have projected values of up to US$1,200 per credit [102]. Further detail about China's current and historical policies are in Appendix B.2. Here we contribute to the literature with an analysis of the impact of the IRA on electric vehicle battery manufacturing, battery technology choice and production location, and we compare the production costs and effect of current national-level policies and incentives in both the United States and China.

3.3 Analysis of impacts of IRA regulatory interpretations

In Figure 3.1, we provide an overview of the electric vehicle supply chain and the relevant policy interventions in the IRA and China's national policy, updating and augmenting Figure 1.2. For each of the five stages of the electric vehicle supply chain (mining and extraction, refining and processing, component materials manufacturing, battery manufacturing and vehicle manufacturing), we display a stacked bar chart depicting the geographic distribution of the production. The scope of these policies with regards the steps of the supply chain they target are shown using the colored vertical bars.

As seen in Figure 3.1, there are several relevant EV-related IRA credits. Detailed information can be found in Appendix B.3. The 30D New Clean Vehicle Credits, made up of a critical minerals and a battery components credit, are the most complicated in terms of restrictions and qualification. These credits (worth US$3,750 each) can only be claimed when an electric vehicle is sold to an end user. This credit replaced a previous version of 30D, active from 2010 to 2022, which provided a flat U$7,500 credit for all manufacturers, up to a total of 200,000 electric vehicles sold after 2010 [103]. We group countries in these supply chain steps into three categories, depending on how they allow manufacturers to qualify for these credits: the United States and countries with which it has a free-trade agreement; 'covered nations' deemed as a 'foreign entity of concern' (FEOC), as explicitly defined in the Infrastructure Investment and Jobs Act (Bipartisan Infrastructure Law) [104]; and countries that are neither. To qualify for the critical minerals credit, a minimum percentage (relative

to the total value of all critical minerals within the battery) must only originate from free-trade countries or domestically. This percentage increases over the period that the credit is claimable (Figure 3.2). To claim the components credit, a similarly increasing minimum percentage of the components must be manufactured in the United States, Canada or Mexico. For both credits, no critical minerals or components can originate from a FEOC throughout the entire supply chain [104, 105]. This designation not only applies to production within any of the statute-defined four covered nations (China, Russia, North Korea and Iran) but also firms that are partially owned or held (25% stake or more) by citizens, other firms or the government of those four countries, or headquartered in those four countries, further increasing the difficulty of qualifying for the credits.

We map the possibility space if any particular supply chain step were moved such that the manufacturing would then qualify for the credits. We examine the effect of the FEOC ruling on the ability for automakers to claim the 30D critical minerals credit. As seen in Figure 3.1, some material supply chains and markets are structured in a manner that may be more conducive to avoiding such FEOCs. We thus determine the makeup of each battery by chemistry in terms of weight (in kilograms) and value of critical minerals in Figure 3.2, building off of the methodology presented in Trost and Dunn [19].

Figure 3.2 shows that there are several pathways to claim the credits at the 80% critical mineral value level. However, each supply chain has varying levels of dependence on FEOCs and free-trade partners (Figure 3.1), so certain supply chains may be easier to move than others. For example, 74% of lithium extraction and 32% of processing currently occurs in free-trade countries, so automakers may find it easier to move that supply chain in comparison to nickel or cobalt, where only 14% and 9% of mining and extraction and 19% and 12% of refining occurs in free-trade-agreement countries, respectively. It appears that LFP may have a strong advantage over the other two chemistries: the long-term 80% requirement can be met by only shifting the lithium and aluminium supply chains, while also not requiring nickel, cobalt or manganese that could be limited by the FEOC ruling.

The 45X Advanced Manufacturing Credit and 45W Commercial Clean Vehicle Credit are simpler in comparison. Under current interpretations of IRA [107], the 45X credits give a 10% discount to production costs for critical minerals processing and electroactive materials manufacturing within the United States, disregarding the cost of input materials. A further US$35 per kWh per qualifying battery cell and US$10 per kWh per battery module can be claimed by domestic battery manufacturers in the United States. The 45W credit allows any electric vehicle that is purchased for commercial use to receive up to US$7,500 in incentives, depending on the 'excess' in purchase price of the electric vehicle over a comparable internal combustion counterpart, regardless of any critical mineral or battery component requirement

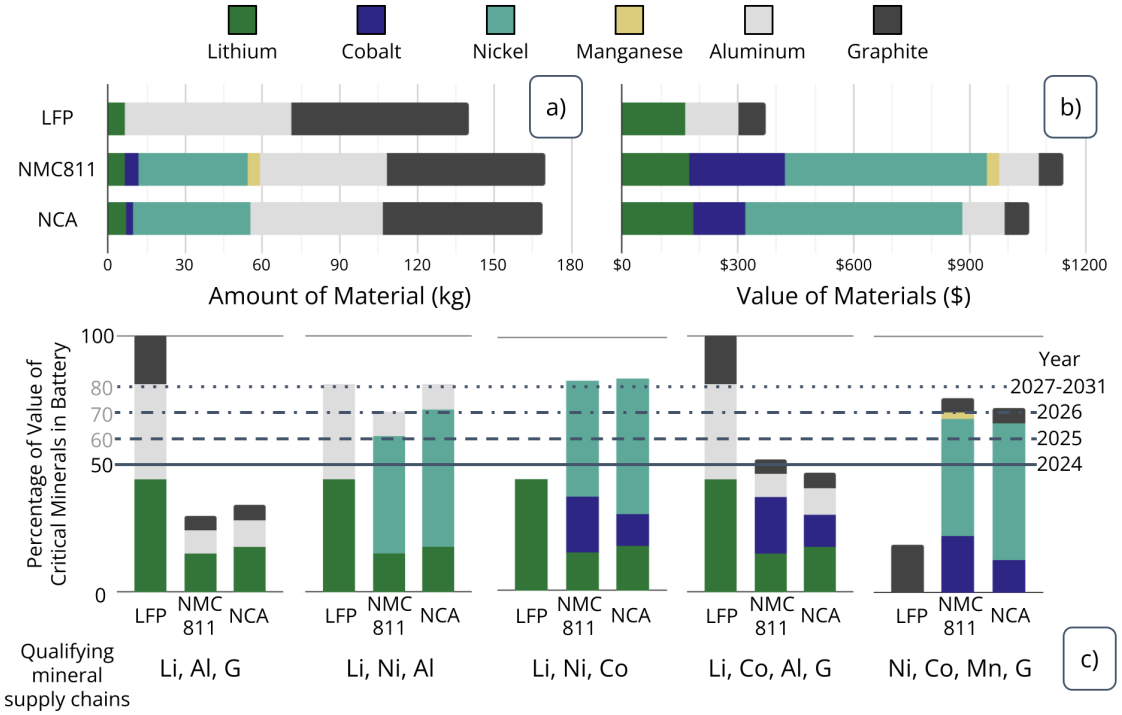


Figure 3.2. Comparison of critical minerals in a 70 kWh battery, by (a) mass and (b) value, and (c) the value–percentage combinations of their supply chains, to meet requirements for the 30D critical mineral credit of the Inflation Reduction Act.

Panels (a) and (b) show the relative mass of critical minerals, and value of those critical minerals, in a 70 kilowatt-hour battery, respectively, across common battery chemistries. Panel (c) describes the amount of critical minerals (as measured relative to the total value of all critical minerals within the battery) required to qualify for the 30D credit. The horizontal lines with increasing amounts of dashes represent that increasing requirement, i.e. 50% of the value of critical minerals must be sourced in the United States or from countries that have a free trade agreement in 2024, increasing up to 80% by 2027. None (0%) of the materials may be sourced from 'Foreign Entities of Concern' (FEOCs) starting in 2025. Critical mineral prices and quantities were gathered from literature and other models [19, 106], see Supplementary Data 2 and Supplementary Data 3 for more information.

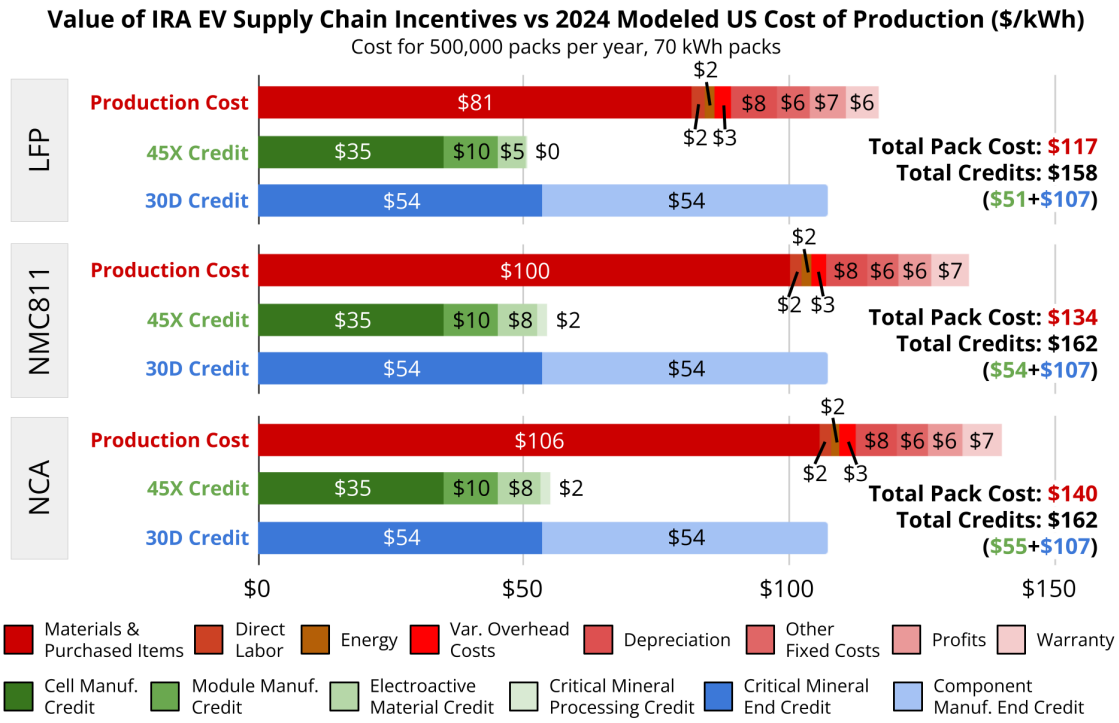


Figure 3.3. Current (2024) modeled production costs for the Lithium Iron Phosphate (LFP), Nickel Manganese Cobalt (NMC), and Nickel Cobalt Aluminum (NCA) battery chemistries, in comparison to the relative value of each Inflation Reduction Act (IRA) credit.

These incentives are designed to simultaneously spur manufacturing, incentivize supply chain shifts, and increase electric vehicle adoption [75–77], and thus the incentives are spread across multiple entities in the supply chain and the end consumer. While the total value of IRA credits exceeds current modeled costs of EV battery production, qualifying for all relevant credits is difficult for EV manufacturers, who also face other challenges in comparison to traditional vehicle manufacturing beyond battery manufacturing. Detailed production costs at other scales (e.g. 40 kWh and 100 kWh; 50,000 and 100,000 packs) are presented in Appendix B.4 and Supplementary Data 5.

and the income and price cap limits. As Treasury has interpreted this to include leases of electric vehicles, this has created a loophole [104] for qualifying for equivalent credit value offered in 30D without the FEOC restrictions. Electric vehicle leasing rates have increased from about 10% in early 2023 to about 25% today [108, 109]. As the lease is not considered a purchase by the end user, a vehicle cannot qualify for both the 30D New Clean Vehicle credit and 45W Commercial Clean Vehicle credit.

3.4 Modeled production costs

With this empirical context, we model battery production costs and the relative impact of each of these policies in comparison to these costs. We use the most recent–as of the date of publication (August 2023)–version of the BatPaC model developed by Argonne National Lab [110]. This model estimates the cost of manufacturing lithium-ion batteries at high volumes (500,000 units annually) after production has reached a mature state of development. Discussion of the use of this model and baseline, and other modelling choices, is found in the Methods section.

Other than a battery capacity of 70 kWh (compared to the standard value of 100 kWh in BatPaC), we use the default values for materials costs, production volume, yield rate, capital and operating costs to generate the production cost results in Figure 3.3 (Supplementary Data 4). The 10% critical minerals production credit and the 10% electroactive materials production credit are applied as discounts to input material prices, whereas the latter four credits (the cell and module manufacturing credits and the two 30D end-user credits) are calculated in terms of US dollars per kilowatt-hour.

We compare the impact of the IRA incentives on the supply chain in comparison to a China-based supply chain. In China, we assume the baseline battery manufacturing scenario production to be the same as the United States, with the exception of 20% lower capital expenditures (for example, machine costs) and materials prices and 75% lower labour costs, based on data gathered from market research [111] (Scenario 0). We then compare the total value of all IRA credits (Scenario 1), a case where the foreign entity of concern (FEOC) ruling prevents the 30D credits (US$7,500 in total) from being claimed (Scenario 2) and the relative effect of the full leasing credit by itself (that is, excluding all other credits) (Scenario 3). These results are compared with modelled production in China under its current Dual Credit system. Each vehicle is assumed to generate the maximum 2.3 credits, with each credit valued at a conservative estimate of US$1,000, as described in the introduction. These results are compared in Table 3.

Table 3.1. Battery Production Costs Less Value of Electric Vehicle Policy Incentive Credits across Chemistry and Production Location Choices.

Current (2024) production costs modeled for 70 kWh batteries; values are in 2024 USD per kWh. For the leasing scenario, the maximal value of the credit is used. Bolded and underlined text indicates the lowest cost option in each scenario; text that is only bolded represents the second lowest cost option.

Scenario	**0. Baseline** No Policy	**1. Full IRA** Mfr. Qualifies for All IRA Credits	**2. FEOC** Mfr. Supply Chain Includes FEOC	**3. Leasing** Relative impact of leasing credit only
US Policy	None	IRA	IRA	IRA
Chinese Policy	None	Dual Credit	Dual Credit	Dual Credit
US-LFP	**<u>$117</u>**	**<u>-$41</u>**	**<u>$66</u>**	**<u>$10</u>**
US-NMC811	$134	**-$28**	$79	**$27**
US-NCA	$140	-$22	$85	$33
China-LFP	**<u>$103</u>**	$70	**$70**	$70
China-NMC811	$119	$86	$86	$86
China-NCA	$124	$91	$91	$91

The results indicate that LFP is the least-cost chemistry regardless of production location, though the impact of the current policies causes production in the United States to be favoured in comparison to a scenario with no policies. Where a firm can claim both 30D credits or the 45W leasing credit, the next best option is to manufacture in the United States with the NMC chemistry, but the nickel-based chemistries are not favoured if neither of the 30D credits are included due to FEOC limitations. These results hold in sensitivity analyses involving land costs, labour rates and the cost of energy (Appendix B.4). These results are also largely consistent across battery sizes, as seen in Table 4, though for smaller batteries under Scenario 2, the production location decision flips back to favouring production in China, and for larger batteries, LFP's lower energy density can make packaging impractical, depending on the vehicle design. Whereas only LFP is presented in Table 4, this result is consistent across the three chemistries (Appendix B.4). These modelled results—both the baseline scenarios and those with the included credits—are largely consistent with reported prices in the United States and China [112–114] though we note battery prices are not always cleanly related to production cost [115].

Overall, we find that the full value of IRA credits exceeds battery production cost. We estimate that the minimum subsidy needed to bring subsidized US production costs in line with subsidized Chinese production costs is about US$50 per kWh for a 70-kWh battery, and total value of IRA credits is more than three times that value, though we emphasize that it is difficult to qualify for all of the IRA credits and that large, temporary subsidies are sometimes needed to overcome frictions in the system.

Table 3.2. Lithium Iron Phosphate (LFP) Battery Production Costs across Battery Size, Location, and Policy Impact.

All values are in costs (2024 USD) per kWh. For the leasing scenario, the maximal value of the credit is used. Bolded text indicates the lower cost choice of production geography for each battery size in each scenario. *100 kWh LFP packs currently only fit in larger form factor vehicles (e.g. trucks and large SUVs) due to their lower energy density. See Appendix Table B.12 for the full table with all scenarios, chemistries, battery sizes, and locations.

Battery Size	**Scenario**	**0. Baseline** No Policy	**1. Full IRA** Mfr. Qualifies for All IRA Credits	**2. FEOC** Mfr. Supply Chain Includes FEOC	**3. Leasing** Relative impact of leasing credit only
	US Policy	None	IRA	IRA	IRA
	Chinese Policy	None	Dual Credit	Dual Credit	Dual Credit
100 kWh	US*	$106	**-$20**	**$55**	**$31**
	China*	**$93**	$70	$70	$70
85 kWh	US	$111	**-$28**	**$60**	**$23**
	China	**$97**	$70	$70	$70
70 kWh	US	$117	**-$41**	**$66**	**$10**
	China	**$103**	$70	$70	$70
55 kWh	US	$127	**-$61**	$76	**-$10**
	China	**$111**	$70	**$70**	$70
40 kWh	US	$148	**-$91**	$97	**-$40**
	China	**$131**	$73	**$73**	$73

3.5 Discussion and conclusions

Prior studies have found that previous electric vehicle incentives were sufficient to promote adoption of electric vehicles beyond an unregulated market equilibrium. We find that the full value of available IRA credits exceeds battery production cost, though some IRA credits may be difficult to qualify for. The IRA's 45X advanced manufacturing credits for domestic cell and module production (totalling US$45 per kWh) are alone nearly sufficient to cover the gap between current production costs in the United States and subsidized production costs in China. However, historical Chinese subsidies in its Phase 1 and Phase 2 periods have far exceeded today's IRA subsidies, controlling for inflation (Appendix Table B.3), so a Chinese industrial policy response could shift incentives again. Indeed, analyses of China in the current dual-credit system indicate that further policy adjustments are required to meet their stated sales and automotive emissions peak goals [109], which may spur further electric vehicle manufacturing subsidy regimes. China could also choose to respond with other policies: between October 2023 and August 2024, China restricted exports of graphite [116] and related critical minerals gallium and germanium [117] and later antimony [118]. These choices provide additional geopolitical incentives to 'on-shore' or 'friend-shore' production of critical materials and materials processing.

We find that the IRA credits focus on the 'downstream' battery and vehicle manufacturing stages of the supply chain (Figure 3.1, Figure 3.3), leaving the comparatively 'upstream' stages of material extraction and refining vulnerable to supply disruption risk [17]. Indeed, most new battery supply chain development in the United States since passage of the IRA has focused on the battery and vehicle manufacturing stages (Figure 3.4). The FEOC-based restrictions on IRA's 30D New Clean Vehicle Credits restrict qualification for IRA credits and create incentives to develop a battery supply chain that excludes China from all stages, but the so-called 'leasing loophole' in IRA's 45W Commercial Clean Vehicle Credit [104, 109] may limit effectiveness. Furthermore, whereas it only takes about 24 months to build a battery plant, the average mine or refining plant can take 5 to 7 years to plan, fund, permit and deploy (plus additional time for prospecting) [121, 122]. These project time horizons do not necessarily explain the lack of upstream projects in Figure 3.4, which includes sites that are announced, under construction or already operating. Rather, because such projects have not yet been announced, even if domestic or friendly natural resources were not relatively limited, EV manufacturers may face challenges in meeting the strict requirements defined in the IRA. As a result, it may be easier for manufacturers to ignore those requirements and instead manufacture in lower-cost locations while focusing on promoting leases for consumers.

When considering technology choices, the IRA incentives are slightly larger for NMC and NCA chemistries compared with LFP (Figure 3.3); however, LFP has fewer supply chains

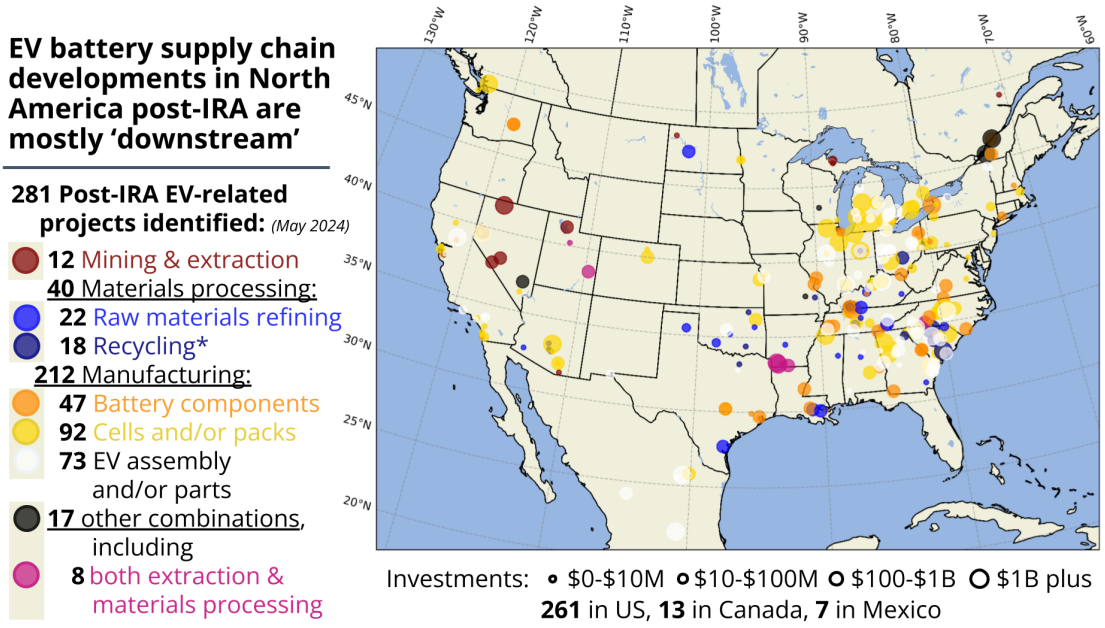


Figure 3.4. The IRA appeared to spur the development of the EV battery supply chain in North America, but mostly in the battery and vehicle manufacturing segments. Data aggregated as of May 2024 [119, 120]. *3 projects were identified as having both recycling of used materials and refining of virgin materials. See Supplementary Data 9 for the full dataset, which is not necessarily fully comprehensive, especially for projects in Mexico and Canada. Not pictured: one graphite project in Alaska, a lithium project in Canada, and a battery pack and an EV manufacturing plant in Mexico. Projects without known investment amounts were added to the "$0-$10M" size. Projects without an announced location were not included on the map or in the counts.

that are currently dependent on China and Russia [17] (Figure 3.1), making it probably easier to avoid FEOC restrictions. Thus, LFP may more easily qualify for the 30D New Clean Vehicle Credits (Figure 3.2). LFP has key advantages in production cost (Figure 3.3), life-cycle emissions12 and resilience to fast charging [123], though LFP can also be more sensitive to cold weather [124], and lower energy density can limit application in some long-range vehicles [125]. Nearly all high-energy-density LFP technology is currently being developed by Chinese companies such as BYD and CATL. However, these companies have been somewhat open to licensing this technology [126], and licensing is currently not subject to FEOC guidelines [127], which may allow opportunities for technological catch-up. As a result, we suggest that on balance, LFP chemistries may be more favoured in the current policy environment.

The IRA's electric vehicle manufacturing incentives provide lessons for nations looking to develop and respond to clean energy manufacturing industrial policy. These findings can inform understanding on how incentive design can meet national goals. In this case, direct subsidies appear to fulfil the goal of onshoring manufacturing while reducing the cost of electric vehicles and diversifying the EV battery supply chain, though the design of these incentives focus on diversification of battery and vehicle manufacturing stages. The success of reducing supply disruption vulnerabilities across the entire supply chain may depend on how automakers navigate FEOC regulations and the vehicle leasing loophole, in the face of other nations' responses to these incentives, particularly China and its evolving industrial strategy.

3.6 Methods

3.6.1 Methodological approach

The analyses in this article rely on two models: the Supplementary Data file and the BatPaC model developed by Argonne National Laboratory. This analysis was conducted as a snapshot of current (2024) battery manufacturing in comparison to the IRA's set of incentives and current national incentives in China.

3.6.2 Supply chain production geographies

To generate the diagrams in Figure 3.1, we use production data from a number of sources [2, 7, 96–98] and aggregate the production depending on their categorization as a US or US free-trade country, a designated FEOC or neither (Supplementary Data 0).

3.6.3 Measuring the effect of IRA credits on material prices

To calculate the effect of credits on input material prices, we use the weight percentage of critical minerals calculated from chemical formulas (Supplementary Data 1) and battery weights from the Greenhouse Gases, Regulated Emissions and Energy Efficiency in Technologies (GREET) model [106] to calculate the total weight of each critical mineral in the battery in Supplementary Data 3. These weights and relative amounts are then multiplied by material prices sourced from empirical data and literature [19, 110, 111] (as collected and aggregated in Supplementary Data 2) to get the cost for critical minerals per battery and per kilogram of input material. We note that the credits are applied in the context of production cost and not sale price. In the absence of exactly knowing the cost of input materials, we use the default prices found in BatPaC as a starting point. We then apply 45X production credit for critical materials processing, bounding the maximum impact of this credit to 10% of the processed material price. Then, where applicable, the effect of the additional 10% electroactive component manufacturing credit is also applied. We note that this is an overestimate in the impact of these production credits, as the price of the processed materials or components is not equal to the cost of materials processing or component manufacturing. However, because this credit was found to be small compared with the other IRA credits, variation in raw material prices would have a modest impact on overall incentives. Thus, we did not conduct an analysis around the sensitivity of material costs to fluctuations in raw material prices.

3.6.4 Modeling with the BatPaC model

Previous literature has used various models and methodologies to analyse battery production, including within the American and Chinese contexts [49, 128]. We choose to use the BatPaC model developed by Argonne National Laboratory. This model is a standard because of its comprehensiveness and consistent history of updates since its release in 2011 to match industry standards and trajectories, often being used as a source of data for manufacturing parameters (Appendix B.1.6) [128].

The model works by taking many input parameters (for example, pack energy capacity, power rating, battery chemistry, material prices, machine capital costs, yield rates and so on), optimizing a battery design to fit these parameters (including the battery management system and thermal design) and then calculating the cost of manufacturing such a battery and its components. Each input parameter has a default value, which can be altered. The effect of each of these alterations is usually mediated by a scaling power factor approach relative to that default value, an approach that is also adopted by other independent battery

manufacturing models [9, 129].

We use the standard manufacturing system defined in BatPaC, which assumes 500,000 units being produced per year (with a default battery pack size of 100 kWh) and calibrates capital expenditures to match this baseline. This 50-GWh plant is on par with largest factories around the world (which can be up to 100 GWh or more) but larger than the average manufacturing plant, existing or planned (Supplementary Data 8). Other literature has identified that economies of scale are reached at around 30,000 packs per year [130], 7.8 GWh of capacity per year [131] or below 2 GWh of capacity per year currently and up to 15 GWh in the future [132], using bottom-up methodologies, with unit costs that are fairly insensitive to production volume above that level. Thus, we believe that the baseline values used in BatPaC are probably still reasonable approximations despite economies of scale for battery manufacturing. Other previous critiques of BatPaC have included, for example, that energy costs were not separated out and instead assumed as a fixed percentage of final production costs or that the cells being modeled were cylindrical instead of the more modern prismatic pouch type [131], but several of these elements specifically were rectified in the most recent version [110].

In our analysis, we make one key change: the default battery pack size used in BatPaC is 100 kWh, which is large in comparison to the typical battery size of 40 kWh–100 kWh [106]. We use a 70-kWh battery as our baseline, which is effectively the battery capacity needed for a 300-mile vehicle in the GREET model (69–74 kWh for batteries produced between 2022 and 2030) [106], as 300 miles is commonly considered to be the threshold for which electric vehicles become competitive from a range perspective in the United States [15, 125].

3.6.5 Modeling policy scenarios

We use the default input values in BatPaC as a representation of electric vehicle battery manufacturing within the United States and then vary these inputs (Supplementary Data 4) to represent the various scenarios presented in Tables 1 and 2. For the US-based scenarios, we use the different input material costs calculated in Supplementary Data 3. Variations on these default values for modelled production in China are done through a comparative analysis of empirical [111] and modeled [131] data between the United States and China, where material costs are taken to be 10% less in China, labor costs are estimated to be 75% less and other primary costs (for example, energy, land, capital equipment, indirect costs) are assumed to be 20% less. To change these values in BatPaC, cells in the 'Power and Storage Requirements,' 'Material Costs' and 'Plant Specifications' sections of the Dashboard sheet and 'Cost Rates for Manufacturing and Cost Breakdowns' and 'Battery Assembly Costs (baseline reference)' sections of the Cost Input sheet were altered through a series

of Excel scripts. Because of these scripts, generating sensitivity analyses for important parameters was relatively straightforward, including sensitivity analysis on battery pack size, battery capacity, labour rates, energy costs and land and building costs (Appendix B.4 and Supplementary Data 5). In particular, additional sensitivity analysis was conducted at production scales of 50,000 and 150,000 packs and other battery capacities (40 kWh, 55 kWh, 85 kWh, 100 kWh; Appendix B.4). These scenarios and inputs resulted in various production costs as calculated by the model, with results presented on the Dashboard, Summary of Results and the 'Cost of Cells,' 'Cost of Module-Level Additions' and 'Cost of Pack-Level Additions' sections of the Manufacturing Costs sheets of BatPaC. These output values are recorded in Supplementary Data 5 and constructed into Figure 3.3, Table 3 and Table 4 in Supplementary Data 6 and Supplementary Data 7.

3.7 Data Availability

The source and processed data generated in this study are attached as Supplementary Data. The model used in this study (BatPaC) [110] is publicly available via the Argonne National Lab. We only use publicly available production and critical mineral price data for the production analysis, primarily from the BatPaC model and Trost and Dunn [19]. Estimates of production costs in China are estimated from Krishna [111]. Data used to generate supply chain data in Figure 3.1 are from publicly available Survey [97], Sun et al. [2], Endo et al. [96] and Agency [7] datasets and a Frost and Sullivan report [98] that is not in the public domain. Additionally, we have created an interactive tool to allow users to adjust model assumptions around mineral prices and see how they affect key results in Figure 3.2 and Figure 3.3 at https://acheng98.shinyapps.io/IRAMineralPriceEffectsSim/.

Chapter 4

Economic and environmental implications of onshoring US lithium production

Ongoing research.

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4.1 Abstract

Rapid growth in batteries for electric vehicles and energy storage has intensified global demand for critical minerals, particularly lithium. Although geological lithium resources are widely distributed, incumbent supply chains are highly dependent on China for refined lithium materials production, raising concerns about supply chain vulnerability and environmental impacts. While most Chinese lithium chemical production relies on imported ores, domestic production from lepidolite feedstocks currently appears to operate on or near the margin of the global supply curve, as inferred from recent price-sensitive entry and exit behavior. Yet, its cost and environmental performance are poorly characterized in the open literature, limiting assessment of whether alternative production pathways can compete economically and environmentally. We develop an integrated, empirically grounded and fully transparent bottom-up technoeconomic and life cycle greenhouse gas emission model for three lithium projects representing distinct resource-pathway combinations: U.S. evaporative brine at Silver Peak, the only active domestic producer; U.S. sulfuric acid-leach clay at Thacker Pass, currently under construction; and Chinese sulfate roasting lepidolite at Jianxiawo, a large near-marginal producer on the global supply curve. We find that resource-pathway pairings create structural cost and emissions differences that persist across geographies and dominate over other location-specific factors such as reagent costs, labor costs, or electricity grid impact factors. Silver Peak ($5,000 [$4,100—5,900]/t Li_2CO_3; 5.2 [4.5—5.8] t CO_2eq/t) and Thacker Pass ($8,750 [$7,400—12,300]/t; 17.0 [14.9—19.0] t CO_2eq/t) compare favorably to Jianxiawo ($10,400 [$9,500—12,300]/t; 22.6 [17.0—33.0] t CO_2eq/t), where ranges reflect plausible variation in the dominant cost and emissions drivers. These estimates are made under conservative project-specific designs for Thacker Pass and an optimistic ore grade for Lepidolite, suggesting the true gap is likely wider. Contrary to conventional wisdom that Chinese operations benefit from lax environmental regulation, we find that high environmental compliance costs of lepidolite production in China, driven by hazardous waste disposal requirements for roasting residues, further disadvantage the pathway relative to US sources. These results suggest that growth of U.S. lithium production has potential to economically displace marginal lepidolite production in China while offering lower climate change externalities.

4.2 Introduction

Lithium is an essential component of nearly all current electric vehicle batteries and large-scale battery energy systems, yet it is projected to experience an increase in demand by a factor of 20 to 40 between 2020 and 2040 [3, 133, 134]. While materials from battery recycling could dampen this demand, production from raw materials extraction will still need to produce more than 80% of future demands under most scenarios [3]. Natural endowments of lithium are relatively widespread, but current production is geographically concentrated, while also mismatched with demand sources (Figure 4.1). Australia, Chile, and Argentina dominate primary extraction while China dominates refining and materials production, and yet most demand for lithium products, such as for electric vehicles, is in China, the U.S., and Europe [17, 135]. This concentration has raised concerns about supply chain resilience for countries that rely on lithium for key energy technologies [10, 17, 133]. These concerns span both geopolitical risks such as export restrictions and resource nationalism [136, 137] and physical risks from natural disasters, labor strikes, or environmental impacts [8, 41], leading to an increasing interest in "reshoring" and "friend-shoring" supply chains.

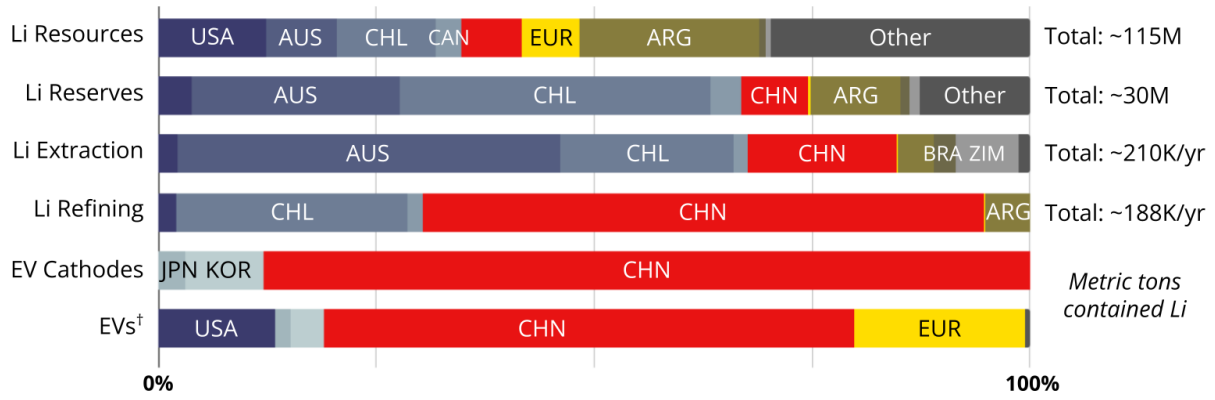


Figure 4.1. Geographic distribution of lithium (Li) across resources, production, and demand (2023).

There is a significant mismatch between the location of lithium resources, reserves, lithium production, lithium refining, and end lithium demand. The following codes are used to represent countries or regions: USA, United States of America; AUS, Australia; CHL, Chile; CAN, Canada; CHN, China; EUR, Europe; ARG, Argentina; BRA, Brazil; ZIM, Zimbabwe; JPN, Japan; KOR, South Korea; Other any other country not explicitly listed here. Data for the year 2023, with resources, reserves, extraction, and the quantity of refined lithium from the USGS Mineral Commodity Survey 2025 [138], refining percentages from the IEA Global Critical Minerals Outlook 2024[133], and cathode and EV percentages from the IEA Global EV Outlook 2024[139]. Note that lithium mining totals in the United States are not disclosed by USGS, but are reported in other sources[140].

Global lithium production currently draws on two primary resource types: brines and hard rocks, with sedimentary clay deposits representing a significant but underdeveloped third source. Spodumene hard rock deposits, mined primarily in Australia and refined in China, currently represent the largest share of overall lithium production, and sit toward the lower-to-middle of the cost curve at roughly $5,000—8,000/t LCE [141, 142]. Continental brine deposits found in desert salars in Chile and Argentina represent the lowest-cost production globally, with high brine grade and high aridity enabling solar evaporation-based concentration at costs as low as $3,000—5,000/t LCE [141]. However, with the high water usage of these resources in water-stressed regions [143, 144], alternate 'direct lithium extraction' (DLE) technologies are emerging as a lower-footprint alternative for processing these brine-based resources, though these are currently only operating in China and at relatively limited scale. A range of other 'unconventional' resources, such as geothermal brines, jadarite, zinnwaldite, and others, currently contribute negligibly to global supply.

Of these other resources, sedimentary clay deposits, most prominently found in the western United States, represent the largest undeveloped unconventional lithium source. Largely because of these clays, the United States held about 12% of world resources in 2023 Figure 4.1, which has since increased to 20% as of 2026 [145]. However, regulatory timelines, depressed prices, uncertainty in financing, and community and Indigenous opposition have led to delays in developing these resources; currently, only two projects have been permitted: Thacker Pass and Rhyolite Ridge, both clay-based resources. With the latter suffering from a collapse in funding [146], Thacker Pass remains the only U.S. project under construction, with a target operational date of 2028 - two decades after significant exploration activity started in 2007 [147–149]. However, its target annual production capacity of 160,000 tonnes LCE by 2044 [150] would be roughly 10% of world production in 2025 [145]. Given only that the Silver Peak evaporative brine facility in Nevada has been the only lithium producer in the United States since the beginning of the 21st century, with a current nameplate capacity of roughly 6,000 tonnes, this marks a significant increase in US production. In sum, if all Nevada clay projects operated by publicly listed companies were to come online, that could result in a total of 320,000 tonnes LCE of annual production (including Thacker Pass) [150–155]. Thus, while other Direct Lithium Extraction sites in the United States are under construction, their relatively limited total annual production capacity of 40,000 tonnes [156–158] and relatively low reported operating costs ($5,600-7,400 per tonne LCE) point towards an initial focus on better understanding U.S. clay resources.

On the other hand, China's domestic lithium production has grown roughly tenfold between 2017 and 2025, from about 37,000 to an estimated 330,000 tonnes LCE, driven primarily by rapid expansion of low-grade *lepidolite* hard rock mica deposits [145, 159]. At costs

of roughly $10,000—16,000/t LCE [160–162], or roughly the 90th percentile of the global cost curve [163], lepidolite is the source most sensitive to price fluctuations and the relevant marginal producer against which alternative suppliers must compete.

A key question for countries seeking to reduce their dependence on incumbent supply chains is therefore whether domestic production can undercut this marginal cost and compete on the global market. Lepidolite production has contracted during periods of low prices, with disruptions to lepidolite supply from Jiangxi province repeatedly triggering significant volatility in Chinese lithium futures markets [159, 164]. While the dynamics for these projects are examined in more detail in Section 4.4.3, this price-driven entry and exit behavior, combined with the scale of existing installed capacity, suggests that lepidolite is likely to remain the relevant marginal producer as demand grows, making it an appropriate competitive benchmark.

However, as demand for lithium continues to grow, displacing lepidolite production on the cost curve would require the production of new alternative sources, at a competitive cost. Whether that investment is justified from a financial perspective depends on the expected return relative to risk and alternative uses of capital, typically evaluated via net present value (NPV) or internal rate of return (IRR) metrics [165, 166]. However, from a competitive market-based supply security perspective, the more relevant threshold is whether a project's average lifetime operating cost is competitive with the current marginal producer: the price at which an incumbent's production becomes uneconomic sets the ceiling that a new entrant must undercut to capture market share [167]. It is this long-run cost competitiveness, rather than project-level returns or the exact cost curve of non-marginal production, that determines whether domestic supply development could durably compete on the global scale.

Production characteristics such as costs and emissions are not separable from the resource itself: the chemistry and geology of the deposit — whether brine, hard rock, or clay sources — determine the extraction and processing pathway, which in turn drives both the cost structure and also externalities such as emissions and environmental damages. This means that geographically shifting production is not simply a matter of replicating existing processes, but also that these production-resource pairings could inherently provide advantages or disadvantages. This resource-pathway relationship creates opportunities for policy to shape supply chain configurations in ways that could simultaneously address security, cost, and environmental objectives, rather than trading one off against another [168–170]. Some studies [171] have identified that communicating the national security and domestic sourcing benefits of decarbonization projects can help navigate local public opposition. The extent to which such 'win-win-win' outcomes are achievable depends on the relative cost and emissions profiles of competing pathways across production locations, which this work aims

to characterize.

Assessing whether these outcomes are achievable requires characterization of the cost and emissions profiles of these competing pathways. Yet, the literature remains incomplete, particularly for the Chinese lepidolite operations that define the current benchmark for marginal production. While a substantial body of literature has examined models of lithium production through life cycle assessment with varying levels of detail (see the review of over 40 such studies in Appendix C.4.1), process-based technoeconomic analyses are comparatively sparse (e.g. [142, 172]), and studies that jointly evaluate both dimensions across multiple resource types are rarer still, with only Mousavinezhad et al. [173], Gao et al. [174] to our knowledge. Mousavinezhad et al. [173] is the only study to evaluate multiple production pathways along the full mine-to-product scope in both dimensions, but focus solely on four projects representing US-based pathways (U.S. evaporative brine, U.S. direct lithium extraction (DLE) from brine, U.S. clay, and spodumene mined in Canada and processed in the US). On the other hand, lepidolite has received limited attention and highly varied results: Liu et al. [175] estimated its global warming potential (GWP) to be the highest amongst production pathways in China at roughly 24.5 tonnes CO_2eq/t Li_2CO_3. In contrast, Wang et al. [176] estimated a much higher GWP impact of lepidolite at 66.0 tonnes CO_2eq/t Li_2CO_3. Its production costs are even more poorly characterized in the open literature, with only Gao et al. [174] reporting that the production cost of lithium carbonate from lepidolite concentrate via the sulfate method was about $4,500 per ton Li_2CO_3 - excluding the direct production costs of lepidolite concentrate production from ore, and not providing an open, examinable model. In contrast, news reports and grey literature have identified lepidolite production costs to vary widely between USD$10,000 and USD$16,000 [160–162]. We plot these values, in addition to literature estimates of costs and emissions for the Silver Peak and Thacker Pass operations as dashed lines in Figure 4.3, noting that each of these estimates come from different studies with different scopes and assumptions.

We aim to assess whether the cost and emissions advantages suggested by resource type hold, and what this implies for the competitiveness of lithium production in the United States relative to lepidolite in China as the current marginal producer.

4.3 Scope and Methodology

As a first step towards addressing these literature gaps, this work evaluates and compares the production costs and cradle-to-gate greenhouse gas emissions of three lithium projects representing distinct resource-pathway combinations, using a transparent, empirically-grounded technoeconomic-life cycle assessment framework and consistent assumptions across US and

Chinese sites. This process-level framework directly interlinks production costs and emissions, providing a static treatment of lithium production but empirical grounding of production, laying the groundwork for more richly represented dynamic analyses, such as responses to price signals, technological learning, policy feedbacks, or geopolitical shifts. Furthermore, given that the emissions literature is substantially richer and more methodologically mature than the technoeconomic literature for lithium, consistency between modeled emissions profiles and reported life cycle assessment values can thus provide an independent cross-check on the underlying process model, and therefore the cost estimates derived from it.

In order of increasing complexity of mineral resource and processing pathway, we define flowsheets for each project as shown in (Figure 4.2), based on publicly available project disclosures, patents, and other literature. In particular, Silver Peak and Thacker Pass are owned and operated by Albemarle and Lithium Americas, respectively, two companies that are publicly traded on US stock exchanges. As a result, under SEC Regulation S-K 1300[177], they are required to disclose mineral resources and reserves through reports written by an independent Qualified Person, from which high-level flowsheets can be determined [150, 178]. On the other hand, publicly available detailed representations of lepidolite projects in China are limited, given the lack of similar disclosure requirements in China. We focus on Jianxiawo (Yichun, China, Figure 4.2c), the largest lepidolite project in China by production volume: its integration with CATL, the world's largest battery manufacturer, means that its operational changes, including production suspensions and restarts, attract substantial coverage in industry press and commodity market reporting, providing more contextual data than is available for most Chinese lepidolite operations (e.g. [159, 162, 164, 179]). Furthermore, its usage of the sulfate roasting pathway, the dominant processing route for lepidolite in China [180], makes it representative of the pathway, beyond being the largest individual project. These three projects represent key competing pathways relevant to supply chain security; the full set of generic lithium carbonate production pathways is described in Appendix Figure C.1.

The primary methodology we employ in this analysis is a series of process-based technoeconomic models, from which environmental impacts can be estimated and policy interventions can be applied. The modeling framework is structured hierarchically across four levels of resolution, enabling consistent comparison across projects while preserving sensitivity to pathway-specific technology requirements. At the *supply chain* level, facilities are linked by *transportation routes* into a directed network from raw material extraction to refined Li_2CO_3 product. Each *facility* aggregates co-located production steps operating under shared geographic and operational parameters (e.g. electricity grid composition, labor costs, utility prices, and plant schedule), ensuring that location-specific cost and emissions factors are ap-

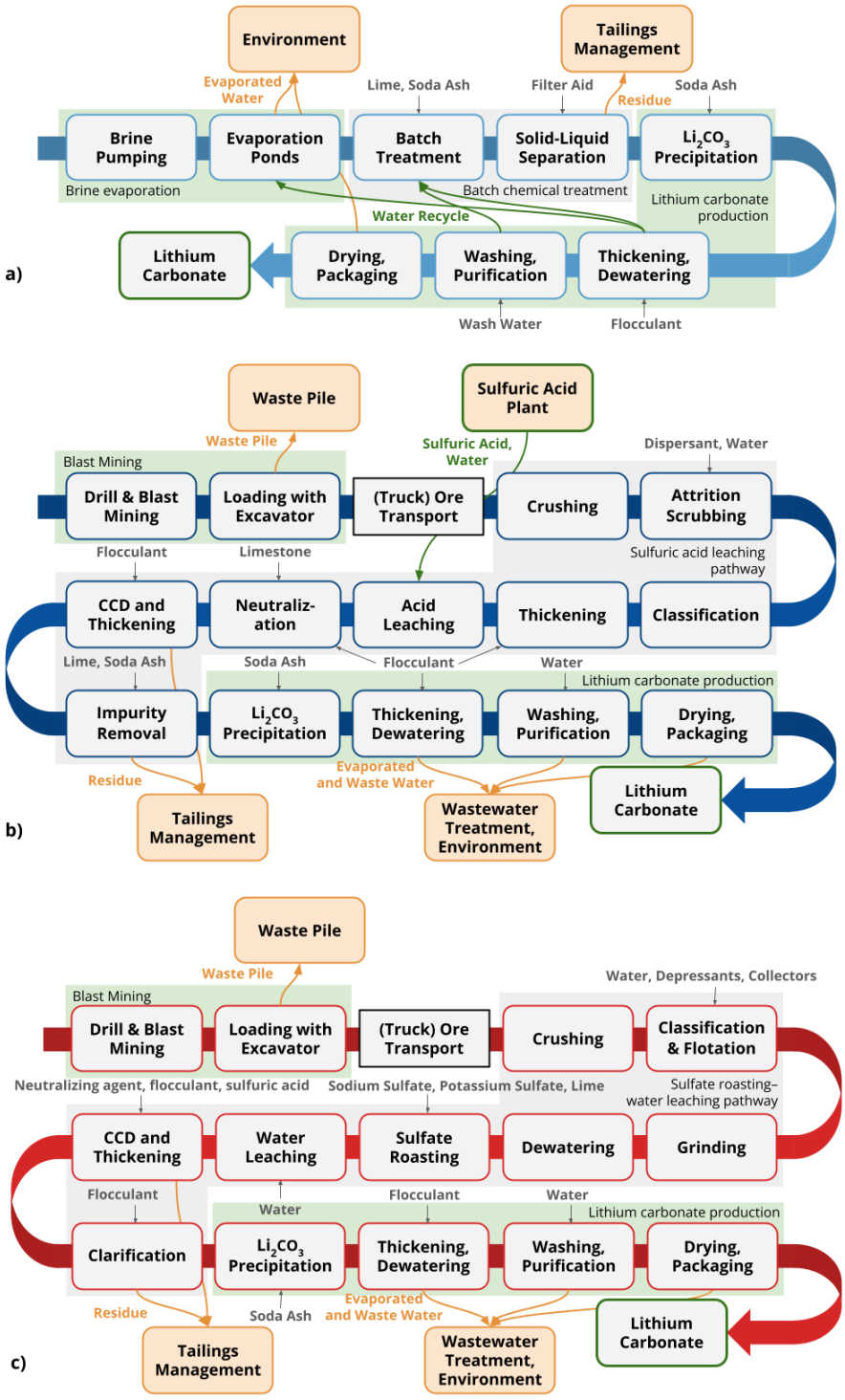


Figure 4.2. Process overview diagrams for each project modeled: (a) Silver Peak; (b) Thacker Pass; (c) Jianxiawo

Each subfigure shows the sequence of production steps, key reagents consumed (arrows inward) and coproducts or waste streams produced (arrows outward). Each shaded box in the background represents a facility, where production steps (rounded boxes on top of shaded boxes) are co-located, with the same sitewide operating parameters.

plied consistently. Within each facility, *production steps* represent individual unit operations (e.g., acid leaching, impurity removal, lithium carbonate precipitation), each defined by a primary input and output stream, secondary reagent inputs, and coproduct or waste streams routed to defined sinks. At each step, three material volume quantities are tracked: the primary input, the process basis (the internal scaling quantity, such as m^3 slurry processed or tonnes solids crushed), and the primary output, which are related using two conversion factors (from input to step basis and step basis to output) that accommodate the diversity of natural reporting units across feedstock types and process steps. Reagent consumption is calculated as a function of constituent concentration in the process stream, allowing stoichiometric or empirical dosing ratios to be applied directly to the relevant constituent rather than to bulk throughput. Each production step is parameterized using *machine blocks*, which specify equipment sizing, capital cost scaling relationships, utility intensities, and labor requirements. This separation between process chemistry (defined at the production step level) and equipment characteristics (defined at the machine block level) allows alternative equipment configurations or technology variants to be substituted without altering the underlying mass balance, enabling systematic sensitivity and scenario analysis. Costs and emissions are computed at the production step level and aggregated upward; because material flows are propagated from a terminal production target back through the full process chain while taking into account process unit conversions and yield rates, throughput, reagent consumption, and equipment sizing at every step are directly linked to the final output volume. For further detail on this hierarchical modeling system and a formal mathematical formulation, see Appendix C.5.

Six overarching facility types are defined, as shown in Figure 4.2: brine evaporation, blast mining, pregnant leach solution (PLS) production from batch chemical treatment, PLS production from sulfuric acid leaching, PLS production from sulfate roasting and water leaching, and lithium carbonate production (which is parameterized differently depending on if the PLS is a chloride or sulfate-based solution). Each of these facilities consist of multiple production steps, for a total of thirty production steps modeled (Figure 4.2). Production volumes are modeled at approximately 6,000 t Li_2CO_3/yr for Silver Peak [178], 66,000 t/yr for Thacker Pass (consistent with the prior feasibility disclosure [181]), and 46,000 t LCE/yr for Jianxiawo (the most commonly cited production figure [141, 182]).

For each pathway, we compute average lifetime operating costs of lithium carbonate production in USD per metric tonne Li_2CO_3. As discussed in the introduction, this metric is more appropriate for assessing whether a project can sustainably compete with current or future marginal producers than a full NPV or IRR analysis. For each production step, operating expenditures (OPEX) are computed as the sum of variable costs (materials, la-

bor, utilities) and fixed operating costs (maintenance and overhead), expressed per unit of annual Li_2CO_3 production. Capital expenditures (CAPEX), such as equipment, buildings, and auxiliary systems, are estimated using installation cost factors and equipment scaling relationships. For continuous processes, overall equipment costs are scaled with throughput using power law exponents (e.g. the six-tenths rule) while batch processes are expanded by adding parallel lines [183–185]. These capital costs are not included in the primary cost comparison given the focus on operating competitiveness, but are reported for reference and cross-comparison with the literature.

These results are extended to capture the environmental impacts of production, where impact factors for reagent use, energy consumption, and utilities are multiplied by the mass and energy flows through the system (see Appendix C.5.0.5). Scope 1 emissions result from specified process emissions (which are limited in our three cases) or direct combustion of fuel, primarily for heating process throughputs. Scope 2 emissions result from electricity usage, which is then multiplied by the local grid emissions factor. Scope 3 emissions are calculated by multiplying each reagent's annual consumption by its cradle-to-gate emissions factor, reflecting the upstream production burden of that reagent as a purchased commodity input. This methodology is applied uniformly across all pathways, with geography-specific emissions factors applied where production location affects this upstream burden (e.g., soda ash from trona mining versus the Solvay process).

For both costs and emissions, conservative, midpoint, and optimistic values are determined for each machine block design and material input cost, yielding sensitivity ranges that represent plausible operating variation across the dominant uncertainty drivers for each project. Process chemistry (e.g. reagent stoichiometry, conversion factors, and yield rates) is held fixed within each pathway, since these are generally determined by the underlying mineralogy and cannot be varied independently of the process design. Similarly, utility prices, labor rates, and land costs are not varied in the sensitivity analysis, as these are set by location rather than by operational choices, and their cross-pathway differences are instead captured through the location-specific parameters applied at the facility level.

While Silver Peak and Jianxiawo are modeled using flowsheets derived from project-specific disclosures and patents respectively, the clay pathway is represented using a generic acid-leach flowsheet grounded in the fundamental process chemistry of Nevada sedimentary clay deposits rather than the specific design of Thacker Pass. Several other clay projects in Nevada share similar mineralogy and processing fundamentals (e.g. Rhyolite Ridge [151], Bonnie Claire [155], Tonopah Flats [152], TLC [186]) and a generic flowsheet provides a more transferable baseline for this class of resource than a project-specific model would. The key distinction is in impurity management: because sulfuric acid dissolves both lithium

and magnesium simultaneously from magnesium-rich hectorite, the generic flowsheet precipitates all magnesium with quicklime, whereas the Thacker Pass project-specific design employs a magnesium crystallization route that recycles magnesium compounds as reagents, reducing lime consumption to approximately 2,630 kg/t at the cost of higher thermal energy demand [150]. This omission renders the generic flowsheet conservative relative to the Thacker Pass design specifically, but representative of what a first-generation clay operation without project-specific optimization would require. The cost and emissions implications of this choice are discussed in Section 4.4.2.

4.4 Results and Discussion

Our results indicate that US-based lithium production from evaporative brine and sulfuric acid-based clay pathways have lower production costs and life cycle emissions than production from Chinese-based lepidolite sulfate roasting pathways. The Silver Peak project occupies the lowest-cost, lowest-emission frontier across both our estimates and the literature, at an estimated cost of about $5,000 [$4,100-$5,900] per tonne LCE, and 5.2 [4.5-5.8] tonnes CO_2eq per tonne LCE. Modeled production costs and emissions from the clays in the generic version of the Thacker Pass flowsheet are closer to modeled production from lepidolite from Jianxiawo, with lower midpoint costs ($8,750 [$7,400-$12,300] vs. $10,400 [$9,500-$12,300]) and emissions (17.0 [14.9-19.0] tonnes CO_2eq vs. 22.6 [17.0-33.0] tonnes CO_2eq).

Our modeled ranges for all three projects are broadly consistent with reported literature values (Figure 4.3), confirming the cross-check between emissions and cost model structure described in Section 4.3. The substantially wider ranges for lepidolite across both the literature and our study also reflects the data scarcity in reported production costs and emissions for this pathway, underscoring the importance of better understanding this major source of lithium production.

From this, we see that Silver Peak's cost and emissions advantage is robust across all modeled ranges, but the ranges aiming to represent plausible operating variation for Thacker Pass and Jianxiawo overlap, particularly when taking into account a scenario where hazardous wastes generated by Jianxiawo's production process are treated as if they were non-hazardous. This indicates that the relative competitiveness of U.S. clay projects versus Chinese lepidolite is directionally favorable, but this is sensitive to project-specific assumptions explored in the sections below.

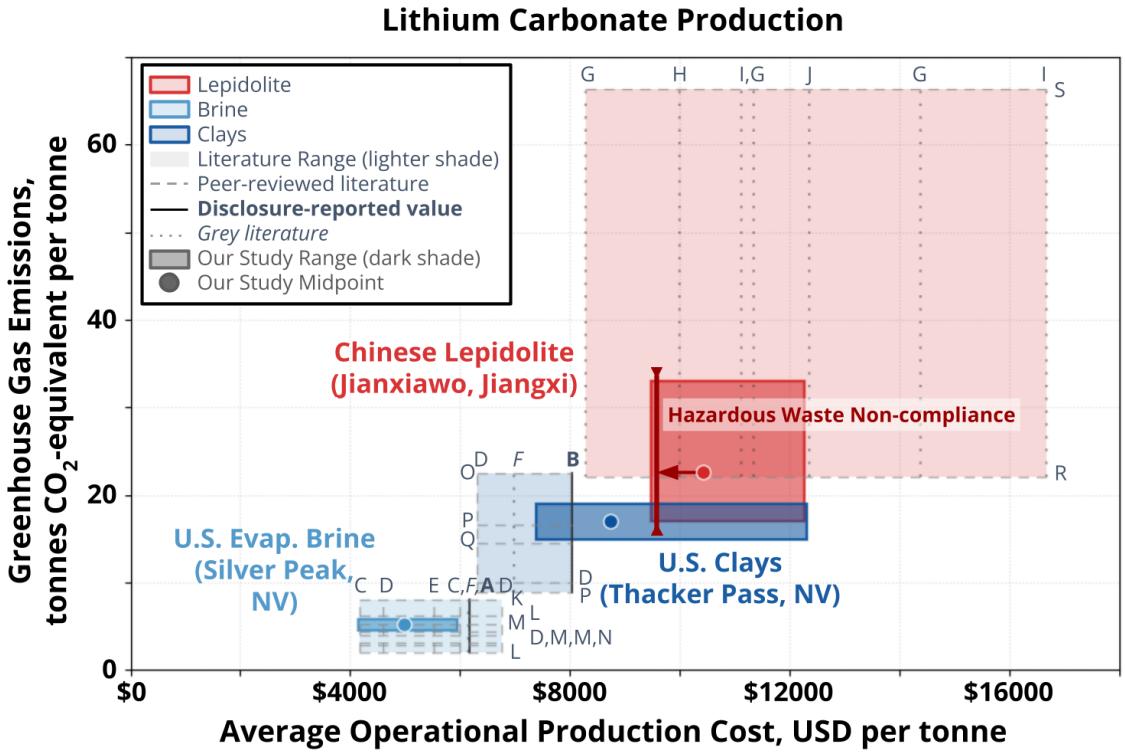


Figure 4.3. Production cost ($/t Li_2CO_3) versus life cycle greenhouse gas emissions (t CO_2eq/t Li_2CO_3) for three key lithium projects representing distinct resource-pathway combinations: Silver Peak (evaporative brine, United States), Thacker Pass (clay, United States), and Jianxiawo (lepidolite, China)

Darker shaded rectangle span the range from this study, with the central point indicating the study midpoint; these ranges reflect variation in the three largest sources of cost and emissions uncertainty for each project, identified through sensitivity analysis (Figure 4.5, Figure 4.7, Figure 4.9). The complete uncertainty range incorporating all parameters simultaneously is shown in Figure C.2; the ranges shown here are intended to represent plausible operating variation rather than absolute worst-case bounds. The dark grey spur extending from the Jianxiawo midpoint shows the modeled cost reduction under a hazardous waste non-compliance scenario, in which lepidolite roasting residues are disposed of as non-hazardous waste rather than under current Chinese regulations (see Section 4.4.3). Lighter rectangles span the range of values reported across grey and peer-reviewed literatures, with each line representing one data point. The lighter shaded rectangles span the full range of values reported across grey and academic literatures, with each line representing a data point, with A: SRK Consulting [178]; B: SGS Canada et al. [150]; C: Ambrose and Kendall [187]; D: Mousavinezhad et al. [173]; E: Fleming et al. [141]; F: Wesselkaemper et al. [142]; G: Chen [188]; H: Atkinson [161]; I: Liu and Hayley [160]; J: Li [189]; K: Sun et al. [143]; L: Manjong et al. [190]; M: Ambrose and Kendall [191]; N: Schenker and Pfister [192]; O: Khakmardan et al. [193]; P: Iyer and Kelly [194]; Q: Roy et al. [195]; R: Liu et al. [175]; S: Wang et al. [176]. See Table C.1 for a summary of all literature studies.

4.4.1 Results for the Silver Peak Project

The Silver Peak project, located in a historic silver mining town in southwestern Nevada, is the only active operating lithium producing facility within the United States. Currently owned by Albemarle, it has produced lithium since 1966 through a traditional evaporative brine, batched-chemical process, with a current production target of roughly 6000 tonnes of lithium carbonate each year [178]. In this pathway, lithium-bearing brine is pumped from subsurface aquifers and concentrated through solar evaporation in large, shallow ponds before downstream chemical purification yields battery-grade Li_2CO_3 (Figure 4.2a). Most desert salar brines are composed of sodium chloride solutions with lithium, magnesium, calcium and potassium ions [196]; as water evaporates from brines, many of these ions naturally precipitate from the solution. However, magnesium and lithium are typically soluble, and magnesium removal is typically the most chemically demanding purification step for lithium brines[141, 197]. As opposed to South American salars such as Salar de Atacama (~2,000 mg/L), Silver Peak's brine resource is relatively dilute at approximately 95 $mg\,L^{-1}$, requiring extensive evaporation before chemical processing can begin; in our model, approximately 95% of pumped water evaporates as the brine concentrates to 6,750 PPM strong brine. However, Silver Peak's relatively low magnesium content allows for an all-chemicals, batched precipitation process with lime and soda ash, a simpler production method than solvent extraction or ion exchange processes necessary for other salar brines [178, 198]. Thus, in spite of this resource disadvantage, Silver Peak's cost and emissions profile is the most favorable of the three projects modeled here, a consequence of its structural reliance on solar evaporation for lithium concentration and said low magnesium concentration, shifting its cost and emissions profile towards maintenance and upkeep of the solar evaporation system rather than the more chemically intensive ore-based production pathways.

Evaluating the step-by-step costs breakdown (Figure 4.4a), we see that three steps dominate: evaporation ponds (~$1,350/t), batch treatment (~$1,300/t), and brine pumping (~$1,000/t). The evaporation pond cost is predominantly labor and the "Other" category, here represented primarily by maintenance operations on the large pond footprint. Pond-level pumping electricity is assumed to be included with the pumping step, another major cost contributor. On the other hand, lithium carbonate precipitation is the primary step with significant material costs, making soda ash the primary reagent cost driver. The overall cost structure is labor- and capital-intensive rather than reagent-intensive, reflecting an operating cost that is relatively insensitive to global markets and more sensitive to local operating conditions.

From an emissions perspective, brine pumping (~2.0 t CO_2eq/t), batch chemical treatment (~1.9 t CO_2eq/t), and lithium carbonate precipitation (~1.1 t CO_2eq/t) are by far

the dominant emissions steps (Figure 4.4b). The former is Scope 2, driven by electricity consumption for pumping using an emissions factor based on the Nevada grid[199], while the latter two are Scope 3, from embodied emissions in lime and soda ash production. While soda ash production in the United States is generally cleaner (∼0.3-0.7 t CO_2eq/t Na_2CO_3) due to its direct production from Trona rather than the heat-intensive Solvay process used in China (∼0.7-1.2 t CO_2eq/t Na_2CO_3) [106, 195, 200], the heavy use of lime for magnesium removal (∼0.83-1.7 t CO_2eq/t CaO at 1.5 tonnes CaO per tonne Li_2CO_3) and soda ash to remove the calcium introduced by the lime and precipitate lithium carbonate (2.4 tonnes Na_2CO_3 per tonne Li_2CO_3) results in significant impacts from the embodied emissions associated with the production of these upstream inputs. However, this emissions profile does also indicate that pumping is by far the most significant electricity consumer, and that decarbonization of the Nevada grid would have a direct and substantial impact on Silver Peak's emissions intensity.

The cost and emissions sensitivities shown in Figure 4.5 tell somewhat different stories. For costs, the evaporation pond capital and operating assumptions dominate, followed by soda ash price and brine pump costs — meaning the largest uncertainty is in the pond sizing and cost model rather than reagents, even though maintenance and labor costs are relatively small (0.3-0.5% of total capital costs as annual maintenance operations costs, 8-12 laborers managing salt removal and dike upkeep). For emissions, lime and soda ash embodied emissions factors dominate, followed by brine pump electricity intensity, meaning emissions uncertainty is primarily driven by upstream supply chain assumptions rather than on-site operations. This divergence is worth noting explicitly: the parameters that drive cost uncertainty are not the same as those that drive emissions uncertainty, which has implications for how policy interventions would affect the two dimensions differently.

Several modeling simplifications limit this study. Most substantively, several machine blocks (particularly thickening and dewatering, and washing and purification) are parameterized for throughputs characteristic of much larger operations (i.e. Thacker Pass and Jianxiawo); at Silver Peak's modest production scale of about 6,000 t Li_2CO_3 per year, these steps currently contribute negligibly to modeled costs and emissions, which may explain why our central cost estimate sits under the lower end of the literature range in Figure 4.3. Additionally, other modeling simplifications are made given limited site-specific engineering data, such as incorporating pond circulation pumping costs into brine extraction pumping costs, and combining the liming pre-treatment step into the batch chemical purification step. That said, the relative consistency between our modeled range and the literature range on both cost and emissions axes, and the relative narrowness of that literature range compared to the other two projects, suggests that these simplifications do not substantially distort the

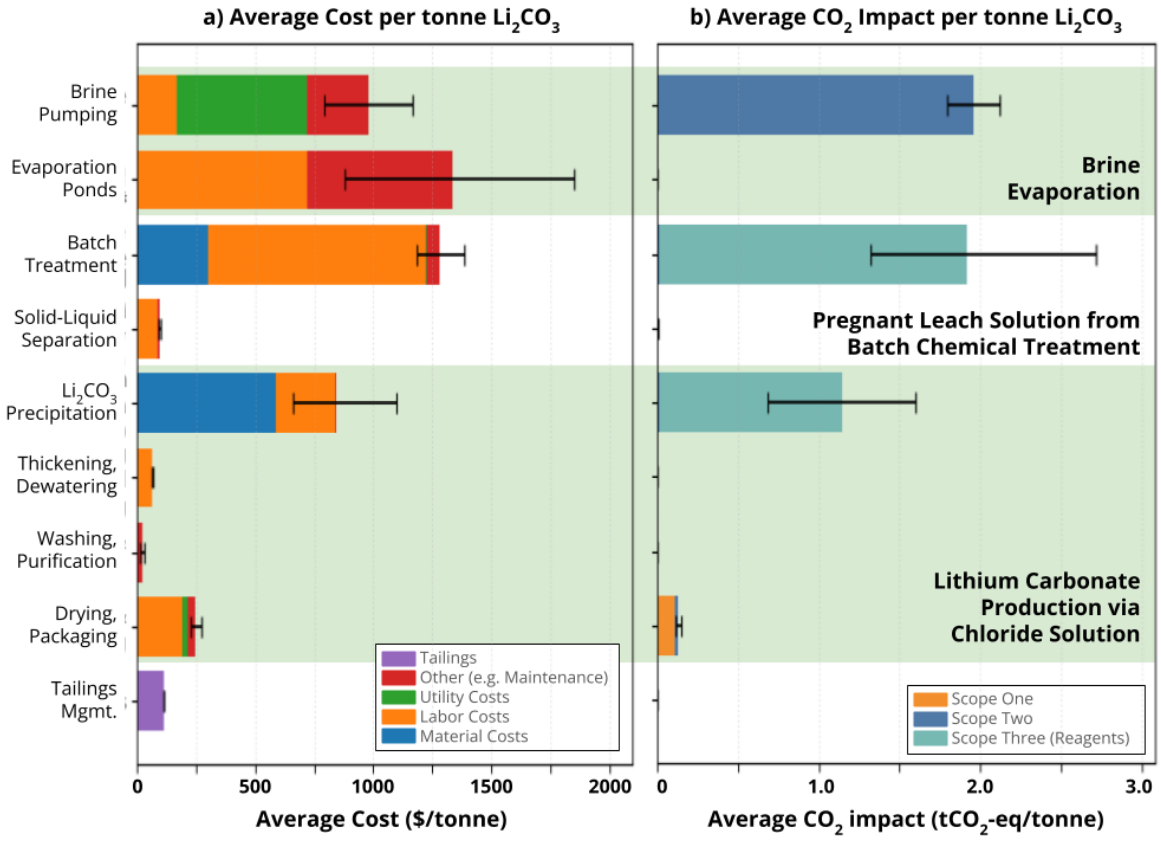


Figure 4.4. Costs(a) and emissions(b) for each step of the Silver Peak Project. Error bars represent a sensitivity envelope formed by simultaneously applying conservative and optimistic values for all machine design parameters and material input costs that carry defined scenario variation; they do not reflect variation in process chemistry, reagent stoichiometry, or steps for which no scenario variation is defined (e.g., tailings management). As a result, the envelope is wider for steps with more varied parameters and narrower or absent for steps with few or none. Error bars should therefore be interpreted as indicative of the dominant sources of operating uncertainty rather than as statistical confidence intervals.

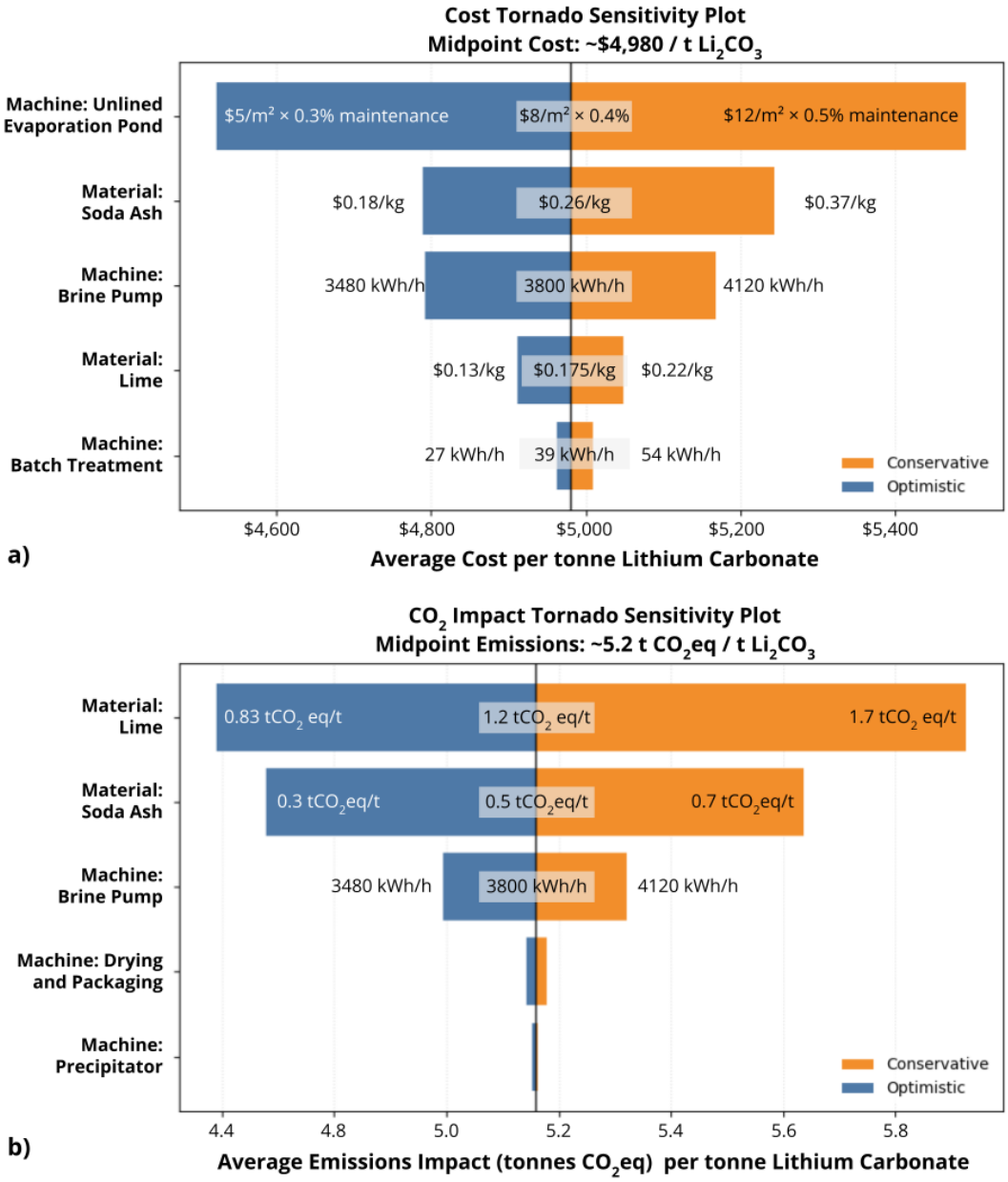


Figure 4.5. Tornado sensitivity plots for operating costs (a) and carbon dioxide emissions (b) for the Silver Peak Project.

The top five inputs with the greatest influence on each outcome are shown. The central vertical line indicates the midpoint estimate (~4,980/t Li_2CO_3 and ~5.2 $t\,CO_2eq/t\ Li_2CO_3$, respectively). Bars extend left and right in optimistic and conservative using estimates of input costs or impact factors for materials and operating designs for machine block models, respectively. Machine inputs represent design scenarios rather than continuous parameter variation, with labels representing the most significant design variation relating to the measured output; machine labels represent actual use or throughput-adjusted consumption at each scenario. Inputs whose scaled variation produced limited cost or emissions impact are not labeled.

overall cost and emissions competitiveness picture for a project that has been operating and reported on for nearly six decades.

With Silver Peak's relatively limited resources, its limited ability to expand due to water withdrawal limitations, and the overall lack of similar brine resources in the United States that can be exploited through the traditional evaporative pathway [142], clay resources represent a credible pathway for expanding domestic lithium production capacity, and are the next focus of this analysis.

4.4.2 Results for the Thacker Pass Project

The Thacker Pass project, located in Humboldt County, Nevada, is a pre-production, under-construction lithium clay deposit operated by Lithium Americas Corporation. The project targets production of lithium from hectorite clay, a magnesium-rich sedimentary mineral in which lithium and magnesium occupy the same structural positions in the clay lattice [201, 202]. Unlike hard rock ores such as spodumene or lepidolite, which require high-temperature roasting or calcination to break down the mineral structure before lithium can be extracted [174], the clay structure at Thacker Pass is amenable to direct acid leaching at moderate temperatures, avoiding the most energy-intensive step of those pathways [195]. However, because sulfuric acid dissolves both lithium and magnesium simultaneously, large quantities of magnesium enter the leach solution and must subsequently be removed, making magnesium management the central processing challenge of the clay pathway — analogous to its role in brine processing, but at substantially higher concentrations. Lithium Americas' latest project disclosure has defined a staged production ramp-up across four phases targeting 40,000, 80,000, 120,000, and 160,000 t Li_2CO_3 per year from a resource averaging 2,538 PPM lithium [150]. This analysis models Thacker Pass at 66,000 t Li_2CO_3 per year, consistent with the production volume assumed in the prior feasibility disclosure [181], to facilitate direct comparison with previous studies. While this is not the lifetime average across the staged production plan, we aim to represent a useful intermediate snapshot of the project's operating economics.

As described in Section 4.3, this project is modeled using a generic acid-leach clay flowsheet rather than the project-specific Thacker Pass design. The project-specific design for the Thacker Pass project incorporates a magnesium crystallization route that enables recycling of magnesium compounds as reagents, an important optimization given the magnesium-rich composition of the ore body, at the cost of additional energy usage. Excluding this route renders our estimate to be somewhat higher than the disclosure, given our central estimate of ~\$8,750/t Li_2CO_3 compared to the full life-of-mine cost of \$8,039/t Li_2CO_3 reported in the disclosure [150], but much higher than other estimates reported in the literature (\$6,300-

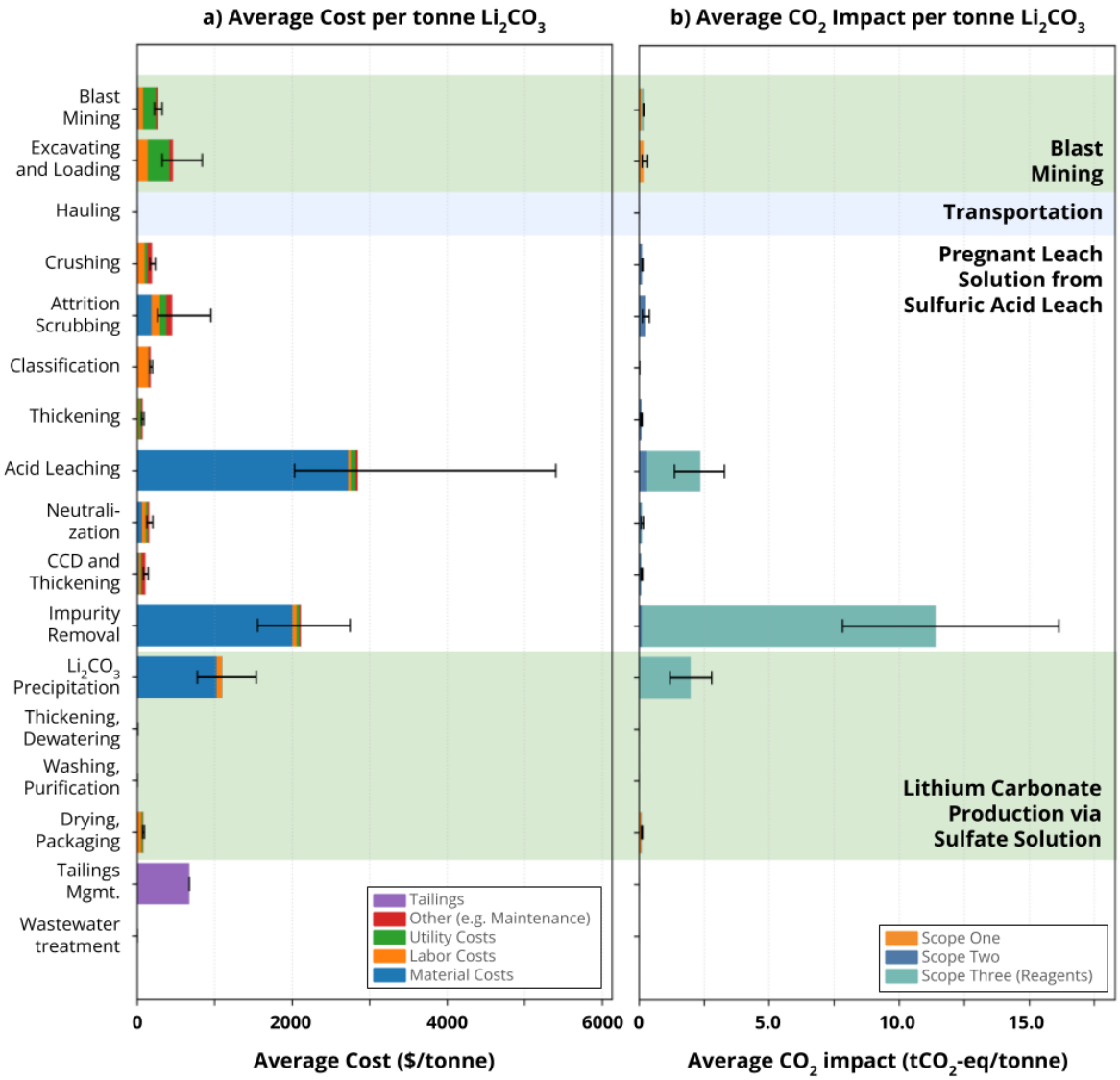


Figure 4.6. Costs(a) and emissions(b) for each step of the Thacker Pass Project. Error bars represent a sensitivity envelope formed by simultaneously applying conservative and optimistic values for all machine design parameters and material input costs that carry defined scenario variation; they do not reflect variation in process chemistry, reagent stoichiometry, or steps for which no scenario variation is defined (e.g., tailings management). As a result, the envelope is wider for steps with more varied parameters and narrower or absent for steps with few or none. Error bars should therefore be interpreted as indicative of the dominant sources of operating uncertainty rather than as statistical confidence intervals.

$6,986) or the 2022 disclosure ($7,198). Notably, the first 25 years of operating costs, when ore grades are the highest, are estimated at $6,743 and $6,238 in the 2022 and 2024 disclosures, respectively; since our analysis did not account for ore grade degradation, our estimate is likely still an overestimate. This gap is attributable primarily to lime consumption: the generic flowsheet fully precipitates magnesium with quicklime, consuming approximately 9,200 kg CaO/t Li_2CO_3, compared to 2,630 kg/t in the project-specific design where the magnesium crystallization route recycles magnesium compounds as reagents instead. Refining the model to incorporate this route represents a priority for future work.

The step-level cost breakdown (Figure 4.6a) confirms that costs are concentrated in the hydrometallurgical processing circuit rather than mining: acid leaching, impurity removal, and lithium carbonate precipitation together account for approximately $6,100/t Li_2CO_3 in the midpoint baseline - over two thirds of the overall costs - with material costs dominating all three steps. Tailings disposal is the fourth most important category at ∼$680/t, reflecting the large volumes of leached clay residue generated by the acid-leach process; with reagents producing these waste volumes structurally linking to handling costs in this pathway, mining operations are comparatively minor, suggesting cost reduction efforts are most productively focused on the processing circuit and its reagent use.

The emissions breakdown (Figure 4.6b) mirrors this concentration, with most emissions coming from the embodied emissions of reagents, but with a difference in ordering: impurity removal is the single largest emissions contributor at ∼11 t CO_2eq/t Li_2CO_3, exceeding acid leaching (∼2.5 t CO_2eq/t) despite ranking second in cost, reflecting the high embodied emissions of impurity-removal reagents, primarily lime. Scope 2 emissions appear in electricity-intensive mechanical steps such as crushing and attrition scrubbing, while Scope 1 emissions are largely confined to mining operations (diesel for vehicles), but these are far more minor in scope. This emissions structure implies that decarbonization of the Thacker Pass pathway depends primarily on reducing the embodied emissions of key reagents, especially lime and soda ash, rather than decarbonizing on-site electricity or fuel use.

The dominant source of cost uncertainty in the Thacker Pass Project is sulfuric acid (Figure 4.7a). Based on project disclosures [150, 181], we model an optimistic case of $88/t H_2SO_4 reflecting on-site production costs, a midpoint of $120/t reflecting anticipated increases in raw sulfur feedstock costs, and a conservative upper bound of $220/t corresponding to current spot market prices, yielding a total cost range of ∼ $8,100—$10,800/t Li_2CO_3. To partially mitigate this exposure, the Thacker Pass project includes an on-site sulfuric acid plant, which not only reduces both costs and price volatility relative to market procurement, but also generates additional electricity due to the exothermic nature of that production process. This plant was not explicitly modeled in this simplified generic spreadsheet. Soda ash

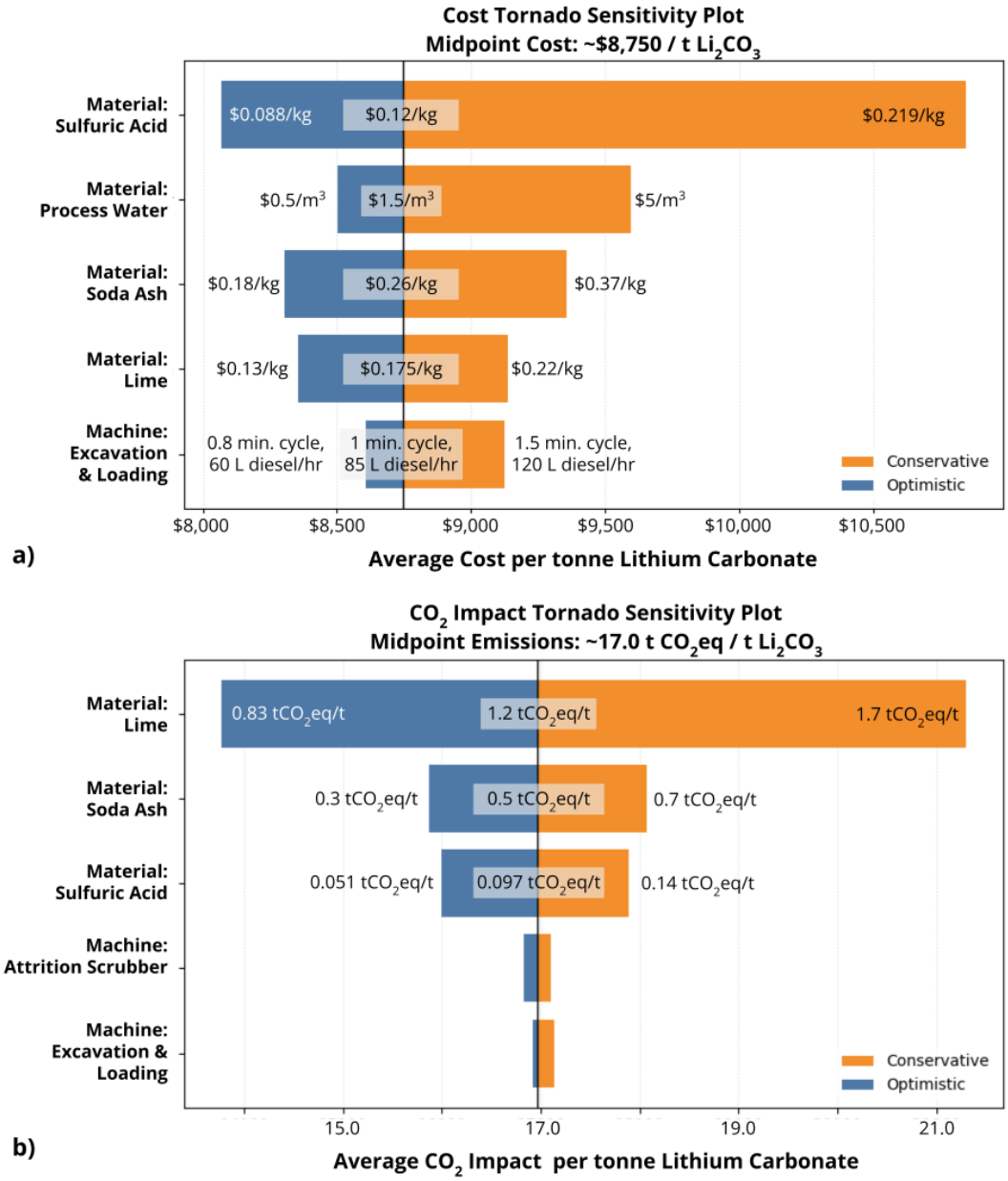


Figure 4.7. Tornado sensitivity plots for operating costs (a) and carbon dioxide emissions (b) for the Thacker Pass Project.

The top five inputs with the greatest influence on each outcome are shown. The central vertical line indicates the midpoint estimate (~8,750/t Li_2CO_3 and ~17.0 t CO_2eq/t Li_2CO_3, respectively). Bars extend left and right in optimistic and conservative using estimates of input costs or impact factors for materials and operating designs for machine block models, respectively. Machine inputs represent design scenarios rather than continuous parameter variation, with labels representing the most significant design variation relating to the measured output; machine labels represent actual use or throughput-adjusted consumption at each scenario. Inputs whose scaled variation produced limited cost or emissions impact are not labeled.

and lime introduce secondary cost uncertainty of similar magnitude to one another, though both are smaller in range than sulfuric acid.

Of particular note is the influence of process water costs, which represent the second largest cost driver overall. Process water in Nevada is relatively expensive compared to other mining regions, and our assumed range of \$0.50—\$5.00/m^3 produces meaningful cost variation at the volumetric consumption implied by the generic flowsheet. Critically, the generic flowsheet does not yet track or recycle water, which likely overstates consumption relative to the disclosed design: our model reports 16.2 million m^3/yr at 66,000 t/yr production, compared to 16.5 million m^3/yr at 160,000 t/yr in the Phase 4—5 disclosure [150], a factor of approximately 2.4× higher on a per-unit basis. Given local opposition to the project's water usage [203], explicit modeling of water recycling loops represents both a technical priority and a necessary refinement before drawing conclusions about Thacker Pass's water intensity relative to the disclosed design.

From an emissions perspective, lime overwhelmingly dominates emissions uncertainties, with a range between ∼13.8–21.3 t CO_2eq/t Li_2CO_3 (Figure 4.7b). Because approximately two-thirds of lime's embodied emissions are process emissions chemically unavoidable from calcination [204], reducing the clay pathway's emissions exposure to lime requires either substituting lime in the impurity removal step with process innovation (e.g. the magnesium recycling process described above) or sourcing lime from producers deploying carbon capture, rather than simply decarbonizing grid electricity. This makes the clay pathway's emissions trajectory dependent on upstream industrial decarbonization decisions in the lime sector in a way that is structurally distinct from the electricity-dominated emissions profile of the brine pathway.

Several other modeling simplifications are also worthy of note as limitations to this study. Water recycling is not currently modeled, particularly from tailings handling and lithium carbonate production: the tailings model uses generalized dewatering cost curves rather than the project-specific lined facility described in the disclosure, which is designed to capture and recycle process water, and the lithium carbonate precipitation step likely overstates water losses relative to a design with explicit recycling loops. Furthermore, ore is reported to be slurried on-site prior to transport to the processing facility in the latest disclosure [150]; we instead model slurrying after the Crushing step, in the Attrition Scrubbing step. This omission likely understates both transport costs and the water balance at the mine face.

Taken together, the generic flowsheet used here is conservative relative to the disclosed Thacker Pass design in both cost and emissions, primarily due to the absence of the magnesium crystallization route and on-site sulfuric acid production. The choice to incorporate these production steps into the Thacker Pass-specific flowsheet appears to be beneficial not

only in terms of economics, but also carbon emission externalities. Despite these limitations, the modeled results suggest that Thacker Pass remains broadly competitive with Chinese lepidolite production even under conservative assumptions — a finding that is likely to strengthen as the model is refined toward the project-specific design.

4.4.3 Results for the Jianxiawo Project

The Jianxiawo lepidolite project, located in Yichun, Jiangxi Province, is operated by Contemporary Amperex Technology Co. Limited (CATL), currently the world's largest battery manufacturer. As China's largest lithium source, Jianxiawo provides around 3% of global supply and 14% of China's supply [145, 164, 182], making it the single most consequential source of marginal lithium supply on the global cost curve.

Lepidolite is a fluorine-rich lithium mica mineral with a low bulk lithium grade, with Jiangxi lepidolite ore at approximately 0.27—0.4% Li_2O [163], or roughly 1000 to 1850 PPM Li. Lepidolite's complex mineralogy requires high-temperature sulfate roasting to liberate the lithium. The high fluorine content in lepidolite can also generate hydrofluoric acid as a byproduct during roasting, necessitating additional reagents (e.g. lime) and generating hazardous waste streams that contribute to both costs and environmental liabilities [180, 205]. But this low grade is precisely what has allowed lepidolite projects to rapidly expand in recent years: mines operating under local kaolinite permits were not required to register as lithium mines as long as stated ore grades remained below 0.4% Li_2O[206]. Our model deliberately assumes the upper bound of this range (0.4% Li_2O), to represent the most favorable ore grade at which these mines could operate under the kaolinite permit threshold, and therefore a conservative estimate of lepidolite's cost disadvantage relative to US sources.

CATL's vertical integration, which spans lithium mining through cell manufacturing, is a critical context for interpreting Jianxiawo's economics: the mine has continued operating during periods when lepidolite production costs exceeded spot lithium prices because downstream refining, cathode active material, and battery segments remained profitable [162]. Yet the decision to idle the mine between September 2024 and February 2025, despite this integrated structure[179], suggests that lepidolite's standalone production costs exceeded the internal transfer price at which CATL's downstream operations could profitably absorb the material — providing indirect evidence that the operation is genuinely near marginal. The standalone costs modeled here can therefore be interpreted as a meaningful benchmark for the cost at which lepidolite production enters and exits the global supply curve.

The step-level cost breakdown (Figure 4.8a) is defined by a single dominant feature: the sulfate roasting step accounts for approximately \$5,000/t Li_2CO_3, comprising material costs (primarily sodium sulfate), utility costs from high-temperature kiln operation (primarily

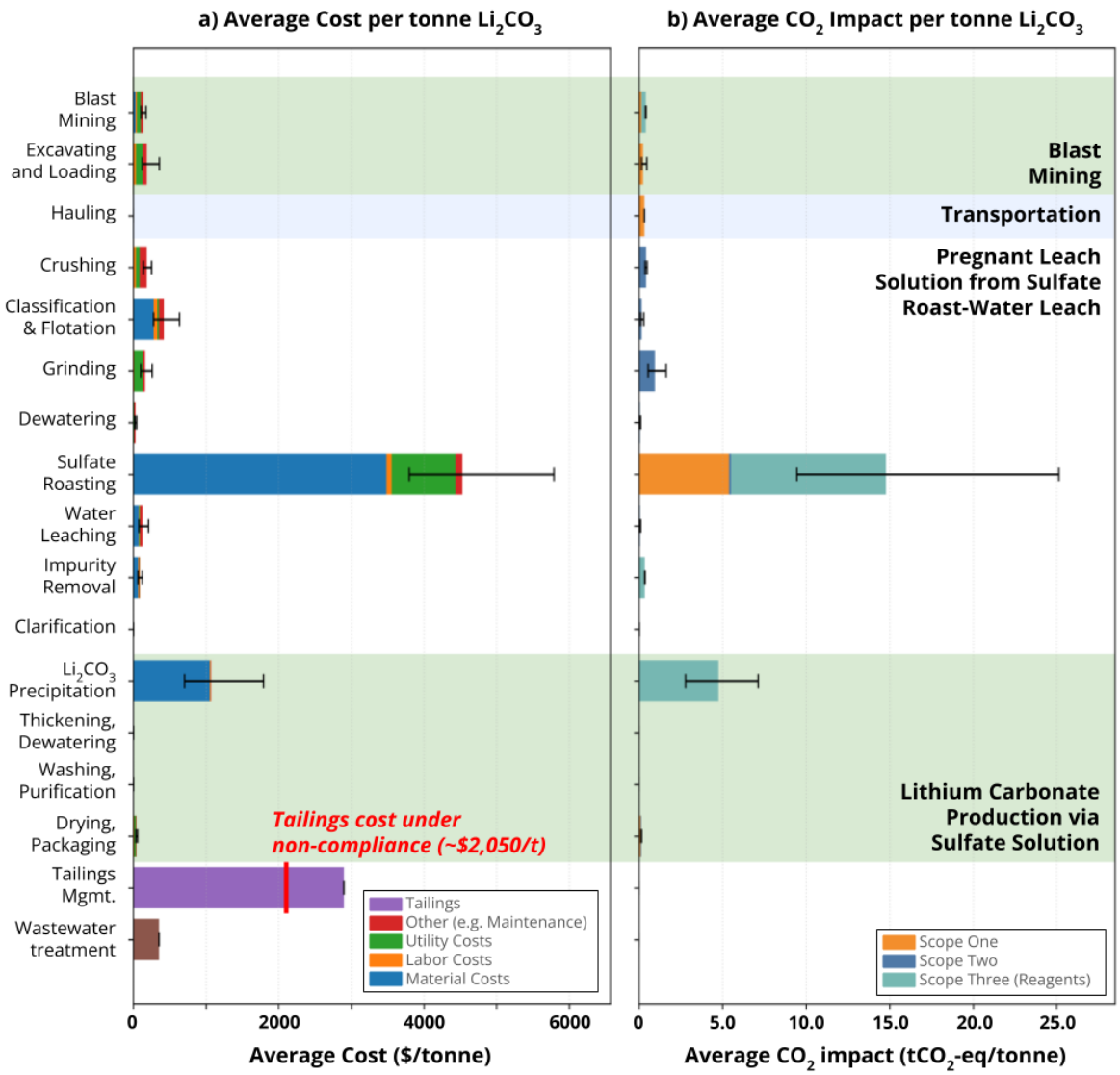


Figure 4.8. Costs(a) and emissions(b) for each step of the Jianxiawo Project. Error bars represent a sensitivity envelope formed by simultaneously applying conservative and optimistic values for all machine design parameters and material input costs that carry defined scenario variation; they do not reflect variation in process chemistry, reagent stoichiometry, or steps for which no scenario variation is defined (e.g., tailings management). As a result, the envelope is wider for steps with more varied parameters and narrower or absent for steps with few or none. Error bars should therefore be interpreted as indicative of the dominant sources of operating uncertainty rather than as statistical confidence intervals. The red vertical marker on the tailings bar indicates the modeled tailings cost under a hazardous waste non-compliance scenario (see Section 4.4.3).

natural gas combustion), and maintenance costs for the capital intensive, corrosion-prone roasting train, altogether representing nearly half of total modeled costs. The second largest cost is tailings disposal at ~$2,900/t, reflecting the classification of impurity sludge as hazardous waste under Chinese regulations and its associated secure landfill disposal costs [207]. Outside of these two major categories, lithium carbonate precipitation contributes a further ~$900/t in material costs (soda ash), while all other steps, including mining, are comparatively minor.

To assess the sensitivity of this cost to regulatory compliance, we modeled a scenario in which impurity sludge is handled as non-hazardous waste, i.e. non-compliance with Chinese hazardous waste classification requirements. Under this scenario, tailings disposal costs fall to approximately $2,050/t, a reduction of roughly one-third. Thus, even under non-compliance, tailings therefore remain a substantial cost driver, because the flotation and classification tailings streams are generated at very large volumes relative to product output, in spite of low direct unit disposal costs. This tailings cost is notable in the context of the prevailing assumption that Chinese environmental regulations permit structurally lower production costs than Western jurisdictions: rather than conferring a cost advantage, hazardous waste compliance at Jianxiawo adds a cost burden largely absent from the US projects modeled here. If the cost of tailings were somehow eliminated entirely, Jianxiawo's costs could possibly undercut Thacker Pass's costs, even under the latter's project-specific magnesium recycling and on-site sulfuric acid route, but the fundamentals of processing low-grade ores and the waste materials set a floor that no regulatory environment can fully offset.

The emissions breakdown (Figure 4.8b) is similarly dominated by the roasting step at ~14.8 t CO_2eq/t Li_2CO_3, but with an important structural distinction from the Thacker Pass pathway: roasting emissions are divided roughly equally between Scope 1 direct combustion emissions from high-temperature kiln operation and Scope 3 embodied emissions from sodium sulfate and lime reagents [205, 208]. This means that even full decarbonization of the natural gas combustion component would leave roughly half of roasting emissions unaddressed, and that emissions reductions require both fuel switching and upstream reagent supply chain decarbonization simultaneously. Outside of the crucial sulfate roasting step, lithium carbonate precipitation is the only other notable emissions contributor at ~4.8 t CO_2eq/t, driven by the embodied emissions of soda ash. It is important to note that this step likely has a higher environmental impact in China than in the United States, given that the soda ash used in this step is likely produced from the heat-intensive Solvay process (~0.7-1.2 t CO_2eq/t Na_2CO_3) [106, 195, 200]. All other steps have much smaller impacts in comparison.

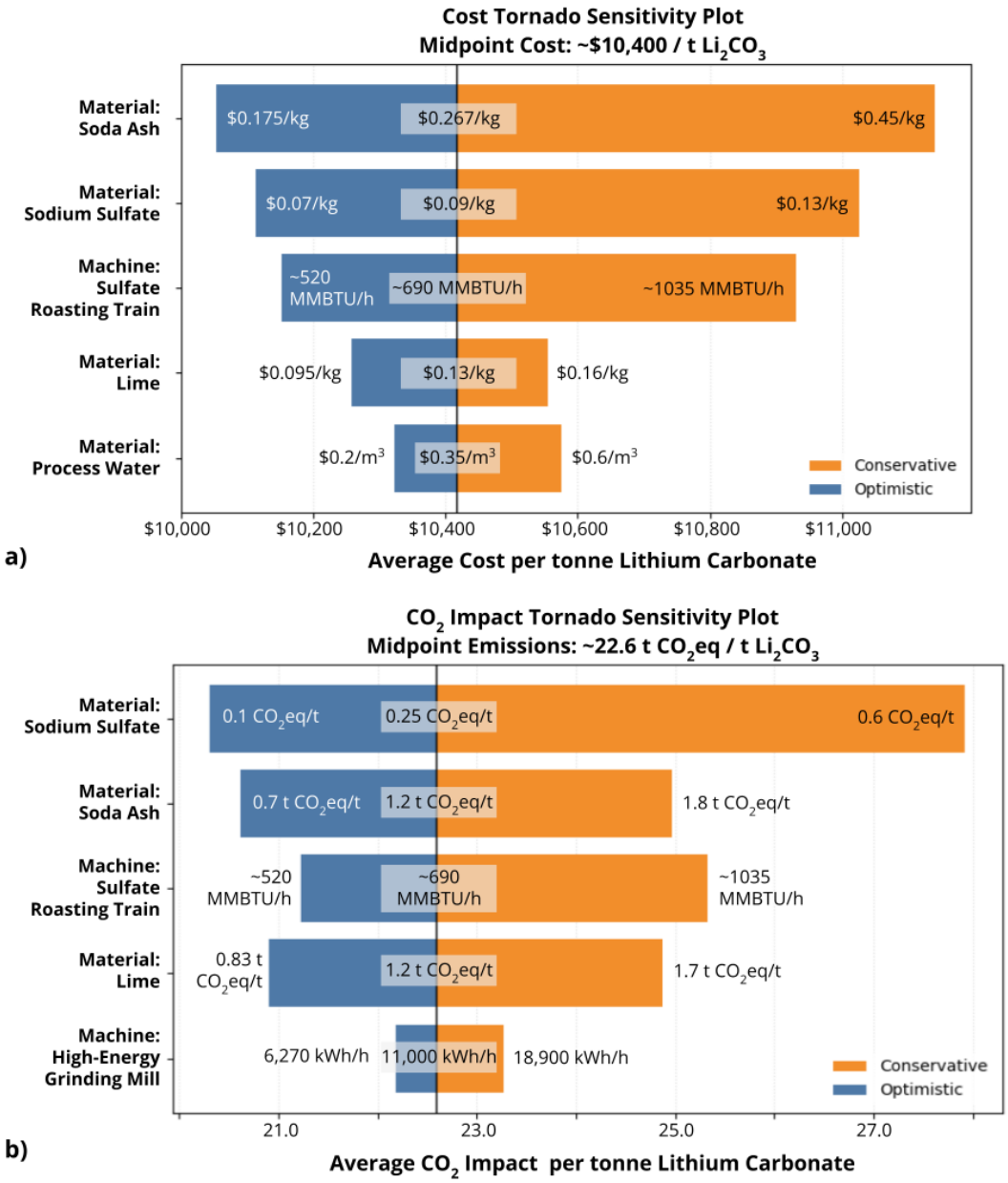


Figure 4.9. Tornado sensitivity plots for operating costs (a) and carbon dioxide emissions (b) for the Jianxiawo Project.

The top five inputs with the greatest influence on each outcome are shown. The central vertical line indicates the midpoint estimate (~10,400/t Li_2CO_3 and ~22.6 $t\,CO_2eq/t\ Li_2CO_3$, respectively). Bars extend left and right in optimistic and conservative using estimates of input costs or impact factors for materials and operating designs for machine block models, respectively. Machine inputs represent design scenarios rather than continuous parameter variation, with labels representing the most significant design variation relating to the measured output; machine labels represent actual use or throughput-adjusted consumption at each scenario. Inputs whose scaled variation produced limited cost or emissions impact are not labeled.

The cost sensitivity for the Jianxiawo lepidolite pathway (Figure 4.9a) is shaped by three large contributing variables. Soda ash price produces the widest range at approximately \$1,100/t, reflecting both the large stoichiometric requirement of soda ash for lithium carbonate precipitation and the wide spread between Chinese Solvay-process and US trona-derived soda ash prices. Sodium sulfate price and the sulfate roasting train design (the latter driven primarily by varying combustion efficiency assumptions in the roasting process) follow at approximately \$900/t and \$800/t, respectively. This is structurally different from Thacker Pass, where a single reagent dominated, and implies that cost reduction at Jianxiawo requires simultaneous optimization across the roasting circuit and multiple reagent supply chains rather than a single-lever intervention. Lime and process water contribute secondary uncertainty of smaller magnitudes.

In contrast to the cost sensitivity, where uncertainty is distributed roughly equally across three drivers, emissions uncertainty is dominated by a single factor: sodium sulfate embodied emissions span ∼20—28 t CO_2eq/t Li_2CO_3 (Figure 4.9b). This range is notably wider than the next three largest drivers: the roasting machine energy intensity, soda ash, and lime follow with roughly equal ranges of ∼4 t CO_2eq/t, consistent with the step-level findings discussed above.

Several limitations specific to the Jianxiawo model warrant elaboration. Most fundamentally, lepidolite processing is considerably less documented in the open literature than either brine or clay pathways: the sulfate roasting flowsheet is reconstructed from patents filed in China and lab-scale studies [180, 205, 209–213] rather than from engineering disclosures or feasibility studies comparable to those available for Thacker Pass and Silver Peak, which are not available given the lack of disclosure requirements for Chinese-listed mining companies. This data gap is itself a small finding, reflecting the limited public transparency of Chinese lithium production operations, but also means that process parameters such as reagent ratios, roasting temperatures, and conversion efficiencies carry greater uncertainty than for the US projects, and that the modeled flowsheet may not accurately represent CATL's proprietary process design. Similarly, the tailings model is relatively simple: impurity sludge is assigned a hazardous disposal cost based on published Chinese regulatory gate fees [207], but the volume and composition of sludge generated are derived from the generic flowsheet rather than measured plant data, introducing uncertainty in what is the second largest cost category. Additionally, the production volume modeled here, based on the most commonly reported production figure of 46,000 tonnes LCE, is less than half of the highest reported capacity of 111,000 t LCE [159, 182], and thus carries more uncertainty than projects owned by US or Canadian-listed companies, which are anchored to stock market-required engineering disclosures. Given the presence of economies of scale in the roasting circuit, cost and emis-

sions estimates are sensitive to this assumption in ways that are difficult to bound without plant-level data. Finally, as noted above, the standalone costs modeled here do not capture the cost-sharing or transfer-pricing arrangements that may apply within CATL's vertically integrated operations, and thus may not fully reflect the true 'marginal' lithium producer in China.

Our bottom-up model suggests that neither the cost nor the emissions profile of Jianxiawo is competitive with Silver Peak, and the overlap with Thacker Pass in the cost-emissions space is partially attributable to conservative generic flowsheet assumptions applied to the latter and the most favorable ore grade permissible under the kaolinite permit threshold (0.4% Li_2O) for the former, suggesting the true gap between US clay and Chinese lepidolite production is likely wider than our midpoint estimates indicate. At the same time, our process-based estimates fall below figures reported in the literature and industry reports [160, 161, 174], suggesting that prior high estimates likely reflect data scarcity and contextual limitations. As a result, lepidolite's cost and emissions disadvantage, while real and structural, may be somewhat less severe than what has been commonly reported. Both conclusions point in the same direction: more detailed, process-grounded modeling of Chinese lepidolite operations is needed to bound these estimates, and represents a priority for future work.

4.5 Limitations

Several important scope limitations of the present analysis should be addressed in future work, beyond those that were discussed for each individual project. First, mine closure and reclamation costs are not modeled here. These costs can be substantial, particularly for open-pit mining operations like Thacker Pass and Jianxiawo, and their exclusion likely understates the full lifecycle cost of all three projects — though the relative magnitude of this omission may differ across jurisdictions given differences in reclamation requirements between the United States and China.

Second, this analysis reports production costs at the mine gate and does not model cost, insurance, and freight (CIF) delivery costs to end markets. Chinese producers may retain a competitive advantage in delivered cost to Asian battery manufacturers even where their production costs are higher, given proximity to demand centers and established logistics infrastructure. US producers serving domestic battery manufacturers could benefit from localized supply chains, but establishing these logistics channels from rural Nevada to battery manufacturing plants primarily located in the US automotive corridor [73] may be more costly in comparison to China. A full delivered-cost analysis, incorporating transportation,

tariffs, and insurance, would provide a more complete picture of competitive positioning under different trade policy scenarios.

Further, water consumption is an important dimension that has only been partially addressed in this analysis. While a water recycling loop was incorporated into the model for Silver Peak, where solar evaporation dominates water consumption, water flows are not yet explicitly tracked or recycled in the Thacker Pass or Jianxiawo models. Explicit water balance modeling for all three projects is a priority refinement, both for accuracy and because water usage is increasingly a determinant of the local social license to operate, especially in water-stressed regions such as Nevada [143, 214].

Additionally, tailings and residue management costs are modeled using a simplified piecewise empirical dewatering cost function, parameterized by solids fraction, combined with project-specific disposal routes [150, 178] and gate fees for hazardous and non-hazardous waste [207]. While the dominant cost drivers are directly observed, particularly the hazardous disposal burden at Jianxiawo, which represents the paper's largest single finding on environmental compliance costs, some parameters within the model remain as order-of-magnitude placeholders, including on-site tailings facility operating costs and wastewater treatment costs. These are unlikely to affect the paper's comparative conclusions materially, given that disposal costs at the US projects are small relative to reagent costs, but represent a priority refinement for site-specific validation. More substantively, the environmental impacts of waste disposal are not currently modeled. For the ore-based pathways, where large volumes of leach residue and impurity sludge are generated and need to be dewatered, this omission likely understates total Scope 1 and 2 emissions modestly, though the dominant emissions drivers identified in the results (reagent embodied emissions and fuel combustion) are unaffected by this gap.

Beyond these refinements to existing models, several extensions would broaden the contribution of this work. The three projects modeled here are representative but not comprehensive: several other clay projects exist in the United States at various stages of development [142], and other lepidolite and spodumene operations in China could be modeled. In particular, the brine-direct lithium extraction (DLE) pathway, with projects under development in both the United States and China, appears unlikely to be marginal at current prices, but would be important for characterizing the full domestic cost curves of both countries, and understanding how the competitive landscape shifts under higher-price scenarios. Altogether, these additions would enable a richer analysis of how cost curves shift under different demand growth scenarios and how individual projects enter and exit production as prices move, providing an empirical foundation for a more complete treatment of supply curve dynamics.

All of these extensions described above, like this current analysis, are static in nature, characterizing production costs and emissions under a given set of assumptions and uncertainties. However, with project-level cost curves established for a broader set of pathways, they can serve as inputs to dynamic models that can capture firm- and market-level responses to policy interventions, technological learning, and shifting market conditions. The progression from static project baselines to dynamic supply curve modeling is therefore a natural next step, and the framework developed here is designed with that extension in mind.

4.6 Conclusions, Policy Implications, Future Work

This study developed bottom-up technoeconomic-life cycle assessment models for three lithium production projects representing distinct resource-pathway combinations: Silver Peak (evaporative brine, United States), Thacker Pass (sulfuric acid-leach clay, United States), and Jianxiawo (sulfate roasting lepidolite, China). The results demonstrate that the cost and emissions profiles of lithium production are not separable from the resource from which it is extracted: the thermochemical and hydrometallurgical requirements of each pathway create structural differences in production costs, emissions intensities, and sensitivity to price volatility that are challenging to overcome with project-specific optimizations. Under central assumptions, both US projects produce lithium carbonate at a lower cost and life cycle emissions than the Chinese lepidolite operation that appears to set the marginal price on the global supply curve, with Silver Peak being distinctly favorable in both dimensions, and Thacker Pass remaining broadly competitive even under conservative generic flowsheet assumptions. Crucially, while Silver Peak has limited ability to expand due to limited land and water rights, the opening of the Thacker Pass project, which when operating at maximum production capacity could supply 10% of current global lithium demand [145], could establish the United States as a major producer of lithium as demand continues to grow.

The United States holds some structural natural resource advantages relevant to lithium production costs and emissions, though these should not be overstated given China's extensive and low-cost chemical processing industry. One example is soda ash: US producers source it from mined trona in Wyoming's Green River Basin, the world's largest natural soda ash deposit [145], at substantially lower cost and emissions intensity than the Solvay process predominant in China [106, 195, 200], conferring a structural advantage across all three pathways modeled here. More broadly, the abundant solar and wind resources of the arid Southwest where US lithium deposits are concentrated, and relatively low natural gas prices represent potential further advantages as energy costs become more significant at

larger production scales.

On the other hand, a common assumption in supply chain security discourse is that Chinese production benefits from lower environmental regulatory costs. Our results complicate this narrative: hazardous waste disposal requirements for lepidolite roasting residues contribute approximately \$2,900/t Li_2CO_3 (more than 25%) to Jianxiawo's modeled costs — a burden largely absent from the US projects, which remains notable even under a hypothetical non-compliance scenario in which hazardous tailings were treated as non-hazardous (\$2,050/t Li_2CO_3). Furthermore, there is evidence that Chinese environmental compliance costs for lepidolite operations are likely to increase rather than decrease: as noted in Section 4.4.3, many lepidolite mining permits in Jiangxi Province were originally issued as kaolinite permits, and ongoing regulatory tightening around this permitting gap [179, 206] could raise operating costs and restrict capacity in ways not captured in current cost curve estimates.

These project-level baselines also could provide the foundation for a systematic analysis of policy interventions. Building on the models developed here, future work could analyze the relative competitiveness of each pathway under existing policies such as including production tax credits under the IRA and OBBBA, Made in China 2025 industrial subsidies, and Chinese export restrictions, or prospective interventions such as a carbon border adjustment mechanism (CBAM), currently being implemented in Europe. Such analyses would allow policymakers to identify which interventions most efficiently close the competitiveness gap between US and Chinese production, and under what conditions domestic lithium supply development becomes self-sustaining without ongoing policy support.

More broadly, this framework for integrated process-level technoeconomic and life cycle emissions assessment across geographic production pathways does not need to be specific to lithium. While the specific projects and economic landscape analyzed here reflect a particular moment in the geopolitics of critical minerals, the framework itself is material-agnostic: wherever production costs and environmental outcomes are jointly shaped by resource type and production location, the same approach could apply. Applied to other critical mineral systems, this approach can contribute to a more rigorous and consistent basis for evaluating supply chain security, environmental performance, and the effectiveness of industrial policy across the materials and manufacturing systems underpinning the energy transition.

Chapter 5

Conclusion

This dissertation reveals how supply chain vulnerability, production costs, and environmental performance differ across countries, technology choices, and production processes in the supply chains for electric vehicle batteries. Pathway-aware analysis can reveal structural advantages, such as the implicit advantage of lithium iron phosphate (LFP) batteries across many dimensions, that aggregate analyses may not resolve. In practical terms, current US policy incentives are weighted toward downstream battery manufacturing, leaving upstream supply chain stages – where vulnerabilities are concentrated – with comparatively modest support. This is unlikely to shift investment decisions, particularly where domestic production already appears competitive, as Chapter 4 suggests for US lithium. Recent removal of significant federal EV demand incentives has further left upstream development without indirect demand signals, suggesting that more direct demand support for specific critical mineral production could be more effective.

Underlying these policy implications are three structural pathway-aware features identified in this dissertation, each resting on foundations that hold regardless of the specific case studies through which they were identified. Two follow from the chemical makeup of cathode chemistries and the resource mineralogy-process chemistry combination, respectively, that constrain outcomes regardless of market conditions or operational conditions. The third follows from policy design and statutory interpretation, robust to assumptions about scale and cost, but contingent on the continued use of these incentive structures.

First, LFP batteries require only lithium as a critical mineral in the cathode, compared to lithium, nickel, cobalt, and manganese or aluminum for nickel-based chemistries. While Chapter 2 found LFP's base-case vulnerability to a disruption in China was higher than NMC's, due to current concentration of cathode production, LFP's vulnerability is concentrated in a single critical-mineral supply chain rather than distributed across four, narrowing the scope of action required to reduce it. In Chapter 3, this same narrower footprint allows LFP batteries to qualify for Inflation Reduction Act (IRA) credits by shifting that smaller set of supply chains. Combined with Chapter 4's finding of favorable economics for US lithium

production, LFP benefits from structural advantages at multiple supply chain stages.

Second, the aforementioned differences in lithium cost and emissions across US and Chinese production are driven by the mineralogy of each resource and the processing chemistry required to liberate lithium. Low-grade fluorine-rich lepidolite found in China requires high-temperature sulfate roasting and produces hazardous residues, whereas brine and clay pathways found in the US avoid both (Chapter 4). Because these are physical features of the resource and processing pathway, they are robust to process engineering optimization or country-level factors, such as lower Chinese labor costs or more stringent US environmental regulation. Project-specific innovations, such as on-site sulfuric acid production or mineralogy-specific reagent recycling, could improve individual projects within a pathway but do not appear to change the ranking across pathways.

Third, in Chapter 3, the IRA's 45X Advanced Manufacturing Credit allocated \$45/kWh to downstream cell and module production, compared to upstream critical minerals and electroactive materials credits structured as 10% of production costs, translating in our model to roughly \$5-10/kWh. This design choice produces an order-of-magnitude gap between downstream and upstream credit values that is robust to plausible variation in production costs. Furthermore, the Foreign Entity of Concern (FEOC) requirements previously attached to the now-repealed 30D credits have been reapplied to 45X under stricter standards [215], so the FEOC framework continues to shape US battery supply chain qualification.

There are a number of limitations to the analyses in this dissertation, addressed in part within each chapter. We surface additional limitations that become visible across chapters, in light of the synthesis above. Which limitations matter most depends on which outcomes are at stake: the first set of limitations bears on how accurately the dissertation's findings reflect empirical reality, while the second and third bear on which questions the dissertation's framework can and cannot address.

First, the input data quality and process representations across these analyses can limit the accuracy of the dissertation's empirical claims. We mitigate this through uncertainty analyses targeted at the specific outcomes most at risk in each chapter: network flow optimization in Chapter 2 to bound the vulnerability index under assumptions about missing trade, and triangular sensitivity analyses (low-midpoint-high inputs) in Chapters 3 and 4 to bound point estimates against plausible parametric variation. Chapter 3 faces an additional limitation: calibrating US-China cost differences requires simplifying assumptions about how those differences distribute across input categories. We applied uniform fixed-percentage cost reductions across materials, labor, electricity, land, and equipment costs, given the goal of representing generic rather than firm-specific production in each country, since differential adjustments would have required firm-level data outside the chapter's scope.

Multiple reports place real-world US production costs above our modeled values and Chinese costs below them, suggesting the actual gap may be wider than modeled. The chapter's qualitative finding that IRA credits meaningfully close the US-China production cost gap holds across these divergences even when including Chinese national-level incentives, though this competitiveness depends on the underlying cost calibrations and assumptions about unit volumes, battery capacities, chemistry, and process maturity. Chapter 4 demonstrates a fully traceable variant of the methodological approach adapted in Chapter 3, building our own bottom-up integrated technoeconomic-life cycle assessment framework in which key equipment is modeled explicitly and each cost and emissions output can be traced to location- and process-specific physical and economic parameters, producing results that are largely consistent with reported literature and empirical values. Improvements in data quality, such as more comprehensive trade statistics and plant-level process resolution, would further strengthen these analyses.

Second, the system-boundary choices made in each chapter bound the set of questions the studies in this dissertation can answer. Chapter 2 stops at cathode material manufacturing, omitting vulnerability at the battery cell and manufacturing stages while not considering the impact of recycling or circularity; Chapter 3 focuses on the battery pack and omits recycling and vehicle-level costs; and Chapter 4 stops at the mine gate and represents some unit operations at lower engineering resolution than others. Expanding boundaries to include circularity, delivered cost, and non-GHG externalities, along with deepening engineering representation of specific unit operations and integrating across the analyses themselves (for example, applying the policy interventions from Chapter 3 to the production pathways in Chapter 4) remains for future work.

Third, the static framing of these analyses limits forward-looking and dynamic analyzes, particularly for policy questions that depend on actor and market responses, as the work provides descriptions of systems at a point in time (cathode supply chains in 2020, battery manufacturing circa 2024, and lithium production circa 2026) without modeling how they could evolve or respond. Dynamic elements such as learning-by-doing, capacity build-out timelines, and policy evolution within analysis windows are not included, nor are market equilibrium, strategic interactions, or supply-demand feedback. But, the models in this dissertation can instead serve as baselines for dynamic and behavioral extensions across system dynamics [216], partial equilibrium [217, 218], general equilibrium [219, 220], and game-theoretic or agent-based [221, 222] approaches, each of which benefits from pathway-resolved, empirically-grounded representations.

Pathway-aware analysis of the kind developed in this dissertation is a starting point, not an endpoint. The frameworks developed here contribute in two directions: the combined

material flow analysis and vulnerability index can be applied to any supply chain with sufficient trade and production data, and the bottom-up technoeconomic-life cycle assessment framework extends to other critical minerals and their production systems. These pathway-based engineering analyses, coupled with dynamic and behavioral modeling, offers a way to align policy with the material and chemical realities of the systems it is trying to shape.

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Appendix A
Supplementary Text for Chapters 1 and 2

A.1 Further Context for Choices in Electric Vehicle Battery Chemistry

A.1.1 Battery Materials Supply Chain Structure (Theoretical and Simplified), and Current Electric Vehicle Manufacturer Decision Making in these Supply Chains

This supplementary section contains two figures that describe the theoretical and modeled supply chains for electric vehicle battery materials, respectively. The primary difference is the removal of stockpiling and manufacturing losses, which are typically not explicitly considered in material flow analysis studies [28, 30, 43, 58, 223]. Instead, we note that any stockpiling or manufacturing losses are incorporated into our uncertainty bounds when we split our refining and manufacturing into battery-related and non-battery-related refining and manufacturing, respectively. This split is identified based on our trade code aggregations as described in Appendix A.3.3. While there may be some lag in production processes, such as lithium brines requiring up to a year to process via evaporation [122], USGS production data is reported in terms of final production quantities (e.g. after lithium concentration), as described in Appendix A.3.2. Using this model, we then map current electric vehicle manufacturer decision making onto the modeled supply chain.

By definition, all companies producing their own battery cells, either in-house or through joint ventures or partnerships, are actively deciding the battery chemistries they have in their vehicle batteries, and thus where they source their battery cathode material from and the risks they might face due to global trade. Additionally, some companies (e.g. Ford [226], GM [57], Tesla [227]) have signed agreements with raw materials producers to supply their joint ventures and/or in-firm production, but may or may not directly control the exact sources of these materials, which may slightly muddy the clear delineations described in the table above. Some companies with lesser control may still choose to source batteries specifically because of the battery chemistry, but since they have no direct control, they also

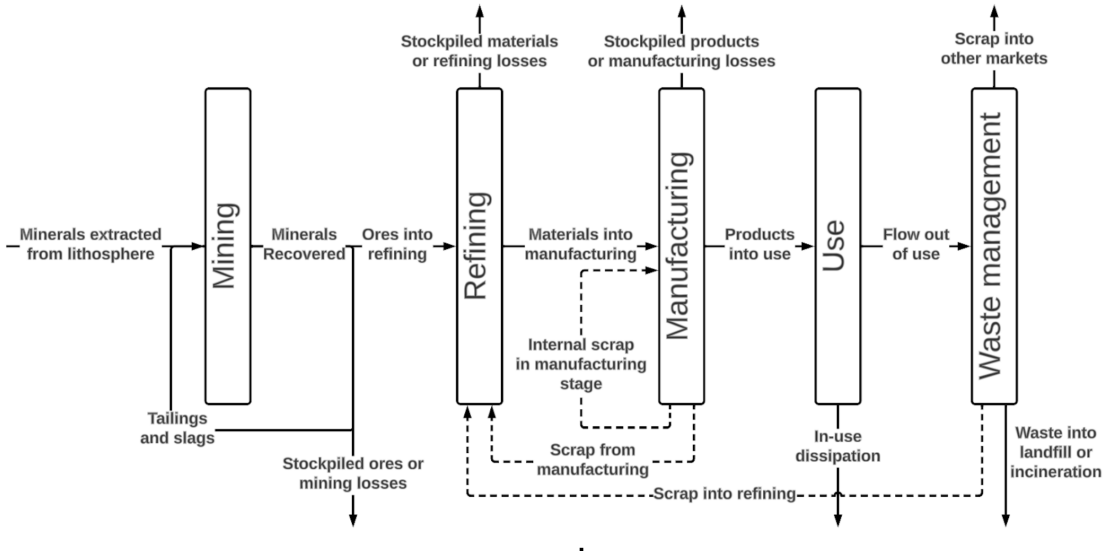


Figure A.1. The theoretical complete supply chain flows for a battery material.

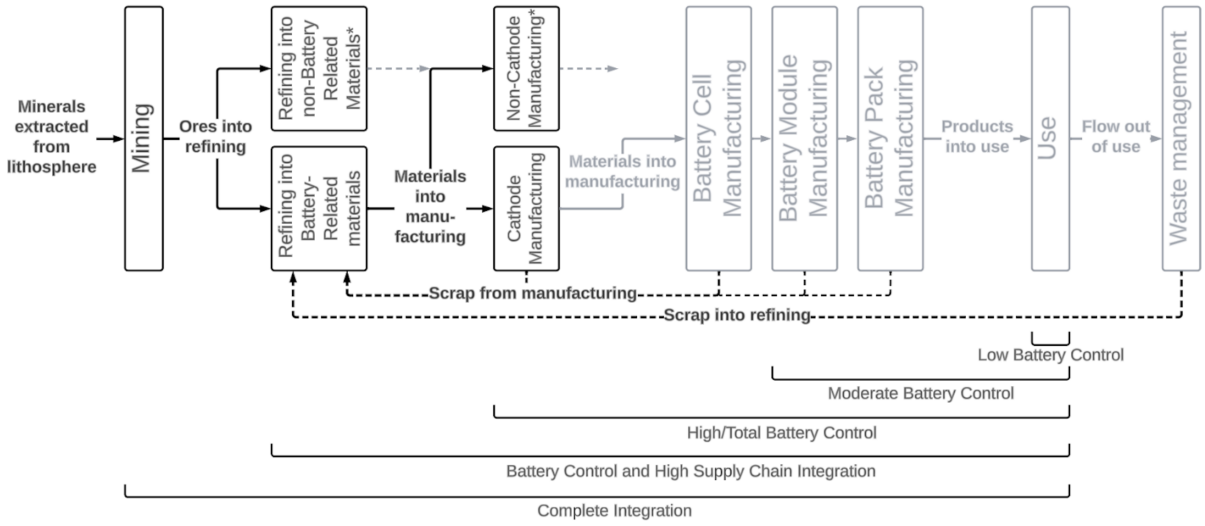


Figure A.2. The simplified supply chain flows for battery materials considered in this study (in solid or dashed black lines).

Manufacturer firm boundaries are described in Appendix Table A.1. Note the steps in light gray are not modeled in this analysis; and the portions of the supply chain relating to non-battery related materials(*) would include any stockpiling of materials or manufacturing losses.

have less direct control over their exposure to risk from the upstream material supply chains.

Table A.1. EV manufacturer decision making in their supply chain, circa 2020.

Arrows indicate general movement of firms towards increasing direct control over their supply chains. Adapted and augmented from [224, 225] and company press releases.

Supply Chain Control	**Examples**
Low Battery control: Cell, module, and pack outsourced to 3rd party supplier, usually a battery manufacturer (e.g. CATL)	Many EV startups: NIO, Lucid, Fisker, etc.
Moderate Battery control: Cell production outsourced. In-house module and pack design and manufacturing	BMW, Renault, Daimler, VW↓
High Battery control: Cell production through joint ventures/partnerships, and/or in-house module and pack design and manufacturing	Nissan, Mitsubishi, PSA (Stellantis), Toyota, Geely/Volvo, GM↓
Total Battery control: Firm manufacturing of cell, module, and pack design	BYD↓, Tesla↓, Ford↓
Battery control and high supply chain integration: Firm raw materials processing	
Battery control and complete integration: Integrated mining and raw materials extraction	

A.1.2 Mineral Criticality Discussion and Other Trade-offs in Battery Design Choices

We analyzed the current vulnerability of the supply chains for four primary battery cathode materials: lithium, nickel, cobalt, and manganese, which are often considered as the primary 'critical' elements for EV batteries. [8–10] However, other materials that are sometimes considered 'critical' can be used cathode materials and/or batteries writ large, such as graphite, aluminum, phosphorus, copper, and fluorine. We summarize various perspectives on mineral criticality in Appendix Table A.2, with a more general discussion of mineral criticality and how to measure it in Appendix A.2. The primary reason for this non- or less-critical categorization seems to be because the relative demand for such materials in batteries is small relative to the overall size of the market, and the number of countries that supply the material is high, as seen in Table A.3. In general, if demand for the aforementioned minerals – or new minerals due to development of novel chemistries – becomes significant and/or are included on critical mineral lists, especially those related to batteries or energy technologies, then more detailed analysis would be useful in better understanding the vulnerabilities present in their supply chains. While we focus on lithium, cobalt, nickel, and manganese due to a general understanding that these are the most critical materials at the current moment and the fact that there exists sufficient data for these materials, likely due to their status as the most critical minerals, other materials that are not yet considered as critical typically have data issues, making it challenging to apply the full analysis method we have suggested here.

Table A.2. A review of the criticality of electric vehicle battery cathode materials.

	Li	Co	Ni	Mn	Al	P	Fe	Notes
[18]Xu et al. 2022 [EV Li-Ion]	**Y**	**Y**	**Y**	**Y**	na	N	na	Specifically responds to Spears et al.'s critique about the criticality of phosphorus.[15] Based on their analysis, they explicitly state that phosphorus is not as critical a raw material from a known reserves perspective as other battery elements.
[19]Valero et al. 2021 [EV Li-Ion]	**Y**	**Y**	**Y**	**Y**	N	N*	N	*Phosphorus is only mentioned in the fact that phosphate rock is one of the six most common minerals throughout the 20th century and beginning of the 21st century.

Table A.2. Continued.

	Li	Co	Ni	Mn	Al	P	Fe	Notes (continued)
[20]Greenwood et al. 2021 [EV Li-Ion]	**Y**	**Y**	**Y**	**Y**	na	na	na	NMC specific analysis only.
[21]Ballinger et al. 2019 [EV Li-Ion]	**Y**	**Y**	**Y**	N	N	N	N	Describes 3 key elements which are significant supply risks.
[22]Sun et al. 2021 [Li-Ion]	**Y**	**Y**	**Y**	**Y**	na	na	na	Describes 15 commodities that include Li, Co, Ni, and Mn as the key LIB commodities that countries compete over.
[23]Scott and Ireland 2020 [Li-Ion]	**Y**	**Y**	**Y**	na	na	na	na	Identifies four key LIB (these three plus graphite) materials based on various indicators (see Table 3).
[24]Matos et al. 2020 [Li-Ion]	**Y**	**Y**	N	**Y**	na	na	na	Defines these three materials as priority raw materials.
[13]Wentker et al. 2019 [Li-Ion]	**Y**	**Y**	**Y**	**Y**	N	**Y**	N	Primarily evaluated elements of supply risk and environmental risk. Phosphorus was identified as critical because of its low substitutability and high global supply concentration, being more critical than Mn but less than Li, Co, and Ni.
[25]Sun et al. 2019 [Li-Ion]	**Y**	**Y**	**Y**	**Y**	na	na	na	Identify these four materials as being generally considered as the core elements for the LIBs.
[12]Helbig et al. 2018 [Li-Ion]	**Y**	**Y**	**Y**	**Y**	Y*	**Y**	**Y**	All are named as critical by construction; see Fig. 4 and 7. *Only aluminum is notably less critical in comparison.
[14]Olivetti et al. 2017 [Li-Ion]	**Y**	**Y**	**Y**	**Y**	na	na	na	Focuses on two primary measurements: static depletion index and production concentration of top three countries.
[26]Granholm 2021 [Energy Storage]	**Y**	**Y**	**Y**	**Y**	na	na	na	Specifically only calls out these four as critical.
[27]Hund et al. 2020 [Energy Storage]	**Y**	**Y**	**Y**	**Y**	**Y**	na	Y	See Table 3.1 “Energy Storage”.

Table A.2. Continued.

	Li	Co	Ni	Mn	Al	P	Fe	Notes (continued)
[28]Bauer et al. 2023 [Energy]	**Y**	**Y**	**Y**	N	**Y**	N	na	See Figure 5.1 in the paper. Phosphorus is assigned low supply risk (1–2) and low importance to energy (1) scores in both the short and medium terms, the lowest combined scores of any material analyzed with the methodology.
[29]USGS critical minerals list 2022	**Y**	**Y**	**Y**	**Y**	**Y**	N	na	The most general study, but this methodology is one of the standards for determining mineral criticality.

We take for example the mineral phosphorus, which in Table A.2 is considered to be critical in some contexts, like other minerals excluded from the main body of the paper. Given that the criticality of phosphorus in the electric vehicle lithium ion battery context is uncertain, we provide a Sankey diagram in Appendix Figure A.3, though data limitations prevent us from understanding country-level production of phosphoric acid that could be used as a precursor to LFP. We find demand for phosphorus for LFP production was 17,500 metric tons, as compared to the roughly 30 million metric tons of phosphorus mined in 2020 (223 Mt of phosphate rock) [228] - roughly 0.06% of total worldwide phosphorus demand. Even when narrowing down the phosphorus supply chain to phosphoric acid production, the primary precursor for the phosphorus used in LFP batteries, [229] this corresponds to roughly 0.075% of phosphoric acid demand. According to the USGS, [1] roughly 23.5 million tons of phosphate rock, or about 2.3 million metric tons of phosphorus were mined in the United States in 2020; it is clear the United States alone could supply the world's LFP-Phosphorus demand hundreds of times over. For further context, the USGS identified 71 billion metric tons of phosphate rock reserves (i.e. currently reasonably economically extractable material) around the world in 2020. While we recognize it is currently non-economical to convert most of these phosphorus resources (and indeed, nickel, cobalt, and manganese, to a certain extent) to the high level of quality and purity required for LFP batteries, in the vein of the Simon-Erlich wagers, we suggest that massive growth in market demand for such materials will encourage a shift in developing lower-grade resources for such applications and technological advancement to allow for such development. With the massive projected growth of all minerals due to the expansion of battery supply chains and the overall energy transition, we suggest that further efforts to gather better data and make such data available to researchers and analysts will be necessary to address future concerns.

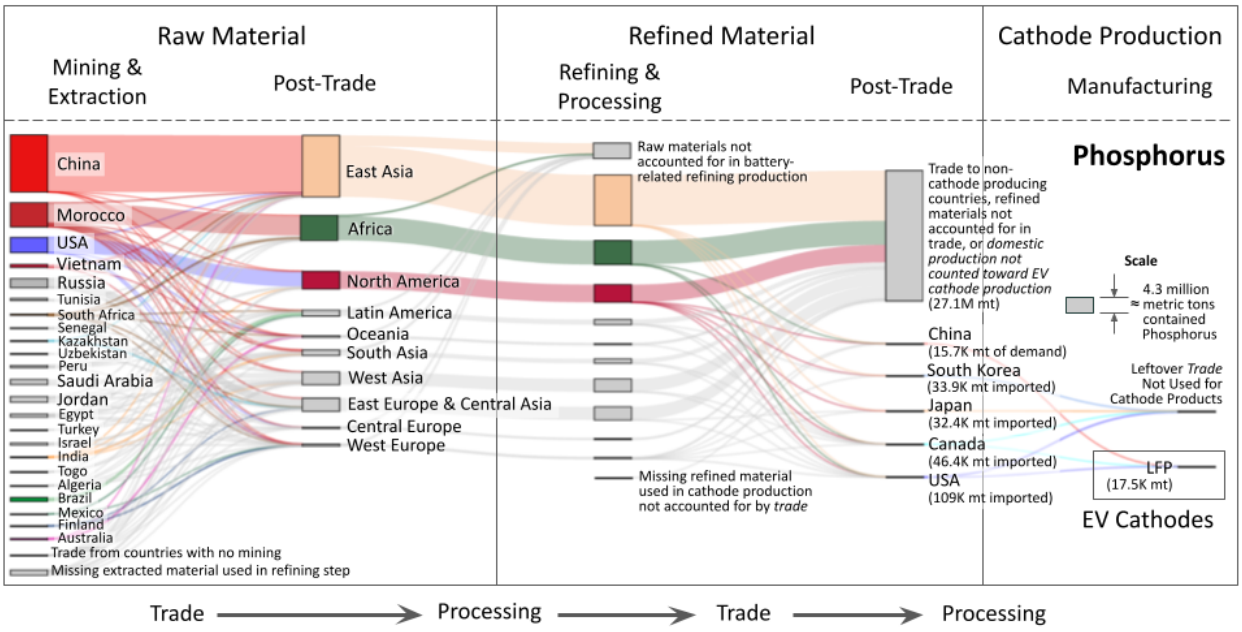


Figure A.3. A Sankey diagram for global flows of phosphorus that available data suggest are involved in battery material supply chains.
See Appendix A.3.2 for further details and data sources. Note the scale of production - approximately 28 million tons of phosphorus (contained phosphorus) was extracted in 2020.

We note Appendix Figure A.3 displays regional production data of phosphoric acid, the primary precursor used in LFP battery cathodes, as this level of aggregation was the most detailed that was publicly available. [71] As a result, we can only compare demand phosphorus used for LFP production to trade of phosphoric acid, and also cannot calculate a full vulnerability index in the style of the other battery materials considered in this article. If we were to assume all production of refined phosphorus and phosphoric in East Asia occurs in China (which is approximately true based on discussion with representatives of the International Fertilizer Association), and only look at observed import data of those materials (rather than including domestic production), we can apply the methodology and see the results found in Appendix Table A.4. We see that the measured vulnerability index values are not altogether that different from that of Lithium for LFP: slightly lower in the proportional case and going up to 100% in the 'known trade' pessimistic case. However, we cannot be certain of any measurements beyond the observed production of LFP in the cathode step, as we only observe inter-regional trade and thus cannot be sure if this is actually the vulnerability index for phosphorus and China.

Table A.3. Comparison of Production Geography and EV-related Material Demand, 2020 † Demand figure applies collectively to Aluminum (Alumina), Aluminum (Bauxite), and Aluminum (smelted). [1, 2].

Material	**Countries specifically listed in USGS Mineral Commodity Survey (MCS) [Number of countries]**	**Smallest amount produced by named country in MCS**	**Total Global Production**	**Material Demand for EVs**
Lithium	United States, Argentina, Australia, Brazil, Chile, China, Portugal, Zimbabwe [8]	348 metric tons (Portugal)* (*US production withheld)	82,500 metric tons	38,200 metric tons (46.3%)
Cobalt	United States, Australia, Canada, China, Dem. Rep. Congo, Cuba, Indonesia, Madagascar, Morocco, Papua New Guinea, Philippines, Russia [12]	600 metric tons (USA) (Second — Madagascar — 850 metric tons)	142,000 metric tons	87,800 metric tons (61.8%)
Nickel	United States, Australia, Brazil, Canada, China, Indonesia, France (New Caledonia), Philippines, Russia [9]	16,700 metric tons (USA) (Second — Brazil — 77,100 metric tons)	2,510,000 metric tons	145,800 metric tons (5.8%)
Manganese	Australia, Brazil, Burma, China, Côte d'Ivoire, Gabon, Georgia, Ghana, India, Kazakhstan, Malaysia, Mexico, South Africa, Ukraine, Vietnam [16]	121,000 metric tons (Vietnam)	18,900,000 metric tons	74,200 metric tons (0.3% of Mn demand)

Table A.3. Continued.

Material	**Countries specifically listed in USGS Mineral Commodity Survey (MCS)** [Number of countries]	**Smallest amount produced by named country in MCS**	**Total Global Production**	**Material Demand for EVs**
Aluminum (Alumina)	United States, Australia, Brazil, Canada, China, Germany, Guinea, India, Indonesia, Ireland, Jamaica, Kazakhstan, Russia, Saudi Arabia, Spain, Ukraine, United Arab Emirates, Vietnam [18]	439,000 metric dry tons (Guinea)	136,000,000 metric tons	2,730 metric tons (5.2e-04% of raw Al production and 4.2e-03% of smelted Al production)†
Aluminum (Bauxite)	United States, Australia, Brazil, China, Guinea, India, Indonesia, Jamaica, Kazakhstan, Russia, Saudi Arabia, Vietnam [12]	3,500,000 metric dry tons (Vietnam)	391,000,000 metric tons	
Aluminum (smelted)	United States, Australia, Bahrain, Canada, China, Iceland, India, Norway, Russia, United Arab Emirates [10]	860,000 metric tons (Iceland)	65,100,000 metric tons	
Phosphorus (Phosphate Rock)	United States, Algeria, Australia, Brazil, China, Egypt, Finland, India, Israel, Jordan, Kazakhstan, Mexico, Morocco, Peru, Russia, Saudi Arabia, Senegal, South Africa, Togo, Tunisia, Turkey, Uzbekistan, Vietnam [23]	577,000 metric tons (Mexico) ≈ 74,000 metric tons contained P	219,000,000 metric tons ≈ 28,000,000 metric tons contained P	17,500 metric tons contained P (0.06% of P demand)

Table A.4. China-based Vulnerability Index Calculations for lithium and phosphorus in the LFP supply chain.

*Indicates that this number is possibly a slight overestimate, as the middle stage of refining production only has regional data and we have assigned all production in "East Asia" to China.

	LFP – Li	**LFP – P***
Optimistic Case (Minimum)	89.9%	89.9%
Base Case (Proportional)	91.6%	90.5%*
Pessimistic Case A (Known trade only)	93.2%	100%*
Pessimistic Case B (Extreme bound)	100%	100%*

It is important to acknowledge that the risks and vulnerabilities associated with each of our four chosen critical minerals varies. In the context of Lithium, for example, concerns have arisen over water extraction in South America and indigenous land rights in North America. Cobalt has been associated with child labor and heavy environmental damage in the Democratic Republic of the Congo. [10, 230] Thus, while we simplify calculations of vulnerability to one dimension of supply availability, we recognize that tradeoffs exist in multiple dimensions across these four materials.

Lithium ion batteries themselves are composed of four primary components: a lithium compound cathode, graphitic carbon anode, liquid electrolyte, and polymer-based separator, as well as various other structural elements (e.g. copper and/or aluminum current collectors). There are some possible alternatives for electrolyzers, separators, and anode materials, but the diversity of choices in these are much fewer (e.g. anodes are currently made with graphite, with only some possibility for partial substitution with silicon), [13] so the opportunity for design alternatives is highly limited.

Few other battery cathode material alternatives exist for rechargeable batteries that have both sufficient energy density and power density at low enough cost, with the sodium-ion batteries being the only prominent non-lithium based cathode. Other traditional battery chemistries (e.g. Lead Acid, Zinc-Manganese Dioxide, Nickel Cadmium, Nickel Metal Hydride) are either non-rechargeable or not sufficiently energy dense. While relatively exotic chemistries like Lithium Metal, Lithium-Sulfur, and Lithium-Air batteries have been in various stages of development, they currently are not deployed for electric vehicles. [11, 13, 231] While Sodium Ion batteries have been developed for electric vehicles in China, [232] market penetration is currently low and the cathode materials often contain some mix of nickel, cobalt, and/or manganese, as well as other minerals such as titanium or copper, which does

not fully remove the risks discussed in this article.

Beyond the critical materials required for each chemistry, other trade-offs exist – LFP battery pack energy densities are anywhere from 50 to 85% of that of NMC battery packs [11, 233, 234] while being 105% to 85% of the cost of production. [9, 110] Thus, to achieve a battery with a certain amount of energy, differing amounts of critical materials would be required to build a battery of each type.

A.2 Literature Review

To answer our research question, we need to build a model that maps the battery material supply chains at a national level and measures interactions between trade flows and production capacity, while accounting for uncertainty and missing data. Several streams of literature have investigated relevant concepts and metrics, including literature on energy security, materials criticality, material flow analysis, input-output analysis, and supply chain disruption propagation. The relationship between our model and these literature streams is summarized in Appendix Table A.5.

Energy security debates have long focused on where energy resources are sourced, with definitions often made in terms of access to fuels or security of supply, largely in the context of oil and other fossil resources. [246, 247] Numerous studies of past and potential disruptions of oil supply and the threat of the 'oil weapon' have considered specific regions and countries. [235–238, 248] However, transitions to renewable and electricity-based systems have introduced new geographical dependencies that differ from those raised by fossil fuels. [242, 249, 250] The materials criticality literature was in part developed to address these nuances.

The literature on materials criticality attempts to assess the relative risks and vulnerabilities associated with the supply of minerals and materials. These assessments often aggregate geological, technological, and economic measurements with social, regulatory, and geopolitical indicators into overarching summary metrics. [41, 251, 252] These aggregated metrics have been applied across multiple battery supply chains and battery chemistries, particularly from the perspective of import dependence, rather than the relative impact of various countries' involvement in supply chains. For example, some studies [8, 33, 34] measure concentration of production with the Herfindahl-Hirschman Index and then weight them (using indices such as the World Governance Indicators, material demand, import reliance, etc.) to account for different levels of risk. For example, the first of these studies [8] aggregates four measures of risk for battery-related critical minerals in the context of various cathode material compositions, finding that in a simple arithmetic mean aggregation, LFP is the only battery chemistry with "measurably lower supply risk compared to the other battery types." While these metrics can capture a broad range of concerns, they may mask the dimensions that are important for ensuring supply availability throughout a specific product's entire supply chain, [40, 253] including obfuscating the vulnerability of risk caused by specific countries. In the case of battery materials, while we learn that lithium and cobalt tend to be 'more critical' than other battery minerals, we suggest that these measurements do not adequately describe the importance of specific countries and their trade inter-dependencies, especially in the context of battery material chemistry choices and their supply chains.

The material flow analysis literature maps and characterizes the production and flow of materials using international trade data and various sources of production data to better understand geographical relationships of supply chains. In the context of battery materials, some of this literature focuses on specific stages of the value chain, e.g. raw materials and mining, while others encompass all steps. Most of these trace specific materials at a global context, such as lithium, [28] cobalt, [29, 30] nickel, [31] and manganese. [32] Additional studies have considered all of the cathode materials in a regional context, [22] though there is little explicit inclusion of the cathode material production step in these material flows outside of one study. [28] One recent material flow analysis-adjacent study [19] evaluates the feasibility of meeting the recently passed IRA's goals for minimum critical mineral requirements across all battery materials for the United States, finding that achieving the market value-based target may be possible with NCA batteries but not necessarily for LFP or NMC batteries. We provide an analysis in the context of potential mass-based targets and identify a need for a global perspective. In general, while studies describing the state of supply chains are useful for describing the structure of material flows and supply chains, allowing for analysis of potential vulnerabilities, none of these studies quantitatively measure the interdependencies between specific technology choices and material mass requirements on specific countries.

Two types of studies have considered different ways to measure relationships between steps of supply chains between production geographies, though not necessarily in the specific definition of vulnerability we have used in this article. The input-output (I/O) literature measures flows of materials, emissions, and other quantities through interconnected industries, sometimes through the perspective of a specific material's or product's supply chain. These models can calculate measures of vulnerability through matrices of relationships between industries or steps of a supply chain, making the assumption that input proportions between each stage or step are fixed. [69] While input-output models are typically described in terms of one unit (e.g. dollars), a subset of this literature, known as a 'mixed-unit' input-output technique, have focused on combining I/O models with material flow analysis, using mass balance to measure physical flows of materials and sometimes energy or emissions.68 These studies track material flows in detail across steps of the supply chain, but they rely on proportionality constants and thus can only provide point-estimate snapshots of vulnerability on certain geographies. Furthermore, there has been little study of battery material supply chains with input-output analysis, with existing studies focusing on either just lithium-ion battery manufacturing [243] or in non-lithium-ion contexts. [244, 245]

Other studies have focused on 'supply chain disruption propagation' for certain battery materials, such as natural graphite, [37] cobalt ores, [38]lithium carbonate and lithium bat-

teries, [35] and nickel. [36] These studies calculate the 'avalanche size' of a disruption – the number of countries affected by the removal of supply or trade from one country – while assuming that disruptions occur as proportional losses of trade from the disrupted country that are propagated to other countries. Most of these studies underscore the significance of China as a critical node in the supply chain though these studies are limited to specific trade codes or materials. Furthermore, all of these studies except the study on nickel do not consider the fact that there are relationships between countries at multiple steps of the supply chain, and the nickel study uses a correlation instead of considering the actual physical quantities of materials that are converted from one step of the supply chain to the next. While this measure of vulnerability may be helpful in understanding linkages between countries, we suggest that the lack of analysis across multiple materials and the physical quantities of materials needed constitutes a gap in the literature.

As summarized in Appendix Table A.5, no existing models incorporate these six primary qualities, so we create a new model, leveraging prior work and bridging these gaps, to answer this question. We work backwards from lithium ion battery cathode material requirements to map the entire supply chain on a country-by-country basis, with awareness of the specific amount of constituent materials needed for each cathode material. We measure overall vulnerability on any given country, going beyond the proportional measurements found in the input-output and supply chain disruption propagation literatures by bounding the possible range of disruption on any given region of interest in the supply chain.

Table A.5. A summary of existing key literature on EV battery material supply chain vulnerabilities.

We recognize that studies with these methods and themes could be used in the context of EV battery materials, but in some cases have not been applied in such a manner.

Literature stream	**Use of countries as unit of geographic resolution**	**Explicit consideration of cathode material production**	**Quantifies flows of battery-related materials**	**Scope of studies includes multiple materials**	**Metrics of merit for quantifying relative importance of materials**	**Characterizes uncertainty**	**Units used**	**Example studies**
Energy security	Often	No (not in scope)	N/A	N/A	N/A	N/A	N/A	[235–238]
Materials criticality	Often	Small number of studies	No	Often	Yes, various indices	Yes	Mass, indices	[8, 33, 34, 239–242],
Material flow analysis	Often	Small number of studies	Yes	Unusual, usually one material	No, focus on mapping flows	Unusual, usually a 'snapshot'	$, mass	[26, 28–31, 43]
Input-output analysis	Sometimes, sometimes subnational or regional in scope	Could be used for this, but no related studies explicitly do	Yes, but with proportionality assumption	Sometimes, usually considers materials in one industry	No related studies	Often	$, mass	No Li-ion battery material studies, some battery-related work [243–245]
Supply chain disruption propagation	Often	Not explicitly found in these studies	Yes, but with proportionality assumption	No, focus on single materials	No, calculation of number of countries affected by disruption	Yes, by varying probabilities of disruption	Mass	[35–38]
This study	Yes	Yes	Yes	Yes	Yes, relative vulnerability	Yes, via bounding analysis on missing data	Mass	—

A.3 Full Methodology and Data

A.3.1 Methods

A.3.1.1 Data Aggregation

We model the supply chain for each chemistry by considering supply of each material for each battery chemistry, focusing on mining, refining, and cathode material production. Each country can produce, import, and export material at multiple steps in the supply chain, and thus the portion of end cathode material that involves a given country in the supply chain varies depending on the amount of end material being produced and the trade relationships between all countries.

While the supply chain for each material can be complex, we make a number of simplifying assumptions given the data available (see Appendix Figure A.1 and Appendix Figure A.2). We do not incorporate stages after battery cathode material manufacturing, but include the movement of scrap, with the caveat that little of it is currently involved in the electric vehicle battery supply chain (most nickel and manganese scrap is used in the iron and steel market, [22, 32, 60] while cobalt scrap is largely from superalloys [29, 61]). Additionally, we generally make the assumption that no firm-level buffers/national stockpiles of these materials exist or are minimal in size relative to trade and/or production.

We aim to represent the flow of materials via trade and processing as a two-dimensional network. We do this by assuming that countries with known production are the only countries with the technological production capability to produce materials at each step, and that no one else has this capability and thus must trade from these producers in order to get these materials.

We track five battery material supply chains: the lithium in LFP cathode materials, and the lithium, nickel, cobalt, and manganese in NMC cathode materials. For each step of the supply chain, we compute a mass balance:

$$D_{mi} + \sum_{j \in J} T_{mij} = S_{mi} + \sum_{j \in J} T_{mji} \qquad \forall m \in M, \forall i \in J \tag{A.1}$$

Supplementary Equation A.1. Total Demand (Demand + Exports) = Total Supply (Production + Imports), for all materials and countries.

where D_{mi} is the mass of material m used to make all potential products in country i, T_{mij} is the amount of material m traded from country i to country j, J is the set of all countries, and S_{mi} is the mass of material m produced in country i. This equation holds for each material $m \in M = \{\text{Li, Ni, Co, Mn}\}$ and each country $i \in J$.

Total trade between countries may involve some trade that is observable and some trade that is unobservable due to smuggling, unintentional misclassification, tariff avoidance, and other reasons [29, 39, 43]. We represent these as follows:

$$\sum_{j \in J} T_{mji} = \left(\sum_{j \in J_{\text{prod}}} T_{mji}^{OBS} - \sum_{j \in J \setminus J_{\text{prod}}} T_{mji}^{OBS} \right) + T_{mi}^{UNOBS} \qquad \forall m \in M,\ \forall i \in J \tag{A.2}$$

Supplementary Equation A.2. Total Imports = Imports from Producing Countries + Imports from Non-Producing Countries + Unobserved Imports, for all materials and countries.

where T_{mji}^{OBS} is the observed trade from country j to country i available in the data, J_m^{PROD} is the subset of countries that are known to produce material m, and T_{mi}^{UNOBS} is unobserved trade of material m to country i that is missing in the data. We compute the net value of T_{mi}^{UNOBS} (positive or negative) needed to satisfy Supplementary Equation A.1 for each material, country pair, effectively assuming that any mismatch between supply and demand in the data is explained by unobserved trade. The countries that export relevant materials but are not known to have production may obfuscate the true 'original source' of the material, assuming the countries listed with production capacity are the only ones that have the technological capability. This creates additional uncertainty.

$$D_{mi} = \sum_{p \in P} d_{pi} \cdot c_{pm} \tag{A.3}$$

Supplementary Equation A.3. Total material D demanded = Supply d of product p × material needed per product.

$$S_{mi} = \sum_{\alpha \in A} s_{\alpha i} \cdot c_{\alpha m} \tag{A.4}$$

Supplementary Equation A.4. Supply of contained material S = Supply s of compound supplied α × contained material per compound.

$$T_{mji} = \sum_{\beta \in B} t_{\beta ji} \cdot c_{\beta m} \tag{A.5}$$

Supplementary Equation A.5. Trade T between countries j and i = Trade t of items β × contained material per item.

Furthermore, we cannot directly observe the movement of material m in its pure state (e.g. contained nickel compared to nickel ores and concentrates, nickel mattes, nickel sulfates, unwrought nickel alloys, etc.). Assuming no losses in the production processes, we use

conversion factors c to convert from input materials α with supply $s_{\alpha i}$, imported goods β at amount $t_{\beta ji}$, and demand from produced products d_{ip} to their contained material amounts, S_{mi}, T_{mji}, and D_{mip}, as seen in Supplementary Equations A.3 through A.5. These conversion factors c are described below in the *Data Sources* section (Appendix A.3.2) and compiled with associated trade codes in Appendix A.3.3. Values for $t_{\beta ji}$ are associated with relevant trade codes in Appendix A.3.3.

As a result, combining all unobserved trade into one variable and trade from and to non-producing countries into another, recognizing that quantity can be positive or negative, we describe our total supply-demand balance in Supplementary Equation A.6:

$$\sum_{p \in P} d_{ip} \cdot c_{pm} = \sum_{\alpha \in A} s_{\alpha i} \cdot c_{\alpha m} + \sum_{\beta \in B} \left(\left(\sum_{j \in J_{\text{prod}}} t^{OBS}_{\beta ji} \right) - \left(\sum_{j \in J \setminus J_{\text{prod}}} t^{OBS}_{\beta ij} \right) \right) \cdot c_{\beta m} + T^{UNOBS}_{mi} \quad \text{(A.6)}$$

Supplementary Equation A.6. Total demand for each material = Material supply + (Material imported − Material exported) + Unobserved Trade.

We can measure the variables in Supplementary Equation A.6 in terms of demand for cathode materials that require refined minerals. These refined minerals in turn require raw minerals, so we have a two-step process, where the total supply of refined materials S_{mi} in the refining–cathode material flow analysis is a subset of the products considered in the total demand for refined materials D_{mip} in the mining–refining material flow analysis.

A.3.1.2 Vulnerability Analysis and Uncertainty Bounds

We calculate the total vulnerability of cathode material supply for a set of countries, which we designate as the countries of focus. Clearly, any cathode production in these countries would count towards vulnerability due to those countries. However, other countries that produce cathode material could potentially also be vulnerable to the countries of focus for refined materials, while the countries that supply those cathode material producing countries with refined materials could also be dependent on the countries of focus for raw materials. Taking into account this 'ripple effect' across multiple supply flows, we determine the quantity of cathode material dependent on our countries of focus across cathode material production, materials refining, and raw materials production.

To bound this analysis, we consider both uncertainty due to observability of trade between countries that goes to battery end-products, and uncertainty due to unobserved trade data: trade that should exist in order to produce battery products but otherwise is unobserved in the trade data. Using these, we measure the amount of vulnerability based on a country in four ways.

In a **proportional trade** case, we find the total amount of internal supply less exports, and the total amount of imports for each country with production capability at each step. With this data, we can calculate the percentage of supply at each step that depends on the countries of focus. By aggregating these percentages at each stage of production, we can calculate an overall percentage of the end cathode production that becomes unavailable. In this scenario, we assume any unobserved trade follows the same patterns as observed trade. This follows methodologies described in input-output and supply chain disruption propagation literature. [35–38, 69, 244, 254–257]

The other three vulnerability calculations use the concept of maximum flow [258] to calculate the minimum and maximum vulnerability on any set of focus countries. We translate our Sankey diagrams into network flows. Nodes of the network correspond to the vertical bars in each Sankey diagram at each of the five stages of the diagram – mining, mining post-trade, refining, refining post-trade, and cathode production, while connections in the network correspond to the flows in each Sankey diagram between the stages, representing either trade (internal or international) or material processing between each stage. Furthermore, we add a 'supersource' node where all mined and raw material originates from, with capacity equal to the amount of production claimed by each raw material producing country.

In a **maximum known trade** case, we try to optimize the flow of material across our network, such that as much of it passes through the countries of focus as possible. By connecting the supersource node to all of the countries of focus at each step of the supply chain, with effectively infinite capacity in these flows, we can apply a maximum flow algorithm (in this case the Edmonds-Karp algorithm [70]) to calculate the maximum vulnerability based on the countries of focus. Through careful analysis of the distribution of flows using the outputs of the algorithm, we can determine the vulnerability at each step of the supply chain.

To understand the impact of **unobserved trade in the maximum trade** case, we assume it could all come from the countries of focus. In practice, we connect the nodes that represent uncertain supply (MMRS, TCNM, UARP, TCNR, MRMC) to the supersource, to include them as potential sources of material. However, not all of these materials can necessarily come from our country of focus if they are not actually known to produce materials in the steps upstream of the uncertain supply, e.g. Russia is not known to be a producer of raw or refined lithium, so none of the uncertain lithium supply should come from Russia. Thus, we include the (still relatively naive) assumption that the maximum flow from our supersource to any of these uncertain supply nodes cannot exceed the sum of the production across all countries of focus at the step that the uncertain supply node joins the material flow network.

A **minimum trade** case is the same measurement as the pessimistic scenario, but instead chooses the countries that import the least quantities of materials, and further assumes unobserved trade does not come from the country at all (minimal vulnerability based on the country in question). The practical implementation of this is to calculate the inverse – determining the maximum flow through the network where all flows into and out of nodes in the countries of focus are not included in the network, and subtracting that amount from the expected demand at the last (cathode-producing) step to determine the amount of material that at minimum must come from the countries of focus.

A.3.2 Data Sources

A.3.2.1 Production Data ($s_{\alpha i}$ and d_{pi})

We choose to use 2020 USGS mining data [1] like many other studies in the critical materials and material flow analysis literature. [10, 29, 66] These data are largely reported in terms of contained minerals (e.g. metric tons of lithium equivalent or thousands of tons of nickel equivalent), which reduces uncertainty from needing to use conversion factors, as described below. While there may be some lag in production processes, such as lithium brines requiring up to a year to process via evaporation, [122] USGS production data is reported in terms of final production quantities (e.g. after lithium concentration). We use refined materials and cathode materials production data (as well as US lithium production) reported in Sun et al. 2021, [2] which is largely from 2020. For the analysis of phosphorus included in Appendix A.1.2, given that granular production data for the refining step (phosphoric acid) was not found, we used regional production statistics from IFASTAT, the International Fertilizer Association's statistics website. [71]

A.3.2.2 Trade Data (t_{ij}^{OBS})

Following methodologies from the material flow analysis literature, [27–32, 43, 58–63] we trace inter-country trade from country to country by identifying relevant Harmonized System (HS) codes. These codes, internationally standardized by the World Customs Organization to 6 digits, allow for importing countries to levy tariffs and monitor compliance with regulations (e.g. rules of origin), and as such are designed to classify and cover all internationally traded items. [65] Most sources of trade data are ultimately based off of the UN Comtrade database, which aggregates reports submitted by UN member countries on their trade at a high level, which is the database used by nearly all global material flow analysis and supply chain disruption propagation studies reviewed in this article. [27, 29–32, 35–38, 43, 58, 59, 61–63, 66, 259] Other national and regional-level trade databases, such

as the US Census's database or Eurostat, are much more detailed, but do not cover the entirety of trade around the world, and as such are less useful when considering whole supply chains. These national and regional databases could be used to supplement data gathered from international sources, which is what is done in the data source we use. It is also important to note that both export and import data exist in these databases. Nearly all reviewed studies in the material flow analysis and supply chain disruption propagation literatures that specified a choice between the two used import data [29, 30, 35, 38, 43] rather than export data, [32] as it tends to be more complete and accurate, as many countries impose various import tariffs and are concerned about what materials enter their borders, [29, 30, 64, 65] so we also use reported import data figures.

We use **TradeMap** data from IntraCen [65] as it provides the UN Comtrade data while additionally supplementing it with national and regional trade where import and export data may be conflicting or missing, as some countries do not report data to UN Comtrade. It is the opinion of the authors that this incorporation of national and regional trade data positively augments the UN Comtrade database upon which most material flow analyses base their analyses on. We use 2020 data to synchronize with our 2020 production data, and as a result rely on the 2017 HS code nomenclature. Databases based on the UN Comtrade dataset report their data in both dollar value and quantities. While using either has its drawbacks (commodities have constantly fluctuating prices and materials imported under trade codes are not homogeneous and not all shipments are accurately weighed and contained amounts of the critical material vary), we chose to use the latter to minimize the impact of uncertainty and issues of propagated errors. This necessitates the use of *Conversion Factors* (as described in the next subsection) to change the units of the broad categories of each trade stream into terms of contained materials. 28 HS codes were selected to represent trade of materials at each stage of the supply chain between mining and cathode manufacturing (see Appendix Table A.7). We reference a variety of literature to fully aggregate across all trade codes that may include battery material supply. [2, 27, 29, 30, 34, 63, 66–68]

We make the assumption that the traded raw and refined material in the selected codes is representative of materials that could actually end up in battery cathode materials. Globally, about 74% of mined lithium [1] and 57% of mined cobalt [20] is used in lithium ion batteries, but only a portion of lithium ion batteries are used in electric vehicles. Furthermore, only about 11% of nickel [21] and 2% of manganese [22] is used to make batteries of any kind. Different grades of ores, refining processes, and precursor materials, including recycled and scrap materials, make it more economically efficient to produce certain types of materials, not all of which can be used for battery cathode material production, and not all battery cathode materials are used in electric vehicles. However, as it is possible to use any of these

materials in the electric vehicle battery material supply chain with enough processing, we choose to include all of these trade data in our analysis. The relevant supply chain flows included in our dataset are found in Appendix A.1.1 and Appendix Figure A.2. Additionally, while other studies account for missing trade by exploring country-specific trade codes and known inter-country trade relationships, other than the export of Australian lithium, we do not account for any of these potential clarifications of the missing data problem.

We recognize that this snapshot of a dynamic system is likely not to be perfectly accurate; our inclusion of uncertainty analysis around our primary point estimates attempts to account for these challenges. Additionally, with improved data, we suggest the power of this methodology only increases.

Table A.6. Flow Constraints in the maximum trade case that includes uncertain (unobserved and indirect) trade data.

Included uncertain data source	**Flow Constraint**	**Reasoning**
MMRS (Missing (extracted) material used in refining step) *plus* TCNM (Trade from countries with no mining)	Amount countries of focus that is TCNB (Trade to countries with no battery-related refining, or raw materials not accounted for in trade) *plus* amount countries of focus that have RMRP (Raw materials not accounted for in battery-related refining production)	TCNB and RMRP are our materials in raw form that could have been re-imported or miscategorized / misrepresented as not feeding into the next step.
UARP (Unaccounted-for additional refining production) *plus* TCNR (Trade from countries with no refining) *plus*MRMC (Missing refined material used in cathode production)	TCNB + RMRP *plus* the amount countries of focus that is TNPC (Trade to non-cathode producing countries, or refined materials not accounted for in trade) or assigned as Non-Cathode Products	We assume any of the TCNB and RMRP could have been converted at some point. TNPC is any 'missing' refined materials, while any refined material not recorded as being used in cathode material manufacturing could have been re-imported or miscategorized.

A.3.3 Trade Codes and Conversion Factors

Table A.7. Trade Codes and Conversion Factors used in this analysis. The codes and conversion factors in **bold** below are the ones included in the analysis. Explanations for the exclusion of codes are noted in the corresponding labeled table notes. S: Self-calculated or estimated.

	Code	Name	Source	Conv.	Source
Raw lithium	**253090**	**Lithium concentrates** [a]	[27]	**0.03**	[58]
Refined lithium	280519	Alkali and alkali earth [b]	[27]	0.03	[58]
	282520	**Lithium oxide and hydroxide**	[2, 27, 34]	**0.165**	[58], S
	282690	Fluorosilicates of sodium or of potassium [b]	[27]	—	—
	282739	Lithium Chlorides [b]	[27]	0.164	[58], S
	283691	**Lithium carbonates**	[2, 27, 34, 68]	**0.188**	[58], S
Raw nickel	**260400**	**Nickel ores and concentrates**	[2, 34, 62, 67, 68]	**0.015**	[31]
Refined nickel	**282540**	**Nickel oxides and hydroxides**	[34, 66, 67]	**0.787**	[31]
	282735	**Nickel chlorides**	[66]	**0.454**	[31]
	283324	**Sulphates of nickel**	[34, 66–68]	**0.224**	[31]
	720421	Ferrous waste & scrap of stainless steel [c]	[66]	0.024	[31]
	750110	**Nickel; nickel mattes**	[34, 66, 67]	**0.75**	[31]
	750120	**Nickel oxide sinters**	[34, 66, 67]	**0.75**	[31]
	750210	**Nickel, not alloyed, unwrought**	[63, 67]	**0.995**	[31]
	750220	**Unwrought nickel alloys**	[63, 67]	**0.5**	[31]
	750300	**Nickel; waste and scrap**	[66, 67]	**0.5**	[31]
	750400	**Powders and flakes, of nickel (excluding nickel oxide sinters)**	[63, 67]	**0.995**	[31]
Raw cobalt	**260400**	**Nickel ores and concentrates**	[30]	**0.001**	[30]
	260500	**Cobalt ores and concentrates**	[2, 27, 29, 30, 34, 62]	**0.15**	[30]

Table A.7. Continued.

	Code	Name	Source	Conv.	Source
Refined cobalt	**282200**	**Cobalt oxides and hydroxides; commercial cobalt oxides**	[2, 27, 29, 30, 34, 62, 68]	**0.329**	[30]
	282734	Cobalt chlorides [d]	[2, 30]	0.454	[30]
	283329	**Sulphates**	[30]	**0.01**	S
	283699	Carbonates, Others; Peroxocarbonates (percarbonates) [e]	[30]	—	—
	284190	Other salts of oxometallic or peroxometallic acids [e]	[30]	—	—
	810510	Cobalt mattes and other intermediate products of cobalt metallurgy [f]	[29, 30, 34]	—	—
	810520	**Cobalt mattes and other intermediate products of cobalt metallurgy; unwrought cobalt; cobalt powders**	[2, 27, 29, 30, 62]	**0.6**	[29, 30]
	810530	**Cobalt waste & scrap**	[2, 30]	**0.6**	[30]
	810590	**Articles of cobalt, n.e.s.***	[2, 29, 30]	**0.6**	[29, 30]
Raw manganese	**260200**	**Manganese ores and concentrates**	[2, 34, 67, 68]	**0.3**	[32]
Refined manganese	**282010**	**Manganese Dioxide**	[2, 34, 67]	**0.632**	[32]
	282090	**Manganese oxides; excluding manganese dioxide**	[34, 67]	**0.774**	[32]
	283321	Magnesium sulfate [g]	[68]	—	—
	283329	**Sulphates**	[68]	**0.02**	S
	811100	**Manganese; articles thereof including waste and scrap**	[67]	**0.004**	[32]
Raw Phosphorus	**2510**	**Phosphate, Ground or Unground**	[260, 261]	**0.119**	[260, 261] [h]
Refined Phosphorus	**280470**	**Phosphorus**	[260, 261]	**1**	[260, 261] [h]

Table A.7. Continued.

	Code	Name	Source	Conv.	Source
	280920	**Phosphoric Acid**	[260, 261]	**0.273**	[260, 261] [h]
LFP	**N/A**			**0.0440**	S [i]
NMC	**N/A**			**0.0716**	S [i]

[a]: While most lithium is produced from brines that are refined into lithium hydroxide or lithium carbonate forms near the mining sites, lithium as an ore (which is currently essentially all found as spodumene) is presently only produced in Australia and China [1] and not tracked as a specific commodity in the HS system. [65] Noting that only Australia and China have any lithium spodumene ore production, [1] we assume all Chinese production is domestically consumed for refining, and refer to Australia's country-specific export data of "25309011 Lithium concentrates", using the "ores" conversion factor from Sun et al. [58] to convert it to quantity of contained lithium, to estimate Australian lithium exports.

[b]: The codes 280519, 282690, 282739 are used in lithium-ion batteries, sometimes as lithium alloys, but primarily as inputs for battery electrolytes. [27] Because materials covered under these trade codes have an unclear relationship with cathode material production (and cathode materials are already traded under other trade codes), we disclude these from our trade analysis.

[c]: Excluded as it primarily refers to stainless steel waste and scrap

[d]: Excluded as it removed in 2007 HS code revision [65]

[e]: Excluded as unclear which of cobalt compounds listed would be included, and cobalt content is highly uncertain

[f]: Excluded as it removed in 2002 HS code revision [65]

[g]: This study [68] mistakes magnesium for manganese and thus this is not included; this code is for magnesium sulfate [65]

[h]: Li et al.87 identify the conversion factor for Phosphorus rock (ground or unground) to be 9.8%. Chen and Chen [260] assume 32% of the weight of phosphate rock is P_2O_5, which, using a chemical formula, results in about 14% of the rock as phosphorus. We take the average of these two numbers. We assume the phosphorus content in phosphorus is pure (100%), while for phosphoric acid, we take the average of the values of phosphorus in phosphoric acid calculated from chemical formula (0.316), Li et al. [261] (0.268), and Chen and Chen [260] (0.236).

[i]: Using chemical formulas for LFP ($LiFePO_4$) and NMC622 ($LiNi_xMn_yCo_zO_2$, $x+y+z \approx 1$), we find conversion factors of 0.0440 and 0.0716, respectively. Other sources [58]

list them as 0.07 and 0.04, respectively, instead; it is possible that they were accidentally swapped in their analysis. We choose to use NMC622 as it is quite close to the average chemical composition based on a predicted market size-weighted blend of NMC technologies, as seen in Appendix Table A.8. This calculation is based on a simulated share of choices presented in a projected LFP-dominant scenario, [3] which is what is currently empirically happening as the electric vehicle market is developing.

Table A.8. NMC simulated share of choices and representative chemistry analysis. [3]

We bold NMC622 and the Weighted Average rows to highlight their similarity; we use NMC622 as the rough approximation of the average NMC chemistry.

	Molecular Weight	**Chemistry**	**Ni**	**Co**	**Mn**	**Total Mole. Weight**	**%Li**	**%Ni**	**%Co**	**%Mn**	**Predicted Share**
Lithium	6.941	NMC333	0.333	0.333	0.333	96.46	7.2%	20.3%	20.4%	19.0%	4.1%
Nickel	58.6934	NMC523	0.500	0.200	0.300	96.55	7.2%	30.4%	12.2%	17.1%	10.7%
Cobalt	58.9332	**NMC622**	**0.600**	**0.200**	**0.200**	**96.93**	**7.2%**	**36.3%**	**12.2%**	**11.3%**	12.2%
Manganese	54.938	NMC721	0.778	0.222	0.111	103.79	6.7%	44.0%	12/6%	5.9%	0%
Oxygen	15.9994	NMC811	0.800	0.100	0.100	97.28	7.1%	48.3%	6.1%	5.6%	2.9%
Iron	55.845										
Phosphate	30.9738	**Weighted Ave.**	**0.547**	**0.209**	**0.244**	**96.77**	**7.1%**	**33.2%**	**12.7%**	**13.9%**	

A.3.4 Qualitative Discussion of Limitations

We recognize that this analysis has a number of limitations. We only consider a first-order measurement of vulnerability based on specific countries or sets of countries, rather than considering a more realistic analysis where a disruption occurs, such as where a country's production partially shuts down, or trade in a specific material is restricted. This analysis does not consider any market or government responses, in the form of supply increases, price shocks, contracts (either upheld or broken), or drawing upon stockpiles. Furthermore, we recognize that a static snapshot of the market does not necessarily tell us how the market may evolve as additional production capacity comes online and different policies and restrictions affect trade. We do note that most countries have very limited amounts of refined and cathode materials stockpiled, and that it takes 5-15+ years to prospect, plan, fund, permit and deploy a new mining site [122] and about 7 years to build a mine and refining plant, [262] so market responses to increase supply outside of using any possible slack supply will not occur in the short term. In general, market reports seem to indicate very little to no slack capacity in materials refining and processing and cathode manufacturing plans.

We caution against using these figures as exact measurements, given the age of the data (2020) and the number of assumptions needed to be made about the type of data. For example, because NCA and NMC were not separated in our source data, we make an assumption about the relative market shares of each based on projected market shares,88 and assume production is homogeneous across China, Japan, and South Korea, the three NCX manufacturers. While we recognize that these numerical estimates are from 2020 data, which may not necessarily be representative of trade in a "business as usual" scenario given the COVID-19 pandemic and subsequent restrictions in global trade, we do argue that the rapidly developing nature of this industry means the most recent available data is most appropriate for characterizing a snapshot of potential risk. Furthermore, several projections indicate even higher potential future reliance on China in the refining and cathode production steps given current growth trends, [7] such as its share of overall cathode production increasing from 78% to 87% by 2030, [263] as well as Russia, DRC, and South Africa in the raw material production steps, so the result with the least missing data uncertainty may still be underestimating the amount of current and future risk. Of course, this methodology could be extended to analyze the change in vulnerability over multiple years or smaller time-frames, assuming the ability to gather both production and trade data within the time interval in question. This analysis could also be extended with further awareness of country-based ownership of production capability, given data about production quantities by individual firms, which currently has not been aggregated for all materials in the battery material supply chain. The fact that other countries' companies control assets in other countries may result in different risk factors that these vulnerabilities create. While these issues of firm choice and control may be less 'strict' bounds on market dynamics, these firms likely could maintain some market power in choosing the customers they sell to. This work begs the question about market responses as the markets grow and evolve. These trade relations are not static (though long-term supply contracts are potentially significant), and neither are the production capabilities. We recognize that market structures can and have changed in response to economic factors (price shocks, etc.) and political disruptions (export bans, tariffs, etc.). Describing scenarios of disruption as well as understanding firm best responses to such disruptions and production displacement is a natural next step in this work.

A.4 Further Results

A.4.1 Additional Vulnerability Index Calculation Visualizations for NMC Cathode Supply, for Cobalt, Nickel, and Manganese supply

Note that, while the example solutions in the below figures and Fig. 5 show a significant involvement of uncertain flows in the minimum case, there exist other possible solutions, as the optimizer only presents one solution. There likely exist other solutions with less involvement of uncertain flows, but the optimizer was not coded to minimize use of uncertain flows. Additionally, note that sums may not add evenly due to rounding. Table 2.1 in the main body of the paper defines the four sensitivity cases. The amount of overall vulnerability at the cathode manufacturing step, added at each upstream step, is noted with a plus (+) sign. The solid horizontal black lines are visual aids to indicate separation of countries; semi-transparent red bars represent vulnerability that is propagated downstream. Indirect trade and unobserved trade do not count towards the measurement of vulnerability in the optimistic case, but are counted towards the pessimistic calculation in the case that includes uncertain data. For the minimum and maximum cases, one possible solution is presented, but other distributions are possible.

A.4.2 Further results for multiple countries and potential trade blocs

In Appendix Figure A.7 and Appendix Figure A.8, we present additional illustrative results, similar to Figure 2.4, that describe minimum, proportional, and maximum vulnerability to other sets of countries and regions.

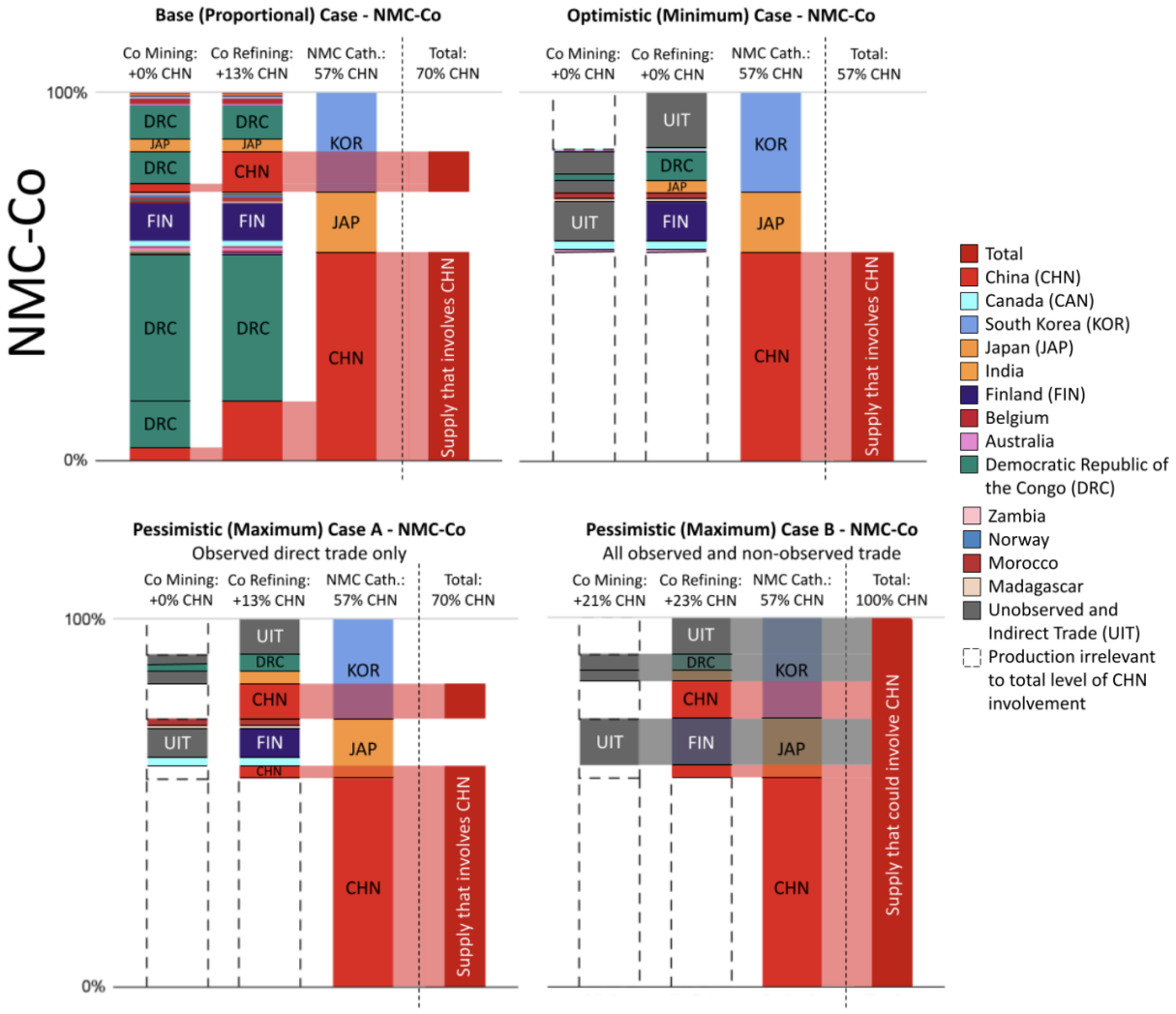


Figure A.4. Visualization of a vulnerability index for the global NMC (Lithium Nickel Manganese Cobalt) cathode supply, for a cobalt supply chain disruption in China. Note that India, Belgium, Australia, Zambia, Norway, and Madagascar are represented on these diagrams, as identified in the color legend on the right hand side, but the bars are too small to label.

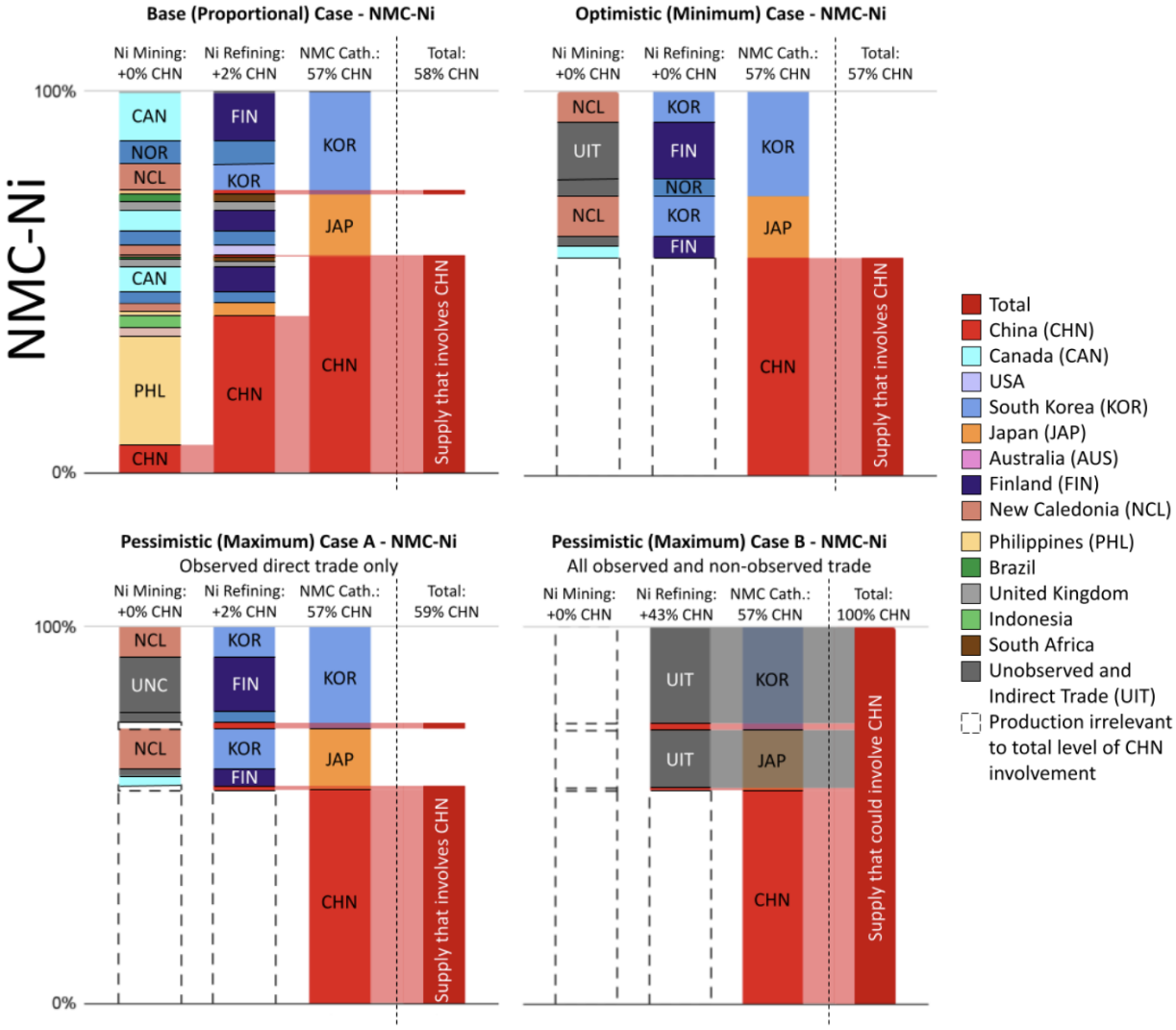


Figure A.5. Visualization of a vulnerability index for the global NMC (Lithium Nickel Manganese Cobalt) cathode supply, for a nickel supply chain disruption in China. Note that Brazil, the United Kingdom, Indonesia, and South Africa are represented on these diagrams, as identified in the color legend on the right hand side, but the bars are too small to label.

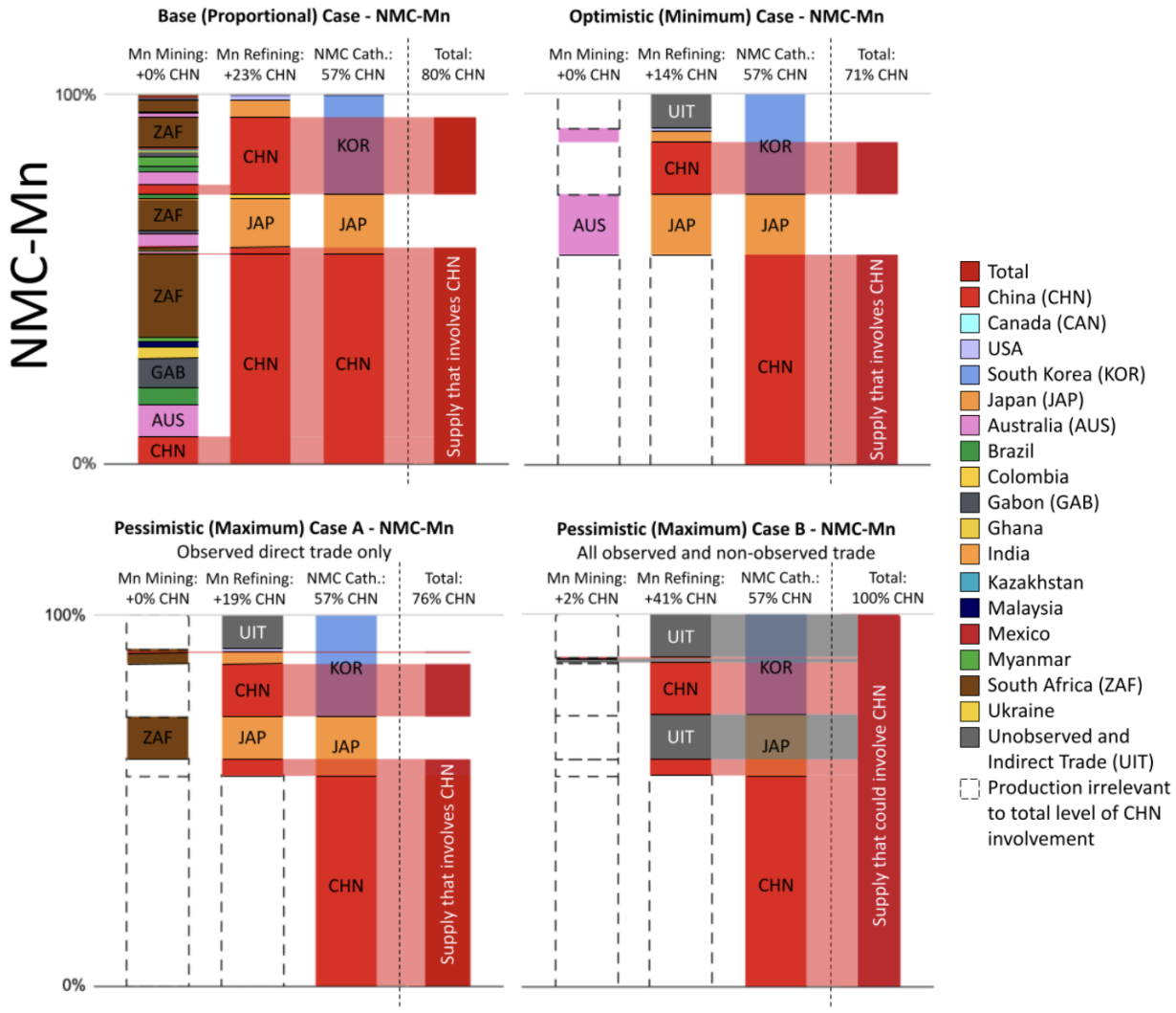


Figure A.6. Visualization of a vulnerability index for the global NMC (Lithium Nickel Manganese Cobalt) cathode supply, for a manganese supply chain disruption in China. A constraint placed on the uncertain flows at the mining step results in less than 100% dependence, even though the uncertainty would otherwise allow the vulnerability index to approach 100%. This is because the maximum amount of uncertain trade assigned to the country of focus at each step is limited to the maximum amount of trade that the country of focus exports to countries that are not known to be involved in the next step of the supply chain. Note that the United States, Brazil, Colombia, Ghana, India, Kazakhstan, Malaysia, Mexico, Myanmar, and Ukraine are represented on these diagrams, as identified in the color legend on the right hand side, but the bars are too small to label.

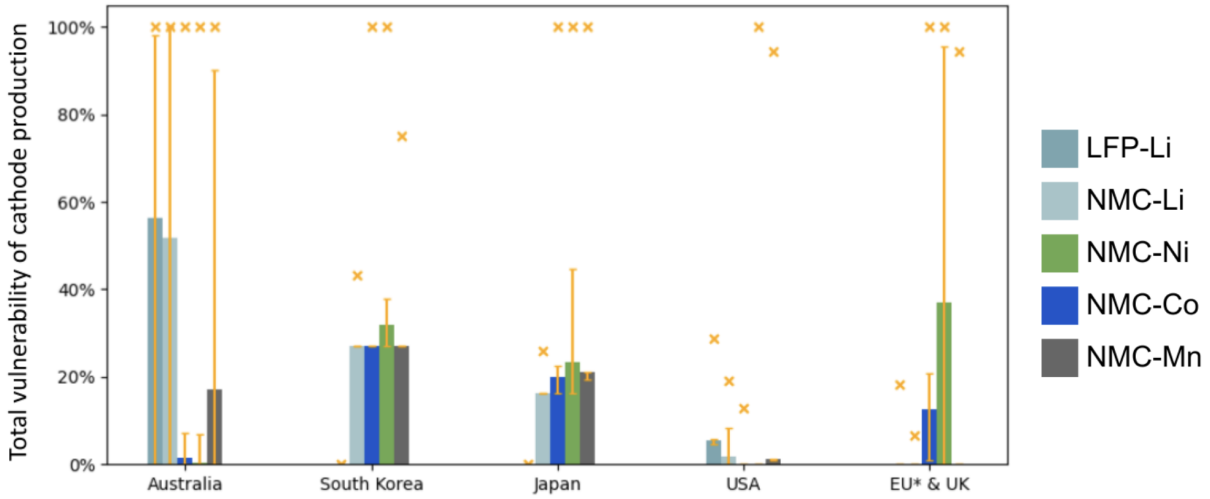


Figure A.7. Calculated vulnerability indices for a set of "western-aligned" countries and regions.

Note EU* represents countries in the European Union and European Free Trade Association. Bars show results using proportionality assumptions, error bars (in yellow) show range from optimistic (minimum) and pessimistic (maximum) based on uncertainty about intracountry flows, and the yellow × symbols show extreme upper bounds assuming all missing data originate from the focal country.

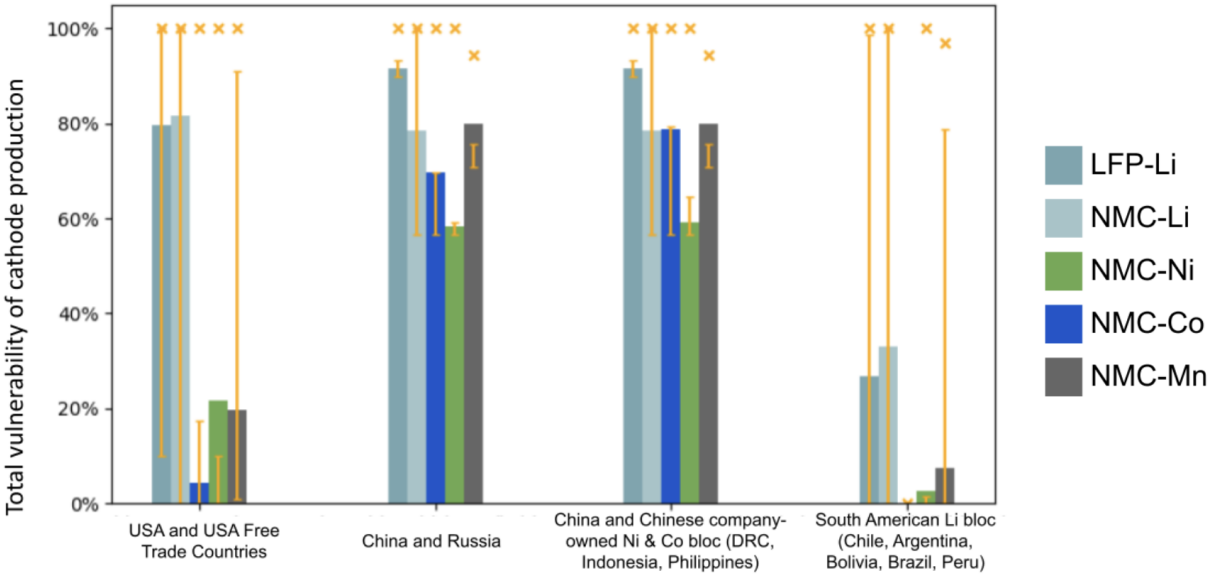


Figure A.8. Calculated vulnerability indices for a set of potential future trade blocs.

Bars show results using proportionality assumptions, error bars (in yellow) show range from optimistic (minimum) and pessimistic (maximum) based on uncertainty about intracountry flows, and the yellow × symbols show extreme upper bounds assuming all missing data originate from the focal country.

Appendix B
Supplementary Notes for Chapter 3

B.1 Literature Review

B.1.1 An Introduction to Industrial Policy

The intersection of technology choice, geography of production, and production cost in the context of policy interventions would be typically categorized as industrial policy. The term "industrial policy" is generally used to describe any deliberate and proactive policies by a government to shape and support specific industries, firms, and/or economic activities. [264, 265] While the term is generally recognized to be controversial, as internationally recognized 'best practices and 'free trade' rules suggest protectionism and government subsidies are not fair (though this viewpoint is disputed), [266, 267] it is generally recognized that significant market failures exist in markets such as labor, knowledge, and climate-related emissions, especially in industries with nascent technologies or nascent industries that are developing in a country. [268, 269]

Thus, to a certain extent, it is clear governments should create policy to help correct these failures and help industry overcome barriers where practical difficulties with implementing such policies can also be overcome. [268] Drawing on Rodrik [264]'s list of ten principles for designing industrial policy, it should incentivize "new" activities, establish clear benchmarks for success and failure, incorporate sunset clauses, target activities rather than sectors, and focus on activities with potential spillover effects. [264]

It is important to note that many studies of industrial policy have historically been in the context of developing nations and development economics, especially in the context of the "Asian Tigers" of the latter half of the 20th century, especially in the context of Japan (the "Japanese Miracle"). [270, 271] However, recent literature has identified that countries in the 21st century are pursuing 'smart' growth with a focus on innovation, beyond just 'fixing' market failures or system coordination issues, [272] particularly within contexts such as climate change and 'green growth,'[269] as already industrialized countries race to claim shares of new markets that appear to have the chance to (or in some cases, to avoid climate

impacts, must) substitute for existing technologies and industries. By leveraging existing and developing new country-specific capabilities and advantages and creating new markets, countries can ensure economic growth and social well-being with well-designed industrial policy. Subsequently, determining the 'correct' policies to implement are dependent on national (and perhaps subnational and/or regional) characteristics of each industry, motivating the study of specific scenarios and interventions whose lessons and/or models may be generalizable to similar situations.

B.1.2 Industrial Policies with Local Content Requirement

In the specific context of the IRA, the primary challenge in meeting the EV Credit requirements is the local (or 'friend-shored') content requirement, for critical mineral production, battery component production, battery production, and final vehicle assembly. With the rapid scale-up of electric vehicle (EV) battery production also implying a massive increase in demand for critical minerals, with projected increases of 5 to 40 times 2020 demand in 2040, depending on the material, [3, 6, 7] it makes sense that policymakers in the United States would like to onshore or at least friendshore as much of the entire supply chain as possible.

These types of requirements have been studied within the automotive context, such as Veloso [82]'s study on content requirements. While the study is primarily in the context of foreign direct investment in developing countries, Veloso finds that the resources invested by foreign investors and local suppliers generates spillovers and learning effects in addition to the private value generated by the economic activity, and that local content requirements are preferable to policies such as tariffs and subsidies as a "means to increase domestic purchases."

In the context of renewable energy systems, Kuntze and Moerenhout [83] find that local content requirements are frequently used in public financial support programs but these choices are largely made for political reasons (i.e. to curry political support) rather than based on economic and empirical analyses. They find that, under some conditions, these requirements "may be able to facilitate the development of a global innovator that is able to compete on the international market and push down technology costs," but that the conditions are highly variable and country- and technology-specific. They find that local content requirements cause short-term costs to the industry and likely inflate retail power prices, but medium-term benefits of increasing competition and international innovation may help offset these costs. Overall, they suggest that positive spill-over effects have not been empirically demonstrated.

In the specific case of electric vehicles in China, local content requirements (at minimum one of the battery, motor, or drivetrain had to be produced locally) and the requirement

that all foreign direct investment had to partner with a local joint venture, in contrast to local institutional support (market protection and subsidies) for domestic electric vehicle firms, have contributed to the proliferation of innovative technology and firm strategy in the country's nascent electric vehicle sector. [48] The most successful of these firms (such as CATL, BYD, SAIC, XPeng) have gone on to become China's national champions in the space.

B.1.3 Technology-forcing policy

As identified by Mazzucato [272], industrial policies can be used to deliberately shape technological and industrial advancement beyond what the market would naturally achieve. Such "technology forcing" policies can not only address market failures of R&D expenditure (as identified by e.g. Arrow [273] and Hall and Lerner [274]) but also create alignment with broader national goals such as national security, energy independence, or environmental sustainability. However, such policies can also create market failures and unintended consequences, due to the misallocation of resources (e.g. investing in the 'wrong' technologies) or improper market distortions. A few examples of technology-forcing policy in the automotive sector revolve around automobile emissions control technology and CAFE (corporate average fuel economy) standards, as exemplified by Lee et al. [78] and Whitefoot and Skerlos [79]. Lee et al. [78] show that more stringent regulations in the period from 1970 to 1998 induced a leapfrog innovation in the catalytic converter, rather than allowing automakers to focus on incremental engine design improvements. On the other hand, Whitefoot and Skerlos [79] identified that the CAFE standards implemented for the time period of 2011 to 2016, could actually encourage manufacturers to produce larger overall vehicles. While the policy was obviously intended to reduce the overall emissions from each automotive manufacturer's fleet of vehicles, due to the different requirements for emissions levels depending on vehicle size (footprint), and the fact that designing a policy to avoid such design changes is challenging to implement across the wide variety of vehicle designs and manufacturers, we see a failure of a similar market-forcing policy.

B.1.4 The Relationship between Technology Development and Production Location

It is also important to consider the third dimension - that technology development is heavily influenced by the location of production, and vice versa. In the automotive context, Clark et al. [275] establish that international differences in operations and product development explain differences in productivity and competitiveness. But Fuchs and Kirchain [276] extend

this argument by noting that the geographical placement of the manufacturing process has a significant impact on technological development. This influence arises from the tradeoffs that occur between improving the manufacturing process and gaining knowledge through practical experience, particularly in the context of technologies with lower production costs that are considered "low-tech." The authors suggest that the location-related decisions in manufacturing can affect the balance between achieving cost efficiency for simpler technologies and investing in more advanced but costlier technologies. Thus, we establish that it is important to determine whether or not industrial policy will have an effect on both the location of manufacturing but also whether or not such policies prefer one technology choice over another, as the design of the policy will lead to differing end outcomes due to the effect of production location and technology choice on each other.

B.1.5 Literature involving IRA-Specific Analyses

The literature on the effects of IRA is burgeoning, with most literature focusing on the macroeconomic and macro-environmental impact of the policy change at a high level. [90, 92, 93, 277–289] As seen in Appendix Table B.1, only some of these analyses have been peer reviewed, with a substantial number of publicly oriented analyses and whitepapers from consultancy groups and think tanks. We evaluate this body of literature along six dimensions of focus: economic analysis, emissions impacts, jobs, technology choice, policy choices and scenario analysis, and production location. The fourth and the sixth columns include some nuance: we define studies of technology choice to also encapsulate the dimension of technology adoption (i.e. choosing to adopt a substitute or complement), while studies of 'production location' are also defined to include analyses of international competition or trade arguments. Additionally, we briefly summarize the primary takeaway of each study in a seventh column.

Many studies have begun to evaluate the various impacts of the IRA in the context of specific clean technologies, such as clean hydrogen and other fuels, [84–88, 290] carbon capture and sequestration, [86, 89] solar panels, [90, 91, 291] wind power, [90, 91, 292] and the electric grid as a whole. [293–295] These studies incorporate various levels of engineering detail to evaluate the cost-competitiveness of the technology with and without the IRA and/or previous incentives (such as an Investment Tax Credit or Production Tax Credit). However, the analyses of electric vehicles do not include these dimensions - nearly all of the studies that include electric vehicles focus primarily on projections of electric vehicle adoption, [92, 93, 280, 281, 296] including additional considerations such as vehicle chargers. [297] They identify that the IRA provisions accelerate sales of new light-duty vehicles in 2030 by about 10 percentage points more than scenarios without IRA (eg. 32-52% vs 22-43% [92]),

but do not explicate the types and characteristics of such vehicles. It is important to note that the adoption of electric vehicles is highly dependent on characteristics such as travel range, cost, charging speed, and battery life53 which are attributes that are mostly dependent on the battery chemistries used in the electric vehicle. [298] In the specific context of battery chemistries and their constituent materials and the IRA, only one study has evaluated any sort of technology choice between current options. [19] However, this study focused on the possibility of different meeting the criteria established in the IRA to qualify for tax credits, and not the economic or production location implications of such policy interventions. As different chemistries will be affected differently by the subsidy qualification requirements of IRA (different chemistries are composed of different critical minerals), further analysis should be done to evaluate the choice between electric vehicle battery chemistries and how that is affected by IRA.

Additionally, the literature on production location, as defined previously, has up to this point largely focused on potential trade policy violations of and international responses to the IRA, with particular emphasis on the European and South Korean responses, [286, 299–305] and one study discussing the challenges around friendshoring. [306] These studies do not evaluate how the IRA will quantitatively affect the choice of production location, given that different production locations will have differing costs, especially when accounting for local incentives and policy interventions. As a result, this research aims to address both the effect of IRA on technology choice and adoption, with a specific focus on electric vehicle battery chemistries, but also bring in the additional dimension of production location, all within the larger context of economic analysis with regards to production costs and incentives.

Table B.1. Review of IRA-related literature (May 2024).

Green: Present / True. Yellow: Working Paper intended for Peer Review or Under Peer Review. White: Not present, or 'gray' literature in the 'peer reviewed' column.

Study (peer review status)	Economic (Costs, Incentives)	Emissions	Jobs	Policy choices and adoption scenarios	Technology Choice and/or Adoption	Production Location (inc. Competition)	Main finding(s) (Paraphrased or quoted)
Shan and Kittner 2024 [293]	ReEDS, optimize system costs	Impacts on emissions		Impact of IRA '[reconciling] competing incentives', impacts of state policies	Adoption of energy storage across different US regions/states		Energy storage can reduce costs and grid emissions, but tradeoffs can occur. The IRA can help balance these, but more policies (esp. at the state level) are needed to more quickly increase deployment while maximizing emissions reductions.
Aldeman et al. 2023 [294]	Least-cost generation and transmission expansion planning model	Consideration of emissions reductions		Impact of ITC and PTC under a range of Clean Energy Generation Targets	Comparison of wind, solar, and storage vs other energy generation	Analysis of production across different states	Due to the ITC and PTC extensions by the 2022 Inflation Reduction Act, there is little additional investment in renewable energy generation assets at low CEGT levels. Significant additional installations become important at CEGT levels above approximately 30%.
Bracci et al. 2023 [86]	Hydrogen production costs	Net zero target		Impact of 45Q and 45V tax credits	Many H_2 production pathways (electricity sources, electricity vs fossil based)	3 locations in the US (California, Texas, New York); regions outside US	Model results indicate next decade hybrid electricity-based solutions are lower cost ($2.02–$2.88/kg) than fossil-based pathways with natural gas leakage greater than 4% ($2.73–$5.94/kg). These results also apply to regions outside the US with a similar climate and electric grid.
Jordan et al. 2023 [279]	System costs, carbon tax	Emissions goals		Carbon taxes and standards	Choice of technologies to achieve goals		The IRA alone is insufficient to meet US climate goals; additional modeled policy interventions can help get closer. To meet goals, rapid & sustained deployment of zero-emission technologies is required.
Bistline 2023a [92]	Technology costs and incentives	Emissions goals		IRA vs counterfactual(s)	EV adoption, clean electricity, carbon capture		Nine state-of-the-art models show economy-wide emissions reductions between 43–48% below 2005 by 2035. This can serve as a baseline to which other studies of further policy intervention can compare.

Table B.1. Continued.

Study (peer review status)	Economic (Costs, Incentives)	Emissions	Jobs	Policy choices and adoption scenarios	Technology Choice and/or Adoption	Production Location (inc. Competition)	Main finding(s) (Paraphrased or quoted)
Cheng et al. 2023 [85]	Hydrogen production costs & credits	Levelized subsidies for carbon mitigation		IRA vs counterfactual(s)	Clean hydrogen & liquid fuels		With IRA credits, green and blue hydrogen are competitive with gray hydrogen; biomass-derived H_2 is not competitive. Synthetic liquid fuels would need additional support to compete. IRA subsidies are cost-comparative with social cost of carbon estimates.
Grubert and Sawyer 2023 [89]	Carbon sequestration costs	Carbon sequestration		IRA vs counterfactual(s)	Carbon sequestration technologies in coal & nat. gas		Over six scenarios, \$0.4–\$3.6 trillion in 45Q tax credits could alter greenhouse gas emissions by −24%; deployment of CCS is an extremely costly and likely ineffective GHG abatement strategy compared to alternatives.
Steinberg et al. 2023 [295]	Electric system costs	Projects decline in emissions		Analysis of No New Policy vs IRA-BIL	Clean electricity as share of system; CCS adoption		Clean electricity shares could increase substantially with IRA and BIL; annual power sector CO_2 emissions could decline to 72%–91% below the 2005 level by 2030; net decrease in total and average annual bulk power system costs; wind and solar deployment rates could grow rapidly (44–93 GW per year).
Larsen et al. 2022 [93]	Fuel and hydrogen prices, household energy costs	Emissions reduction		IRA vs counterfactual(s)	Clean electricity share, carbon capture adoption, EV adoption, oil & nat. gas production		The IRA as a whole drives US net GHG emissions down to 32–42% below 2005 levels in 2030, compared to 24–35% without it. Incentives targeted toward nascent green industries help support them.
Krupnick and Bergman 2022 [290]	Hydrogen production costs	Emissions reduction		IRA vs counterfactual(s)	Clean hydrogen		IRA subsidies should enable the use of clean H_2 where it is currently used. Even with the full \$3/kg-H_2 subsidy, green H_2 is not competitive with blue H_2 when purchasing power directly from the grid; coordinating with renewables or low prices will reduce production capacity.

Table B.1. Continued.

Study (peer review status)	Economic (Costs, Incentives)	Emissions	Jobs	Policy choices and adoption scenarios	Technology Choice and/or Adoption	Production Location (inc. Competition)	Main finding(s) (Paraphrased or quoted)
Mahajan et al. 2022 [280]	GDP growth, capital & fuel expenditure, household energy costs	Emissions reduction		IRA vs counterfactual(s)	Clean electricity share, EV & carbon capture adoption, oil & natural gas production		Using the US EPS model, findings of: 2030 reduction of US GHG emissions 37 to 43 percent below 2005 levels, though the impact is limited in the transportation and buildings sectors and strongest in electricity generation. The bill greatly encourages domestic clean energy manufacturing and jobs while providing public health co-benefits.
Cook and Hagen 2024 [87]	Hydrogen production costs	Measured to determine qualification for tax credits		Selected locations based on local policies		3 locations in the US (Oregon, Colorado, Massachusetts)	The cost of transported hydrogen in a 40 metric ton/day facility with [and without incentives] was calculated to be $3.47 [$4.07]/kg H2 in Klamath County, OR, $4.11 [$4.71]/kg H2 in Park County, CO, and $3.63 [$4.23]/kg H2 in Middlesex County, MA.
Delgado et al. 2023 [88]	Modeled e-fuel production with nuclear	Measured to determine qualification for tax credits		Analysis under base, IRA, and Hydrogen shot scenarios		Consideration of costs from four example US bioethanol plants	The life cycle GHG emissions are estimated to be 7 and -25 gCO_2e/MJ, without and with steam coproduct credit, respectively. The process economics were evaluated for three scales of nuclear power plant capacities (100, 400, and 1000 MWe), estimating a minimum fuel selling price of $3.61/gal, $2.82/gal, and $2.66/gal of fuel, including the IRA's $3/kg H2 tax credit.
Voigts and Paret 2024 [282]	Macroeconomi Model for the Energy Transition, fiscal costs	Emissions reduction		Analysis of permitting delays, other adoption scenarios			Application of large model in adoption scenarios to measure fiscal cost of IRA and its subsequent emissions implications.
Jiang et al. 2024 [283]	Autocorrelatio of economic indicators	Impacts on emissions					Use of 'Dynamic Autoregressive Distributed Lag Model (DYNARDL)' to 'promote sustainability'.

Table B.1. Continued.

Study (peer review status)	Economic (Costs, Incentives)	Emissions	Jobs	Policy choices and adoption scenarios	Technology Choice and/or Adoption	Production Location (inc. Competition)	Main finding(s) (Paraphrased or quoted)
Min et al. 2023 [90]	Solar and wind production costs			IRA vs counterfactual(s)	Solar panels, wind turbines; construction material demand	Comparison of component manufacturing with cost of imports	All solar and some wind components are now likely cheapest to manufacture in the US under 45X vs imports. The ITC and PTC credits make many more renewable projects viable; labor requirements for ITC/PTC bonuses are more than compensated for.
Pollin et al. 2023 [284]	Input-output model						Estimates creation of 9.1 million job-years, including approx. 760K from sections 13401, 13403, and 13502 (though 13502 includes all sectors the credit applies to).
Celsa and Xydis 2023 [91]	Solar and wind production costs			IRA vs ITC & PTC history	Wind power		In comparison to historical values and 'bumpy investment environments', the bill's PTC value of 2.5 cents/kWh is too high in comparison to a 30% ITC.
Chyong, Italiani, Kazantzis 2023 [84]	Ammonia production costs & credits			IRA vs counterfactual(s)	Ammonia production pathways		Comparison of ammonia production pathways under IRA. Two (CCS and BH2S) are economically viable, but electrolysis (AEC) is not; more investment would be needed.
Cole et al. 2023 [297]	EV demand modeling and charger supply			IRA vs counterfactual(s)	EV vs ICE adoption, EV chargers		Charging station subsidies are the most efficient of all policies to achieve 50% sales share of zero-emissions vehicles by 2030. Significant heterogeneity exists across all studied policies in terms of impact of EV penetration per dollar invested.
Mirza et al. 2023 [285]	Inflation rates, GDP growth, trade balance			Relative effectiveness of fiscal & monetary policies			Monetary policy (e.g. changing interest rates) is more effective than fiscal policy (e.g. subsidies and stimulus packages) in achieving the IRA objectives [inflation, growth, and employment], especially in the post-covid period.

Table B.1. Continued.

Study (peer review status)	Economic (Costs, Incentives)	Emissions	Jobs	Policy choices and adoption scenarios	Technology Choice and/or Adoption	Production Location (inc. Competition)	Main finding(s) (Paraphrased or quoted)
Bistline 2023b (NBER) [277]	System technology costs and incentives			IRA vs counterfactual(s)			Substantially larger investments than expected imply that costs may be larger than projected, but, even at the high end, IRA provisions remain cost-effective. Electricity rates will generally be lowered. The subsidy approach has expansionary supply-side effects relative to a carbon tax, and is preferred only if a strong learning-by-doing externality exists.
Woodhouse et al. 2024 [291]	Solar PV production costs & credits				Comparison of different PV tech	Comparison across prod. locations	Evaluation of US PV manufacturing LCOE under IRA, where PTC and ITC may be more favorable.
Trost and Dunn 2023 [19]	Mineral-weighted averages to meet IRA requirements				Material requirements between EV battery chemistries		The IRA 30D requirements may be achievable for NCA chemistries, but less so for LFP and NMC, in terms of long-term material supply. Design of the incentives doesn't sufficiently account for externalities or gamesmanship.
Cheng and Fan 2024 [286]	Global trade input-output model; Trade Indicators					'Restructuring of the Asia-Pacific energy industry' + 'China's response strategies'	The IRA's policies are found to promote the US clean energy industry, but worsen the trade environment for other Asia-Pacific countries. Implications for China (i.e. strengthening domestic manufacturing, promoting technological innovation) are further discussed.
Yue et al. 2024 [287]	Statistical inference between economic indicators						Use of asymmetrical relationship analysis, namely 'Nonlinear Autoregressive Distributed Lag (NARDL) approach' finds 'a strategic amalgamation of [natural resources, sustainable finance, and innovation] is recommended to maximize [impact]'.

Table B.1. Continued.

Study (peer review status)	Economic (Costs, Incentives)	Emissions	Jobs	Policy choices and adoption scenarios	Technology Choice and/or Adoption	Production Location (inc. Competition)	Main finding(s) (Paraphrased or quoted)
Li et al. 2024 [288]	Measurement of 'Environmental Kuznets Curve'						The results confirm the existence of the EKC curve, indicating that environmental degradation initially increases with [Natural Resource] usage but diminishes as inflationary pressures decrease pollution levels.
Suarez-Cuesta et al. 2024 [289]	Computable general equilibrium model						Scenario analysis with model suggests that 'the federal funding increase positively impacts GDP', and that the IRA 'enhances productivity, slightly reducing employment demand but significantly increasing real wages and long-term productive capacity, ultimately curbing inflation by the decade's end.'
Zhao et al. 2022 [281]		Emissions reductions		Implementat of policy beyond federal level	EV adoption		An all-of-society strategy (including IRA) may enable the US to meet its target of 50–52% emissions reductions by 2030 from 2005 levels; collaboration can reduce emissions well beyond what the federal government can do alone.
Slowik et al. 2023 [296]		Meeting climate goals		IRA vs counterfactual(s)	EV adoption		On average over the period 2023–2032, the IRA tax credits will reduce light-duty EV purchase costs by $3,400 to $9,050. By 2030, a range of 48%–61% EV sales share in the light-duty sector is found. However, electrification rates are not sufficient to meet Paris or nationally set goals.
Glicksman 2023 [278]		GHGs + general air pollution		IRA vs counterfactual(s)			The emissions reductions from GHGs and associated co-benefits, in addition to other specific public-health programs in IRA, will have a significant and positive effect.

Table B.1. Continued.

Study (peer review status)	Economic (Costs, Incentives)	Emissions	Jobs	Policy choices and adoption scenarios	Technology Choice and/or Adoption	Production Location (inc. Competition)	Main finding(s) (Paraphrased or quoted)
Kleimann et al. 2023 [299]		Emissions goals				Global trade, competition w/EU	The IRA contains trade-distortive subsidies, including local-content requirements prohibited under WTO rules. US and EU subsidies are similar in size, except in renewables (EU's are still larger), but EU restrictions are less discriminatory, more focused on innovation rather than deployment, and more fragmented.
Hansen et al. 2024 [292]						Discussion of locational effect on permitting & interest rates	Identification of 'five grand challenges' for offshore wind financing in the United States; the IRA is shown to largely solve one of them due to the transferable nature of the credits.
Vishnoi and Meena 2024 [306]					Focus on EVs / EV supply chains	Focus on 'friend-shoring'	Largely qualitative discussion on the implications of the IRA in terms of friendshoring.
Bown 2023a [300]					EV vs ICE adoption	Competition w/EU	The EU and US have diverging approaches: the EU prefers to tax carbon while the US subsidizes energy; the IRA increased frictions. The leasing interpretation provides some relief for EU automakers, but considerable disagreement remains.
Bown 2023b [301]					EV vs ICE adoption	Competition w/South Korea	South Korea felt 'betrayed' when the IRA was passed into law. However, Treasury implementations of IRA provisions (namely the leasing provision) have assuaged these concerns; Korea adjusted its own consumer EV tax credits and domestic EV manufacturing subsidies.

Table B.1. Continued.

Study (peer review status)	Economic (Costs, Incentives)	Emissions	Jobs	Policy choices and adoption scenarios	Technology Choice and/or Adoption	Production Location (inc. Competition)	Main finding(s) (Paraphrased or quoted)
Bernoth and Meyer 2023 [303]						Competition w/EU	The IRA could have far-reaching negative effects for Europe as an industrial location, as critical raw materials from reliable suppliers could become scarce. The consequences of the IRA are not to be underestimated by the EU Commission; the announced strategic countermeasures should be taken swiftly.
Nam 2023 [305]						Competition w/South Korea	Though IRA provisions may violate WTO rules, the current system is 'toothless'; it could be prudent to adapt, e.g. increasing North American EV assembly plants, or developing the South Korean critical mineral processing industry while pursuing redress in the WTO.
This study	Production costs			IRA vs counterfactual(s)	Choice of battery chemistries	US vs China	—

B.1.6 A Review of Battery Manufacturing Models

Table B.2. A non-comprehensive review of recent* or significant** battery manufacturing models.

*published within last five years; **≥100 citations per Google Scholar, as of March 2024. We choose to not include most models and articles before 2019 because of the publication of the 2020 review paper by Duffner et al.[64]

Paper / Reference	**Style**	**Source of Data**	**Main takeaway(s)**
Knehr et al. 2023 (BatPaC 5.1) [110]	Exponential scaling manufacturing model, bottom up cost model	Experimental / published data, transport or process models, and industry feedback on costs and manufacturing procedures	None; achieves goal of creating a flexible and relatively representative model. An additional release (version 5.1) was made in August 2023 but no updated manual was.
Degen et al. 2023 [307]	Machine-based energy consumption model	Interviews with 60 experts at workshops, 18 additional interviews (via Degen and Schütte and Degen and Krätzig [308])	Energy consumption for the global production of LIB and Post-LIB cells will be 130,000 GWh if no measures are taken. Yet, it is possible to optimize future production and save up to 66% of this energy demand.
Orangi and Strømman 2022 [131]	Process-based Cost Model	BatPaC, Jinasena et al., Patry et al.	Production costs for ten battery cell chemistries on large scales in nine locations are estimated. For a case study plant of 5.3 GWh/year producing NMC333 cells, cost of production can vary from 106 $US/kWh in China to 153 $US/kWh in Norway. The minimum efficient battery production plant is 7.8 GWh/year for this case study.
Liu et al. 2021 [309]	General process model	BatPaC, Yuan et al., various chemistry and process-specific literature estimates	Most manufacturing innovations have been reported, but very limited adoption has occurred in industry. Nine research areas are identified based on processes reviewed, and their associated costs and energy requirements.

Table B.2. Continued.

Paper / Reference	Style	Source of Data	Main takeaway(s)
Mauler et al. 2021 [132]	Process-Based Cost Model	BatPaC, Kwade et al., Duffner et al., Wentker et al.	Minimum viable plant sizes are demonstrated to be below 2 GWh/year but may exceed 15 GWh/year in the future.
Scheller et al. 2021 [310]	'Master production and recycling scheduling model'	LithoRec, London Metal Exchange, Berckmans et al.	Focus on interactions and optimization between recycler and manufacturer, who achieve positive margins in all scenarios. Inefficiencies always occur due to opportunistic behavior; reuse is only performed in the case of centralized planning.
Jinasena et al. 2021 [311]	Process model for energy consumption	BatPaC, Life Cycle Assessment literature,[73] various process-specific literature estimates	For a case study of a 2 GWh plant, energy usage for a variety of cell types has low variation 47.23 ± 13.03 $Wh_{in\ production}/Wh_{cell\ capacity}$. Energy use is dominated by electrode drying and the dry room.
Duffner et al. 2021 [312]	Process-based cost model	Multiple BatPaC versions, Wentker et al., Ciez and Whitacre, Satki et al., process-specific literature estimates	NMC811 have the largest single effects on cost. Location-based parameters are 'of minor importance'. Costs in the base scenario exceed BatPaC but in the optimized scenario are lower.
Wentker et al. 2019 [9]	Bottom-up direct manufacturing cost model (CellEst)	London Metal Exchange, metalary.com, BatPaC, Schmuch et al. 2018, Gopalakrishnan et al. 2017, Blomgren 2017, Placke et al. 2017, Meister et al. 2016, Berg et al. 2015, Brodd 2013	A model is created to calculate the cost of cathode materials from metal prices; these are combined with cell component costs in order to calculate full cell costs, which are in turn scaled up to full battery pack costs. Cell costs are estimated to be between \$40/kWh and \$90/kWh depending on production capacity and battery chemistry.

Table B.2. Continued.

Paper / Reference	Style	Source of Data	Main takeaway(s)
Kwade et al. 2018 [313]	Varies; review article that includes production cost modeling	BatPaC, Diekmann et al. 2017, Schuenemann 2015, various process-specific literature estimates	An introductory summary of the state-of-the-art production technologies for automotive LIBs. The key relationships between process, quality and performance are discussed, and the impact of materials and processes on scale and cost is explored.
Schmuch et al. 2018 [314]	Aggregation of battery material production costs	Numerous process-specific literature estimates for battery cell component material	Review article that focuses on production processes for battery materials and their subsequent implications on energy density and cost.
Berckmans et al. 2017 [129]	Process-Based Cost Model, Exponential learning model	BatPaC, Chung et al. 2016, Patry et al. 2015, Nykvist and Nilsson 2015, Wood et al. 2015, Gaines and Cuenca 2000	Estimation and projection of silicon-based and NMC lithium-ion battery manufacturing costs: the $100/kWh sales barrier will be reached between 2020 -2025 for silicon based lithium-ion batteries and 2025–2030 for NMC batteries.
Ciez and Whitacre 2017 [315]	Process-Based Cost Model	BatPaC, USGS Mineral Commodity Summary, interviews with equipment manufacturers	Although cost savings are possible from increasing cell dimensions and electrode thicknesses, economies of scale have already been reached (at a production volume of less than 1 GWh), and future cost reductions from increased production volumes are minimal. Prismatic cells can offer further reductions than those possible for cylindrical cells.
Yuan et al. 2017 [316]	'Comprehensive manufacturing energy analysis', Gabi 7.3	BatPaC, Ecoinvent database, direct measurement in a pilot-scale facility	For a 24 kWh battery pack with 192 prismatic cells, 29.9 GJ of energy is embedded in the battery materials, 58.7 GJ energy consumed in battery cell production, and 0.3 GJ for the final battery pack assembly.

Table B.2. Continued.

Paper / Reference	Style	Source of Data	Main takeaway(s)
Sakti et al. 2015 [130]	Process-Based Cost Model	BatPaC, expert opinion, Kromer and Heywood 2007	Economies of scale in battery manufacturing are reached quickly at a production volume of ∼200–300 MWh annually. Vehicle applications with larger energy requirements are able to utilize cheaper cells due in part to the use of thicker electrodes.
Patry et al. 2015 [317]	Direct manufacturing cost model	BatPaC, Helios project 2013, Gaines and Cuenca 2000	With a primary focus on electrode coating thickness parameter, the cost of these cells is estimated at between $230 and $400 per kWh. Potential savings resulting from doubling the electrode coating thickness from 50 to 100 μm at a given porosity represent roughly 25% of the cell cost.
Brodd and Helou 2013 [49]	Direct manufacturing cost model	Face-to-face interviews with people in R&D and manufacturing; vendors	When produced at large volume using automated equipment, cell materials account for 78–80% of total cell cost; costs for highly trained personnel are similar in China and the United States. Several other major drivers of production costs (such as logistics, land acquisition and energy costs) are higher in China, making total costs for each location roughly comparable.

B.2 History of New Energy Vehicle Incentives in China

Table B.3. Incentive Regimes Phases for Electric Vehicles in China.
[a]minimum range to qualify for credit increased to 100 km / 62 miles in 2016. Yuan converted to USD in same year, USD adjusted to 2024 values using CPI. From sources [99, 318–320].

	Phase 1 ("Unrestricted Direct Subsidy")	**Phase 2** ("Decreasing Direct Subsidies with Increasing Restrictions")			**Phase 3** ("Dual Credit")
Years Active	Jan. 2009 – Sep. 2013	Sep. 2013 – March 2018			April 2018 – Present
Subsidy in ¥, R in km	¥3000/kWh, ¥60,000 max.	80[a]≤R<150 ¥35,000 ('13) reduced to ¥20,000 ('17)	150≤R<250 ¥50,000 ('13) reduced to ¥36,000 ('17)	R≥250 ¥60,000 ('13) reduced to ¥44,000 ('17)	Additional complexity; reduction to $0 by Jan 1. 2023 (See Appendix Table B.6)
Subsidy in $, R in miles	$440/kWh, $8,780 max., 2009 dollars (12,622 '24$)	50[a]≤R<93 5,720 ('13$), ending at 3,170 ('18$) [7,573–3,892 in '24$]	93≤R≤155 8,170 ('13$), ending at 5,705 ('18$) [10,816–7,004, '24$]	R≥155 9,800 ('13$), ending at 6,970 ('18$); [12,974–8,557, '24$]	
Other note(s)	Started in 2009 in a few pilot cities. In 2010, expanded to include all private NEVs.	Various restrictions in this phase: higher range requirements, maximum speed requirements, minimum battery energy density, and maximum energy consumption relative to curb weight.			—

This Supplementary Note describes the direct incentive structures created by the Chinese government between 2009 and 2022 specifically in the context of private, personal battery electric vehicles, as well as the current incentive structure that is in place. Prior to 2009, incentives for "New Energy Vehicles" (NEVs)–electric vehicles, plug-in hybrid electric vehicles, and fuel cell vehicles–took the form of R&D grants, including through the high technology-focused "863 Program", [321] and thus are not included here. While the technology for

lithium ion batteries was first developed in England and the United States in the 1980s and first commercialized in Japan in the 1990s, [322] the technology reached technological maturity in China for personal electronics in the early 2000s, which was then rapidly adopted for electric vehicles. [323]

China was one of the first countries to create significant incentives for NEVs, with electric vehicle technology identified as a priority science research project in the 2001 Five-Year Plan. [324] In general, five key types of policies for NEVs were promulgated: 1. reduced purchase taxes, 2. direct subsidies to manufacturers, 3. mandated government procurements, 4. the Made in China 2025 (MIC2025) industrial policy, and, most prominently, 5. direct consumer subsidies. [99, 318, 325, 326] Additionally, local content requirements (at minimum one of the battery, motor, or drivetrain had to be produced locally) and the requirement that all foreign direct investment had to partner with a local joint venture, in contrast to local institutional support (market protection and subsidies) for domestic electric vehicle firms, have contributed to the proliferation of innovative technology and firm strategy in the country's nascent electric vehicle sector. [48] The Chinese government financially subsidized NEV purchases starting in 2009 for individual consumers in pilot cities, with multiple 'phases' of incentives and regulatory adaptations. [99, 100] As shown in Appendix Table B.3, Hao et al. [99] identify and describe the impacts of the first two phases of the EV subsidy. In Phase 1, active between January 2009 and September 2013, there was a simple ¥3000/kWh (440 2009USD) subsidy for any electric vehicle purchased, with a maximum subsidy amount limited to ¥60,000 (8,780 2009USD). Beginning in September 2013, a more sophisticated range-based subsidy was implemented. An update in 2016 required qualifying vehicles to meet stricter minimum range requirements (100 km minimum) as well as a minimum maximum speed limit of 100 km/h, [325] which was further revised in early 2017 to include a minimum battery energy density (90 Wh/kg) and maximum energy consumption relative to curb weight, as defined in Appendix Table B.4.

Table B.4. 2017 Energy Consumption-Curb Weight restriction.[325]

Curb weight (kg)	**Max. energy consumption (kWh/100 km)**
≤1,000	$0.014 \times m + 0.5$
1,000–1,600	$0.012 \times m + 2.5$
>1,600	$0.005 \times m + 13.7$

In addition to direct subsidies, there was a further exemption of NEVs from government purchase taxes (effectively, a sales tax), up to a certain amount. Vehicles purchased in China face a 10% purchase tax, which was waived starting in 2014. Set to originally expire in 2017, these tax exemptions were extended three times and currently are intended to expire in 2027, though starting in 2024 limitations apply. These exemptions are summarized in Appendix Table B.5.

Table B.5. Timeline of Purchase Tax Exemptions for NEVs in China.[327, 328]

Year(s)	**2014–2023**	**2024–25**	**2026–27**
Exemption, ¥	All vehicles from 10% tax	Up to 30,000	Up to 15,000
Exemption, $		4,170 (‘23$)	2,085 (‘23$)

During this time period, local governments were allowed to set additional subsidy levels, potentially differing from the central governments. For example, NEVs in Shanghai having their their license plate fee waived (plates can auction for 100,000 yuan or approximately $14,500), 3,317 million yuan ($510 million) subsidies offered by central and local fiscal funds since 2012 for BYD, and local cities initially had policies to purchase exclusively local-made cars (e.g. Changchun buying exclusively from FAW, Beijing from FOTON and Shenzhen from BYD). [48, 329]

Direct subsidies also existed at the local level. Shenzhen maintained consumer subsidies that were higher than the central government’s for a few years, with local companies such as BYD having models receiving ¥120,000 in credits (¥60,000 from Shenzhen and ¥60,000 from the central government). These local subsidies, combined with other local measures to spur production and innovation15 allowed for ‘local champions’ such as BYD, Nio, and Wuling Hongguang to emerge with products competing for different niches. However, as instances of fraud and collusion emerged, [52, 321] and the cost of maintaining such subsidies became increasingly prohibitive, direct subsidies were gradually phased out. In 2017, local incentives were capped at 50% of national levels, and removed entirely by June 2019. [321]

As demand for electric vehicles began to increase and the market began to grow, national subsidies were gradually reduced and restricted, as seen in Appendix Table B.6, with further restrictions applied for minimum fuel efficiency and battery density standards. [330] The minimum vehicle range for was raised to 300km in June 2019,100 and in 2020, a price cap of ¥300,000 (US$42,376) was set for battery electric vehicles in order to qualify for these credits. [52] These credits only avoided complete removal in 2020 due to the impact of COVID19, [321] and finally expired without extension at the beginning of 2023. [331] The

expiry of these direct purchase credits caused Tesla to slash prices in China to maintain its market share, leading to a price war that has only recently stabilized. [100]

Table B.6. Direct Subsidy of BEV purchases in China in 'Phase 3' (Subsidy Phase-Out and Implementation of Dual-Credit System).[332]

Active Time Period	**Range (km) Requirement**	**Range (miles) Requirement**	**Subsidy in ¥**	**Subsidy in $, at beginning of time period**
Feb. 2018 – June 2018	100–150	62–93	¥14000	$2220
	150–250	93–155	¥25200	$4000
	≥250	≥155	¥30800	$4890
June 2018 – Mar. 2019	150–200	93–124	¥15000	$2330
	200–250	124–155	¥24000	$3730
	250–300	155–186	¥34000	$5280
	300–400	186–249	¥45000	$6990
	≥400	≥249	¥50000	$7760
Mar. 2019 – June 2019	150–200	93–124	¥9000	$1340
	200–250	124–155	¥14400	$2140
	250–300	155–186	¥20400	$3040
	300–400	186–249	¥27000	$4020
	≥400	≥249	¥30000	$4460
June 2019 – July 2020	300–400	186–249	¥18000	$2600
	≥400	≥249	¥25000	$3610
July 2020 – Dec. 2022	300–400	186–249	¥16200	$2318
	≥400	≥249	¥22500	$3219

Alongside the reduction and eventual removal of direct subsidies, the Chinese government promoted a new supply-side subsidy mechanism in 2017, known as the 'dual-credit' system, based on California's EV incentive system. There's a Corporate Average Fuel Consumption (CAFC) credit, where carmakers gain credits for reducing the average fuel consumption to below a certain level and lose credits for failing to achieve it. The other credit is the New Energy Vehicle (NEV) credit, where carmakers receive positive credits for each EV produced

and negative credits for not producing a specified percentage of EVs in a year. The criteria are designed so most carmakers have a negative total of CAFC credits each year, and the criteria for getting positive credits is supposed to get harder every year. [101] A negative number of CAFC credits can be offset with positive CAFC or NEV credits, but negative NEV credits can only be offset with positive NEV credits; a positive NEV credits are the only tradeable credits. [333] To avoid a(n alleged) penalty in the form of a next-year production cap, carmakers need to either make EVs of their own or purchase NEV credits from other companies.

Each year, China's Ministry of Industry and Information Technology (MIIT) releases annual credit totals for all automobile companies operating in China, with most prominent firms (e.g. joint ventures between Volkswagen, General Motors and their Chinese partners) accumulating hundreds of thousands of negative credits. But Tesla, Chinese company BYD and other EV makers generate large surpluses of NEV credits that they can then sell. As a result, some Chinese companies are even selling EVs below cost, as long as the revenue from selling the credits can make up for the shortfall. While carryover of accumulated credits is allowed from year to year, there is a disincentive to do so - any NEV credit that's not used is discounted by half going into the next year.

Table B.7. NEV Credit Generation under the Dual-Credit Policy, where R = range in km.[100, 328]

Timeframe	**Jan 2019 – Dec 2020**	**Jan 2021 – July 2023**	**August 1 2023 – Present**
BEV Cars	$0.012 \cdot R + 0.8$, upper limit of 6	$0.0056 \times R + 0.4$, upper limit of 3.4	$0.0034 \times R + 0.2$, upper limit of 2.3
Plug-in Hybrids	1 if R $\geq$ 50 2 if R $\geq$ 80	1.6	1

The number of credits generated are calculated based on the range of the vehicle, as described in Appendix Table B.7, with increasing strictness. Every year for a few designated months in the summer, ending in August, these companies are supposed to trade their credits with one other directly. There's no public trading platform, so each participant has to negotiate a price based on their own deal-making abilities. That has motivated most of them to stay silent about the price and the volume of the transactions. But from media reports and numbers released by the government, it's clear that the average price of a NEV credit has changed drastically over the years: it went from about $50 in 2019 to $180 in 2020 to approximately $360 in 2021. [334] Reuters reported that Tesla reached a deal with a

Volkswagen JV in China to sell NEV credits at $460 each in April 2021, [331] and there was some speculation of upwards of $900 per credit in 2022. [101] As a result, production of one EV can equate to thousands of dollars of extra profit in a given year, collectively adding up to a major revenue stream for EV companies.

This policy has had unintended consequences. Though the CAFC system was intended to get carmakers to improve internal combustion engine vehicle efficiencies, because NEV credits were much easier to get relative to the amount of investment needed to receive more positive CAFC credits, most carmakers have not bothered to improve their traditional engine models. Furthermore, as the range requirements for EVs to qualify for the NEV credit are still relatively limited (300 km or 186 miles), companies are incentivized to produce EVs that barely meet the mileage limit rather than cars that far exceed the minimum.

B.3 Details of Electric Vehicle-Related Subsidies in the Inflation Reduction Act

Passed in August 2022, the Inflation Reduction Act (IRA) promotes a wide variety of clean technologies through a series of tax credits, including in the realm of electric vehicles, among a wide range of other provisions and industries. The provisions specifically relating to electric vehicles' manufacture or purchase are Section 13401-Clean Vehicle Credit (Section 30D of the IRS Code), Section 13402-Credit for Previously-Owned Clean Vehicles (Section 25E), and Section 13403-Qualified Commercial Clean Vehicles (Section 45W), while Section 13502-Advanced Manufacturing Production Credit (Section 45X) provides additional incentives further up the supply chain (battery module, cell, and component manufacturing, as well as critical mineral processing and production). We discuss in detail each of these provisions and their requirements.

B.3.1 Clean Vehicle Credit (Section 13401 / 30D)

The first and most complex of these provisions involves Section 13401, the Clean Vehicle Credit, which corresponds to Section 30D of the Internal Revenue Code.105 This section provides two transferable tax credits for producers of new clean vehicles (including fuel cell vehicles, but primarily focusing on battery electric vehicles), as described in Appendix Table B.8. To qualify for either credit, the initial conditions identified in Appendix Table B.9 must be met. The two tax credits are based on the critical minerals and components used in the 'battery from which the electric motor of such [sic] vehicle draws electricity,' respectively. Each tax credit is for up to $3,750, which may be transferred to the vehicle dealer for point of sale credit given the purchaser has any federal tax burden. [335] The provision went into effect in 2023, replacing the existing Section 30D plug-in EV tax credit, and expires for vehicles acquired after December 31, 2032.

Table B.8. Electric Vehicle Credit Details (Section 13401).

	Critical Mineral Credit	**Battery Component Credit**
Amount	$3,750 per qualifying vehicle	$3,750 per qualifying vehicle

Table B.8. Continued.

	Critical Mineral Credit	**Battery Component Credit**
Yearly Targets	The relative share of the total dollar value of minerals used in the battery fulfilling the sourcing requirement must exceed: • 40% before 2024 • 50% by 2024 • 60% by 2025 • 70% by 2026 • 80% between 2027–2032	The relative share of the total dollar value of minerals used in the battery fulfilling the sourcing requirement must exceed: • 50% before 2024 • 60% in 2024 and 2025 • 70% by 2026 • 80% by 2027 • 90% by 2028 • 100% between 2029–2032
Sourcing Requirement	(i) extracted or processed (I) in the United States, or (II) in any country with which the United States has a free trade agreement in effect, or (ii) recycled in North America	Must be manufactured in North America
Additional Requirements	Beginning 2025, no percentage of the material can be extracted or processed in or by an entity of foreign concern	Beginning 2024, no percentage of component manufacturing can occur in or by an entity of foreign concern
	• Final assembly of the vehicle must take place in North America. • The vehicle must be an electric vehicle with battery capacity of ≥ 7 kWh or a qualified fuel cell vehicle. • The amount of the credit cannot exceed 30% of the purchase price of the vehicle. • No credits shall be allowed for vehicles that cost more than \$55,000 for a sedan, \$80,000 for a van, \$80,000 for an SUV, and \$80,000 for a pick up truck. • The credit is not available to individuals earning more than an adjusted gross income of \$150,000, \$225,000 for head of household, or \$300,000 for partners married and filing jointly. There is no phase down on this — consumers are ineligible if they earn any more than those caps.	

Table B.9. Requirements to receive each part of the Clean Vehicle Tax Credit.

Final assembly of the vehicle must take place in North America.
The vehicle must be an electric vehicle with battery capacity of ≥ 7 kWh or a qualified fuel cell vehicle.
The amount of the credit cannot exceed 30% of the purchase price of the vehicle.
No credits shall be allowed for vehicles that cost more than $55,000 for a sedan, $80,000 for a van, $80,000 for an SUV, and $80,000 for a pick up truck.
The credit is not available to individuals earning more than an adjusted gross income of $150,000, $225,000 for head of household, or $300,000 for partners married and filing jointly. There is no phase down on this — consumers are ineligible if they earn any more than those caps.

B.3.1.1 Foreign Entity of Concern Regulation

The two 30D New Clean Vehicle credits can only be claimed when an electric vehicle is sold to an 'end user', but qualification for the credits depend on choices the automaker makes regarding their supply chains. For both credits, no critical minerals or components can originate from a 'Foreign Entity of Concern' throughout the entire supply chain. [105] This designation is applied to firms "owned by, controlled by, or subject to the jurisdiction or direction of a government of a foreign country that is a covered nation", where those covered nations are specifically China, Russia, North Korea, or Iran. [105] In practice, this means any firm that is i) directly owned by the government of these four countries, ii) owned by a citizen of these countries or iii) headquartered in these countries, or subject to the 'control' or 'direction' of any entity that falls under these three categorizations, cannot qualify for the two credits. In particular, no more than 25% of the entity's equity or board seats can be owned by FEOCs, and licensing agreements with other companies cannot let "that other entity to exercise effective control over the extraction, processing, recycling, manufacturing, or assembly (collectively, "production") of the critical minerals, battery components, or battery materials that would be attributed to the entity." These strict limitations add additional limitations beyond just the geographical location of material production.

This FEOC limitation only applies to battery components in the year 2024 but expands to include all critical minerals within batteries by 2025. However, a Final Regulation [336] provided a two-year exemption (to the beginning of 2027) for "graphite contained in an-

ode materials (both synthetic and natural) and applicable critical minerals contained in electrolyte salts, electrode binders, and electrolyte additives", as they were deemed "impracticable to trace" in the current market.

B.3.1.2 Critical Mineral Requirement for 30D New Clean Vehicle Credit

For the **critical mineral requirement** for the first $3,750 of the credit, the *relative share of the total dollar value* of minerals used in the battery fulfilling the sourcing requirement must exceed certain yearly targets: 40% before 2024, 50% by 2024, 60% by 2025, 70% by 2026, and 80% between 2027-2032. The source of these relative shares of material must be either be (i) extracted or processed (I) in the United States, or (II) in any country with which the United States has a free trade agreement in effect, or (ii) recycled in North America. The minerals that are considered critical are listed in Section 13502-Advanced Manufacturing Production Credit, described in Appendix B.3.2.

The 'free trade agreement' clause has been interpreted to not only include countries with traditional free trade agreements (e.g. South Korea, Australia, Chile, Canada), but also those in which US trade representatives have struck specific deals about battery materials, namely Japan and the United Kingdom, with trade representatives are working on a similar deal with the European Union (see Appendix B.1.5). [337] However, these agreements may face legal constitutional challenges as the executive branch may be viewed as making trade decisions without the permission or direction of the legislative branch, [338] and the fact that they may break WTO law. [83]

For the years 2023-2024, treasury guidance has established a '50% value added test' to determine if minerals qualify as being produced in a certain geography. For each procurement chain (geographical location), as long as 50% or more of the mineral extraction, mineral processing, or mineral recycling occurred in the United States or a free trade agreement country, the whole batch of materials is considered to qualify. Summing the dollar values of all qualifying critical minerals and dividing by the total dollar value of all critical minerals in the product. This means in practice that these critical minerals could still pass through (and be wholly dependent on) a processing step in a country of concern, if that step is critical but does not add more than 50% of the added value. Treasury indicates that this requirement may be tightened after 2024 as automakers firm up their supply chains.

B.3.1.3 Battery Component Requirement for 30D New Clean Vehicle Credit

For the **battery component requirement**, a similar *relative share of the total dollar value* of the all of the components of the battery must exceed a stricter set of yearly targets: 50% before 2024, 60% in 2024 and 2025, 70% by 2026, 80% by 2027, 90% by 2028,

and 100% between 2029-2032. Furthermore, the geographical restriction of the source of these components is stricter, as they must be manufactured in North America, similar to the vehicle manufacturing requirement. A similar limitation banning any components from being sourced in countries of concern goes into effect in 2024. It is critical to note that the materials (e.g. cathode and anode powders) that are used in the manufacturing of battery components are classified as battery minerals and not battery components themselves, as currently defined in Treasury guidance. [75] This allows them to be sourced in trade-agreement countries and not manufactured in North America, a significant interpretation that will allow for significantly easier fulfillment of the requirements (given that very little of any battery component, particularly the very costly cathode component, is manufactured in North America). [17] It is also important to note that treasury guidance indicates battery management systems and thermal management systems do not count as components, as they are not components "from which the electric motor of such [sic] vehicle draws electricity." The guidance defines components as including, but not being limited to, "a cathode electrode, anode electrode, solid metal electrode, separator, liquid electrolyte, solid state electrolyte, battery cell, and battery module." Furthermore, to fulfill the requirement, "substantially all of the manufacturing or assembly" would need to occur in North America, "without regard to the location of the manufacturing or assembly activities of the components that make up the particular battery component." This means that current interpretation of Treasury guidance could allow cell production to occur wholly within e.g. South Korea, as long as all battery module production occurs in the United States. As a result of these limitations and promulgated guidances, [338] in 2024, only 23 fully electric and 1 plug-in hybrid electric vehicles models out of 621 electric or plug-in hybrid electric models qualify for both credits. [339] An additional 14 and 6 models, respectively, only qualified for one credit, though which of the credits the models qualify for is not specified publicly. [339] (17 fully electric vehicles qualified for both credits before the exemption was announced, while 5 and 4, respectively, qualified for only one. [339]) With the additional critical minerals requirement, supply chain vulnerability analyses [17] indicate that likely even fewer models will qualify in the future as the requirements become more stringent, primarily based on current supply chain dependence on China, assuming limited changes in supply chains.

B.3.2 Advanced Manufacturing Credit (13502 / 45X)

The second major set of provisions related to electric vehicle battery material supply chains are detailed in Section 13502-Advanced Manufacturing Production Credit, corresponding to Section 45X of the Internal Revenue Code, [340] as described in Appendix Table B.10. These production credits provide (new) incentives for manufacturers for a wide

variety of clean technologies, from wind turbines and solar panels to every step of the electric vehicle battery supply chain, but only applies to manufacturing within the United States. The relevant industries where the 45X credit applies in the context of this analysis are the critical mineral processing, electroactive materials manufacturing, battery cell manufacturing, and battery module manufacturing industries.

The full value of these credits is available for sales between 2022-2029, which then phases down as follows: 75% of the full value in 2030, 50% in 2031, 25% in 2032 and 0% thereafter. This does not include the critical minerals production credit, which retains its full value and is effectively permanent. It is unclear how recycling of materials is considered for these credits, but as it is either a 'critical minerals production process' or production of 'electroactive materials', either 10% credit applies for now.

Table B.10. Advanced Manufacturing Credit Details (Section 13502).

Component	**Amount**	**Restrictions and Notes**	**Phase Out**
Battery Cells	$35 per kWh	Not less than 20 Wh storage capacity and 100 Wh/l energy density	75% of the full value in 2030, 50% in 2031, 25% in 2032, and 0% thereafter
Battery Modules	$10 per kWh	Not less than 5 kWh in size; $45 per kWh if the module does not use cells	”
Electroactive materials (cathodes, anodes, etc)	10% of cost		”
Critical minerals production process	10% of cost	Limited to materials considered critical — in this case Lithium, Cobalt, Nickel, Manganese, Graphite, Aluminum	None

This section also lists all minerals that are considered critical and thus qualify for this incentive (and also count towards the battery critical mineral requirements), but the relevant ones for battery materials are aluminum, cobalt, graphite (natural and synthetic), lithium, manganese, and nickel [17, 19] (and notably *not* copper).

B.3.3 Used Clean Vehicle Credit (Section 13402 / 25E) and Commercial Clean Vehicle (13403 / 45W)

It is also important to note that Section 13402-Credit for Previously-Owned Clean Vehicles provides a new federal tax credit of an amount equal to the lesser of $4,000, or 30% of the sale price, for used EVs priced below $25k, subject to several additional requirements, listed below. A used vehicle qualifies for tax credit only once in its lifetime.

- Lower annual income limits: $150,000 for those filing jointly, $112,500 for head of household, or $75,000 otherwise
- Have a model year at least two years earlier than the calendar year when you buy it
- Not have already been transferred after August 16, 2022 to a qualified buyer
- Have a gross vehicle weight rating of less than 14,000 pounds
- Be an eligible FCV or plug-in EV with a battery capacity of least 7 kilowatt hours (kWh)
- Be for use primarily in the United States
- Be purchased from a dealer as an individual (no businesses)

There is an additional 'loophole' created through Section 13403-Qualified Commercial Clean Vehicles, which allows for the full $7,500 tax credit to be claimed for any electric vehicle that is purchased for commercial use, regardless of any critical mineral or battery component requirement, as well as the income and price cap limits. Treasury has interpreted this to include leases of electric vehicles, which has led to over one-third of electric vehicles purchased in April 2023 to be leased, up from an average of 10% through the first quarter of 2023. [341] This interpretation was heavily pushed for by Korean automakers Hyundai and Kia, who, despite qualifying under the Critical Mineral requirement, currently do not manufacture their vehicles in the United States (though plans are underway to do so) [301]. There are three limitations in this. First, the credit is non-transferrable, so the leasing company does not necessarily need to pass on the full value of the credit, though in practice there is no incentive not to. Furthermore, as most vehicle leases are for two to three years, buyers may be dissuaded from leasing as they do not get to keep the vehicle long term. Lastly, the value of the leasing credit is determined by the 'incremental cost' of a typical electric vehicle in comparison to a 'comparable' internal combustion vehicle. However, given potential battery degradation issues, the continued growth in battery size and vehicle ranges,

and increasingly modern technology in newer vehicles combined with typically lower monthly leasing costs and down payments, this effect may be less muted in comparison to traditional internal combustion engine vehicles.

B.4 Additional Results

Table B.11. Sensitivity Analysis of Modeled Battery Production Costs under different scenarios.

Default values of BatPaC (including 70 kWh and 500,000 packs) are used where input parameters are not specified. **Bold** indicates the default value. Sensitivity ranges were determined using estimated production costs in China described in the main methods, as well as empirical evidence of e.g. land costs in China [49], costs of industrial electricity in the United States [342], and the labor rate for unionized workers [343].

	# Packs/year (thousands)			**Land and building costs** ($/m^2)				**Direct Labor Rate**			**Energy Costs** (cents/kWh)			
	50	100	**500**	0	800	**1000**	3000	$6.50	**$25**	$42	3.2¢	**4.0¢**	8.8¢	15.0¢
LFP	$127	$122	$117	$115	$116	$117	$120	$113	$117	$120	$116	$117	$119	$122
NMC811	$144	$139	$134	$133	$134	$134	$136	$131	$134	$137	$133	$134	$136	$139
NCA	$150	$145	$140	$139	$140	$140	$142	$137	$140	$143	$140	$140	$142	$145

Table B.12. Extended results of modeled battery production costs under different scenarios, and different battery sizes.
Note all NMC results presented here are for the 811 (nickel-cobalt-manganese) chemistry.

Battery Size	**Scenario**	**0. Baseline** No Policy	**1. Full IRA** Mfr. Qualifies for All IRA Credits	**2. FEOC** Mfr. Supply Chain Includes FEOC	**3. Leasing** Relative impact of leasing credit only
	US Policy	None	IRA	IRA	IRA
	Chinese Policy	None	Dual Credit	Dual Credit	Dual Credit
100 kWh	US-LFP*	$106	-$20	$55	$31
	US-NMC	$121	-$8	$67	$46
	US-NCA	$128	-$2	$73	$53
	China-LFP*	$93	$70	$70	$70
	China-NMC	$107	$84	$84	$84
	China-NCA	$113	$90	$90	$90
85 kWh	US-LFP	$111	-$22	$66	$23
	US-NMC	$126	-$9	$79	$38
	US-NCA	$132	-$3	$85	$44
	China-LFP	$97	$70	$70	$70

Table B.12. Continued.

	Scenario	0. Baseline	1. Full IRA	2. FEOC	3. Leasing
	China-NMC	$112	$84	$84	$84
	China-NCA	$117	$90	$90	$90
70 kWh	US-LFP	$117	-$10	$97	$10
	US-NMC	$134	$5	$113	$27
	US-NCA	$140	$11	$118	$33
	China-LFP	$103	$70	$70	$70
	China-NMC	$119	$86	$86	$86
	China-NCA	$124	$91	$91	$91
50 kWh	US-LFP	$127	-$84	$66	-$23
	US-NMC	$147	-$71	$79	-$3
	US-NCA	$153	-$65	$85	$3
	China-LFP	$111	$65	$65	$65
	China-NMC	$130	$84	$84	$84
	China-NCA	$135	$89	$89	$89
40 kWh	US-LFP	$148	-$91	$97	-$40
	US-NMC	$167	-$75	$113	-$20
	US-NCA	$173	-$69	$118	-$14
	China-LFP	$131	$73	$73	$73
	China-NMC	$148	$91	$91	$91
	China-NCA	$154	$96	$96	$96

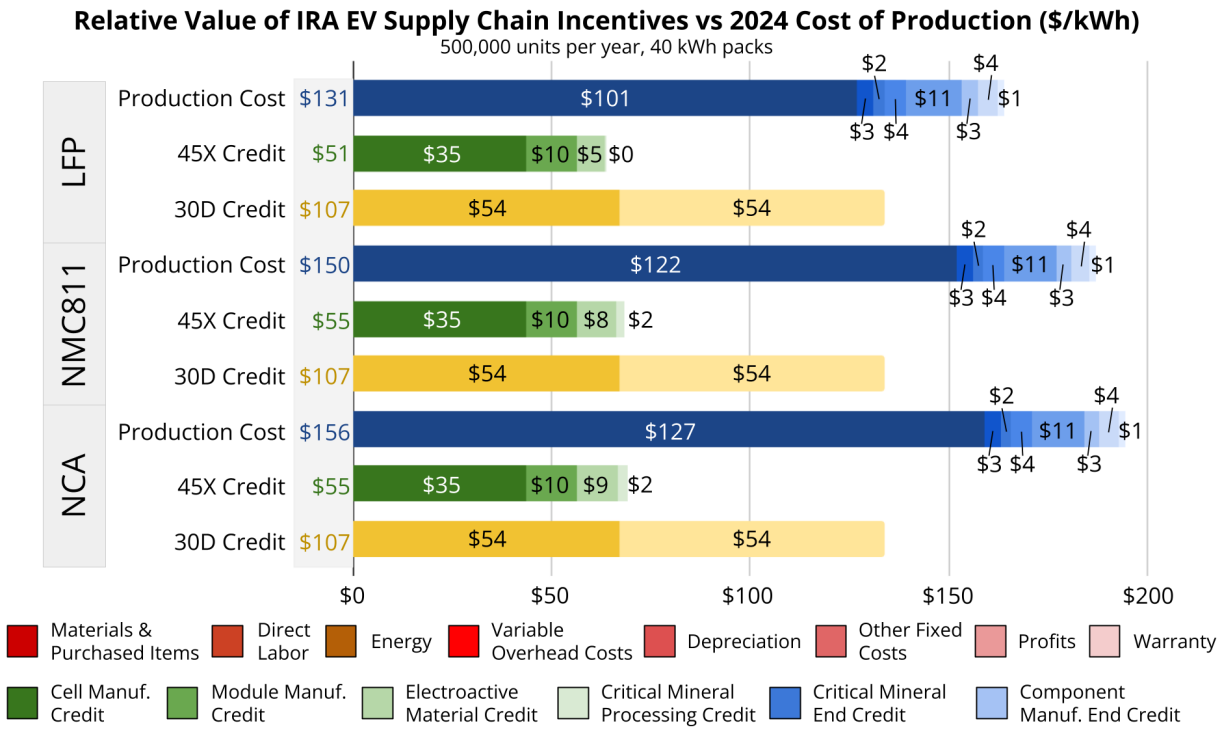


Figure B.1. Breakdown of modeled battery production costs, 40 kWh packs
Please note the different x-axis scale in comparison to Figure 3.3 and the other Appendix Figure B.2.

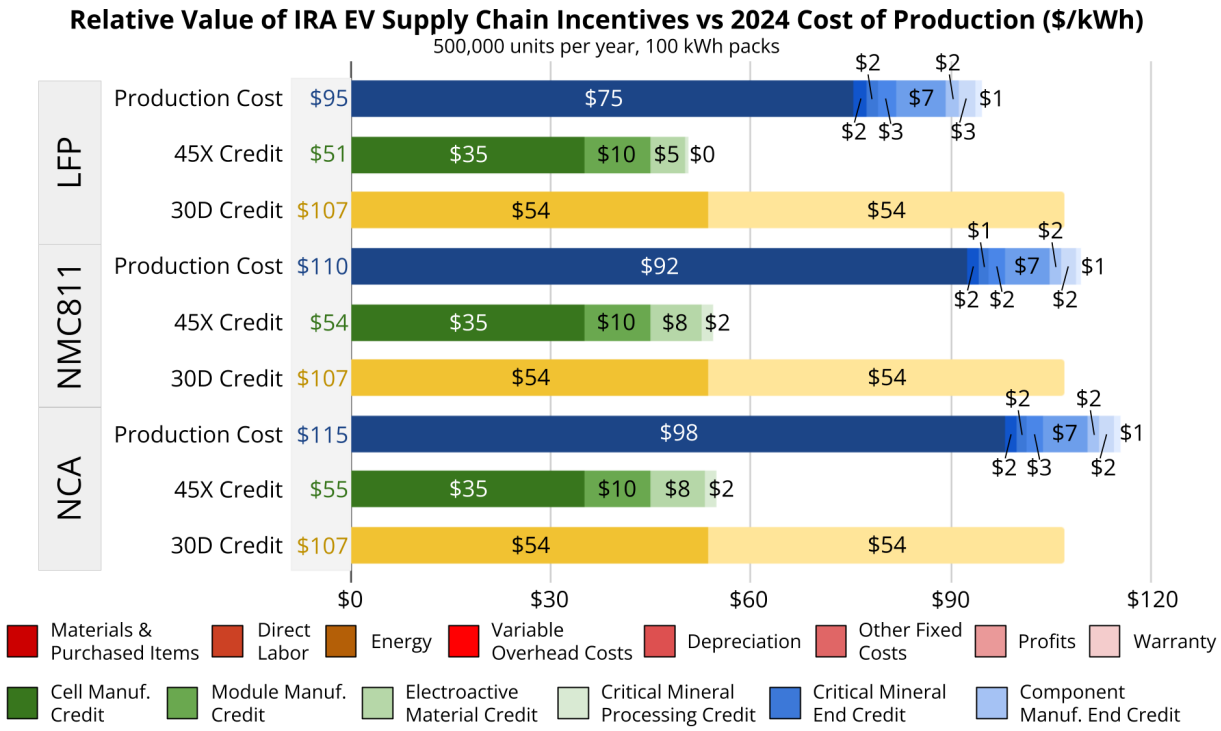


Figure B.2. Breakdown of modeled battery production costs, 100 kWh packs
Please note the different x-axis scale in comparison to Figure 3.3 and the other Appendix Figure B.1.

Appendix C
Supplementary Appendix for Chapters 4

C.1 Literature Review

Many studies of the lithium system and broader lithium battery system have been conducted, evaluating all aspects from costs, environmental impacts of many types, security, process technology choices, and more. Our study aims to answer questions about the onshoring of lithium production in the United States, and compares against currently marginal production – lepidolite resources in China. We suggest that this security and policy-focused framing combined with engineering details provides unique insights in comparison to a very extensive set of previous studies, compiled primarily from the last five years, as shown in Table C.1.

In particular, the two dimensions that most distinguish this work from others are the detailed process modeling and connection between cost and emissions modeling, as very few papers do both. More importantly, even amongst such papers, open, examinable models are not published, making conclusions difficult to evaluate and extend. There are limitations to our scope. For example, we focus on three lithium production steps in this paper, and further modeling of other pathways would be necessary for a complete competitive picture. Multi-material papers do exist with some process-level modeling, usually of downstream steps (e.g. battery manufacturing), but modeling of specific critical minerals production pathways such as lithium is often treated at a lower resolution. Additionally, on the emissions side, this paper only covers greenhouse gas emissions. Other impact categories, such as air quality considerations and water usage, are important elements of the lithium sustainability question, but our framework is extensible to these in future work. Furthermore, studies (e.g. Thomsen and Lux [344]) have indicated that the impacts of transportation on the critical mineral and battery supply chains are undercounted. Although the impacts were found to be relatively negligible in this study given the constrained geographical scope, extensions of this framework to multi-country production networks would necessitate more detailed modeling of these transportation impacts that many studies miss. Finally, the approach towards uncertainty in this study should be noted. A significant limitation in LCA studies (and to a lesser extent,

TEA studies) is that they make a singular point estimate, without considering operating and input data uncertainty. Some studies do make this consideration, as seen in the table below, with treatments ranging from full Monte Carlo and probabilistic uncertainty methods to parametric variation to simple scenario-based analyses. Probabilistic methods can provide a rich picture of output distributions, but require well-characterized input distributions, which can be challenging to measure for emerging lithium production pathways. Parametric variations are more tractable, but applying relatively arbitrary, uniform swings (e.g. ±20%, ±50%) without grounding such variation in what is actually possible can lead to overemphasis on variables that have little real leverage and underweighting those that do. Thus, we use empirically estimated conservative, midpoint, and optimistic estimates for triangular sensitivity analysis to represent genuine operating uncertainty.

Further detailed discussion of our choice of system boundaries and modeling scope is given in the following sections.

Table C.1. Literature Review Scope and Attribute Matrix. Filled cells indicate the attribute is present. The yellow row is the present study; green rows are notable related papers. Blue-tinted headers indicate key contribution dimensions. Grey codes index other studies; see Table C.2 for the legend.

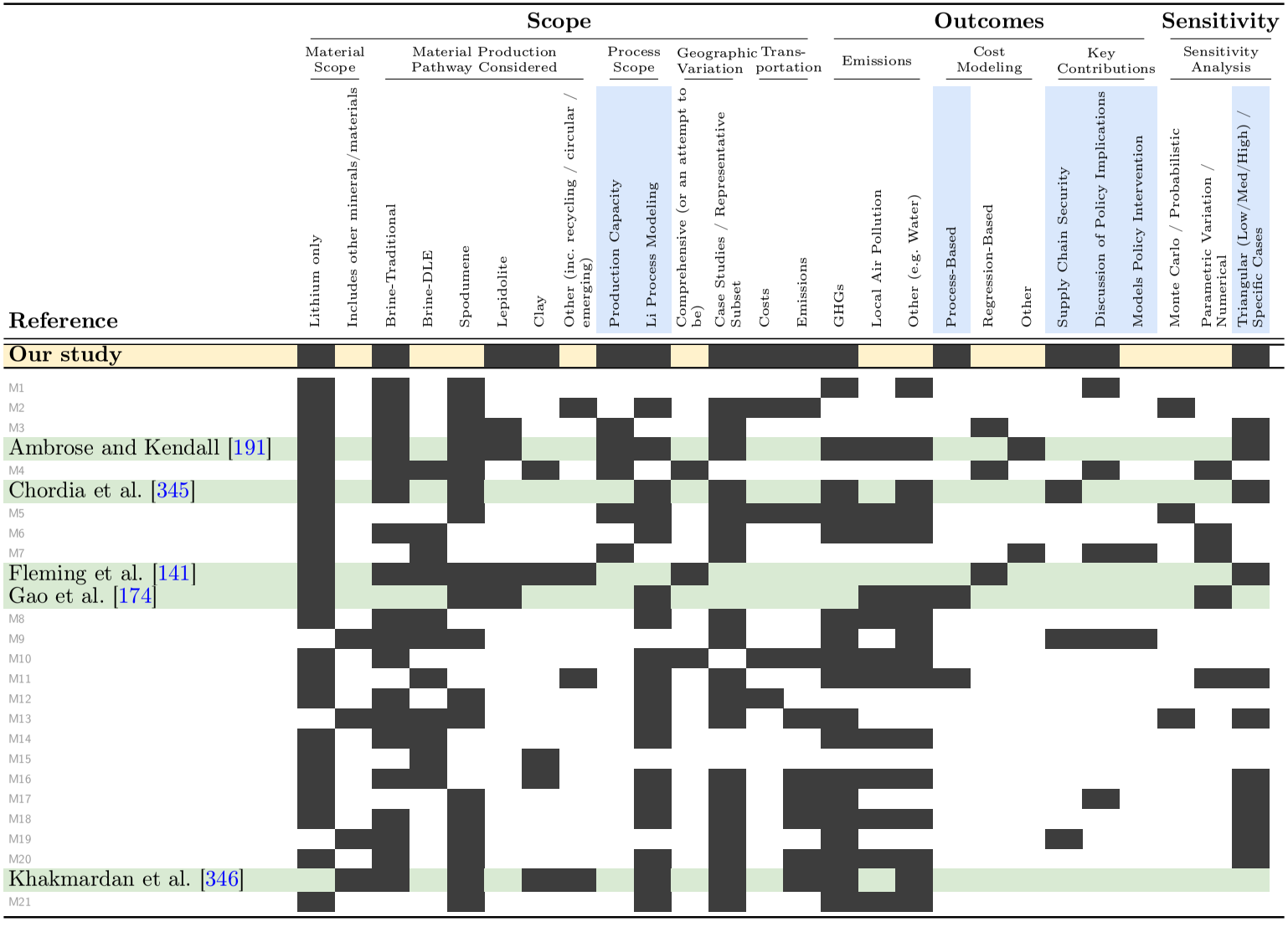


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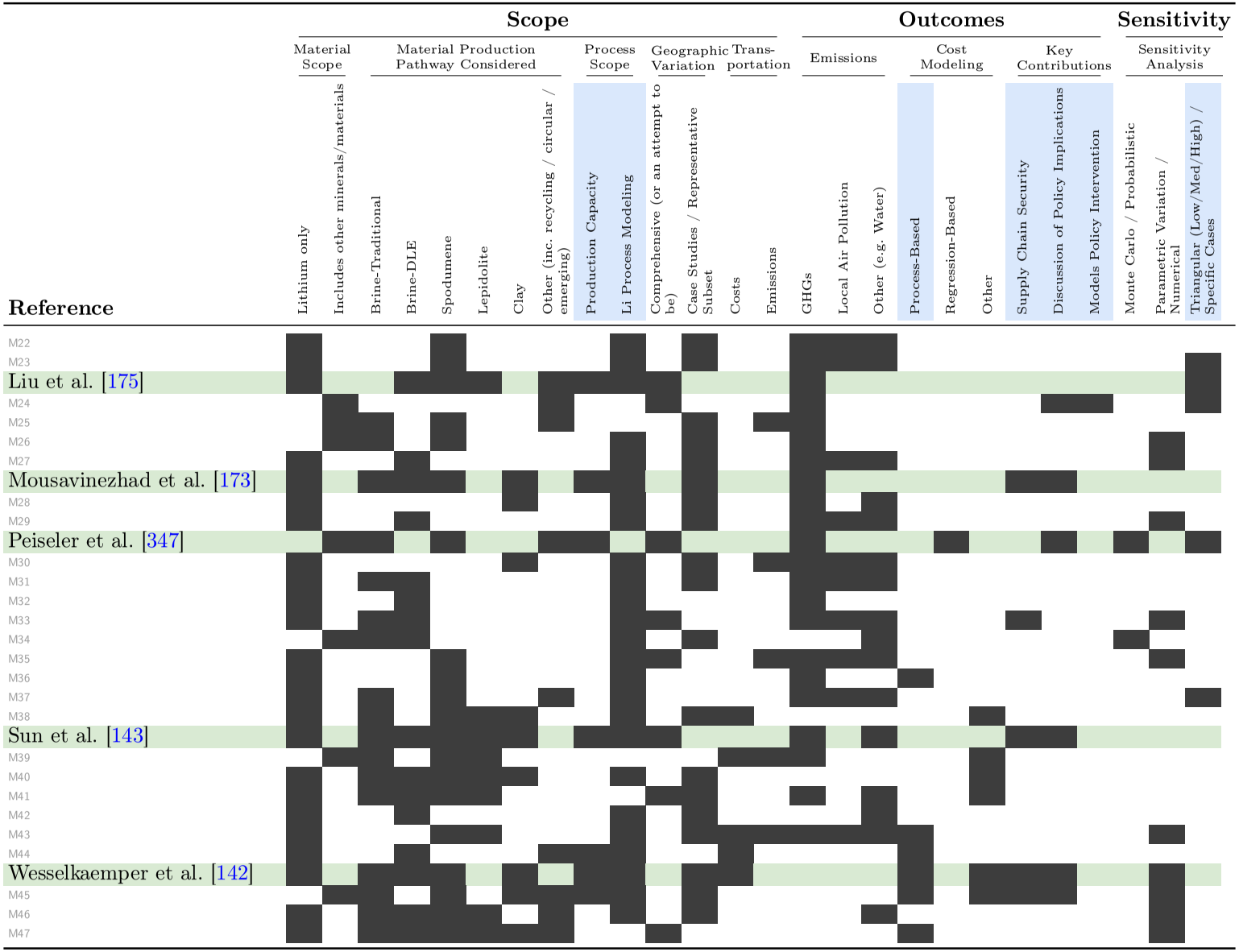


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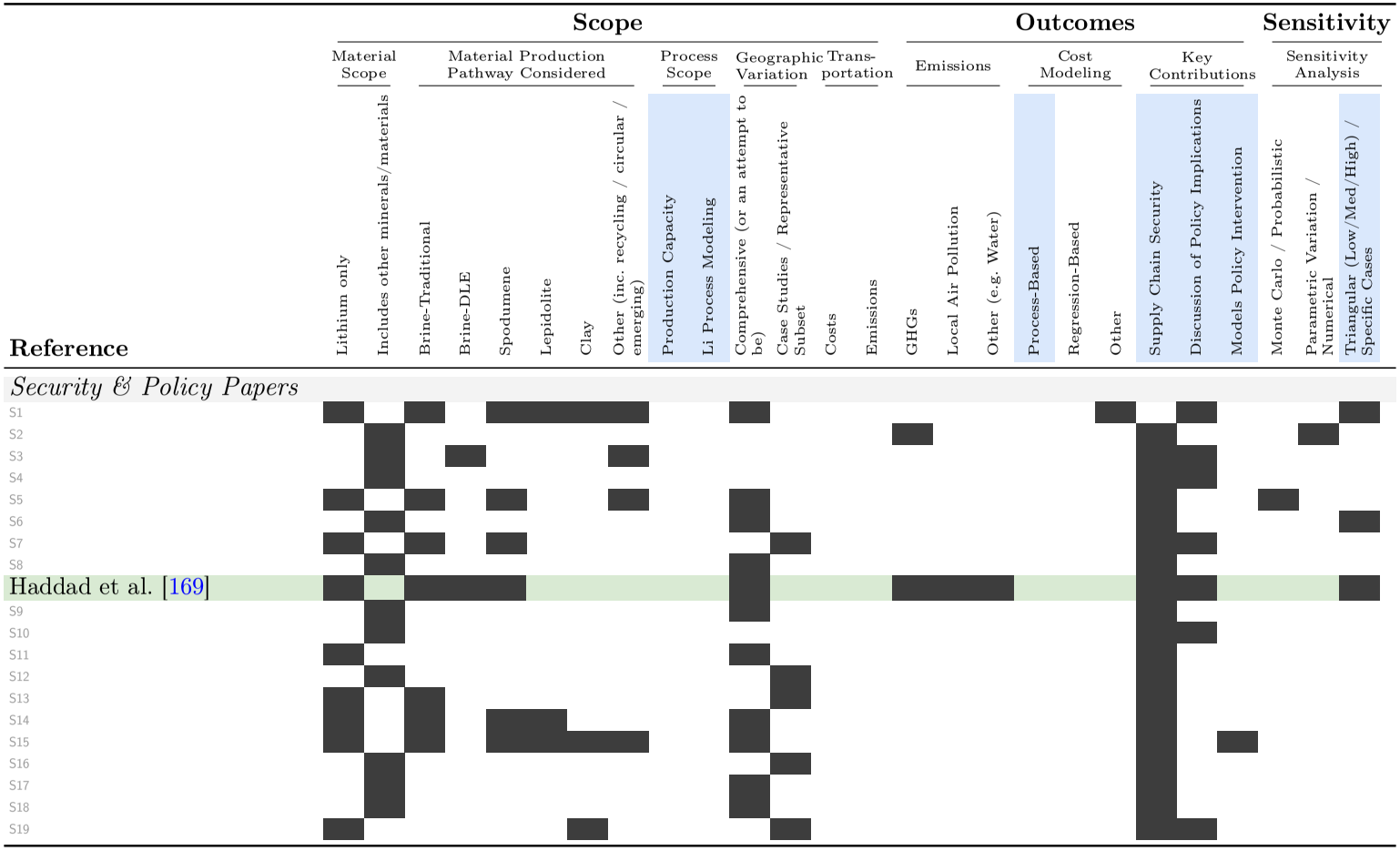


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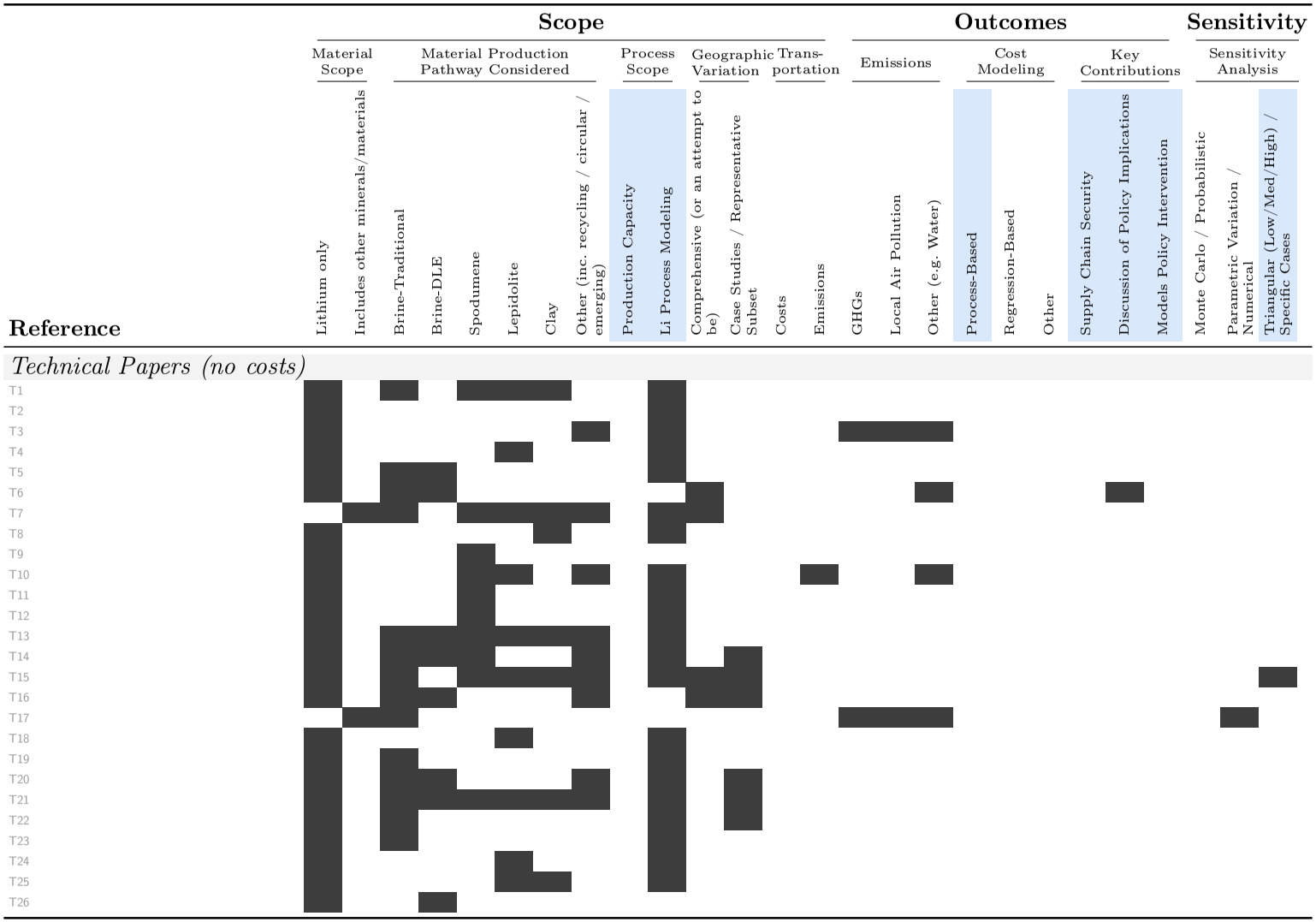


Table C.2. Extended Lithium Literature Review Code Legend.
Codes prefixed: M = main LCA/TEA papers (n=47); S = security & policy papers (n=19); T = technical papers, no costs (n=26).

Code	Reference	Code	Reference	Code	Reference
M1	Agusdinata et al. [348]	M32	Schenker et al. [349]	S16	Ren et al. [350]
M2	Alessia et al. [351]	M33	Schenker and Pfister [192]	S17	Sun et al. [2]
M3	Ambrose and Kendall [187]	M34	Schomberg et al. [214]	S18	Tan and Keiding [352]
M4	Busch et al. [170]	M35	Setu and Strezov [353]	S19	Vivoda et al. [354]
M5	Feng et al. [355]	M36	Mousavinezhad et al. [356]	T1	Averill and Olson [357]
M6	Feng et al. [358]	M37	Stamp et al. [359]	T2	Battaglia et al. [360]
M7	Fitzgerald and Spiller [361]	M38	Sterba et al. [362]	T3	Bonk [363]
M8	Halkes et al. [144]	M39	Thomsen and Lux [344]	T4	Deng and Feng [180]
M9	Hanna et al. [364]	M40	Tran and Luong [208]	T5	Ehren and Alem [365]
M10	He et al. [366]	M41	Vanneste and Van Der Bruggen [367]	T6	Flexer et al. [198]
M11	Huang et al. [368]	M42	Vera et al. [369]	T7	Garrett [196]
M12	Ibarra-Gutiérrez et al. [370]	M43	Wang et al. [176]	T8	Haddad et al. [371]
M13	Istrate et al. [372]	M44	Warren [172]	T9	Kundu et al. [373]
M14	Iyer and Kelly [374]	M45	Wesselkaemper et al. [375]	T10	Li et al. [376]
M15	Iyer and Kelly [377]	M46	Yelatontsev and Mukhachev [378]	T11	Liu et al. [379]
M16	Iyer and Kelly [194]	M47	Zhou et al. [380]	T12	Liu and Azimi [381]
M17	Iyer et al. [382]	S1	Andrade et al. [383]	T13	Liu et al. [384]
M18	Jiang et al. [385]	S2	Ballinger et al. [240]	T14	Marcinov et al. [386]
M19	Kallitsis et al. [387]	S3	Bridge and Faigen [388]	T15	Meshram et al. [389]
M20	Kelly et al. [390]	S4	Busch et al. [170]	T16	Munk et al. [391]
M21	Khakmardan et al. [193]	S5	Calisaya-Azpilcueta et al. [392]	T17	Notter et al. [393]
M22	Khakmardan et al. [394]	S6	Cheng et al. [17]	T18	Park et al. [210]
M23	Lappalainen et al. [395]	S7	Egbue and Long [396]	T19	Perez et al. [397]

Table C.2. Continued.

Code	Reference	Code	Reference	Code	Reference
M24	Llamas-Orozco et al. [398]	S8	Greitemeier et al. [135]	T20	Stringfellow and Dobson [399]
M25	Machala et al. [400]	S9	Helbig et al. [8]	T21	Swain [401]
M26	Manjong et al. [190]	S10	Hensel [402]	T22	Torres et al. [197]
M27	Mousavinezhad et al. [403]	S11	Jin et al. [404]	T23	Wilkomirsky [405]
M28	Mousavinezhad et al. [356]	S12	Manley et al. [406]	T24	Yan et al. [205]
M29	Nikfar et al. [407]	S13	Miatto et al. [408]	T25	Zhao et al. [202]
M30	Roy et al. [195]	S14	Ouyang et al. [409]	T26	Wang et al. [176]
M31	Schenker et al. [410]	S15	Prina Cerai [137]		

C.1.1 Firm-level process modeling, costs, and economic incentives

At a theoretical level, the structure of global value chains and which locations become developed are driven by costs borne by firms, compared to their competitiveness with current and future prices. Comparative cost advantages exist between locations that depend on 'factor endowments' (e.g., labor, capital, resources) [411, 412], mediated by regulation. For a commodity, the structure of the market is generally determined by the price of the commodity, with mines 'turning on' and 'turning off' when the price of the commodity exceeds their marginal operating cost, mediated by uncertainty in future prices. While lithium is not a pure commodity in the sense that significant amounts of production are under contract, we can model 'competitiveness' from a national security perspective at a first order by understanding the relative operating production costs of differing production pathways.

We do so via technoeconomic modeling: a generalized set of methods to determine production costs as a function of location and production pathway (technology choice). While many applications of technoeconomic methods have been applied to battery and electric vehicle manufacturing (e.g. [128, 130, 131]), far fewer works have evaluated production costs through the upper and midstream battery supply chain, particularly in the context of lithium.

This gap may stem from the ongoing pace of technological advancements in lithium extraction and refining, as well as the wide range of technologies at varying stages of maturity, from well-established methods like spodumene processing to emerging processes such as direct lithium extraction (DLE). The diversity of technological readiness, combined with

limited data availability and rapidly shifting market conditions, makes it challenging to develop standardized cost models across the upper and midstream supply chain. As such, many papers in the literature provide reviews of bench-scale or even industrial-scale processes from a scientific, technological perspective (see the non-comprehensive 20 studies listed under 'Technical Review Papers' in Table C.1), but do not fully consider the economic dimension.

For those that do, two primary methods emerge: regression-based models and process-specific models.

Regression-based cost modeling can give a general sense of costs given a certain project's characteristics. Three studies use regression models to predict costs without considering emissions: Ambrose and Kendall [187], Fleming et al. [141], and Zhou et al. [380]. All of these studies rely on white and grey literature databases, often S&P Global Capital IQ cost data, estimating average production costs from input variables for each project (e.g., deposit grade and location, details in company reports, and evaporation rate for brine deposits). These studies are limited when production in different countries and pathways needs to be understood at a detailed level; in particular, the effect of targeted policy interventions cannot be easily well-understood (e.g., reducing cost of capital, subsidies for specific material inputs, etc.). Furthermore, regression-based models suffer when input data is sparse, which is the general case for the lithium supply chain, and cannot adequately predict the costs of developing technologies (e.g., direct lithium extraction). Even some mature projects have production costs that appear to be notably overstated or understated, for example $10,300 for Silver Peak [187, 380] versus the $4,290 midpoint of our model and $4,600—6,750 range reported in other literature, or $1,200 for lithium production from lepidolite in Yichun versus the $10,900 midpoint of our model and $10,000-16,680 range reported in other literature. Thus, while regression-based cost modeling may be sufficient for certain questions, we suggest that an engineering-calibrated model structure may result in more realistic outcomes.

Numerous studies, especially in the last five years (e.g. [194, 345, 346, 349, 385, 390, 403, 407]), have analyzed lithium production pathways from an environmental perspective, especially given lithium's reputation for high water usage in sensitive areas, and the general understanding that mineral processing requires significant amounts of energy and chemical resources. The large majority of this literature originates from the Life Cycle Assessment (LCA) methodology, which not only considers direct and indirect greenhouse gas (GHG) and particulate emissions but also additional environmental impact factors, such as water impacts, ecotoxicity, radiative damage, and many others. These models typically build detailed models of processes throughout the included system ('foreground'), while relying on external models and measurements of the system that is not explicitly modeled ('background').

This literature can be limited because:

- Production in specific locations with firm-specific pathways are modeled, yet measurements from these studies are often used to represent the entire industry without regard for these variations in geography and technology [143, 387];
- Measurements, calculated as emissions factors (emissions per unit output), are necessarily 'attributional' and do not capture the marginal change in emissions if production were to change [413]; and
- Studies are conducted independently, without necessarily trying to have the same scope, so there is no guarantee of equal comparison across studies [414]. (Most studies use the same functional unit of one kilogram or metric ton of lithium carbonate, but boundaries of what are included in the foreground versus background systems are not always specified clearly or explicitly disagree with other studies.)

Of particular importance is the difference in emissions across geographical areas. Kallitsis et al. [387] highlight this across battery supply chains generally, finding that variation in production location across Chinese and North American states and provinces, as well as 19 other countries, can result in a factor of approximately 4 difference between the carbon footprints of different battery material manufacturing processes. Sun et al. [143] highlight this finding for lithium in particular: "environmental impact intensities of battery-grade Li_2CO_3. production from various mine sites differ between 4 times for GHG emissions to 2885 times for land use." Indeed, when we compile reported emissions from 103 pathways across 19 studies, we find widely different emissions factors across geographies and production pathways (Fig. 1). However, due to the depth to which these studies analyze different production pathways, they can provide a foundation upon which process-based integrated technoeconomic–environmental models can be built.

C.1.2 Combining economic and environmental dimensions of lithium production

Several examples within the battery supply chain have combined dimensions of technoeconomic and life-cycle assessment, particularly in the context of battery manufacturing; a non-exhaustive search of recent literature (within five years) in the battery manufacturing space turned up 6 models [415–420]. However, far fewer have focused on the lithium system specifically, and those that have have been limited to a few production pathways. These studies are limited by only modeling a subset of all technical production pathways, with varying levels of detail about cost and environmental impact, focusing on individual pathways (mineral resources [174, 376] and geothermal brines [368]). However, as previously

discussed, not only can we not necessarily compare analyses or interventions across these studies due to possible inconsistencies in system boundaries, functional unit definitions, and other assumptions [414], but standard solar evaporative brines and clay resources (which are significant in the United States) are not modeled.

C.1.3 Empirical Technological Context and Systems Scoping

Two primary compounds are used in lithium-ion batteries: lithium hydroxide and lithium carbonate. Lithium hydroxide is primarily used when producing nickel-rich cathodes (e.g., NMC, NCA) due to its lower decomposition temperature. The complex integration of multiple ions, particularly nickel, requires high-purity inputs and precise temperatures to achieve the proper structure. Lithium carbonate, which is cheaper, is used for LFP and LCO batteries, due to the less stringent thermal and structural requirements for producing those chemically simpler cathodes. Other lithium compounds (lithium fluoride, lithium chloride) are used for the lithium in the electrolytes.

In our analysis, we initially limit our scope to lithium carbonate, as is standard in both industry (e.g. [421, 422]) and literature (e.g. [141, 145, 423]). Discussion of the implications of this scope limitation are included in the Limitations section.

There are three major sources of lithium: brines, hard rock ores, and clays. Each of these can be broken down into additional subcategories: salar/continental brines and geothermal brines; spodumene, lepidolite, and other minerals; hectorite, zinnwaldite, jadarite, and others. Each of these categories has a relatively standard extraction and refining process, though new technologies (such as Direct Lithium Extraction (DLE)) are being developed. While much more detail could be incorporated due to inter-firm differences in production methods, the specific chemical composition of each mineral resource, and nuances in daily operations, we show in Figure C.1 a simplified block diagram of primary production pathways for the production of lithium carbonate from different lithium-bearing resources. This includes the pathways described in the main body of the paper, namely Silver Peak, Thacker Pass, and Jianxiawo, but these models can be adapted with other localized parameters that can allow for comparison with other possible production pathways involving other pathways in China and the United States, pathways in non-aligned countries, multi-nation production pathways, or other outside options.

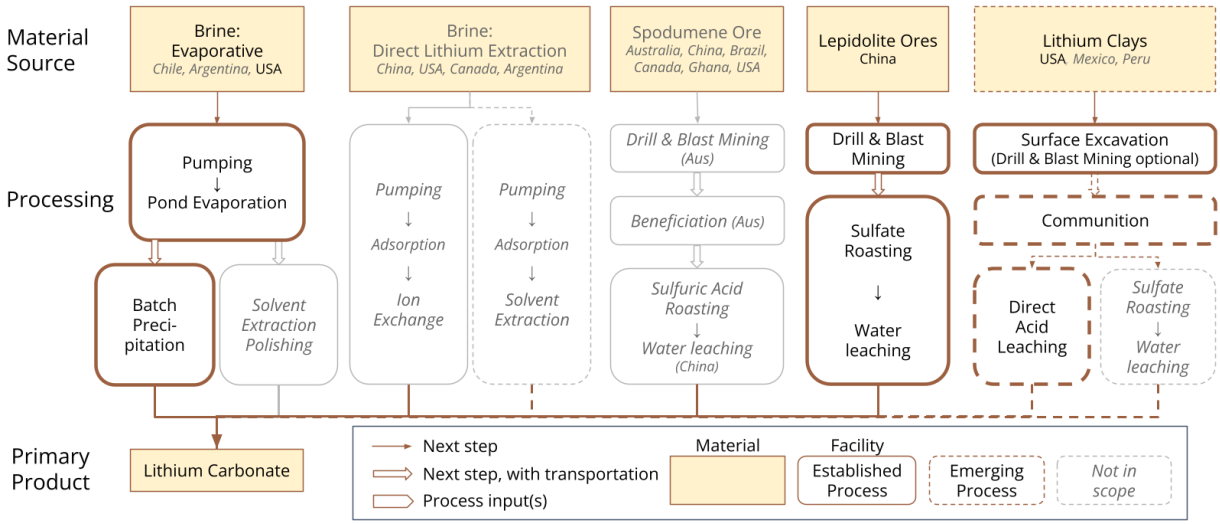


Figure C.1. Existing (solid lines) and future (dashed lines) lithium production pathways

C.2 Limitations with Scoping and Measurement

Several limitations arise due to the scope of this work, which we explore below.

C.2.1 Focus on Lithium Carbonate for Batteries

This study primarily focuses on lithium demand from the battery sector, which represents approximately 88% of total projected demand [145]. While this approach excludes smaller markets such as glass manufacturing, lubricants, or pharmaceuticals, the overwhelming dominance of battery demand means that the findings should still capture the primary driver of overall future lithium demand.

Furthermore, we are presently only considering lithium carbonate as the primary product of comparison. Limiting the scope to lithium carbonate provides a more consistent basis for comparing different production pathways, as all resource types can be processed into lithium carbonate. In contrast, lithium hydroxide is currently primarily derived from lithium carbonate, but could be produced directly from lithium resources in a variety of generally more novel, less mature processes, making direct technoeconomic comparisons more complex. As mentioned above, this scoping limitation is standard in both industry and literature, as some benchmark is needed for price comparisons and academic studies. Future work could explicitly model lithium hydroxide production and assess how technology, geography, and policy interventions impact the relative competitiveness of lithium carbonate versus lithium hydroxide pathways, and subsequent implications for meeting consumer preferences.

C.2.2 Static versus Dynamic Modeling

This analysis employs a static technoeconomic modeling framework. While this limits the ability to represent technological improvements, shifts in overarching market conditions, or responses by firms to interventions occurring in other locations, the static approach provides a transparent, point-in-time comparison of pathways under consistent assumptions. However, we argue that this type of analysis offers the perspective of a first-order response from lithium producing firms, who by and large would react to policies by making changes to their baseline business-as-usual production. These baselines (and first-order responses to policy) can be useful for identifying which production pathways and regions are currently competitive or vulnerable to policy interventions.

C.2.3 Technological Maturity

Lithium production pathways vary significantly in technological maturity, from well-established methods like spodumene conversion and evaporative brine concentration to emerg-

ing technologies such as direct lithium extraction (DLE) or lithium recovery from clay deposits. While uncertainties in cost and performance for newer technologies are unavoidable, including these pathways provides valuable insight into the potential role they could play in diversifying supply chains. Highlighting these uncertainties under best practices for emerging technologies [424, 425] and recognizing that the study will have limitations can make the study useful for identifying where further technological development or data collection could reduce uncertainties-and vulnerabilities.

C.2.4 Focus on Representative, Significant Pathways Rather Than Comprehensiveness

Our analysis aims to construct representative models for each major lithium production pathway across key production pathways, accounting for the most significant current and future pathways identified by the supply chain security and materials demand literature, rather than attempting an exhaustive accounting of every facility or process variant. This approach is necessitated by the substantial variability in cost and emissions data, which stems from differences in process design, energy sources, and operational practices, which are difficult to quantify precisely. While alternative pathways can be expanded upon later with project-specific data, we focus on the pathways define the current state of production in the United States and the marginal production worldwide.

C.3 Additional Results and Model Outputs

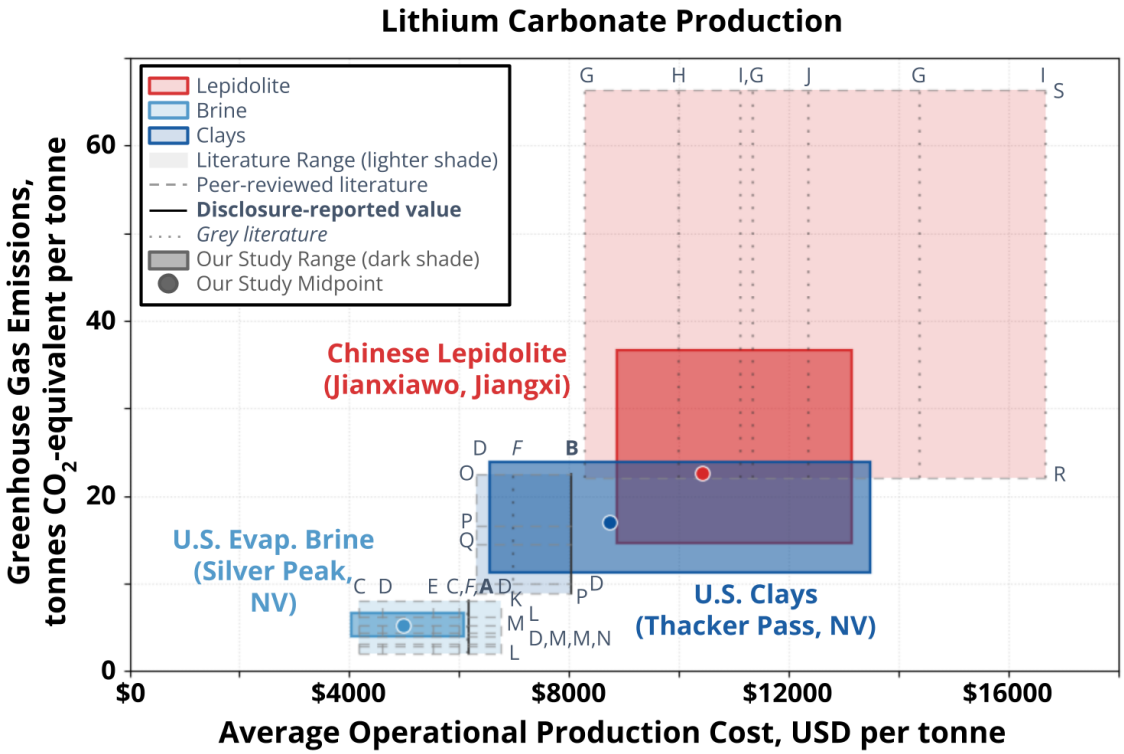


Figure C.2. Production cost ($/t Li_2CO_3) versus life cycle greenhouse gas emissions (tCO_2eq/ /t Li_2CO_3) for three key lithium projects representing distinct resource-pathway combinations, with full uncertainty ranges.

Figure C.2 shows the simultaneous impact of all of the best-case and worst-case scenarios across model input variation - machine design and material costs. Comparing this figure to Figure 4.3, the top 3 restricted ranges capture the majority of the variation for Silver Peak and Thacker Pass, where a small number of drivers dominate; more of the literature range is also covered by this broader interpretation of model inputs. For Jianxiawo, the full uncertainty range is considerably wider in both dimensions, reflecting the larger number of parameters carrying meaningful uncertainty when lepidolite process data is reconstructed from patents and lab-scale studies rather than engineering disclosures. Notably, while the ranges for the Silver Peak project remain clearly separated from the other two projects, the full ranges for Thacker Pass and Jianxiawo overlap more substantially. However, two features of the analysis limit the scope of this overlap. First, it requires simultaneously pessimistic assumptions for the clay pathway and optimistic assumptions for lepidolite — a combination that is unlikely to hold in practice given that the dominant cost drivers for each pathway are

largely independent of one another. Second, the Thacker Pass range shown here reflects the generic flowsheet and high reagent costs, which is already conservative relative to the project-specific design; the overlap would likely narrow or disappear if the magnesium crystallization route and no consideration of offsite sulfuric acid production costs were incorporated.

Across all three projects, the conservative bounds are asymmetrically larger than the optimistic bounds relative to the midpoint, reflecting the fact that cost and emissions overruns are structurally easier to achieve than equivalent improvements, particularly where reagent consumption or equipment sizing are at the high end of their plausible ranges.

C.4 Detailed Methods Writeup

Lithium production systems span heterogeneous resource types (brines, hard rock, claystone, geothermal fluids), processing pathways (physical beneficiation, leaching, solvent extraction, precipitation), and geographic supply chains. These systems differ materially in ore grade, lithium concentration, reagent intensity, energy demand, logistics requirements, and capital intensity. Direct comparison across projects is therefore non-trivial. We use Technoeconomic Assessment (TEA) combined with elements of Life Cycle Analysis (LCA) to identify the 'competitiveness' of these projects.

The purpose of this framework is to:

- Standardize comparison across lithium production pathways using a consistent mass and energy accounting structure.
- Link chemistry-dependent material intensity to equipment sizing, enabling realistic scaling of capital and operating costs.
- Enable parametric sensitivity analysis (e.g., grade, recovery, transport distance, energy price).
- Provide modular extensibility, such that new technologies, detailed process modeling, or site-specific data can be incorporated without restructuring the full model.
- Support implementation and evaluation of engineering-constrained policy interventions, including measures such as carbon pricing, water-use and environmental restrictions, production-based subsidies, and domestic production requirements, where process-level detail is required to understand the impact of such interventions.

C.4.1 Methodological Literature Review

Across the recent lithium supply chain modeling literature, various levels of abstraction are engaged, to answer differing research and policy questions. Many papers cover multiple resource pathways (traditional brines, direct lithium extraction, spodumene, clays), but far fewer simultaneously analyze key parameters around costs and emissions while also being open and transparent about their model forms and assumptions.

Regression and reduced-form cost models [e.g., 141, 170, 191] infer costs from project descriptors and focus on outcomes, rather than process models and manufacturing choices. These models can allow for near-comprehensive coverage of projects, but make counterfactual policy analysis challenging: if a policy changes reagent prices, electricity grid compositions,

recovery, financing terms, or the like, these regression models can only respond indirectly at best.

Process-specific, bottom-up technoeconomic assessment (TEA) and life cycle assessment (LCA) models are on the opposite end: they typically represent a pathway in detail, but often only a specific project or set of projects. The LCA and environmental impacts literature is particularly robust, with over 30 articles since 2020 discussing various environmental impacts (usually, but not exclusively, greenhouse gases) but these are not necessarily consistent with each other given differing system boundaries. Technoeconomic studies are much fewer in number, but their lack of clarity on specific engineering modeling assumptions and lack of publicly available, interactive models makes it challenging to assess conclusions and apply further policy interventions.

This work contributes an openly examinable, modular, process-explicit, machine-level production modeling framework, combining both technoeconomic and life-cycle assessment principles within a single accounting framework. This level of bottom-up detail and open modeling has to the best of the authors' awareness not been applied in the context of lithium production and its distinct production pathways.

This methodology is inspired by parallel methodologies from process-based cost modeling and physical input-output tables, with sufficient detail to remain responsive to policy-relevant engineering variables, while avoiding the data intensity, engineering complexity, and narrow scopes associated with high-fidelity process simulation methods such as Aspen Plus or IDAES [426]. With each step centered around target production volumes, linked together with key operational parameters such as yield rates, reagent intensity, and conversion factors, capital and operating costs remain causally linked across production steps, facilities, and supply-chain representations. By operating at an intermediate resolution, the framework enables standardized comparisons across lithium pathways, while maintaining extensibility to emerging technologies and site-specific data.

C.5 Modeling System

C.5.0.1 System Architecture and Module Hierarchy

The modeling framework is structured hierarchically into four module types:

Production Step Production steps are the fundamental unit operation or process block in which material transformation occurs, in the style of physical input-output analysis. Production steps can be high-level representative blocks, or replaced with sets of specific step-by-step breakdowns, or even integrated with detailed physics-based models.

Facility A facility module is a collection of production steps co-located at a single site, operated as an integrated processing system. Production steps in the same facility are subject to the same geographical (e.g. electricity grid, labor costs) and operational context (e.g. working days per year, shifts per day).

Transportation Transportation route modules are representations of discrete movements of material between facilities, composed of transportation legs to account for the possibility of multiple transportation steps between facilities (e.g. truck to barge to rail).

Supply Chain Supply chain modules are network-level representations linking facilities, with transportation legs connecting them as appropriate. This level also aggregates the statistics from the facility, transportation, and production step level.

C.5.0.2 Production Step Formulation

Each production step is represented as a structural transformation consisting of a primary input, secondary inputs (reagents, utilities, consumables), a process unit (black-box transformation of inputs to outputs), a primary output, and secondary outputs (byproducts, wastes, emissions). For modeling tractability, each production step is constrained to one primary input stream and one primary output stream, enabling consistent mass accounting and simplified scaling relationships.

Modeling of Materials: Inputs, Outputs, Conversion Factors, and Units.

The model explicitly defines three material volume quantities at each production step: primary inputs (e.g., tonnes ROM ore, m^3 brine extracted), the process basis (e.g., m^3 slurry processed, tonnes solids milled, m^3 solution reacted), and primary outputs (e.g., m^3 slurry, tonnes Li_2CO_3 product, tonnes concentrate). These quantities are not inherently equivalent, nor in the same units. Equipment sizing and capital cost estimations are based on the process unit throughput, which serves as the internal scaling metric at each step.

To accommodate the diversity of natural reporting units across process steps and feedstock types, we define two conversion factors to mediate between the three volume quantities:

- **Input-to-Process Conversion Factor (IPCF)** : Converts the primary input from its natural reporting units (e.g., tonnes ore, m^3 brine, tonnes slurry) into standardized process units.
- **Process-to-Output Conversion Factor (POCF)** : Converts standardized process units into primary output units (e.g., tonnes Li_2CO_3, tonnes concentrate, m^3 solution).

These factors are designed to be flexible, allowing for ad hoc adjustments such as volume changes due to reagent addition or recycling loops in the process model. As such, care must be taken to ensure that units are consistent across the three volume quantities (step input, step basis, and step output). Although the model architecture is designed to allow for multiple primary inputs and outputs, the present implementation models a single primary input, representing the output of the previous step. The IPCF then converts this volume into the process basis units without further modifications to its composition.

Secondary inputs, such as reagents and additives, are dosed relative to either the process volume or the concentration of a target constituent, expressed in relative mass units (e.g. kg reagent per kg constituent, or kg per tonne process basis). They are assumed not to materially affect the bulk process volume; where reagents do significantly affect process volume, this is instead absorbed into the IPCF. With the addition of these secondary inputs, the primary and secondary inputs become the process basis, but no interaction is assumed yet; the transformation of the process basis into the primary output occurring at the POCF interface is where any compositional changes are assumed, treated as a black-box operation.

Material constituents are tracked for each input, process basis, and output volume, recorded as mass fractions relative to the process unit (e.g., kg/m^3, kg/kg, etc.). Constituent concentrations $c_{\text{constituent}}$ are always expressed as weights (usually parts per million (PPM)) relative to the constituent-bearing fraction of the volume, such as mg Li per kg dissolved solids in a slurry mixture, or mg Li per kg Li_2CO_3 in a lithium cake. This convention ensures that constituent concentrations are invariant to dilution or concentration of the carrying phase; in this way, when units change, the underlying concentrations of the tracked constituent carrying phase do not need to vary. However, understanding the absolute constituent mass per material flow unit is important, for example to size reagent doses; often, reagents are intended to target a particular constituent of the process stream (e.g., lithium, magnesium, calcium, etc.) rather than the entirety of the production stream. Thus, each volume has an associated Constituent Conversion Factor (CCF), which converts from the constituent units (always mass units) to the volume unit. Reagent use M_r (also always in mass units) is therefore modeled as:

$$M_r = r_{\text{reagent}} \cdot c_{\text{constituent}} \cdot \text{CCF} \tag{C.1}$$

where r_{reagent} is the reagent ratio in kg reagent per kg constituent, $c_{\text{constituent}}$ is the constituent concentration (in this work, always reported as ppm, or mg/kg).

The process basis volume is determined by the volume defining basis, which may either be the process step basis itself or a constituent. The latter is required when equivalent outputs demand different process volumes depending on feed composition. For example, all else equal, more volume of lower lithium content brine will need to be processed than higher lithium content brine, to yield the same mass of lithium carbonate product. In such cases, the process basis is sized by the ratio of constituent mass per process unit to constituent mass per output unit. CCFs are essential to this calculation: since constituent mass must be conserved across this process black box boundary (subject to yield losses), both the process and output concentrations must be converted to the same absolute units before the sizing ratio is formed. This is particularly important when the constituent-bearing material changes physical identity between steps, often resulting in significantly different concentration values — for example, dissolved lithium in concentrated brine becoming lithium in lithium carbonate crystals during precipitation — where concentration values reference different physical bases and cannot be directly compared without conversion through their respective CCFs.

Thus, with a defined output production target, we can determine the process basis volume Q_{process} and input to each step:

$$\sum \text{Inputs} \times \text{IPCF} = Q_{\text{process}} \times r_{\text{yield}} \times r_{\text{conv}} = \frac{\sum \text{Outputs}}{\text{POCF}} \tag{C.2}$$

where r_{yield} is the bulk yield of the process volume (assuming a fraction of everything is lost proportionally). The conversion ratio r_{conv} is equal to one if the volume-defining basis is the process step basis itself. However, when sizing is constituent driven, it is given by:

$$r_{\text{conv}} = \frac{c_{\text{in}} \cdot \text{CCF}_{\text{basis}}}{c_{\text{out}} \cdot \text{CCF}_{\text{output}} \cdot \text{POCF}} \tag{C.3}$$

representing the ratio of absolute constituent mass per process basis unit to absolute constituent mass per output unit, with the POCF normalizing for the unit conversion between process basis and output.

Steps are linked sequentially by passing each step's output as the input to the succeeding step, such that a single terminal production target propagates upstream through the full process chain, determining throughput, input demand, and output volume at every step. With process basis volumes established, equipment can be sized, allowing capital and operating costs to be estimated.

C.5.0.3 Modeling of Operations

The operations modeled in each production step are derived from standard Process Based Cost Modeling [427, 428]. Given the target throughput, we must consider the time utilization of each machine. Some machines are sized for continuous throughput, whereas others are based on batch operations, in which case throughput can be calculated by time per batch. In either case, with target production set to an annual basis, we can calculate the required line time needed to meet a given production target. This quantity can be compared with line time available (LTA), which accounts for plant schedule and downtime, to determine how many machines to run simultaneously, or how much the continuous process needs to be oversized relative to the baseline.

In the present implementation, facility-level operating assumptions define the plant schedule:

- **Working days per year** (d_{py})
- **Shifts per day** (s_{pd})
- **Hours per shift** (h_{ps})
- **Unpaid breaks** (u_b)
- **Paid breaks** (p_b)
- **Scheduled maintenance** (s_{cm})
- **Unplanned downtime** (u_{pd})

From these, the model computes:

$$\text{PLT} = d_{\text{py}} \times s_{\text{pd}} \times (h_{\text{ps}} - u_b) \tag{C.4}$$

$$\text{WLT} = d_{\text{py}} \times s_{\text{pd}} \times (h_{\text{ps}} - u_b - p_b) \tag{C.5}$$

where PLT denotes paid line time and WLT denotes worked line time. Each production step then computes its own LTA by adjusting worked line time for step-specific downtime:

$$\text{LTA} = \text{WLT} \times (1 - u_{\text{pd}} - s_{\text{cm}}) \tag{C.6}$$

This separation preserves the process-based cost modelling distinction between paid labor time and productive equipment time, allowing labor cost and machine capacity to scale differently.

Batch Processes. For batch operations, throughput is governed by discrete processing cycles characterized by a base volume per batch (V_{base}), a batch cycle time (t_{cycle}), and a batch setup time (t_{setup}). The effective production rate per machine is:

$$\frac{V_{\text{base}}}{t_{\text{cycle}} + t_{\text{setup}}} \tag{C.7}$$

Required line time (LTR) is thus:

$$\text{LTR} = \frac{Q_{\text{step}}}{V_{\text{base}}/(t_{\text{cycle}} + t_{\text{setup}})} \tag{C.8}$$

and machines required are:

$$N_{\text{machines}} = \begin{cases} \lceil \text{LTR/LTA} \rceil & \text{if dedicated line} \\ \text{LTR/LTA} & \text{otherwise} \end{cases} \tag{C.9}$$

Labor requirements are scaled analogously: if labor is dedicated, headcount is rounded up to the nearest integer; otherwise, fractional allocation is permitted. Utilities (electricity, gas, water, steam, compressed air) scale linearly with required line time. Thus, in batch systems, capacity expansion occurs discretely via additional parallel lines, labor and utilities scale proportionally to processing time, and setup time explicitly penalizes low throughput operation.

Continuous Processes.

For continuous operations, throughput is parameterized relative to a reference design capacity (V_{base}). Rather than accumulating discrete cycle time, scaling is expressed as a volume ratio:

$$r_{\text{vol}} = \frac{Q_{\text{step}}}{V_{\text{base}} \times \text{LTA}} \tag{C.10}$$

In contrast to batch operations, only a single line is assumed ($N_{machines} = 1$). Total scaled equipment cost scales with a power-law exponent (α_{equip}), consistent with standard chemical engineering scaling laws.

$$C_{\text{equip,scaled}} = C_{\text{equip,base}} \cdot r_{\text{vol}}^{\alpha_{\text{equip}}} \tag{C.11}$$

Utilities (electricity, natural gas, diesel, etc.), measured in per-hour quantities, scale with the same throughput exponent (α_{equip}), multiplied by the operating time (LTA).

$$Q_u = Q_{u,\text{base}} \cdot r_{\text{vol}}^{\alpha_{\text{equip}}} \cdot \text{LTA} \tag{C.12}$$

Labor (L) scales with its own exponent (α_{labor}), with dedicated labor treated in the same manner as with batch processes. This formulation reflects the economic reality that continuous systems (e.g., leach trains, evaporators, large thickeners) are typically resized rather than duplicated, and exhibit sub-linear capital scaling consistent with the six-tenths rule [183].

$$L = L_{base} \cdot r_{\text{vol}}^{\alpha_{\text{labor}}} \cdot \text{LTA} \tag{C.13}$$

Unlike batch systems, setup time does not appear explicitly; instead, downtime and availability are incorporated directly into available operating time, i.e. the line time required is equal to the total line time available

Location-Based Operating Parameters

In this analysis, the United States (Nevada) and China (Jiangxi) share the same scheduling parameters: 350 days/year, three 8-hour shifts, 5% unplanned downtime, and 8% scheduled maintenance. This reflects the assumption that equipment-driven availability is primarily a function of process technology rather than location; the principal location-dependent differentiation is instead captured through factor costs (labor, utilities, capital) as described below. Actual plant availability data for Chinese lepidolite operations are not publicly available to ground a location-specific adjustment.

C.5.0.4 Modeling of Costs

Following operational scaling, costs are computed at the level of each Production Step and then aggregated to the facility level. The cost model preserves the canonical PBCM separation between variable costs, which scale with throughput, and fixed costs, which reflect annualized capital and overhead charges, but also allows for aggregation of costs across the more policy- and investment-relevant split of operational expenditures (OPEX) and capital expenditures (CAPEX).

For each step, total annual cost is decomposed as:

$$\begin{aligned} C_{\text{step}} = {} & C_{\text{materials}} + C_{\text{labor}} + C_{\text{utilities}} + C_{\text{machine}} + C_{\text{tooling}} \\ & + C_{\text{building}} + C_{\text{aux}} + C_{\text{maintenance}} + C_{\text{overhead}} \end{aligned} \tag{C.14}$$

These are grouped as follows:

- **Variable** : Materials, Labor, Utilities
- **Fixed** : Machine, Tooling, Buildings, Auxiliary Equipment, Maintenance, Overhead
- **OPEX** : Materials, Labor, Utilities, Maintenance, Overhead
- **CAPEX** : Machine, Tooling, Buildings, Auxiliary Equipment

Material Costs.

Material costs are derived from reagent usage determined during chemistry propagation. Annual reagent cost is computed from the usage quantity (C.1) multiplied by the unit cost:

$$C_r = M_r \times \text{unit material cost} \tag{C.15}$$

Only secondary inputs (i.e., reagents introduced at the step) are assigned costs. Primary inputs are treated as upstream transfers and are costed in their originating steps or facilities. This structure ensures full traceability between stoichiometric assumptions, impurity removal efficiencies, yield losses, and economic impact.

Labor Costs.

Labor requirements are determined during operational scaling, as described above. Annual labor cost is then computed as:

$$C_{\text{labor}} = w \times L \times \text{PLT} \tag{C.16}$$

where w is the wage (including benefits), L is required labor headcount, and PLT is paid line time per year. Labor cost is tied to paid time rather than productive time, preserving the PBCM-based distinction between compensated hours and machine availability.

Utility Costs.

Utility consumption (electricity, natural gas, process water, cooling water, steam, compressed air) is determined during operational scaling as described above. Utility cost is then:

$$C_{\text{utility}} = \sum_u Q_u \cdot P_u \tag{C.17}$$

where Q_u is quantity consumed and P_u is the facility- (location-) specific price, allowing explicit representation of location-dependent utility prices.

Machine and Tooling Capital Costs.

The number of machines is determined as described in the operational scaling section above. As production is modeled on an annual basis, we compute the annualized machine cost using a capital recovery factor (CRF), and a local machine cost factor CF_{local} where appropriate:

$$\text{CRF}(r, n) = \frac{r(1+r)^n}{(1+r)^n - 1} \tag{C.18}$$

$$C_{\text{machine,annual}} = N_{\text{machines}} \cdot \text{CRF} \cdot C_{\text{equip,scaled}} \cdot CF_{\text{local}} \tag{C.19}$$

If machine lifetime differs from facility lifetime, a step-specific CRF is applied. The local machine cost factor accounts for differences in machine costs across locations. Tooling costs are defined and annualized analogously; if no separate lifetime is provided, tooling costs

are recovered over the facility-level capital recovery period. For this analysis, inclusion of measurements of CF_{local} are not yet made, which is an area for future work.

Building Cost.

Building cost scales with equipment footprint: in batch systems, footprint is proportional to number of machines; in continuous systems, footprint scales by a throughput exponent. Annualized building cost is:

$$C_{\text{building}} = \text{BCRF} \cdot P_{\text{build}} \cdot A \tag{C.20}$$

where P_{build} is building cost per m^2, A is required footprint, and BCRF is the building capital recovery factor, calculated in a similar manner to C.18, but instead using a building lifetime (default 25 years) instead of n.

Auxiliary Equipment, Maintenance, Overhead, and Other Costs.

The cost of additional auxiliary equipment not included in the model blocks and maintenance are modeled as fractions of machine capital. For auxiliaries, since they are capital costs, we multiply by the annualized machine cost:

$$C_{\text{aux}} = f_{\text{aux}} \cdot C_{\text{machine,annual}} \tag{C.21}$$

In contrast to auxiliary equipment, since maintenance costs are often measured as a fraction of total installed machine costs [183, 184], we multiply here by the scaled machine costs:

$$C_{\text{maint}} = f_{\text{maint}} \cdot C_{\text{equip,scaled}} \tag{C.22}$$

Fixed overhead is applied to the sum of machine, tooling, building, auxiliary, and maintenance costs:

$$C_{\text{overhead}} = f_{\text{over}} \cdot (C_{\text{machine,annual}} + C_{\text{tool}} + C_{\text{building}} + C_{\text{aux}} + C_{\text{maint}}) \tag{C.23}$$

Additionally, an optional additional catch-all OPEX adder is defined similar to the fixed overhead costs, to account for any further additional machine or process-specific costs. In this present analysis, this adder is set to zero for all machine blocks.

$$C_{\text{opex_excess}} = f_{\text{opex_excess}} \cdot (C_{\text{machine,annual}} + C_{\text{tool}} + C_{\text{building}} + C_{\text{aux}} + C_{\text{maint}}) \tag{C.24}$$

Location-Based Operating Costs

Several cost parameters used in this analysis above are location-dependent. The locational input data defines unit prices for each utility: electricity (`elec_price`, \$/kWh), natu-

ral gas (`gas_price`, \$/MMBTU), diesel (`diesel_price`, \$/L), propane (`propane_price`, \$/L), cooling water (`cool_water_price`, $\$/m^3$), steam (`steam_price`, \$/kg), and compressed air (`comp_air_price`, $\$/Nm^3$). These are the P_u terms in the utility cost equation. The direct labor wage including benefits (w, \$/hr) and building cost per unit area (P_{build}, $\$m^2$) are also location-specific.

Capital cost parameters set at the location level include the discount rate (`dr`), capital recovery period (`crp`), building recovery period (`brp`), and the machine cost factor (`machine_cost_factor`, i.e. CF_{local}). Fixed cost fractions for auxiliary equipment (`aux_equip`), maintenance (`maint`), and overhead (`fixed_over`) are also defined per location, as are tailings disposal costs for nonhazardous and hazardous waste streams.

The most consequential differences are in labor (\$50/hr in the U.S. vs. \$10/hr in China), electricity (\$0.09/kWh in the U.S. vs. \$0.075/kWh in China), and natural gas (\$5.04/MMBTU in the U.S. vs. \$8.50/MMBTU in China). The machine cost factor is set at 1.0 for the U.S. and 0.75 for China, reflecting lower equipment procurement costs in China, though this parameter is acknowledged as an area requiring further empirical grounding.

In addition to location-level parameters, reagent unit costs and embodied emissions factors are defined per project in the project-specific input data, with conservative, midpoint, and optimistic scenarios. These capture supply-chain-specific procurement conditions (e.g., sulfuric acid cost in Nevada vs. Jiangxi, soda ash sourced domestically vs. imported) that cannot be reduced to a single location profile. During scenario evaluation, the `update_materials` routine applies the appropriate scenario rank to all facility material cost and impact entries, enabling the triangular sensitivity analysis described earlier.

C.5.0.5 Modeling of Emissions

Environmental impacts are computed in parallel with financial costs in each Production Step. Where possible, physically modeled flows (utilities consumed and coproducts routed to sinks) are aggregated, and location-specific impact factors are applied to calculate their environmental impact, in line with standard methodologies from process-based, bottom-up studies in the Life Cycle Assessment (LCA) literature.

Environmental impacts are divided into two primary channels: utility-related impacts (energy and resource consumption), and sink-related impacts (coproducts and waste streams). For each step, impacts are reported as utility impacts (by utility type), sink impacts (by sink and coproduct), Scope 1 and Scope 2 totals, and total step impacts. Facility and supply chain totals are obtained by summing across all steps. Since these emissions are tightly coupled to operational modeling, we can not only observe the environmental impact of process assumptions and design choices, but also allow for relatively straightforward extensions to

externality monetization and the broader connection between environmental and economic outcomes.

Coproduct and Waste Impacts (Scope 1).

The model tracks secondary outputs from each step, representing coproducts or waste streams routed to facility-defined sinks (e.g., landfill, wastewater treatment, air). Each secondary output includes a conversion factor from step production volume, an annual volume, and an assigned sink. During volume throughput calculations, coproduct volumes are computed consistently with yield and throughput.

Impact factors for sinks can be defined in two ways: sink-level factors (uniform across all coproducts), or coproduct-specific factors within a sink, allowing differentiated treatment (e.g., solid tailings vs. brine discharge to wastewater). For a coproduct c routed to sink s, impacts are computed as:

$$I_{c,j} = V_c \cdot f_{s,c,j} \tag{C.25}$$

where V_c is annual coproduct volume and $f_{s,c,j}$ is the impact factor for that sink-coproduct pair and impact category j. At the moment, impact factors for these coproducts and waste impacts are not defined. Given that Scope 1 waste emissions are expected to be small relative to the Scope 2 and Scope 3 terms that dominate all three pathways, this omission is unlikely to affect the comparative results materially, but represents a priority for future refinement.

Utility-Related Impacts (Scope 1 and 2).

Utility consumption is computed during operational scaling, as described above, yielding annual quantities of electricity (kWh), natural gas (MMBTU), propane (L), diesel (L), cooling water (m^3), steam (kg), and compressed air (Nm^3). Each utility u is associated with an impact factor vector $\{f_{u,j}\}$, where j indexes environmental impact categories (e.g., CO_2, water use, NO_x, SO_2). While most of these impacts occur on-site (e.g. combustion emissions from diesel, natural gas, propane) and thus are considered Scope 1, usage of offsite electricity is considered Scope 2. In this analysis, we primarily consider greenhouse gas impacts, though the framework is easily extendable to other impact categories. Utility-related impacts for a step are computed as:

$$I_{u,j} = Q_u \cdot e_{u,j} \tag{C.26}$$

where Q_u is annual utility consumption and $e_{u,j}$ is the impact factor for category j. Impact factors for each of these utilities are specified in Table C.3.

Upstream Material Use Impacts (Scope 3).

Secondary inputs (reagents) consumed in each production step carry embodied emissions from their own upstream production and transport. For each reagent r consumed at step i, the Scope 3 impact is computed as:

$$I_{r,j} = M_r \cdot e_{r,j} \tag{C.27}$$

where M_r is the annual mass consumption of reagent r (C.1), and $e_{r,j}$ is the cradle-to-gate emissions factor for reagent r and impact category j (e.g., kg CO_2eq/kg reagent).

Reagents are treated as purchased commodity inputs unless otherwise specified; their embodied emissions reflect the upstream production processes of the dominant manufacturing route in the relevant geography. For example, soda ash (Na_2CO_3) consumed at US facilities is assigned an emissions factor reflecting trona mining and calcination ($\sim$ 0.3–0.7 t CO_2eq/t Na_2CO_3), while soda ash consumed at Chinese facilities is assigned a higher factor reflecting the energy-intensive Solvay process ($\sim$0.7–1.2 t CO_2eq/t Na_2CO_3) [106, 195, 200]. Soda ash is an intentionally produced commodity chemical rather than a waste product of another process; its emissions factor therefore reflects the full cradle-to-gate burden of dedicated soda ash manufacturing. Similarly, lime (CaO) is assigned an emissions factor reflecting limestone calcination, where approximately two-thirds of emissions are unavoidable process emissions from carbonate decomposition and the remainder reflects kiln fuel combustion [204]. Emissions factors for all reagents are sourced from published life cycle inventory databases and literature, and are summarized below in Table C.3.

Defined Emissions Impacts.

A table summarizing all emissions factors, by location and material, with parametric variation where possible, is found below in Table C.3. This table includes emission intensities of chemical reagent and energy inputs ($\mathrm{kg\,CO_2e\,kg^{-1}}$ for solids/liquids; $\mathrm{kg\,CO_2e\,kWh^{-1}}$ for electricity; $\mathrm{kg\,CO_2e\,MMBtu^{-1}}$ for natural gas; $\mathrm{kg\,CO_2e\,m^{-3}}$ for water; $\mathrm{kg\,CO_2e\,kg^{-1}}$ for steam; $\mathrm{kg\,CO_2e\,L^{-1}}$ for liquid fuels) under optimistic, midpoint, and conservative scenarios. For soda ash, separate rows reflect US trona-based production (applied to Nevada scenarios) and Chinese Solvay-process production (applied to Jiangxi scenarios); electricity grid intensity is location-specific; all other inputs adopt world-average supply-chain factors. Inputs lacking scenario variation use a single point estimate spanning all three columns. Superscripts correspond to the notes below the table.

C.5.0.6 Machine Model Blocks

Each Production Step is further parameterized using one of several possible machine blocks, to represent alternative equipment configurations or operational archetypes. While reagents and coproducts are handled at the Production Step level, machine-specific char-

Table C.3. Emission intensities of chemical reagent and energy inputs

Input	Geographic scope	Emission intensity		
		Optimistic	Midpoint	Conservative
Chemical reagents — with scenario range				
Flocculant (generic)†	World	2.85[a]	3.914[a]	5.38[a]
Lime	World	0.83[b]	1.20[a]	1.70[a,e]
Limestone	World	0.00225[a]	0.004[a]	0.006[b]
Soda ash	USA	0.30[c]	0.50[c]	0.70[c]
	China	0.70[c]	1.20[a]	1.80[e]
Sodium sulfate	World	0.10[a]	0.25[a]	0.60[f]
Sulfuric acid	World	0.0509[a]	0.0969[a]	0.14[b]
Chemical reagents — single point estimate				
Explosives	World		2.5[b]	
Filter aid	World		0.4[a]	
Wash / process water	World		0	
Energy and utility inputs				
Electricity grid	Nevada, USA		0.1932[g]	
	Jiangxi, China		0.5836[h]	
Natural gas	World		53.06[d]⋆	
Steam	World		0.15[i]	
Diesel	World		2.68[d]	
Propane	World		1.51[d]	

[a] ecoInvent 3.8, as compiled in Roy et al. [195].
[b] GREET model (Argonne National Laboratory) [106].
[c] US soda ash production is primarily trona-based; reported emission intensities span 0.3–0.7 $\mathrm{kg\,CO_2e\,kg^{-1}}$ [200]. The China optimistic value (0.70) equals the upper end of the US trona range, used as a conservative floor for Solvay-process production.
[d] EPA Center for Corporate Climate Leadership, *GHG Emission Factors Hub*. Natural gas: $\mathrm{CO_2}$ only ($\mathrm{53.06\,kg\,CO_2\,MMBtu^{-1}}$); diesel: $\mathrm{10.21\,kg\,CO_2\,gal^{-1}}$ converted to $\mathrm{2.68\,kg\,CO_2e\,L^{-1}}$; propane: $\mathrm{5.72\,kg\,CO_2\,gal^{-1}}$ converted to $\mathrm{1.51\,kg\,CO_2e\,L^{-1}}$ [429].
[e] Estimated conservative upper bound; not directly reported in a single source.
[f] Mannheim-process production; the high-temperature, fuel-oil-fired kiln step is identified as the dominant direct emission source [430].
[g] EPA eGRID summary data for the Nevada grid subregion [199].
[h] China Ministry of Ecology and Environment, provincial grid emission factor for Jiangxi [431].
[i] Raw assumption; no published source.
† ⋆ Natural gas factor is $\mathrm{CO_2}$ only; $\mathrm{CH_4}$ and $\mathrm{N_2O}$ contributions are negligible for combustion at this scale but could be added if needed.

acteristics are aggregated into blocks which can then be scaled as appropriate, allowing for heterogeneity in operational performance while preserving consistent chemistry and throughput logic at the step level. As a result, all operational variables other than materials — labor, utilities, machine, tooling, building, auxiliaries, maintenance, and overhead — are handled by machine model blocks. By default, each step uses a specified `machine_default` block, but alternative blocks may be substituted to represent technology variants or scenario analyses (e.g., tornado analysis).

C.5.0.7 Facility Aggregation

A facility aggregates multiple production steps in series or parallel configuration. Intermediate material transfers within a facility are assumed to occur without modeled transportation losses unless explicitly specified (e.g., evaporation ponds, pumping). Facility-level metrics (CAPEX, OPEX, power consumption, water use, emissions) are calculated as the sum of constituent production steps.

C.5.0.8 Transportation Modeling

Transportation legs connect facilities across the supply chain; multiple legs can be combined into a route to account for multi-modal transport (e.g., spodumene shipped via rail, then barge, then truck). For each leg, transported mass or volume equals the primary output stream quantity. With the exception of international marine freight, transportation is assumed to be round-trip, accounting for empty return:

$$\text{Total Transport} = 2 \times \text{Distance} \times \text{Payload} \tag{C.28}$$

Energy use and emissions are derived from transport work and mode-specific intensity factors. Transportation costs and emissions are roughly estimated from road freight literature [432] rather than mine haul references. However, given mine-site haul distances of less than 10 km at both Thacker Pass and Jianxiawo, haulage contributes negligibly to total modeled cost and emissions.

C.5.0.9 Supply Chain Integration

The supply chain model links facilities and transportation legs into a directed network from extraction to refined product delivery.

C.5.1 Facility-level Modeling Choices

Review of publicly available NI 43-101 and SK-1300 technical reports and feasibility studies reveals that lithium projects are typically reported as a sequence of high-level process areas ('flowsheets') rather than individual machines. These areas are often labeled inconsistently across projects, even within the same resource pathways, presenting a distinct modeling challenge. Directly reproducing project-specific structures would limit cross-project comparability, while fully disaggregating to machine-level details for each project would require a level of engineering and economic understanding beyond what is presented in such disclosures. Given the supply chain, facility, production step, and machine block representations described above, we leverage our understanding of each pathway to create common representations of pathway segments, aggregating commonly sequenced production steps into facility representations.

Each of the pathway segment descriptions below represents a modeled facility in our supply chains. Design decisions are described at this hierarchical level, and the blocks necessary to compose each facility or pathway segment are specified. Engineering details, capital cost aggregations, and reagent usage are described within each of the production step and machine block models themselves.

C.5.1.1 Pumping and Evaporation Ponds (Generic Brine Pathway)

This is the generic pathway for solar evaporation-based processes. The pumping section is also relevant as the initial step for direct lithium extraction (DLE).

Lithium concentrations in brines are typically on the order of hundreds to a few thousand ppm, meaning that millions of cubic meters of brine must be extracted, transported, and concentrated to produce a relatively small mass of final product. As a result, hydraulic transport and surface area requirements are the primary determinants of capital and operating costs. Different projects have differing hydrogeologic (e.g. brine source depths, rock porosity) and climatic parameters (e.g. evaporation rate) that can be further developed as needed, so this model aims to preserve sensitivity to these variables while remaining sufficiently transferable across both traditional salar-based production and DLE-based systems.

This facility is composed of the following Production Steps:

- `Pumping`: Extraction and hydraulic transport of lithium brines.
- `Evaporation`: Solar concentration of extracted brine.

Pumping.

The pumping step extracts lithium-bearing brine from subsurface aquifers and delivers it to the evaporation pond system. The primary input is raw brine in the ground (`raw_`

brine_from_ground), expressed in liters per year, which passes through the step without compositional transformation. As a result, the step basis is brine pump throughput, with an input-to-process conversion factor (IPCF) of 1.0. With a process-to-output conversion factor (POCF) of 1.0, it exits as pumped brine (pumped_brine) of identical volume and chemistry. The step therefore functions as a pure transport and energy accounting block: its primary purpose in the model is to assign pumping-related capital and electricity costs and emissions to the extraction stage, rather than to modify the brine composition.

The step is parameterized by the full multi-ion brine chemistry at the point of extraction: constituents Li, K, Mg, Ca, Na, B, Cl, and SO_4 are tracked, all expressed in ppm (mg/kg brine). These concentrations are set from the project-specific data, and for Silver Peak the dilute feed brine is specified at approximately 100 ppm Li, 190 ppm Mg, 450 ppm Ca, and 49,610 ppm Na, as identified in C.5.2.5. The brine density is specified as 1.058 g/cm^3 and is used as the constituent conversion factor (CCF), converting from ppm (kg constituent per kg brine) to the volumetric step basis (liters pumped per year). Because the output chemistry depends on the input chemistry - the pumped brine inherits the composition of the feed - the primary_output_1_chemistry_dependence flag is set to True, ensuring that any upstream modifications to feed composition (e.g., sensitivity analysis on brine grade) propagate correctly to downstream steps.

No reagents are consumed at this step, and no coproducts are generated. Electricity consumption for pumping is the sole utility input and is the dominant cost and emissions driver of this step; its magnitude is set by the machine block parameters rather than by the process chemistry (see the brine_pump model block, C.5.3.1).

Evaporation.

The evaporation step concentrates the pumped brine through solar evaporation until it reaches the target strong brine chemistry required for downstream chemical purification. The process basis is pond area in m^2, and the step is constituent-driven: the volume-defining basis is the Li constituent of the input brine stream, meaning that the required pond area scales with the lithium mass that must be processed to meet the annual production target, rather than with bulk brine volume directly. This is the correct formulation because lower-grade brines require proportionally larger pond areas to yield the same mass of lithium product.

The CCF for the step basis converts from ppm Li (mg Li per kg brine) to m^2 of pond area, and is defined as density divided by evaporation rate: for Silver Peak, with an evaporation rate of 2.44 mm/day and brine density of 1.058 g/cm^3, this yields a CCF of approximately 0.000942 kg $\cdot$ year/m^2. The output is concentrated lithium brine (concentrated_lithium_brine), whose target chemistry is specified in the project-specific data as the strong brine composition at the point of handover to downstream batch chemical purification. For Silver

Peak, this is approximately 6,750 ppm Li, 14,250 ppm Mg, and 140,000 ppm Na at a density of 1.25 g/cm^3 [178]. The difference between input and output Li concentrations (100 ppm vs. 6,750 ppm) implies approximately 95% evaporative water loss. The pond recovery efficiency parameter (0.51 for Silver Peak [178]) accounts for lithium losses during evaporation attributable to co-precipitation and brine seepage.

Evaporated water is tracked as a coproduct routed to the `Air` sink, representing water vapor loss to the atmosphere. No reagents are consumed at this step. Unlike the pumping step, the chemistry of the output is not flagged as depending on the input chemistry, since the output concentration is set directly from the project-specific intermediate concentration target rather than computed from a reaction stoichiometry; this means that the model treats the evaporation step as a concentration target to be reached rather than a direct chemical transformation to be simulated. Capital and operating costs are driven by pond area and are parameterized through the `evaporation_pond` model block (C.5.3.2).

C.5.1.2 Blasting and Excavating

This is the generic pathway for mining in the traditional sense, applied to ore-based extraction of spodumene, lepidolite, and clays. Physical fragmentation and movement of ore (and surrounding material) is strongly dependent on deposit characteristics: hard rock ores (spodumene, lepidolite) typically require drilling and blasting to generate moveable useful material, whereas clay deposits are often weakly bound and may be excavated without blasting. At the system level, blasting and excavation are aggregated to capture the primary energy and cost drivers associated with rock breaking and material movement, without specifying detailed mine parameters such as bench height, block size, or exact fleet composition. Hauling is treated as a project-specific transportation step connecting the mine site to the initial processing plant.

This facility is composed of the following blocks:

- `Blasting` (optional): Drilling and blasting for hard rock fragmentation.
- `Excavation/Loading`: Excavation and loading of fragmented or unconsolidated material.

Blasting. The blasting step represents drilling and explosive fragmentation of in-situ rock prior to excavation. The primary input is `raw_material` (in-situ rock, in metric tonnes), with an IPCF of 1.0, and the step basis is `rock_blast_throughput` in metric tonnes. The step passes material through without compositional transformation; output chemistry is flagged as dependent on input chemistry, propagating the initial ore grade set from the project-specific data. The CCF is 0.001 kg ore per (tonne-ppm), the standard conversion for ppm-in-mass-basis constituents to absolute kg per tonne.

The sole reagent is explosives, dosed against the step basis (total rock moved, including waste) using the powder factor drawn from the project-specific data — for Jianxiawo, 0.25 kg explosives per tonne rock [433]; for Thacker Pass, 0.10 kg per tonne [150]. Explosives are not a constituent-elimination reagent; their function is fragmentation, not chemical removal of any tracked species. No coproducts are generated at this step. The blasting step is bypassed entirely when the project-specific powder factor is absent or marked as '–', as is the case for projects where unconsolidated material requires no blasting (the no-blasting path routes directly to `excav_load`).

The initial ore chemistry is set in `run_scenarios.py` by loading the project-specific initial concentration and applying a dilution correction before passing it to the blasting step. If the reported grade is a ROM (run-of-mine) grade, the in-situ grade is back-calculated as $c_{\text{in-situ}} = c_{\text{ROM}}/(1-d)$, where d is the dilution factor; if the reported grade is an in-situ grade, the ROM grade is computed as $c_{\text{ROM}} = c_{\text{in-situ}} \times (1-d)$. This ensures that the chemistry entering the blasting step reflects true in-situ conditions, while the chemistry exiting the excavation step reflects post-dilution ROM conditions as they would be observed at the processing plant gate.

Excavation/Loading.

The excavation and loading step represents mechanical fragmentation, loading, and preparation of blasted (or directly excavated) material for haulage to the processing facility. The primary input is `blasted_rock` (from the blasting step) or `raw_material` directly (in the no-blasting case), in metric tonnes. The step basis is `loose_rock_throughput` in tonnes of loose rock, with IPCF of 1.0. The output is `rom_ore_feed`, representing the fraction of moved material that constitutes ore rather than waste.

The POCF for the output and the waste coproduct conversion factor are not fixed, but are set programmatically from project-specific mining factors:

$$\text{POCF}_{\text{rom}} = \frac{1}{1+r_{\text{strip}}} \times (1-l) \tag{C.29}$$

$$f_{\text{waste}} = \frac{r_{\text{strip}}+l}{1+r_{\text{strip}}} \tag{C.30}$$

where r_{strip} is the strip ratio (tonnes waste rock per tonne ore) and l is the loss fraction. The POCF thus represents the fraction of total material moved that becomes usable ROM ore feed after accounting for both dilution of ore by accidental waste inclusion, and ore losses during excavation. The waste remainder, f_{waste}, is routed as a coproduct to the `waste_pile` sink. The yield rate on `rom_ore_feed` is set to $(1-l)$, capturing the ore fraction lost during the excavation and loading process independently of the strip ratio.

The ROM ore chemistry (`rom_comp`) is set as a target composition on the `rom_ore_feed` output directly, reflecting post-dilution grade. For projects where ROM grade is reported directly (`ROM feed = True`), the ROM composition is used as-is; for projects where in-situ grade is reported, the dilution correction described above is applied. The step CCF is 0.001 kg ore per (tonne-ppm), consistent with blasting. No reagents are consumed at this step. This production step uses the `excavation_loading` machine block (C.5.3.4).

C.5.1.3 Batch Chemical Purification (Silver Peak Method)

This lithium production pathway is noted for its relative simplicity, being the first developed for lithium extraction from brines; production at Silver Peak, Nevada began in 1966. The method operates primarily on brines with low Mg:Li mass ratios, given the relatively high selectivity of magnesium for carbonate ions. Magnesium, calcium, and other impurities are removed as part of earlier concentration steps, either through selective precipitation during solar evaporation or by chemical treatment with lime and soda ash prior to lithium carbonate precipitation.

Six studies model the batch chemical purification method [142, 143, 173, 191, 192, 358], in which batches of slaked lime ($Ca(OH)_2$) and sodium carbonate (Na_2CO_3) are added to concentrated lithium brines to precipitate lithium carbonate. Mousavinezhad et al. [173] identify the Silver Peak process as using a resin-based ion exchange process, likely following the model structure of Schenker and Pfister [192]. While the latter do not specify what they mean by 'ion exchange' in their model, no mention of resin-based ion exchange methods appears in the other studies or in the company's NI 43-101 disclosure itself. As a result, our model is based primarily on the reported steps in the Silver Peak disclosure and the model structure reported in Wesselkaemper et al. [142], and is composed of the following blocks:

- `batch_treatment`: Batch treatment of concentrated brine with lime and soda ash.
- `sls_step`: Solid–liquid separation and filtration.

Batch Treatment.

The batch treatment step precipitates dissolved magnesium and calcium impurities from concentrated lithium brine, leaving lithium in solution. The primary input is `concentrated_lithium_brine` from the evaporation step, with an IPCF of 1.0. The step basis is `concentrated_brine_throughput` in liters per batch, with a CCF of 1.25×10^{-6} kg/L reflecting the density of concentrated brine and converting ppm constituent concentrations to absolute kg per liter. The output is `treated_slurry`, passed to the separation step with a POCF of 1.0 (volume is conserved) and output chemistry dependent on input chemistry.

Three reagents are dosed against constituent targets:

- **Slaked lime $Ca(OH)_2$ targeting Mg** at a ratio of 3.658 kg $Ca(OH)_2$ per kg Mg, with 95% elimination of Mg. This ratio is derived from the stoichiometric molecular weight ratio $MW(Ca(OH)_2)/MW(Mg) = 74.09/24.305$, multiplied by an operational excess factor of 1.2 to account for reaction inefficiencies. The key reaction is $Mg^{2+} + Ca(OH)_2 \rightarrow Mg(OH)_2(s) + Ca^{2+}$; precipitated $Mg(OH)_2$ reports to the solids phase, while the calcium introduced by the lime remains in solution and must be subsequently removed.
- **Soda ash Na_2CO_3 targeting Ca (calcium introduced by lime)** at a ratio of 3.173 kg Na_2CO_3 per kg Ca introduced by the lime addition that targeted magnesium removal, with 90% elimination of Ca. This is derived from $MW(Na_2CO_3)/MW(Mg) \times 1.2 \times 1.2 = 105.988/24.305 \times 1.44$, accounting for the stoichiometric equivalence of Ca introduced per Mg removed and two excess factors. The reaction is $Ca^{2+} + Na_2CO_3 \rightarrow CaCO_3(s) + 2\ Na^+$.
- **Soda ash Na_2CO_3 targeting Mg (as a secondary Ca removal dose)** at a ratio of 6.279 kg Na_2CO_3 per kg Mg, with 0% elimination of Mg (this dose targets the calcium natively present in the brine, not the calcium introduced by lime). This is derived from $MW(Na_2CO_3)/MW(Ca) \times 1.2 = 105.988/40.078 \times 1.2$.

No coproducts are generated at this step; precipitated solids remain suspended in the treated slurry and are separated in the subsequent step. This step uses the `batch_treatment_generic` machine block (C.5.3.20).

Separation.

The separation step clarifies the treated slurry by removing precipitated $Mg(OH)_2$ and $CaCO_3$ solids, yielding a purified lithium-bearing solution (`pls_post_polishing`) suitable for downstream lithium carbonate precipitation. The primary input is `treated_slurry` in liters, converted to the step basis (`mixture_being_separated` in m^3/h) via an IPCF of 10^{-3} (L to m^3). The CCF is 1.25×10^{-6} kg/L, consistent with the upstream step. The output `pls_post_polishing` has a POCF of 1000 (m^3 to L) and a yield rate of 0.995, reflecting a 0.5% lithium loss to solids during filtration.

One reagent is consumed: a filter aid (precoat, e.g. diatomaceous earth) dosed at 0.005 kg per m^3 of treated slurry against the step basis. This value is a rough engineering estimate; a suitable published reference for diatomaceous earth dosing in mineral brine clarification has not been identified and represents a low-priority refinement.

The residue coproduct is routed to the `tailings_35` sink, representing an on-site filter cake at approximately 35% solids fraction. The coproduct conversion factor of 0.232 m^3 wet solids per m^3 treated slurry is derived from first-principles stoichiometry. The batch treatment reagent doses (3.658 kg $Ca(OH)_2$ per kg Mg at 95% removal; 3.173 kg Na_2CO_3 per kg Ca at 90% removal) yield 36.9 g/L $Mg(OH)_2$ and 72.8 g/L $CaCO_3$ dry precipitate at

the concentrated brine chemistry (Mg: 12,960 ppm; Ca: 1,535 ppm; $\rho = 1.25$ kg/L), totaling 109.7 g/L dry solids. At a filter press cake of 35 wt% solids and wet density of 1.35 g/cm^3, this yields 0.232 m^3 wet cake per m^3 slurry. This estimate is cross-validated against Silver Peak's disclosed lime consumption of 1.3 t CaO per tonne Li_2CO_3. The residue composition is assumed to be as 25% $Mg(OH)_2$, 25% $CaCO_3$, and 50% water by volume. We assume no volume loss here, as water lost is replenished from water from tailings slurry dewatering and recycling from downstream lithium carbonate production steps, though a full water balance is a key future area for improvement.

This step uses the `thickener_generic` machine block (C.5.3.14).

C.5.1.4 Clay Comminution and Acid Leach

Clay-based lithium sources allow for relatively straightforward chemical extraction from fine-grained matrices, and comminution serves primarily to control input particle sizes. Even weakly bound sediments, as in projects such as Rhyolite Ridge, Tonopah Flats, and Bonnie Claire, are typically delivered from the mine as run-of-mine fragments and therefore require simple primary crushing for size control prior to slurry preparation [173, 193]. For harder claystone systems such as Thacker Pass, Clayton Valley, and TLC, additional mechanical processing, modeled here as attrition scrubbing, is required to break down aggregated particles prior to leaching.

Classification and screening are key operations in clay flowsheets to control slurry particle size and ensure consistent leaching performance. Rather than specifying individual screening or hydrocyclone devices, these operations are abstracted into a generic classifier block parameterized by cut size and efficiency. Given the importance of water management in Nevada-based claystone systems, a thickening step is included following classification to regulate solids concentration and recycle process water.

Clay-origin slurries are processed with sulfuric acid leaching, after which the leached slurry undergoes multistage solid–liquid separation, typically represented as a counter-current decantation (CCD) system, to recover pregnant leach solution and minimize soluble lithium losses to residue. Water recovery and solution concentration are key drivers at this stage and materially affect both reagent consumption and downstream purification requirements. Staged neutralization and impurity removal steps are required to control iron, aluminum, magnesium, calcium, and residual acidity prior to lithium recovery. While specific precipitation sequences vary across project disclosures, these operations perform the same functional role: conditioning the sulfate solution to within acceptable impurity tolerances. The resulting purified lithium sulfate solution is then processed into lithium carbonate.

Of note is that the last three steps of this process are not included in the Thacker Pass project flowsheet, where a magnesium recycling route was adopted in which magnesium salts are removed via evaporative crystallization – consuming fewer reagents at the cost of substantially higher thermal energy demand. Our generic model instead fully precipitates all magnesium with quicklime, causing total quicklime consumption to reach approximately 9,200 kg CaO/t Li_2CO_3, compared to the disclosed 2,630 kg/t. This addition of quicklime also necessitates addition of sodium carbonate to polish the additional calcium, significantly increasing costs.

This facility is composed of the following blocks:

- `Crushing`: Primary crushing for particle size control.
- `Attrition Scrubbing`: Attrition scrubbing for harder claystone systems.
- `Classification`: Classification and screening for slurry particle size control.
- `Thickening`: Thickening to regulate solids concentration and recycle process water.
- `Acid Leaching`: Sulfuric acid leaching of classified clay slurry.
- `Neutralization`: Staged neutralization, removing Fe, Al, and others, via pH adjustment and precipitation.
- `Multistage Primary Counter-Current Decantation`: Primary counter-current decantation for pregnant leach solution recovery.
- `Impurity Removal`: Chemical removal of impurities (Mg, Ca, etc.) via pH adjustment and precipitation

Crushing.

The crushing step reduces run-of-mine (`rom_ore_feed`) claystone to a particle size suitable for slurry preparation and attrition scrubbing. The primary input is ROM ore in m^3, converted to the step basis of `crushed_ore_throughput` in tonnes via an IPCF of 1.9, reflecting the in-situ bulk density of Thacker Pass hectorite clay (approximately 1.9 t/m^3) [150]; this value should be loaded dynamically from project-specific data for other clay projects rather than hardcoded. The step CCF is 0.001 kg ore per (tonne-ppm), the standard conversion for mass-basis ppm constituents. Only Li is tracked as a constituent through the mining and comminution circuit, sufficient to propagate the ore grade forward to sizing calculations in downstream steps.

No reagents are consumed at this step. A `rejected_coarse` coproduct stream routed to the `tailings_solid` sink is defined in the data schema with a conversion factor of zero, reflecting the modeling assumption that no coarse material is rejected to waste at the crushing stage. Oversize material is instead recirculated within the crushing circuit and appears in the feed to attrition scrubbing in the following step.

This step uses the `crushing_cycle` machine block (C.5.3.5).

Attrition Scrubbing.

The attrition scrubbing step deagglomerates crushed claystone particles and produces a pumpable slurry at a target solids concentration suitable for classification. The primary input is `crushed_ore` in tonnes, converted to the step basis (`scrubbed_slurry_throughput`, in m^3 of slurry at approximately 30 wt% solids) via an IPCF of 3.40. This factor accounts for both the mass-to-volume conversion of crushed ore into a slurry and an assumed 20% contribution of coarse recycled material from the downstream classification step: for a target slurry at 30 wt% solids using a grain density of 2.6 t/m^3 and water density of 1.0 t/m^3, the mixture density is $\rho_{\text{mix}} = 1/(w_s/\rho_s + (1-w_s)/\rho_w)$, giving a step CCF of $w_s \cdot \rho_{\text{mix}}/10^3 \approx 3.68 \times 10^{-4}$ kg solids per m^3 per ppm. Including the 20% recycle flow from classification yields the effective IPCF of approximately 3.40.

The output `scrubbed_slurry` is a diluted slurry at approximately 20 wt% solids, reflecting water addition during scrubbing. The POCF of 1.613 converts from the 30 wt% step basis to the 20 wt% output basis, derived as the ratio of the inverse CCF values at the two solids fractions: $\text{POCF} = (1/(0.2 \cdot \rho_{20\%}))/(1/(0.3 \cdot \rho_{30\%}))$. The output CCF is correspondingly $w_s \cdot \rho_{\text{mix}}/10^3 \approx 2.28 \times 10^{-4}$ kg solids per m^3 per ppm at 20 wt% solids.

Two reagents are dosed against the step basis volume: a slurry dispersant at 0.001 kg/m^3 of slurry (a rough engineering estimate for clay/silt deflocculation, with no constituent elimination), and makeup process water at 0.4 m^3/m^3 of slurry. The process water ratio reflects new water introduced for slurrying, with the remainder of total water demand (approximately 73 m^3 per 100 m^3 of scrubbed slurry output) assumed to be supplied by recycled thickener overflow and other internal streams not explicitly tracked in the current model. No coproducts are generated at this step.

This step uses the `attrition_scrubber` machine block (C.5.3.6).

Classification. The classification step separates fine-grained particles suitable for acid leaching from coarser material that is recycled back to attrition scrubbing, abstracting the function of hydrocyclones and/or screens into a single sizing operation. The primary input is `scrubbed_slurry` at 20 wt% solids, with an IPCF of 1.0 (the step basis is the same volumetric unit as the incoming slurry) and a CCF of 2.28×10^{-4} kg solids per m^3 per ppm, consistent with the output of the preceding step.

The output `classified_slurry` is diluted to approximately 15 wt% solids by the classification circuit, with a POCF of 1.380 converting from the 20 wt% step basis to the 15 wt% output, and an output CCF of 1.65×10^{-4} kg solids per m^3 per ppm. These CCFs are derived using the same mixture density formulation as the attrition scrubbing step, with grain density of 2.6 t/m^3: $w_s \cdot \rho_{\text{mix}}/10^3$.

The yield rate is set to 0.95, reflecting a 5% lithium loss at this stage attributable to fine particle entrainment in coarse oversize streams. Under a fully closed recycle of coarse material back to attrition scrubbing, the theoretical yield would be 1.0; the 0.95 value represents a practical allowance for incomplete recirculation and mineralogical losses. Note that a total of 8% lithium loss is attributed to coarse gangue mineralisation in the Thacker Pass disclosure [150]; this loss is spread over several steps. No reagents are consumed and no coproducts are generated at this step; coarse oversize returns to `scrubbing` as part of the recycle loop described above.

This step uses the `classifier_generic` machine block (C.5.3.7).

Thickening.

The thickening step concentrates the classified slurry from approximately 15 wt% solids to approximately 55 wt% solids, preparing a dense leach feed for the acid leaching circuit and recovering process water in the thickener overflow for recycle. The primary input is `classified_slurry` with an IPCF of 1.0 and an input CCF of 1.65×10^{-4} kg solids per m^3 per ppm, consistent with the output of the classification step ($w_s \cdot \rho_{\text{mix}}/10^3$ at 15 wt% solids, grain density 2.6 t/m^3).

The output `thick_leach_feed` is at approximately 55 wt% solids, with a POCF of 0.199 converting from the 15 wt% step basis to the 55 wt% output, derived as the ratio of inverse CCFs at the two solids fractions. The output CCF is 8.31×10^{-4} kg solids per m^3 per ppm at 55 wt% solids.

One reagent is consumed: a polymer flocculant dosed at 2.6×10^{-4} kg/m^3 of slurry feed against the step basis, with no constituent elimination. This dosage is calibrated to match the reported combined flocculant consumption across the thickener and decanter circuits in the Thacker Pass disclosure of approximately 260 g/t [150]. The thickener overflow (`thickener_overflow`) is defined as a coproduct with a conversion factor of zero, routed to the `wastewater_treatment` sink to specify that water is assumed to be recycled back into the scrubbing and classification circuit.

This step uses the `thickener_generic` machine block (C.5.3.14).

Acid Leaching.

The acid leaching step dissolves lithium and other soluble species from the thickened claystone slurry using sulfuric acid, producing a leach slurry (`leached_slurry`) containing a pregnant leach solution rich in Li_2SO_4 alongside gangue residue. The primary input is `thick_leach_feed` at approximately 55 wt% solids, converted to the step basis (`leach_slurry_throughput` at approximately 34 wt% solids) via an IPCF of 1.934, reflecting the dilution of the thickened feed upon addition of acid solution and makeup water. This IPCF is derived as the ratio of the inverse CCFs at 55 wt% and 34 wt% solids: $(1/(0.34 \cdot \rho_{34\%}))/(1/(0.55 \cdot \rho_{55\%}))$,

using grain density 2.6 t/m^3 throughout. The input CCF is 8.31×10^{-4} kg solids per m^3 per ppm (at 55 wt% solids), while the step and output CCF is 4.30×10^{-4} kg solids per m^3 per ppm at 34 wt% solids.

The principal reaction is the dissolution of lithium from the clay lattice:

$$\mathrm{Li_2O} \cdot \mathrm{nSiO_2} \cdot \mathrm{mH_2O\,(s)} + \mathrm{H_2SO_4\,(aq)} \longrightarrow \mathrm{Li_2SO_4\,(aq)} + \mathrm{nSiO_2\,(s)} + (\mathrm{m_1}^{+})\mathrm{H_2O} \quad \text{(C.31)}$$

Other gangue elements (e.g. Al, Mg, Ca, Fe) form soluble sulfates or secondary solids; their removal is handled in downstream impurity removal steps.

Two reagents are dosed against the step basis:

- **Sulfuric acid** at 85 kg H_2SO_4/m^3 leach slurry against the step basis volume. This dosage is calibrated to match the disclosure-reported consumption rate of approximately 16 kg acid per tonne Li_2CO_3 [150, 181], with no constituent elimination (acid consumption drives dissolution but the Li constituent is tracked separately via yield rate).
- **Process water** at 0.483 m^3/m^3 of leach slurry against the step basis, representing the additional dilution water needed to transition from the 55 wt% thickened feed to the 34 wt% leach slurry operating density.

The output yield rate is 0.90, reflecting a 10% lithium loss to the solid residue during leaching, consistent with the disclosed leaching recovery range of 85–90% [150]. No coproducts are generated at this step; solid gangue residue remains entrained in the leach slurry and is separated in the subsequent neutralization and CCD circuit.

This step uses the `acid_leach_circuit` machine block (C.5.3.11).

Neutralization.

The neutralization step consumes residual free acid in the leached slurry and initiates precipitation of iron and aluminum impurities prior to solid–liquid separation in the CCD circuit. The primary input is `leached_slurry` at approximately 34 wt% solids, with the input, step, and output CCFs all equal to 4.30×10^{-4} kg ore per (m^3 · ppm), computed as $w_s \cdot \rho_{\mathrm{mix}}/10^3$ (at 34 wt% solids and grain density 2.6 t/m^3).

Two reagents are dosed against the step basis:

- **Limestone ($CaCO_3$)** at 11.2 kg/m^3 of leach slurry for bulk acid neutralization to pH ∼3.5, calibrated to match the reported limestone consumption rate of approximately 2.95 t/t Li_2CO_3 in the Thacker Pass disclosure [150]. The addition of limestone is assumed to not significantly change the throughput volume. No constituent elimination is modeled at this step — precipitated $Fe(OH)_3$ and gypsum remain in the slurry and are separated in the CCD circuit.

- **Flocculant** at 0.03 kg/m^3 of leach slurry, as a settling aid for the nascent precipitates ahead of the CCD. This value is a rough estimate.

The output `neutralized_pls` passes to the CCD circuit with a POCF of 1.0 and a yield of 1.0.

This step uses the `staged_neutralization` machine block (C.5.3.17).

Counter-Current Decantation.

The Counter-Current Decantation (CCD) step separates the pregnant leach solution from the solid leach residue through multi-stage counter-current decantation, recovering dissolved lithium from entrained solution while minimizing lithium losses to the filter cake. The primary input is `neutralized_pls` at approximately 34 wt% solids, with input and step CCFs of 4.30×10^{-4} kg solids per m^3 per ppm, consistent with the neutralization step output.

The output `washed_pregnant_leach_solution` has a POCF of 1.31, reflecting the volumetric expansion from a 34 wt% solids slurry feed to a clarified PLS at approximately 75 g/L total dissolved solids (TDS) and density ∼1.05 t/m^3. This factor accounts for the wash water introduced across the CCD stages and the liquid retained in the filter cake underflow; at a wash ratio in the range 0.75–1.25, the central estimate of 1.31 corresponds to approximately 99% lithium recovery to solution [150]. The output CCF shifts to 1.05×10^{-3} kg per m^3 per ppm, reflecting the transition from a solids-containing slurry basis to a PLS density basis ($\rho_{\text{PLS}}/10^3$). The yield rate is 0.99, consistent with the reported 0.5–1.5% lithium loss to the leach residue [150].

One reagent is consumed: flocculant at 0.04 kg/m^3 of CCD feed. This value is a rough estimate; typical polymer flocculant doses in CCD circuits for sulfate leach slurries range from approximately 20–80 g/t of solids processed [150], and the modeled dose is broadly consistent with the lower end of this range given the solids loading implied by the 34 wt% feed. Wash water addition across the CCD stages is implicit in the POCF of 1.31 but is not modeled as an explicit process water reagent input, on the assumption that wash water requirements are met by internal recycle streams from tailings and other sources. This assumption has not been verified against a closed water balance, and represents a priority refinement alongside the broader water recycling improvements noted in the main body Limitations section (4.5).

The filter cake coproduct (`filter_cake`) is routed to the `tailings_65` sink with a conversion factor of 0.3, representing approximately 0.3 m^3 of dewatered cake per m^3 of CCD feed. This fraction also warrants re-estimation against a water balance calculation once water recycling is explicitly modeled, as it is sensitive to the assumed wash ratio and cake moisture content.

This step uses the `multi_stage_ccd_thickening` machine block (C.5.3.15).

Impurity Removal.

The impurity removal step precipitates dissolved magnesium and calcium from the washed PLS, producing a clarified lithium sulfate solution (`pls_post_polishing`) suitable for soda ash precipitation of lithium carbonate. This step represents the most significant divergence from the Thacker Pass project-specific flowsheet, which uses magnesium crystallization, ion exchange, and zero-liquid-discharge evaporation rather than chemical precipitation, as discussed in Section 4.3. As such, the reagent doses, sludge volumes, and cost implications of this generic all-chemical route are approximate but conservative, and thus should be revisited with a more detailed mass balance in future work.

The primary input is `washed_pregnant_leach_solution` at approximately 75 g/L total dissolved solids (TDS) and density $\rho_{\text{PLS}} \approx 1.05$ t/m^3. The input, step, and output CCFs are all $\rho_{\text{PLS}} \times 10^{-3} = 1.05 \times 10^{-3}$ kg/(m^3·ppm), converting lithium concentration in ppm (mg Li/kg PLS) to kg Li per m^3 of PLS. The POCF is 1000, converting from the m^3 PLS step basis to the litre output units of `pls_post_polishing`. The yield rate is 0.99, reflecting approximately 1% lithium co-precipitation loss with the Mg/Ca sludge.

Two reagents drive the impurity removal chemistry; both are dosed volumetrically against the step basis (m^3 PLS):

- **Lime (CaO)** at 29.1 kg/m^3 PLS for magnesium removal. The PLS contains approximately 11.1 g/L dissolved Mg (as $MgSO_4$), derived from the acid dissolution of Mg-bearing clay at a 20% reactive ore fraction. Lime reacts with dissolved $MgSO_4$ to precipitate $Mg(OH)_2$, with $CaSO_4$ (gypsum) as a solid byproduct:

  $$MgSO_4 + CaO + H_2O \longrightarrow Mg(OH)_2\downarrow + CaSO_4\downarrow$$

  The stoichiometric minimum for Mg removal alone is $11.1 \times (56/24.3) = 25.6$ kg CaO/m^3; the modeled dose of 29.1 kg/m^3 includes an approximate allowance for residual iron and aluminum not fully removed upstream, as well as general reagent excess. This corresponds to approximately 9,200 kg CaO/t Li_2CO_3, substantially exceeding the Thacker Pass disclosed consumption of 2,630 kg/t [150], which reflects their magnesium crystallization route.
- **Soda ash (Na_2CO_3)** at 5.3 kg/m^3 PLS for calcium polishing. The lime addition introduces calcium into the system, most of which co-precipitates as gypsum. The fraction that remains dissolved (approximately 2 g/L, limited by gypsum solubility) must be removed to avoid contaminating the downstream Li_2CO_3 product. Soda ash precipitates this residual calcium as $CaCO_3$:

  $$Ca^{2+} + Na_2CO_3 \longrightarrow CaCO_3\downarrow + 2\,Na^+$$

At $2.0 \times (106/40) = 5.3$ kg/m^3, this dose is approximately stoichiometric for the dissolved calcium at gypsum saturation.

- **Flocculant** at 0.1 kg/m^3 PLS as a settling aid. This is a rough estimate.

The impurity sludge coproduct (`impurity_sludge`) is routed to the `tailings_25` sink. The conversion factor of 0.05 m^3 wet sludge per m^3 PLS feed is a rough placeholder; this conversion factor should be revisited.

This step uses the `staged_neutralization` machine block (C.5.3.17); the same generic machine block is used for both the neutralization and impurity removal steps, as both involve reagent addition to a slurry in agitated tanks.

C.5.1.5 Lepidolite Comminution and Processing

Our model is based primarily on the methods described in Deng and Feng [180] and Yan et al. [205], as the sulfate roasting–water leach pathway is identified as the most common approach for processing lepidolite [174, 210–213].

Hard rock ores such as lepidolite require mechanical size reduction and beneficiation prior to chemical processing. Run-of-mine lepidolite material is first crushed to approximately 150–250 μm, classified by size, and then beneficiated with froth flotation. The flotation step is critical for this pathway: at typical Jiangxi ore grades of less than 0.4% Li_2O ($\sim$ 1,860 ppm Li), direct roasting of ROM material would require processing enormous reagent and energy quantities per unit of lithium product. Froth flotation using amine collectors and phosphate depressants selectively recovers lepidolite minerals into a concentrate at approximately 1.6% Li_2O ($\sim$ 7,600 ppm Li) [196, 434], reducing the mass throughput by approximately 60% [434]. After flotation, the throughput is ground to a product size of approximately 50–75 μm. This latter grinding step uses high-energy mechanical activation milling, which introduces structural defects in the mica lattice to facilitate the solid-state reactions during subsequent roasting. This product is then dewatered prior to entering the roasting circuit.

Following comminution, lepidolite is processed via a sulfate roasting–water leach pathway. Due to the significant presence of fluorine, acid-, chlorine-, or autoclave-based methods can produce corrosive mixtures and/or toxic gases; sulfate-based methods instead allow for the formation of precipitated fluorides (e.g., CaF_2, AlF_3), at the expense of more costly sulfate reagents. The roasting step, conducted at approximately 800–950°C, converts lithium in tightly bound mica lattices to water-soluble lithium sulfate. The resulting calcine is then subjected to water leaching, producing a pregnant leach solution containing lithium, other dissolved metals, and residual sulfate species. Impurities are removed through staged neutralization and pH adjustment to precipitate aluminum, iron, magnesium, and other species. Solid–liquid separation then yields a clarified lithium solution suitable for downstream car-

bonate precipitation via soda ash addition.

This facility is composed of the following blocks:

- `Crushing:` Primary and secondary size reduction.
- `Classification and Flotation:` Classification, screening, and flotation for lepidolite concentration.
- `Grinding:` Mechanical activation milling using a high-energy ball mill grinder to target product size.
- `Dewatering:` Dewatering of activated concentrate prior to roasting.
- `Sulfate Roasting:` Sulfate roasting at 800–950°C.
- `Water Leaching:` Water leaching of calcine to produce pregnant leach solution.
- `Impurity Removal and Neutralization:` Staged neutralization and pH adjustment for impurity precipitation.
- `Clarification:` Clarification of lithium solution for downstream processing.

Crushing.

The crushing step reduces ROM lepidolite ore (`rom_ore_feed`) from run-of-mine fragments to a particle size of approximately 150–250 μ suitable for wet classification and flotation. The primary input is ROM ore in m^3, converted to the step basis of `crushed_ore_throughput` in tonnes via an IPCF of 2.7, reflecting the grain density of lepidolite (2.7 t/m^3). The step CCF is 1.0×10^{-3}, and the input CCF is 2.7×10^{-3} kg constituent per ($m^3 \cdot$ ppm), reflecting the conversion from volumetric input to mass-basis step. Only Li is tracked as a constituent through the comminution circuit. No reagents are consumed. The output `crushed_ore` passes to classification with a POCF of 1.0 and a yield rate of 0.99, reflecting a 1% loss to fines and dust during crushing.

This step uses the `crushing_cycle` machine block (C.5.3.5).

Classification & Flotation.

The classification and flotation step separates fine lepidolite particles from coarser gangue and produces a lepidolite concentrate enriched in lithium, which is the key beneficiation step in this pathway. The step is constituent-driven: the volume-defining basis is the Li constituent of the input stream, meaning that the required flotation circuit volume scales with the lithium mass to be processed rather than with bulk ore throughput. This correctly captures the dependence of flotation circuit sizing on ore grade.

The primary input is `crushed_ore` in tonnes is converted to the step basis (`classified_slurry_throughput` in m^3 of slurry at approximately 30 wt% solids) via an IPCF of 2.7037, derived as $1/(w_s \cdot \rho_{\text{mix}})$ at 30 wt% solids with grain density 2.7 t/m^3: $\rho_{\text{mix}} = 1/(0.3/2.7 + 0.7/1.0) = 1.233$ t/m^3, giving $1/(0.3 \times 1.233) = 2.704$ The step CCF is 3.70×10^{-4} kg

constituent per ($m^3 \cdot$ ppm), computed as $w_s \cdot \rho_{\text{mix}}/10^3$.

The output `classified_slurry` is a lepidolite concentrate at approximately 60 wt% solids, with a POCF of 0.3836 converting from the 30 wt% step basis to the 60 wt% concentrate, using the density mixing formula. The output CCF is 9.64×10^{-4} kg constituent per ($m^3 \cdot$ ppm), derived as $w_s \cdot \rho_{\text{mix}}/10^3$ at 60 wt% solids. The yield rate is 0.70, representing the combined lithium recovery across classification and flotation, decomposed as

$$r_{\text{combined}} = r_{\text{class}} \times r_{\text{flot}} = 0.93 \times 0.75 = 0.70$$

where $r_{\text{class}} = 0.93$ reflects a 7% lithium loss to classification slimes, consistent with the 5–8% range reported for Jiangxi industrial lepidolite desliming operations [**?**], and $r_{\text{flot}} = 0.75$ reflects flotation lithium recovery at the conservative end of the 75–82% range reported for industrial-scale desliming–flotation circuits processing low-grade Jiangxi ore [434**?**]. The remaining 30% of lithium reports to the coarse reject stream (approximately 7% via classification slimes and 23% via flotation tailings). This step performs the key concentration from approximately 1,860 ppm Li (0.4% Li_2O) at ROM to approximately 7,430 ppm Li (∼1.6% Li_2O) in the concentrate — a 4.0 × grade concentration at 70% combined recovery, yielding a ROM-to-concentrate mass ratio of approximately 5.7:1.

Three reagents are dosed against the step basis volume: process water at 0.863 m^3/m^3 for slurrying and flotation circuit operation; depressants (sodium hexametaphosphate) at 0.1 kg/m^3, which suppress non-lepidolite silicate minerals to improve selectivity; and collectors (dodecylamine) at 0.12 kg/m^3, which selectively adsorb onto lepidolite surfaces to render them hydrophobic and recoverable by froth flotation [434]. None of these reagents eliminate tracked constituents from the process stream — their function is physical separation rather than chemical removal.

Two coproducts are generated. The coarse reject (`coarse_reject`) is routed to the `tailings_65` sink with a conversion factor of 0.75 m^3/m^3 of slurry, representing the 82.5% of feed solids and gangue that report to classification slimes and flotation tailings; this large fraction is consistent with the low ROM ore grade and the ∼ 5.7× concentration ratio achieved. Thickener overflow (`thickener_overflow`) is routed to wastewater treatment and is assumed to be recycled internally, but this needs to be validated with a full water balance (4.5).

This step uses the `classifier_flotation` machine block (C.5.3.8).

Grinding.

The grinding step reduces the lepidolite concentrate to a fine product size of approximately 50–75 μ via high-energy mechanical activation milling, breaking down the mica crystal structure to yield an activated concentrate that is reactive in subsequent sulfate roasting.

This mechanical activation is a distinguishing feature of the lepidolite pathway — the fine grinding increases the surface area and introduces structural defects in the mica lattice that facilitate the solid-state reactions during roasting [180, 205].

The primary input is `classified_slurry` at 60 wt% solids, with an IPCF of 1.0 (the step basis is the same volumetric unit). The step and input CCFs are both 0.0009643 kg solids per ($m^3 \cdot$ ppm), consistent with the 60 wt% slurry density basis. No reagents are consumed. The output `ground_slurry` has a POCF of 1.0 and a yield rate of 0.995, reflecting a 0.5% loss during milling. No coproducts are generated.

This step uses the `high_energy_grinding_mill` machine block (C.5.3.9).

Dewatering.

The dewatering step removes water from the ground lepidolite concentrate slurry to produce a filter cake suitable for high-temperature sulfate roasting. Roasting requires a low-moisture feed; excess water would consume thermal energy in evaporation rather than driving the sulfate activation reactions, and could also cause handling problems in the roasting train. The primary input is `ground_slurry` at 60 wt% solids, with input and step CCFs of 9.64×10^{-4} kg constituent per ($m^3 \cdot$ ppm), derived as $w_s \cdot \rho_{\text{mix}}/10^3$ at 60 wt% solids with grain density 2.7 t/m^3: $\rho_{\text{mix}} = 1/(0.6/2.7 + 0.4/1.0) = 1.607$ t/m^3.

The output `roaster_feed` is an 80 wt% solids filter cake, expressed in tonnes rather than m^3, with a POCF of 1.205 converting from m^3 of 60 wt% slurry to tonnes of 80 wt% cake: POCF $= w_{s,\text{in}} \cdot \rho_{\text{mix,in}}/w_{s,\text{out}} = 0.6 \times 1.607/0.8$. The output CCF is 8.0×10^{-4} kg constituent per (tonne $\cdot$ ppm), reflecting the mass basis of the cake: $w_{s,\text{cake}}/10^3 = 0.8/10^3$. No reagents are consumed. Wash liquor (`wash_liquor`) is generated at approximately 0.4 m^3/m^3 of input slurry, routed to the `wastewater_treatment` sink; this represents approximately 0.4 t/m^3 of water removed during pressing, consistent with the reduction in water content from 0.643 t/m^3 at 60 wt% solids to 0.241 t/m^3 at 80 wt% solids. Yield rate is 1.0.

This step uses the `mechanical_dewatering` machine block (C.5.3.16).

Sulfate Roasting.

The sulfate roasting step thermally activates the dewatered lepidolite concentrate by reacting it with sodium sulfate, potassium sulfate, and lime at 800–950° C, converting lithium from the tightly bound mica lattice into water-soluble lithium sulfate. The central reaction is:

$$(Li_2O)(Al_2O_3)(3\,SiO_2)(KF)\,(s) + Na_2SO_4 \longrightarrow Li_2SO_4\,(s) + Al_2O_3\,(s) + 3\,SiO_2\,(s) + KF\,(g/s) \tag{C.32}$$

with fluorine species captured by lime addition as CaF_2 and AlF_3 precipitates rather than volatilizing as toxic HF gas.

The primary input is `roaster_feed` in tonnes of 80 wt% solids cake, with an IPCF of 1.6, representing the total roasting mix mass per tonne of input cake after addition of all reagents and residual moisture. This factor accounts for: dry ore solids (0.8 t per tonne cake), sodium sulfate addition ($0.5 \times 0.8 = 0.4$ t), lime addition ($0.15 \times 0.8 = 0.12$ t), potassium sulfate addition ($0.1 \times 0.8 = 0.08$ t), and residual moisture (0.2 t). The input and step CCF are both 5.0×10^{-4} kg constituent per (tonne · ppm): $w_{s,\text{cake}}/(G_{18} \times 10^3) = 0.8/(1.6 \times 10^3)$.

Three reagents are dosed against the `roasting_mix` step basis:

- **Sodium sulfate (Na_2SO_4)** at 250 kg/tonne roasting mix (equivalent to 0.5 t per tonne dry lepidolite concentrate), derived from Yan et al. [205].
- **Lime (CaO)** at 75 kg/tonne roasting mix (equivalent to 0.15 t per tonne dry concentrate). The base value of 0.1 t/t from Yan et al. [205] is increased by 50% following Luong et al. [209] to ensure complete fluoride capture, suppressing HF volatilization.
- **Potassium sulfate (K_2SO_4)** at 50 kg/tonne roasting mix (equivalent to 0.1 t per tonne dry concentrate), derived from Yan et al. [205], to supplement the alkali sulfate flux and improve lithium extraction efficiency.

The output `roasted_mix` is the calcine after roasting, with water evaporated during the process. The POCF is 0.875, derived as the ratio of the calcine mass (excluding evaporated water: $0.8 \times (1 + 0.5 + 0.15 + 0.1) = 1.4$ t) to the total roasting mix mass (1.6 t). The output CCF is 6.06×10^{-4} kg constituent per (tonne · ppm): $w_{s,\text{cake}}/(0.8 \times (1 + 0.5 + 0.15) \times 10^3)$. A fluorine offgas coproduct (`fluorine_offgas`) is defined with a conversion factor of zero, reflecting the assumption that lime additions during roasting fix essentially all fluorine as solid precipitates before volatilization.

This step uses the `sulfate_roasting_train_lepidolite` machine block (C.5.3.10).

Water Leaching.

The water leaching step dissolves the lithium sulfate and other soluble species from the roasted calcine into aqueous solution, producing a pregnant leach solution for downstream impurity removal. The primary input is `roasted_mix` in tonnes of calcine, converted to the step basis (`leaching_slurry` at approximately 20 wt% solids) via an IPCF of 4.370, derived as $1/(w_s \cdot \rho_{\text{mix}})$ at 20 wt% solids with assumed calcine grain density of 2.7 t/m^3: $\rho_{\text{mix}} = 1/(0.2/2.7 + 0.8/1.0) = 1.144$ t/m^3, giving $1/(0.2 \times 1.144) = 4.370$. The input CCF is 6.06×10^{-4} kg constituent per (tonne · ppm), consistent with the roasting output. The step CCF is 1.51×10^{-4} kg constituent per (m^3 · ppm), derived as the product of the 20 wt% slurry density ($w_s \cdot \rho_{\text{mix}}/10^3 = 0.229$ kg/L) and the mass fraction of original ore solids in the calcine ($0.8/(0.8 + 0.265 + 0.15) = 0.659$), which corrects for the dilution of the ore-derived

Li quantity by reagent-derived solids in the calcine.

Process water is dosed at 0.915 m^3/m^3 of leach slurry basis, derived from the water fraction required to produce a 20 wt% slurry: $(1 - w_s) \cdot \rho_{\text{mix}}/\rho_{\text{water}} = 0.8 \times 1.144/1.0 = 0.915$ m^3/m^3. The POCF is 0.869, computed as the process water ratio scaled by the 95% yield (0.915×0.95), converting from m^3 of leach slurry to m^3 of pregnant leach solution (`pregnant_leach_solution`). The output CCF is 2.29×10^{-4} kg constituent per ($m^3 \cdot$ ppm), computed as $w_s \cdot \rho_{\text{mix}}/10^3$ at 20 wt% solids without the calcine composition correction, since the output is a PLS where Li concentration is referenced to total solution mass. The yield rate is 0.95, reflecting a 5% lithium loss to the leach residue. The leach residue coproduct (`leach_residue`) is routed to the `tailings_65` sink with a conversion factor of 0.131 m^3/m^3 of leach slurry, representing the solid fraction that does not enter solution; residue constituents include Al, Fe, Si, and CaF_2, which are classified as non-hazardous gangue. The assumption of calcine grain density equal to lepidolite density (2.7 t/m^3) is a simplification, as roasting modifies the mineralogical composition of the solids, but is considered acceptable for the purposes of volumetric sizing.

This step uses the `water_leach` machine block (C.5.3.12).

Impurity Removal and Neutralization.

The impurity removal step precipitates dissolved magnesium, calcium, aluminum, iron, and trace impurities from the pregnant leach solution via staged pH adjustment, producing a clarified lithium sulfate solution suitable for downstream carbonate precipitation. The primary input is `pregnant_leach_solution` at approximately 20 wt% suspended solids, with input, step, and output CCFs all equal to 2.29×10^{-4} kg constituent per ($m^3 \cdot$ppm), derived as $w_s \cdot \rho_{\text{mix}}/10^3$ at 20 wt% solids with assumed calcine grain density 2.7 t/m^3. This density approximation introduces some error since roasting modifies the mineralogy of the solids, but we assume it reasonable for volumetric sizing purposes.

Three reagents are dosed against the step basis volume:

- **Neutralizing agent** at 4 kg/m^3 of PLS, modeled as a generic mixed reagent representing a combination of lime, limestone, and/or NaOH for staged pH adjustment to precipitate $Fe(OH)_3$, $Al(OH)_3$, and $Mg(OH)_2$. The dosage is a rough estimate: stoichiometric lime demand for Mg precipitation could be approximately 3–6 kg/m^3, but the actual demand is moderated by the lower impurity concentrations in a water leach PLS compared to a sulfuric acid leach, and by the use of a mixed reagent [180, 205]. This uncertainty in both composition and neutralizing agent combination warrants further refinement against measured PLS impurity data.
- **Flocculant** at 0.005 kg/m^3 of PLS as a settling aid. This is also a rough estimate.

- **Sulfuric acid** at 1 kg/m^3 of PLS for pH re-adjustment after primary neutralization, to avoid overshooting the target pH and precipitating lithium-bearing compounds. This is also a rough estimate.

The output `neutralized_pls` passes to the clarification step with a POCF of 1.0 and a yield rate of 0.95, representing approximately 5% lithium loss to the impurity sludge and entrained liquor — consistent with the total lithium losses across the magnesium crystallization and zero liquid discharge (ZLD) circuits reported in the Thacker Pass disclosure [150], and applied here as a conservative proxy given the different process route. The impurity sludge coproduct (`impurity_sludge`) is routed to the `tailings_20` hazardous secure landfill sink with a conversion factor of 0.05 m^3/m^3 of PLS, estimated from the 5% lithium yield loss as a proportional proxy for sludge volume generation.

This step uses the `staged_neutralization` machine block (C.5.3.17).

Clarification.

The clarification step performs a final solid–liquid separation on the neutralized PLS to remove residual precipitates and suspended solids, yielding a clarified lithium solution (`pls_post_polishing`) suitable for soda ash precipitation. The primary input is `neutralized_pls`, with input and step CCFs of 2.29×10^{-4} kg constituent per (m^3·ppm), consistent with the upstream step. One reagent is consumed: flocculant at 0.005 kg/m^3 of PLS, a rough estimate for polishing-stage clarification where precipitate loadings are low.

The output `pls_post_polishing` is expressed in liters, with a POCF of 1000 (m^3 to L) and an output CCF of 2.29×10^{-7} kg constituent per (L·ppm), derived by dividing the m^3-basis CCF by 1000. The yield rate is 0.995, reflecting a 0.5% lithium loss to residual impurity sludge. The impurity sludge coproduct (`impurity_sludge`) is routed to the `tailings_20` hazardous landfill sink with a conversion factor of 0.005 m^3/m^3 of PLS input, consistent with the small residual solids loading at this polishing stage.

This step uses the `solid_liquid_separation` machine block (C.5.3.13).

C.5.1.6 Pregnant Leach Solution to Lithium Carbonate (Common to All Pathways)

This section applies to the conversion of purified lithium-bearing solution to lithium carbonate product [178, 345]. Although upstream chemistry differs between chloride-based brine pathways (LiCl) and sulfate-based ore pathways (Li_2SO_4), the downstream precipitation and solids handling configuration is broadly similar.

In both cases, lithium carbonate is produced by reaction of dissolved lithium with soda ash. Chloride systems generate sodium chloride as a byproduct while sulfate systems generate sodium sulfate; these differences primarily affect salt management and recycle strategy

rather than equipment configuration. The core precipitation, thickening, washing, and drying sequence remains functionally consistent across pathways.

Precipitation is conducted at elevated temperatures to improve reaction kinetics and reduce lithium carbonate solubility. Thermal requirements are modeled generically, with pathway-specific parameterization to account for differences in reaction temperature, capturing energy demand without prescribing a specific crystallizer architecture. Following soda ash precipitation, lithium carbonate is separated from mother liquor via thickening and filtration. Washing and drying complete product finishing. Rather than specifying individual equipment types, these operations are represented as aggregated solids-handling steps parameterized by recovery efficiency, moisture content, and recycle fraction.

This facility is composed of the following blocks, with pathway-specific parameterization depending on whether the upstream pathway is chloride- or sulfate-based:

- `Lithium Carbonate Precipitation:` Soda ash precipitation, including preheating and cooling.
- `Thickening and Dewatering:` Thickening and dewatering of lithium carbonate solids.
- `Washing and Purification:` Washing and purification, to remove residual Na, K, Mg.
- `Drying and Packaging:` Drying and packaging of final product.

Lithium Carbonate Precipitation.

Lithium carbonate is produced by reacting dissolved lithium with soda ash (Na_2CO_3):

$$2\,Li^+ + Na_2CO_3 \rightarrow Li_2CO_3(s) + 2\,Na^+ \tag{C.33}$$

In chloride-based brine systems the byproduct is NaCl, whereas in sulfate-based ore systems it is Na_2SO_4; these differences affect downstream salt management but not the precipitation equipment configuration. Both pathways use the same underlying production step, with pathway-specific parameterization for the input chemistry, soda ash dosage, and yield rate. This step uses the `precipitator` machine block (C.5.3.18).

Chloride pathway.

The primary input is `pls_post_polishing` (concentrated lithium brine) in liters, with the step constitutent-driven on the Li constituent such that the IPCF is 1.0 by definition — the step volume scales with the ratio of lithium mass in output volume to input PLS, rather than the bulk PLS volume. The input and step CCFs are both 1.25×10^{-6} kg Li per (L · ppm), reflecting the density of the concentrated PLS (1.25 kg/L) divided by 10^6 to convert from ppm (mg/kg) to kg/L. The POCF is 10^{-3} (L to m^3). The output `li2co3_slurry` is a 20 wt% Li_2CO_3 slurry with density $\rho_{\text{mix}} = 1/(0.2/2.11 + 0.8/1.0) = 1.118$ t/m^3, giving an output CCF of $0.2 \times 1.118/10^3 = 2.24 \times 10^{-4}$ kg constituent per (m^3·ppm). The yield

rate is 0.97, reflecting a 3% lithium loss to mother liquor and wash streams, consistent with reported losses from the precipitation circuit in the Silver Peak disclosure [178].

Soda ash is dosed at 11.5 kg Na_2CO_3 per kg Li, targeting the Li constituent. This corresponds to 2.2 t Na_2CO_3 per tonne Li_2CO_3 produced, consistent with the ratio described in the flowsheet of the Silver Peak disclosure [178].

Sulfate pathway.

The primary input is `pls_post_polishing` in liters, with the step constituent-driven on the Li constituent and IPCF of 1.0 by definition. The input and step CCFs are both 2.29×10^{-7} kg Li per (L·ppm), derived as $w_s \cdot \rho_{\text{mix}}/10^6$ at 20 wt% solids with grain density 2.7 t/m^3: $\rho_{\text{mix}} = 1/(0.2/2.7 + 0.8/1.0) = 1.144$ t/m^3, giving $0.2 \times 1.144/10^6 = 2.29 \times 10^{-7}$. This assumes the density of the calcine-derived solids in the PLS approximates that of lepidolite. The POCF is 10^{-3} (L to m^3). The yield rate is 0.80, a conservative rough estimate reflecting the lower precipitation efficiency expected for a sulfate-route PLS relative to a chloride brine. Laboratory-scale literature for lepidolite PLS reports recoveries of approximately 90% under near-optimal conditions [180], with worse performance expected in a generic all-chemical impurity removal route. This parameter represents a priority area for refinement.

Soda ash is dosed at 16.55 kg Na_2CO_3 per kg Li, targeting the Li constituent. This corresponds to 3.11 t Na_2CO_3 per tonne Li_2CO_3 produced (verified: $0.1879 \times 16.55 = 3.11$ t/t), representing a 2.17× stoichiometric excess. The higher excess relative to the chloride route is justified by the elevated ionic strength in sulfate PLS and is consistent with the Thacker Pass disclosure [150]; we assume it is also reasonable for a lepidolite sulfate-route PLS.

Thickening and Dewatering.

The thickening and dewatering step concentrates the Li_2CO_3 slurry from approximately 20 wt% solids to approximately 70 wt% solids via gravity settling and mechanical pressing, recovering mother liquor for recycle or discharge. The primary input is `li2co3_slurry` at 20 wt% solids, with input and step CCFs of 2.24×10^{-4} kg constituent per (m^3·ppm), derived as $w_s \cdot \rho_{\text{mix}}/10^3$ at 20 wt% solids with Li_2CO_3 grain density 2.11 t/m^3: $\rho_{\text{mix}} = 1/(0.2/2.11 + 0.8/1.0) = 1.118$ t/m^3.

The output `li2co3_cake` is at 70 wt% solids with density $\rho_{\text{mix}} = 1/(0.7/2.11 + 0.3/1.0) = 1.583$ t/m^3, giving an output CCF of $0.7 \times 1.583/10^3 = 1.108 \times 10^{-3}$ kg constituent per (m^3·ppm). The POCF is 0.202, derived as $(0.2 \times \rho_{20\%})/(0.7 \times \rho_{70\%}) = (0.2 \times 1.118)/(0.7 \times 1.583)$, representing the volumetric contraction from 20 wt% to 70 wt% solids. One reagent is consumed: flocculant at 0.005 kg/m^3 of slurry feed, a rough estimate.

The mother liquor overflow is treated differently by pathway. For the chloride (brine) route, the overflow is assumed to be recycled back to the precipitation reactor or the evap-

oration pond system, consistent with the Silver Peak disclosure [178]; the conversion factor is therefore set to zero and no wastewater treatment cost is incurred. For the sulfate route, the overflow is discharged to wastewater treatment with a conversion factor of 0.80 m^3/m^3 of slurry feed, consistent with the calculated volume fraction removed during thickening ($\sim$0.80 m^3/m^3). This water balance is a priority area for refinement 4.5, particularly for Thacker Pass, where water recycling is a central design feature of the project-specific flowsheet [150].

This step uses the `mechanical_dewatering` machine block (C.5.3.16).

Washing and purification.

The washing and purification step removes residual mother liquor and soluble impurities from the Li_2CO_3 filter cake by counter-washing with process water, improving product purity. The primary input is `li2co3_cake` at 70 wt% solids, with input, step, and output CCFs all equal to 1.108×10^{-3} kg constituent per (m^3·ppm), consistent with the thickening output. The POCF and yield rate are 1.0; washing does not change the cake volume or lithium content materially at this stage.

Process water is consumed as a wash input; the dosage is a rough estimate. The wash liquor is handled by pathway: for the chloride route it is recycled back to the precipitation reactor or pond system per the Silver Peak environmental disclosure [178], with conversion factor zero; for the sulfate route it is discharged to wastewater treatment at 0.1 m^3/m^3 of cake, reflecting the pass-through of wash water after contact with the cake. The non-recycling assumption for the sulfate route is a conservative simplification and represents a refinement priority alongside the rest of the water balance.

This step uses the `wash_purify_combo` machine block (C.5.3.21).

Drying and packaging.

The drying and packaging step removes residual moisture from the washed Li_2CO_3 cake and prepares the final battery-grade product. The primary input is `lithium_carbonate_cake` in m^3 at 70 wt% solids, converted to the step basis in tonnes via an IPCF of 1.583 t/m^3, equal to the cake density $\rho_{\text{mix}} = 1/(0.7/2.11 + 0.3/1.0)$; this represents the theoretical maximum density of a 70 wt% Li_2CO_3 cake and likely slightly overestimates the actual bulk density in practice. The input CCF is 1.108×10^{-3} kg constituent per (m^3·ppm), consistent with the upstream steps. The step CCF is 7.0×10^{-4} kg constituent per (tonne·ppm), derived as $w_s/10^3 = 0.7/10^3$, reflecting the mass fraction of Li_2CO_3 in the cake on a tonne basis.

The output `lithium_carbonate_product` is dry Li_2CO_3 in tonnes, with a POCF of 0.7 representing the Li_2CO_3 mass fraction of the input cake, and an output CCF of 1.0×10^{-3} kg constituent per (tonne·ppm), the standard conversion for a pure product on a mass basis. No reagents are consumed. Evaporated moisture is released to the atmosphere as a coproduct with negligible cost or emissions impact.

This step uses the `drying_packaging_generic` machine block (C.5.3.22).

C.5.1.7 Tailings and Residue Management

Tailings and residue management are key contributors to the cost and environmental impacts of minerals extraction. We model this step differentially across each pathway type, given spodumene, lepidolite, clay, and brine-hosted deposits each have their own typical mineralizations and byproducts [195, 345, 346, 353]. Tailings management practices vary across projects, ranging from conventional slurry impoundments to filtered dry-stack systems, and depend on factors such as water availability, seismic risk, climatic conditions, and regulatory constraints. Ore-based processing generates solid residues from comminution, beneficiation, and leaching operations. These residues must be dewatered, transported, and permanently stored in accordance with site-specific geotechnical and environmental requirements. Furthermore, in the production of lithium from pregnant leach solutions, additional residues are generated, which need to be managed.

Due to the fact that modeling tailings inherently deals with the management of co-products rather than primary outputs, in this work, we do not (currently) model this as a model block to fit within the `ProductionStep` paradigm, but rather as a custom function parameterized by residue tonnage, solids concentration, and water recovery fraction. Tailings handling, thickening, filtration, and disposal infrastructure are therefore aggregated, preserving sensitivity to ore grade, recovery efficiency, and leach residue generation while avoiding over-specification of storage facility design.

We parameterize each residue stream independently by its dry mass (derived from the process stoichiometry at each production step), a unit disposal cost c_i (\$/t dry), and a solids fraction s governing transport volume and handling cost. Total tailings cost for each residue stream is then

$$C_{\text{tail}} = m_{\text{chem}} \cdot \left(c_{\text{dew}}(s) + c_{\text{disp}} \cdot \frac{1}{s} \right) \tag{C.34}$$

where m_{chem} are annual dry solid masses (t/yr), and c_{disp} reflects disposal costs for the chemical residue fraction, and $1/s$ converts the gate fee from \$/t wet to \$/t dry. The solids correction factor s scales cost to the wet mass actually handled. $c_{\text{dew}}(s)$ is the unit dewatering costs, derived from estimated dewatering operating costs for sludge from the thickener and mechanical dewatering block models.

For wet sludge streams ($s < 0.55$), dewatering costs range from approximately \$8—17/t DS for gravity thickening ($s \in [0.08, 0.15]$), \$17–30/t DS for belt filter press ($s \in [0.15, 0.25]$), and \$30–48/t DS for plate-and-frame filter press ($s \in [0.25, 0.50]$). For dry-stack streams ($s \geq 0.55$), stacking costs range from \$3–8/t DS for paste pumping and

conveyor stacking ($s \in [0.55, 0.75]$), declining to \$1.5–4/t DS for dry conveyor-only handling ($s \in [0.75, 1.00]$). Dewatering electricity consumption is estimated within the cost model, ranging from approximately 3–8 kWh/t DS for gravity thickening to 35–60 kWh/t DS for plate-and-frame pressing, and the associated electricity cost is incorporated into the total sink handling cost via the facility-level electricity price. However, the corresponding Scope 2 greenhouse gas emissions from dewatering electricity are not currently computed, as the electricity consumption estimate is not passed to the emissions accounting framework.

Disposal gate fees are drawn from China's National Development and Reform Commission for hazardous secure landfill costs in Jiangxi province [207]. GHG emissions from waste disposal sinks are not currently modeled; this represents a limitation discussed in Section **??**.

Type 1: Evaporative Residue (Brines)

Waste streams from brine production primarily consist of salts generated through precipitation during the evaporative phase, and brine-derived chemical residues generated during concentration, chemical treatment, and selective lithium extraction. Solid volumes are from evaporation low relative to ore-based pathways, but still significant given the volume of brine that needs to be processed per tonne output, with a strong dependence on brine composition, especially the ratio of Magnesium to Lithium in the feed.

Firstly, as brine is concentrated across a series of solar ponds, major salts precipitate sequentially in approximate order of decreasing solubility, and are harvested periodically from the pond floors [196, 198, 435]. Depending on the brine composition, this order is typically halite (NaCl), gypsum ($CaSO_4 \cdot 2\,H_2O$), sylvite (KCl), and carnallite ($KCl \cdot MgCl_2 \cdot 6\,H_2O$). These evaporation salt precipitates are physically stable and chemically non-reactive; at operations they are typically restacked adjacent to ponds, and thus incur primarily mechanical haulage costs. We thus model these as primary maintenance costs of the evaporation ponds, and do not explicitly model them using this tailings model.

For the chemical residues, primarily slurries of $Mg(OH)_2$ and $CaCO_3$, the stream is modeled at $s = 0.35$, for dewatering to a stackable, nonhazardous filter cake composed of these precipitated salts. This stream is routed to an on-site stack [178], with an on-site containment facility operating cost adder applied.

Type 2: Leach Residue (Clay)

Clay leaching generates two primary solid waste streams: an impurity sludge from the neutralization and impurity removal steps, and a coarse leach residue from the counter-current decantation (CCD) circuit. The impurity sludge, consisting primarily of magnesium and calcium hydroxide precipitates with residual sulfate, is modeled at $s = 0.25$. This stream is routed to an on-site lined containment facility at Thacker Pass rather than off-site

disposal, consistent with the project disclosure [150]; the containment facility operating cost is applied as a flat adder of \$5/t DS, which is a placeholder pending more detailed permit and operational data. The coarse CCD filter cake, consisting of spent claystone with low residual lithium and reagent content, is modeled at $s = 0.65$, and routed to an on-site dry stack.

Type 3: Complex Reactive Ore Residue (Lepidolite)

Lepidolite processing generates three residue streams with distinct handling requirements. The impurity sludge from staged neutralization — containing metal hydroxide precipitates including aluminum, iron, and fluoride-bearing compounds — is classified as hazardous waste under Chinese regulations due to elevated concentrations of thallium, rubidium, and cesium [163, 436]. This stream is modeled at $s = 0.20$, representative of thickener underflow prior to mechanical dewatering, and routed to an off-site hazardous secure landfill with a gate fee of \$280/t wet (~\$1,400/t DS at this solids fraction). This stream drives the majority of the tailings cost reported for Jianxiawo. The classification and flotation coarse reject (dilute gangue slurry at approximately $s = 0.30$) is routed to an on-site conventional wet tailings storage facility (TSF), which does not require lining under Chinese standards for non-hazardous lepidolite gangue; operating costs are estimated at \$4/t DS for earthwork, monitoring, and water reclaim, plus 2 kWh/t DS for slurry pumping. These numbers are also placeholder estimates pending more detailed permit and operational data, especially given ongoing regulatory tightening around Lepidolite permitting [179, 206]. Finally, the leach residue from the CCD circuit is modeled at $s = 0.65$ and routed to on-site dry stacking, consistent with the coarse clay residue handling described above.

C.5.2 Machine Model Block Methodology

C.5.2.1 Model Block Data Aggregation Methodology

This study employs a structured bottom-up aggregation methodology to translate heterogeneous sources of industry data (technical reports, specification sheets, regulatory filings, and interviews where appropriate) into internally consistent technoeconomic model blocks, with each `ProductionStep` object having various model block options. The objective is not necessarily to replicate any singular plant configuration, but to construct reasonable baselines for representative, transparent, and sensitivity-aware unit operations suitable for high-level comparative analysis. Using scaling factors or parallel production lines, these blocks can then be sized to meet desired demand.

For each model block, two distinct but co-defined components are specified:

- **Machine block options** Determines equipment sizing and associated capital and operational requirements.
- **Input and output materials** Specifies the mass streams entering and leaving the block.

These components are interconnected but decomposable: in the absence of detailed engineering...

Four major steps go into the creation of each block: (1) definition of bill of equipment and key inputs and outputs; (2) raw data collection; (3) aggregation into model blocks, including scenario development (low/mid/high); and (4) validation and cross-checking.

Step 1: Define a Bill of Equipment and Key Inputs and Outputs.

The machine block is composed of a Bill of Equipment (BOE), which identifies:

- **Primary process equipment** Singular or a key set of core process machines.
- **Required secondary systems** e.g., handling, dosing, recirculation, discharge.
- **Balance-of-plant auxiliary elements** e.g., piping, controls, structural and access.

There are three types of inputs:

- **Primary inputs** (e.g., ore, brine, slurry) defined with a calculated flow rate or batch size and associated cycle time.
- **Secondary inputs** i.e., reagents (e.g., lime, soda ash, sulfuric acid), defined with calculated dosages based on stoichiometric analyses with operational excess factors, inefficiencies, and possible recycle simplifications.
- **Utilities** (e.g., process water, compressed air, natural gas) calculated as inputs that are fully consumed unless otherwise specified as a co-product.

There are likewise two types of outputs:

- **Primary outputs** (e.g., excavated ore, pumped brine, dewatered slurry) the assumed result of the reaction of primary inputs, secondary inputs, and additional machine-specific processing.
- **Secondary outputs** i.e., co-products (e.g., CO_2 emissions, wastewater, tailings) calculated based on stoichiometric analyses.

As discussed in the production step formulation, primary and secondary inputs and outputs can have constituents defined. While calculation of reagent usage and co-product production are carried out in Step 3, the definition of key inputs and outputs is performed here, as they affect the engineering design and assumptions made in the model.

Step 2: Raw Data Collection.

Raw data are compiled from:

- **Feasibility studies and technical reports** SEC Technical Report Summaries (SK-1300) and NI 43-101 reports.
- **Vendor specification sheets and package data**
- **Standard engineering references** e.g., the Australasian Institute of Mining and Metallurgy's *Cost Estimation Handbook*; Peters and Timmerhaus [183].
- **Literature sources** where detailed data exist.
- **Interviews and technical discussions** with operators, vendors, and other modelers.

The primary sources for each modeling block are listed at the top of each modeling block.

Step 3: Aggregate into Model Blocks.

Raw data are aggregated into standardized model blocks using a consistent structure:

$$\text{Block output} = f(\text{equation of merit},\ \text{volume defining basis}) \tag{C.35}$$

Each model block provides a baseline for: the volume defining basis (i.e., the design variable, such as slurry flow rate, solids throughput, working volume, or other key sizing metric); core physical relationships as a function of the volume defining basis (e.g., mixing intensity, stoichiometric demand); installed capacity with oversizing margin; and CAPEX and OPEX components.

Given the heterogeneous and uncertain nature of raw data, we develop an envelope approach to estimate conservative (pessimistic), midpoint, and optimistic values for key parameters, anchored to reported values with uncertainty bounds introduced transparently. These envelopes help capture extremes of performance and encompass the possible space of models. Assumptions are stated explicitly to maintain traceability.

CAPEX Aggregation.

While capital equipment costs are not a primary driver of this analysis given the focus on operational competitiveness, they are nonetheless important as a point of comparison against established studies, and operational maintenance costs are often sized as a percentage of capital costs. We use an anchor equipment approach, consistent with standard chemical and mineral processing practice:

$$C_{\text{total}} = C_{\text{anchor}} + C_{\text{auxiliary}} \tag{C.36}$$

where C_{anchor} is primary equipment cost derived from vendor data, scaling relationships, and/or reported feasibility values, and $C_{\text{auxiliary}}$ represents explicit adders for integration and balance-of-plant elements.

Auxiliary CAPEX adders typically cover systems such as pumps and piping, structural steel and platforms, instrumentation and controls, electrical distribution, and reagent handling and dosing. Adders are applied as either explicit percentage ranges derived from standard cost breakdowns, or discrete subsystem packages when vendor costs are available. Standard default estimates are given in Table C.4 drawn from Table 4 and Table 6 of Chapter 6 of Peters and Timmerhaus [183]. Where possible, equipment-specific adders from Peters and Timmerhaus [183] were used.

Table C.4. Standard auxiliary CAPEX adder ranges by equipment class and scenario.

Scenario	**Mechanically simple, pump-dominated**	**Mechanically intensive, multiple-drive systems**	**Thermal systems, fans/ducts/controls**
Conservative	50%	60%	55%
Midpoint	35%	45%	40%
Optimistic	20%	30%	25%

Maintenance Estimation.

Maintenance is estimated as an annual cost, as a percentage of all-in fixed-capital investment costs, depending on the complexity of operation. We directly quote maintenance costs for chemical processes from Peters and Timmerhaus [183], but as not all of the processes modeled in this system are necessarily chemical in nature, we construct a similar table for physical processes (e.g., bulk ore crushing and processing, material movement, pumping and other auxiliaries), based on the SME Mining Engineering Handbook [433].

Labor Requirements Estimation.

Operating labor requirements are estimated using published operator-per-shift factors and a staffing multiplier; the full method is described in C.5.2.4.

Table C.5. Maintenance cost as % of fixed-capital investment (annual basis)

Type of operation	**Wages**	**Materials**	**Total**
Simple chemical process	1–3	1–3	2–6
Average chemical processes with normal operating conditions	2–4	3–5	5–9
Complicated chemical processes, severe corrosion, or extensive instrumentation	3–5	4–6	7–11
Materials movement only	0.5–1	0	0.5–1
Simple mechanical processes	4	1–3	5–7

Electricity Estimation.

Electrical demand for each block is decomposed analogously to CAPEX:

$$P_{\text{total}} = P_{\text{anchor}} + P_{\text{auxiliary}} \tag{C.37}$$

The maximum electrical demand for each machine is typically reported in specification sheets; load factors (LF) are estimated to approximate average power consumption during operation. Auxiliary electrical loads are estimated using an adder structure analogous to CAPEX, but with different ranges reflecting the distinct physical meaning of electrical auxiliaries. Standard default estimates are given in Table C.6.

Table C.6. Standard auxiliary electrical load adder ranges by equipment class and scenario.

Scenario	**Mechanically simple, pump-dominated**	**Mechanically intensive, multiple-drive systems**	**Thermal systems, fans/ducts/controls**
Conservative	25%	55%	65%
Midpoint	15%	40%	50%
Optimistic	10%	25%	35%

Other Utilities.

The use of other utilities (water, compressed air, natural gas, steam, etc.) varies significantly across blocks depending on the process and equipment being modeled. Utility usage is derived from project disclosures, mass balance or thermodynamic calculations, and vendor specifications. Where appropriate, the same anchor-plus-adder approach is applied: identifying the dominant utility consumer, anchoring to reported usage, and applying auxiliary adders as needed.

Step 4: Validation and Cross-Checking.

Model outputs are validated through cross-comparison with published feasibility studies, academic literature, and grey literature; internal consistency checks across unit operations; sensitivity testing of dominant parameters. This iterative validation ensures that no block is isolated from industrial reality, while preserving transparency about sources of data and uncertainty. While interviews and discussions with industry professionals and other modelers informed initial design choices and some machine block designs, additional discussions could further validate the model inputs and results.

C.5.2.2 Methods for estimating capital costs

Throughout this analysis, capital costs are estimated using a structured 'anchor equipment + adders' methodology, consistent with standard mining engineering and chemical processing guidance (e.g. Peters and Timmerhaus [183], AusIMM [184], Turton et al. [437], Ulrich [185], Darling [433]).

For each model block, a primary process equipment or set of primary process equipment are selected as the anchor, based on vendor specifications and published references. Installed capital costs are then estimated by applying installation factors, using optimistic, midpoint, and conservative values. Adders are included for auxiliary equipment and support subsystems that are required for operations, but not explicitly listed in vendor quotations or specifications.

These adders typically include pumping and piping, structural steel and access, electrical and instrumentation systems, reagent handling subsystems, and other balance-of-plant requirements. For clarity and transparency, rather than embedding these costs within a singular multiplier, we explicitly define each as additional adders on top of the installed equipment cost.

Percentage ranges were selected based on the aforementioned guidance as well as published cost breakdowns, and adapted to each specific block model context. Midpoint values aim to reflect a baseline plant layout; conservative and optimistic values represent extreme values that are believed to be possible.

C.5.2.3 Methods for estimating operating costs

Beyond reagents and other chemistry-related production costs, certain operating costs are tied to machines, such as the amount of labor necessary to operate a machine, the average electricity draw, or direct fuel consumption.

C.5.2.4 Method for estimating labor intensity

Operating labor requirements are estimated at the equipment level, consistent with standard chemical process engineering guidance (Ulrich [185], Turton et al. [437]). For each model block, operator requirements per shift are assigned to each key piece of equipment or equipment train based on published operator-per-unit-per-shift values from Table 6-2 of Ulrich [185]. Annual full-time equivalent (FTE) operators are then derived by applying a staffing factor of 4.5, accounting for the mostly continuous nature of chemical processes, as well as vacation and sick leave ($168/40 = 4.2$). Ulrich's method implicitly assumes labor intensity is invariant with capacity; where machines require scaling factors (i.e. continuous machines), we differentiate by process type. For comminution and materials handling blocks, we apply a scaling exponent of 0.5, consistent with AusIMM guidance [184] for treatment plant labor. For downstream chemical processing blocks (e.g. leaching, precipitation, solids-liquid separation), we apply a lower exponent of 0.2, consistent with fluid processing guidance in Peters and Timmerhaus [183], reflecting the higher automation and lower solids-handling intensity of these unit operations. Maintenance, laboratory, support, and supervisory staff are excluded and treated as separate maintenance and overhead adders.

C.5.2.5 Method for estimating chemical compositions

Brine Concentration Estimations.

For Silver Peak, Garrett [196] reports various analyses of Clayton Valley Brine. The Silver Peak disclosure indicates a concentration of 95 mg/L (Table 1-2)[178] but does not report other ionic concentrations; we use Garrett's 1996 measurements for non Li compounds, as well as reported density ($\rho = 1.058$ g/cm^3) as a baseline, resulting in a concentration of approximately 90 ppm Li. While Garrett reports higher lithium concentrations, we use the number stated in the disclosure under the assumption that lithium grades in brine have declined over time, whereas other major compounds have remained generally constant. The disclosure notes that at about a concentration of 0.54 wt% Li, the strong brine is then processed in the lithium carbonate plant. At an assumed density of $\rho = 1.25$ g/cm^3, this results in a concentration of 6,750 ppm Li, or a brine concentration factor F of 75×. We generalize to assume this concentration factor applies to conservative species that stay in solution (Li, Mg, B, Rb), while other compounds largely precipitate out (e.g. about 95% of Ca as calcite; Na, K, Cl, CO_3, SO_4 buffered by saturation points [198, 366].

Thacker Pass Concentration Estimations.

For the Thacker Pass hectorite clay deposit, only lithium concentration is tracked through the processing circuit; detailed multi-constituent leach feed and PLS compositions are not

Table C.7. Assumed Brine Compositions, Silver Peak
*Reported concentration, [178]; all others from Garrett [196] reporting from Garrett (1996).

		Li*	Na	K	Mg	Ca	Rb	Cl	SO_4	CO_3	B
Raw Brine	wt%		4.69	0.40				7.26	0.34		
	ppm	90*	49610	4232	190	450	21	76811	3597	74	67
Strong Brine	ppm	6750	140000	20000	14250	1688	1575	220000	10000	1000	5025

publicly disclosed in the SK-1300 technical reports [150, 181], and the reagent dosages in the generic clay flowsheet are calibrated directly to disclosed per-tonne consumption rates rather than to constituent-specific stoichiometry (see Appendix C.5.1.4). This represents a modeling simplification relative to the Silver Peak and Jianxiawo pathways, where multi-ion brine and ore compositions are available from the literature.

The ROM ore lithium grade of 3,542 ppm is drawn from Table 14-6 of the 2024 SK-1300 disclosure [150], representing the proven and probable reserve-weighted average feed grade for the initial production phases. At input to the acid leach circuit (after crushing, attrition scrubbing, and classification), the lithium concentration is estimated at 2,538 ppm, derived by applying the upgrading ratio observed in the 2022 feasibility disclosure [181] to the 2024 ROM grade. In the 2022 disclosure, the feed entering the acid leach circuit carried 3,270 ppm Li from a ROM ore grade of 4,563 ppm, implying a loss factor of $3,270/4,563 = 0.716$ attributable to gangue separation and attrition losses in the comminution circuit. Applying this ratio to the 2024 ROM grade yields $3,542 \times 0.716 = 2,538$ ppm.

The lithium concentration in the pregnant leach solution at input to the impurity removal step is estimated at 1,107 ppm Li, derived from the leach circuit operating conditions. Starting from the ROM ore grade (3,542 ppm) as a proxy for the Li loading entering the leach circuit, converting to absolute Li mass per unit volume of the 34 wt% leach slurry via the mixture density CCF, dividing by the CCD POCF of 1.31 to account for volumetric expansion during counter-current washing, and normalizing by the PLS density of 1.05 t/m^3:

$$c_{\text{PLS}} = \frac{3,542 \times \left(\frac{0.34}{\frac{0.34}{2.6}+0.66}\right)/10^3}{1.31 \times 1.05} \times 10^3 = 1,107 \text{ppm Li} \tag{C.38}$$

The ROM ore grade is used as the starting point rather than the upgrading-adjusted leach input grade (2,538 ppm) as a modeling simplification; the two values reflect the same lithium at different reported concentration bases, and a full reconciliation would require the multi-species mass balance that is not available for this pathway.

Lepidolite Concentration Estimations.

Table C.8. Assumed Lithium Concentrations, Thacker Pass
Lithium grades from SGS Canada et al. [150] and Roth et al. [181]; only Li is tracked in the generic clay flowsheet. Other ionic species are not modeled due to the absence of publicly disclosed multi-constituent leach feed and PLS compositions.

	Li (ppm)
ROM Ore	3,542
Acid Leach Circuit Input	2,538
PLS (Impurity Removal Input)	1,107

For the Jianxiawo lepidolite deposit, we use an initial ROM ore Li concentration of 1,858 ppm (0.40% Li_2O), consistent with the upper end of CRU's reported range for kaolinite-rich Jiangxi lepidolite deposits (0.27–0.40% Li_2O) and with the regulatory threshold below which local governments have historically issued kaolinite rather than lithium mining permits [206]. Other major ionic species in ROM ore (K, Na, Mg, Ca, Si, Rb, F) are drawn from Garrett [196] and the peer-reviewed lepidolite mineralogy literature (e.g. [174, 210]).

The intermediate concentration represents the classified slurry entering the sulfate roasting step, after crushing, classification, flotation, and high-energy mechanical activation milling to a product size of approximately 50-75 μm. At this stage, fine grinding has liberated the lepidolite mineral from gangue; the Li concentration of 7,576 ppm (1.63% Li_2O) therefore reflects a mechanical enrichment factor of approximately 4.1× relative to ROM ore. The flotation step is critical for this pathway: at typical Jiangxi ore grades of less than 0.4% Li_2O (∼$1,860 ppm Li), direct roasting of ROM material would require processing enormous reagent and energy quantities per unit of lithium product. Froth flotation using amine collectors and phosphate depressants selectively recovers lepidolite minerals into a concentrate, reducing the mass throughput to the energy-intensive roasting circuit by approximately 60% [434]. The implied mass yield of approximately 13-18% (6-8 tonnes feed per tonne of classified slurry, depending on assumed recovery) is broadly consistent with CRU's industrial benchmark of 8 tonnes of 0.40% Li_2O ore per tonne of 2.20% Li_2O concentrate [436], noting that our intermediate is lower grade (1.63% vs. 2.20% Li_2O) is based on reporting from Garrett [196].

After sulfate roasting at 800-950° C and water leaching, the pregnant leach solution (PLS) Li concentration is set at 21,630 ppm [196]. This represents a further 2.9 × concentration relative to the classified slurry feed and an 11.6 × factor relative to ROM ore. The associated major species in the PLS (K, Na, Rb, F, Si, Cs) reflect the solubilisation of alkali and fluoride constituents during roasting and leaching, consistent with the phase transformations reported

in Yan et al. [205] and Deng and Feng [180].

Table C.9. Assumed Lepidolite Compositions, used for Jianxiawo
*Reported concentration, Zhang and Jackson [206]; all others from [196].

	Li	**K**	**Mg**	**Ca**	**Na**	**Si**	**Rb**	**F**
ROM Ore*	1,858	—	—	—	—	—	—	—
Classified Slurry	7,576	24,734	60	429	38,070	321,000	3,016	18,600
PLS	21,630	69,305	0	0	8,384	258,800	10,785	11,100

C.5.3 Machine Model Block Definitions

C.5.3.1 `brine_pump`

Sources.

SRK Consulting [178], Becerra and Riquelme [435], Consulting [438], Limitada and Gunn [439], Brooker and Gunn [440]; See Brine Evaporation Datasheet in Supplementary Data Spreadsheet.

Process description.

The brine pump block captures extraction of lithium-bearing brine from subsurface aquifers and delivery to the evaporation pond system. The block is parameterized per well-pump unit; total system capacity is obtained by multiplying by the number of active wells. Capacity scaling is therefore discrete (additional parallel units) rather than continuous, and no size-scaling exponent is applied to primary equipment cost.

Equation of merit.

$$\dot{V}_{\text{total}}\ (\text{m}^3/\text{h}) = \dot{V}_{\text{pump}}\ (\text{m}^3/\text{h}) \times N_{\text{wells}}$$

where $\dot{V}_{\text{pump}}$ is the flow rate of a single well-pump unit and N_{wells} is the number of active extraction wells.

Bill of Equipment.

Table C.10. Bill of equipment for the brine pump block.

Equipment	**Existing machine data used**	**Source(s)**
Pumping package with piping, electrical, and other support	Brine pumping reported system analysis	Brine Evaporation Datasheet; company disclosures

Sizing basis.

The base volume is set to **72,000 L/h** (72 m^3/h) per well-pump unit, derived from an average reported extraction rate of 20 L/s. Silver Peak reports individual well extraction rates ranging from 5.2 m^3/h to 187.5 m^3/h; the 20 L/s (72 m^3/h) value is used as a central representative rate for a single active well, calibrated to other reported average pump rates (see Brine Evaporation Datasheet in Supplementary Data Spreadsheet).

The block is assigned `process_type = batch` with `batch_cycle_time = 1 h` and zero setup time. This is a modeling convenience: brine pumping is physically a continuous operation, but the hourly batch representation is consistent with the model's time-step structure and produces equivalent annual throughput. It does not imply any actual batch cycling of the pump.

CAPEX.

Capital cost per well-pump unit is derived from a linear regression of total reported well-field capital expenditure against number of wells across seven lithium brine project disclosures. The Salar de Atacama-SQM disclosure (438 wells, $370.76M) was excluded as a clear outlier — its scale is more than an order of magnitude larger than all other data points and would dominate the regression. The fitted relationship is:

$$C_{\text{total}}\ (\$\text{M}) = 0.903 \times N_{\text{wells}} + 9.07 \qquad R^2 = 0.784$$

The slope ($0.903M/well) represents the marginal installed cost of each additional well-pump unit. The intercept ($9.07M) captures fixed site mobilisation and infrastructure costs — pump house, electrical switchgear, surface manifolding, and SCADA installation — that are shared across the well field regardless of well count. In the per-unit model block, the slope is rounded to approximately $1M/well as a central estimate; scenario variation ($0.8–1.3M/well) spans the observed scatter in reported total-cost-per-well values across the regression dataset ($0.89–1.91M/well), and reflects uncertainty in drilling depth, casing specification, and pump selection across the wide reported head range (90–550 m):

Table C.11. CAPEX scenario values for the `brine_pump` block (per well-pump unit).

Scenario	`prim_equip_price_base` (USD)
Conservative	$1,300,000
Midpoint	$1,100,000
Optimistic	$800,000

Equipment life is set to 20 years (`prim_equip_life = 20`), consistent with standard submersible pump replacement intervals in brine service. No throughput scaling exponent is applied; additional capacity is obtained by adding parallel well-pump units at the same unit cost. The fixed site infrastructure captured by the regression intercept is not separately modeled; it is implicitly absorbed into project-level CAPEX estimates for the well-field as a whole.

Electricity.

Electrical demand per well-pump unit is derived from the hydraulic power relation:

$$E = \frac{\rho g H}{\eta \cdot 3.6 \times 10^6} \quad (\text{kWh/m}^3)$$

For Silver Peak, with brine density $\rho = 1,058 \text{kg/m}^3$, pump efficiency $\eta \approx 70\%$, and a central head estimate of $H = 300\text{m}$ (reported range: 90–550 m):

$$E = \frac{1,058 \times 9.81 \times 300}{0.70 \times 3.6 \times 10^6} \approx 1.235 \text{kWh/m}^3$$

At a flow of 72 m^3/h, this gives a base motor demand of approximately $1.235 \times 72 \approx \mathbf{90}$kW per well-pump unit.

An additional load factor is applied to account for intra-playa brine transfer pumping and the elevated specific gravity of increasingly concentrated brine at later evaporation stages — loads not captured by the single-well extraction head alone. The factor is applied as a multiplier on the base motor demand:

Table C.12. Electricity scenario derivation for the `brine_pump` block.

Scenario	**Flow** (m^3/h)	**Base motor (kW)**	**Playa adder**	`electricity_base_total`
Conservative	72	90	×1.3	**117 kWh/h**
Midpoint	72	90	×1.2	**108 kWh/h**
Optimistic	72	90	×1.1	**99 kWh/h**

Other utilities.

None. Natural gas, diesel, propane, cooling water, steam, and compressed air are all zero for this block.

Labor.

Labor is assigned per well-pump unit in FTE, reflecting the monitoring and intervention requirements of a single operating well. At Silver Peak's 6 active extraction wells, the scaling exponent of 0.5 yields a total staffing of approximately 0.10–0.15 FTE for well monitoring and pump intervention — consistent with automated pump operation requiring only periodic site visits rather than continuous attendance.

The `labor_scaling_exponent` of 0.5 is the AusIMM default. For well monitoring, which scales sub-linearly with the number of wells (shared site visits, centralized monitoring), this should be physically reasonable.

Table C.13. Labor base values for the `brine_pump` block (per well-pump unit).

Scenario	`labor_base` (FTE)	**Implied FTE at 6 wells**
Conservative	0.06	$0.06 \times \sqrt{6} \approx 0.15$
Midpoint	0.05	$0.05 \times \sqrt{6} \approx 0.12$
Optimistic	0.04	$0.04 \times \sqrt{6} \approx 0.10$

Footprint.

Each well-pump unit adds a fixed 5 m^2 surface footprint (scaling exponent 1.0) for the wellhead, electrical switchgear, and access pad.

Availability.

`proc_avail_factor` = 0.90, with `unplanned_downtime` = 0.05 and `scheduled_maintenance` = 0.05 (10% total downtime, split equally between planned and unplanned), held constant across scenarios.

Maintenance.

Maintenance costs apply to the pump and surface infrastructure only; wells are not practically maintainable once installed and are excluded from the recurring maintenance budget. Costs are specified via `maint_override` as an annual fraction of CAPEX, reflecting the pump-only maintenance scope:

Table C.14. Maintenance override fractions for the `brine_pump` block.

Scenario	`maint_override` (fraction of CAPEX/yr)
Conservative	0.020
Midpoint	0.015
Optimistic	0.010

Notes and flags.

Several parameters carry meaningful uncertainty that is not fully captured in the conservative–optimistic range and should be noted in the methodology:

The central head estimate of 300 m is used as an average across Silver Peak's reported range of 90–550 m. This range is unusually wide and the distribution of active well depths is not disclosed; a more detailed well-field characterisation could substantially tighten the electricity estimate.

The playa adder (10–30%) is intended to capture intra-playa transfer pumping and brine concentration effects but is not directly calibrated to a disclosed value. It is a modeling assumption and represents a source of systematic uncertainty in the electricity estimate.

The CAPEX regression is based on seven project disclosures after outlier exclusion, with $R^2 = \sim 0.784$. The fixed intercept ($9.07M) is not separately modeled as a site infrastructure line item; this is a limitation for small well-field configurations where the intercept represents a proportionally larger share of total cost.

The `process_type = batch` designation with a 1-hour cycle is a modeling convenience only and has no physical meaning for this block. It should not be interpreted as implying that brine pumping operates intermittently.

C.5.3.2 `evaporation_pond` (Lined and Unlined)

Sources.

He et al. [366], Fleming et al. [141], Stamp et al. [359], SRK Consulting [178], Becerra and Riquelme [435], Consulting [438], Limitada and Gunn [439], Brooker and Gunn [440]

Process description.

Evaporation ponds concentrate lithium brine through solar evaporation in large, shallow, lined or unlined basins. The primary design variable is pond surface area, sized to remove water based on the aridity and weather conditions of the region in which it is installed — hence its typical usage in high-desert environments. In operation, ponds require ongoing maintenance to repair the lining, harvest accumulated salts, and conduct regular monitoring.

Equation of merit.

$$A_{\text{pond}}\ (\text{m}^2) = \frac{\dot{V}_{\text{brine}}\ (\text{m}^3/\text{yr})}{\dot{E}\ (\text{m/yr})}$$

where $\dot{V}_{\text{brine}}$ is the brine processing plant intake and $\dot{E}$ is the effective evaporation rate. The evaporation rate is estimated from the reported freshwater evaporation rate, multiplied by derating factors (<1) that account for the reduced vapour pressure of brine due to its salinity, and for practical efficiency losses at industrial-scale pond sizes. For example, the Sal de Vida disclosure applies a derating factor of 0.7 to the reported freshwater evaporation rate [439]. Evaporation rates are typically reported in mm/day and range between 2.4 and 7.4 mm/day depending on location, for example, He et al. [366] report monthly raw (unadjusted) evaporation rates of 4.6–8.6 mm/day for Salar de Atacama, one of the driest and sunniest lithium-producing regions in the world.

Given the strong sensitivity of evaporation rate to geographic location, the specific rate reported in each project disclosure is used where available, rather than a generic default. Silver Peak reports a value of 89 cm/year on average, or 2.44 mm/day [178].

Evaporative precipitation chemistry.

As brine concentrates through evaporation, increasing ion concentrations cause progressive precipitation of dissolved salts. With salar brines, precipitation typically occurs in approximate order of decreasing solubility: calcium compounds (calcite, gypsum — $CaSO_4 \cdot 2\,H_2O$), sodium chloride (halite — NaCl), potassium chloride (sylvite — KCl), and potassium-magnesium compounds (carnallite — $KCl \cdot MgCl_2 \cdot 6\,H_2O$; bischofite — $MgCl_2 \cdot 6\,H_2O$). Lithium, being highly soluble, generally remains in solution through this phase. The order and quantity of precipitation depend strongly on brine composition and therefore vary across projects.

Bill of Equipment.

Table C.15. Bill of equipment for the evaporation pond block.

Equipment	Existing machine data used	Source(s)
Installed ponds (earthworks, lining, civil)	N/A — ponds only	Lithium brine evaporation pond analysis (project disclosures)

No electricity is assumed in this step; pond circulation pumping is included in the `brine_pump` block.

CAPEX estimation — lined ponds.

Installed pond costs include land purchase, earthworks, and lining (HDPE or compacted clay liner). Based on analysis of capital cost data from brine production disclosures, costs vary between approximately \$12/m^2 and \$30/m^2, with one outlier at \$52/m^2.

Table C.16. Lined pond unit CAPEX (`evaporation_pond_lined`).

Scenario	`prim_equip_price_base` (\$/m^2)
Conservative	\$30
Midpoint	\$18
Optimistic	\$12

CAPEX estimation — unlined ponds.

Costs for unlined ponds are estimated from the Silver Peak 2023 disclosure, in which the expansion includes 8 new "unlined earthen dike structures" at a total estimated expenditure of \$48.9M. The Phase 3 pond (1 large pond, 200 acres, \$7.1M) yields a unit cost of approximately \$8.77/m^2, used as the central estimate.

Electricity.

Zero; no electricity is consumed at this step. Pond circulation and inter-pond transfer pumping are included in the `brine_pump` block (C.5.3.1).

Other utilities.

None. All thermal and mechanical utilities are zero for this block.

Table C.17. Unlined pond unit CAPEX (`evaporation_pond_unlined`).

Scenario	`prim_equip_price_base` ($/m^2)
Conservative	$12
Midpoint	$8
Optimistic	$5

Labor parameterization.

The `labor_base` values are expressed in FTE per m^2 of pond area and are calibrated to the Silver Peak disclosure, which reports approximately 8–12 FTE managing salt removal and dike upkeep [178]. Because pond management is a fundamentally area-maintenance task — involving periodic lining inspection, salt harvesting from pond floors, brine level monitoring, and dike upkeep — labor requirements scale approximately linearly with pond area. The values are therefore derived by dividing target FTE directly by a reference pond area:

$$\texttt{labor_base} = \frac{\text{Target FTE}}{A_{\text{ref}}} \qquad (\text{FTE}/\text{m}^2)$$

Using a reference area $A_{\text{ref}} = 26 \times 10^6$ m^2 (≈6,400 acres), consistent with the active evaporation pond footprint at Silver Peak, target FTE values of 10, 8, and 6 for the conservative, midpoint, and optimistic scenarios respectively yield:

Table C.18. Labor base values for evaporation ponds, calibrated to Silver Peak disclosure.

Scenario	**Target FTE at A_{ref}**	`labor_base` **(FTE/m^2)**	**Equivalent fraction**
Conservative	10	3.846×10^{-7}	$10/26,000,000$
Midpoint	8	3.077×10^{-7}	$8/26,000,000$
Optimistic	6	2.308×10^{-7}	$6/26,000,000$

The midpoint value (8 FTE) corresponds to the lower bound of the disclosed range; the conservative value (10 FTE) sits near the midpoint of that range. The optimistic value (6 FTE) falls modestly below the disclosed range, representing a best-case labor efficiency assumption. The `labor_scaling_exponent` is set to 0.5, consistent with the standard template default from AusIMM processing plant guidance; however, for a passive area-maintenance task, linear scaling is the physically appropriate interpretation. Since `base_volume` is set

to 1 m^2, the formal expression `labor_base` $\times A^{0.5}$ is not equivalent to the intended linear relationship, but the values above remain the correct calibration reference for the Silver Peak scale.

Footprint.

Footprint is not parameterized separately for this block. The pond surface area is the sizing basis itself (`base_volume_unit` = m^2), and `prim_equip_scaling_exponent` = 1.0, reflecting that area scales linearly with throughput.

Availability.

Not applicable in the mechanical sense. Evaporation ponds are passive civil structures with no rotating equipment; the block is assigned `process_type = batch` as a modeling convenience representing the seasonal pond management cycle. No availability derating is applied.

Maintenance cost fractions.

Annual maintenance is parameterized as a fraction of installed CAPEX (`opex_fraction_of_capex`). Two considerations motivate the use of substantially lower fractions than the 1–3% range commonly cited for chemical processing equipment.

First, evaporation ponds are passive civil structures with no rotating equipment, no chemical dosing systems, and no instrumentation networks requiring calibration or replacement. Maintenance activities consist of periodic liner inspection and targeted patch repairs (lined ponds only), salt harvesting from pond floors, dike inspection and minor earthworks, and brine level monitoring. These activities are labor-intensive but not capital-intensive relative to the total asset base.

Second, applying a standard chemical-plant maintenance fraction to a large pond area produces implausibly high implied costs. At Silver Peak's approximate pond area of $\sim$12 km^2 and a midpoint lined CAPEX of \$18/m^2, a fraction of 1.5% would imply \$3.2M/yr in maintenance expenditure. The Silver Peak disclosure notes approximately 8–12 FTE managing salt removal and dike upkeep [178]; at a loaded labor cost of roughly \$80,000/FTE/yr, the labor component alone is approximately \$640,000–\$960,000/yr. Total maintenance inclusive of materials and equipment is unlikely to exceed twice this figure — placing a reasonable order-of-magnitude estimate at \$1.5–2M/yr, consistent with a fraction of approximately 0.6–0.9% rather than 1.5%. Lower fractions for unlined ponds reflect their reduced maintenance burden: the absence of a liner largely eliminates significant inspection and repair activity, leaving only dike, berm, and salt management.

The adopted fractions, calibrated to this cross-check, are:

Key measurements and cross-checks.

Table C.19. Annual maintenance cost fractions for lined and unlined evaporation ponds.

Scenario	**Lined** (`opex_fraction_of_capex`)	**Unlined** (`opex_fraction_of_capex`)
Conservative	0.010	0.005
Midpoint	0.0075	0.004
Optimistic	0.005	0.003

- Effective evaporation rate (mm/day), including derating factors applied
- Lithium recovery efficiency through the evaporative phase (accounts for co-precipitation losses and seepage; 0.51 for Silver Peak [438])
- Salt accumulation rate and harvest frequency (primary driver of maintenance cost)
- Pond lining integrity for lined variants (leak rate, repair interval)

Notes and flags.

The evaporation rate is the dominant sizing driver and is highly site-specific; the model preserves sensitivity to this variable by accepting it as a project-specific input rather than applying a generic default. At Silver Peak, the comparatively low evaporation rate (2.44 mm/day versus 4–7 mm/day at South American salars), in addition to lower grade brines C.5.2.5, results in proportionally larger required pond area for a given production target, which is the primary reason evaporation pond costs dominate the Silver Peak cost structure.

The distinction between lined and unlined variants is material: lined ponds are appropriate where regulatory requirements mandate containment or where substrate permeability necessitates a barrier; unlined earthen structures are used where the subsurface is sufficiently impermeable and permitting conditions allow. The Silver Peak expansion uses unlined structures, reflecting both the site geology and the relatively benign chemistry of evaporation residues in a low-magnesium brine system.

C.5.3.3 `blasting_generic`

Sources.

Darling [433] (Ch. 4.9 Cost Estimating for Surface Mines; Ch. 7.3 Explosives and Blasting); Warren et al. [441]; Inc. et al. [442]; Sandvik Leopard DI550 specification sheet (Sandvik [443]) and Cat C13 Diesel Engine (Caterpillar [444]) for diesel derivation.

Process description.

This block models the combined drilling and blasting operation that converts in-situ rock to fragmented material suitable for excavation and haulage. The block is parameterized per single production drill rig; total system capacity is obtained by multiplying by the number of active drills, analogous to the `brine_pump` block. The dominant OPEX driver is explosive consumption, which is set at the production step level via a project-specific powder factor and is therefore not a machine block parameter; the machine block captures equipment CAPEX, diesel consumption, and labor for the drill rig and ancillary blasting equipment.

Equation of merit.

The equation of merit relates single-drill throughput to the drilling rate and blast pattern geometry:

$$\dot{m}_{\text{rock}}\ (\text{t/h}) = R_{\text{drill}}\ (\text{t/h}) \times f_{\text{rock}}\ (\text{t/m})$$

where R_{drill} is the effective drilling rate in meters of hole per operating hour (including fixed time for repositioning, rod changes, and collar setup) and f_{rock} is the rock broken per meter of hole drilled, which consolidates bench height, blast pattern spacing and burden, sub-drill allowance, and in-situ rock density into a single productivity factor.

Sizing basis.

The step basis is `rock_blast_throughput` in tonnes per hour and `base_volume` = 650, representing 650 t/h of in-situ rock broken per single production drill. This value is derived from the Whabouchi drilling disclosure as follows:

- Hole diameter: 115–140 mm; bench height: 12 m
- Blast pattern: 3.3 m spacing × 2.8 m burden
- In-situ rock density: 2.7 t/m^3
- Rock broken per hole: $12 \times 3.3 \times 2.8 = 110.9$ m^3; $110.9 \times 2.7 = 299$ t ≈ 300 t
- Pure drilling rate: 35 m/h; fixed time per hole: 7 min (0.117 h)
- Effective cycle time per hole: $12/35 + 0.117 = 0.460$ h/hole
- Single-drill throughput: $300/0.460 \approx 652$ t/h, rounded to 650 t/h

Table C.20. Bill of equipment for the `blasting_generic` block.

Equipment	Machine data used	Source(s)
Production drill rig	Composite of Whabouchi and Georgia Lake drill parameters; Sandvik DI550 class	NI 43-101 disclosures; Sandvik specification sheet
Bulk explosives delivery truck (ANFO/emulsion)	Ancillary	Not explicit → inferred
Drill steel, bits, and down-the-hole hammers	Consumables → maintenance fraction	—

Bill of Equipment.

Unlike fixed chemical processing equipment, drilling and blasting equipment is mobile and arrives pre-assembled. As a result, the standard Lang-type installation factors (2.3–3.3×) used for fixed-plant process equipment are not applicable. Instead, a delivery, commissioning, and initial tooling factor of 1.15× is applied to bare equipment cost, consistent with mobile mining fleet practice.

CAPEX.

Anchor equipment is a single production down-the-hole (DTH) drill rig in the 5-inch hammer class with a bare purchase price of $3.5–4.5M, reflecting the market range for rigs of this capacity (e.g. Sandvik Leopard DI550/DI650i, Epiroc FlexiROC D65). The bulk explosives truck is estimated at $0.8–1.2M bare. Both are multiplied by 1.15× for delivery and commissioning.

Auxiliary and indirect costs (engineering, site infrastructure for explosive storage, maintenance facilities, and contingency) are applied as explicit percentage adders on the installed equipment subtotal, following the standard three-scenario structure:

Table C.21. CAPEX derivation for the `blasting_generic` block (single drill unit).

Scenario	Drill inst.	Truck inst.	Equip. subtotal	Aux. %	`prim_equip_price_base`
Conservative	$5.18M	$1.38M	$6.56M	30%	**$8,500,000**
Midpoint	$4.60M	$1.15M	$5.75M	20%	**$6,900,000**
Optimistic	$4.03M	$0.92M	$4.95M	15%	**$5,700,000**

Note that drill rig and truck bare purchase prices are drawn from the Georgia Lake disclosure. The 30%/20%/15% auxiliary adder range is consistent with the "mechanically intensive, multiple-drive systems" column in Table C.4, though at the lower end, reflecting the simpler integration requirements of mobile equipment relative to fixed-plant installations.

Equipment life is set to 15 years, reflecting heavy mobile equipment replacement cycles in hard rock mining service. The CAPEX scaling exponent is 0.60 (six-tenths rule).

Electricity.

`electricity_base_total` = 0. The production drill rig is entirely diesel-powered. No grid electricity is consumed at the blasting step.

Diesel.

Diesel is the sole thermal utility for this block; electricity, natural gas, and propane are zero (the drill rig is entirely diesel-powered). Diesel consumption is derived from the Sandvik DI550 engine specification: a Caterpillar C13B rated at 328 kW, with a brake-specific fuel consumption (BSFC) of 0.235 kg/kWh. At a diesel density of 0.84 kg/L, full-load consumption is:

$$\dot{V}_{\text{full}} = \frac{328 \times 0.235}{0.84} = 91.8 \text{ L/h}$$

Effective fuel consumption depends on the operating load factor, which varies with rock hardness, compressor demand, and the fraction of time spent on non-drilling activities (tramming, repositioning). A load factor range of 0.65–0.85 is applied across scenarios. The bulk explosives truck operates intermittently (charging loaded holes while the drill repositions) and is pro-rated to drill operating hours at 8–15 L/h:

Table C.22. Diesel derivation for the `blasting_generic` block at 650 t/h base volume.

Scenario	**Load factor**	**Drill (L/h)**	**Truck (L/h)**	`diesel_base_total`
Conservative	0.85	78	15	**95 L/h**
Midpoint	0.75	69	12	**80 L/h**
Optimistic	0.65	60	8	**65 L/h**

We note that the drill fuel consumption is derived from rated engine power and a standard BSFC; actual fuel consumption depends on rock competence, drilling depth, and site-specific operating practices. The explosives truck pro-rated values (8–15 L/h) are estimates based on intermittent operation. The derivation does not apply the ÷0.80 combustion efficiency adjustment used for thermal process equipment, because diesel here is a direct mechanical

fuel input (powering the compressor and hydraulics), not a thermal heating source with stack losses.

Other utilities.

Natural gas, propane, steam, cooling water, and compressed air base totals are all zero. Compressed air for the DTH hammer is generated by the rig's onboard diesel-driven screw compressor and is captured within the diesel consumption above. No external compressed air supply is expected to be required.

Consumables.

The primary consumable is explosives (emulsion or ANFO), which is dosed at the production step level via the project-specific powder factor and is therefore not a machine block parameter. Drill bits, DTH hammers, and drill steel are significant wear items; their cost is captured implicitly within the `opex_fraction_of_capex` maintenance fraction rather than as a separately tracked consumable at this level of model granularity.

Labor.

Labor is 4.5 FTE at base scale with a scaling exponent of 0.5 and `dedicated_labor` = FALSE. The drill rig requires a single operator; the explosives truck is operated by the same crew during non-drilling intervals. The 0.5 exponent reflects the discrete nature of drill fleet staffing but shared supervisory and support staff:

$$\text{FTE} = 1.0 \times 4.5 = 4.5$$

Footprint.

No fixed plant footprint is assigned. The drill rig and explosives truck operate within the active pit area, which is determined by mine planning rather than process layout.

Availability.

Process availability (`proc_avail_factor`) is set to 0.90, comprising approximately 5% scheduled maintenance (bit changes, hammer rebuilds, engine service) and 5% unplanned downtime (mechanical failures, weather delays, blast clearance). This is held constant across scenarios; throughput variation is driven by the production step level, not the machine block.

Maintenance.

Maintenance is parameterized as `opex_fraction_of_capex`. Drilling and blasting equipment falls between the "materials movement only" (0.5–1%) and "simple mechanical processes" (5–7%) categories in Table C.5. The drill rig has significant wear items (DTH hammers, drill bits, drill steel) that drive replacement costs well above the materials-movement baseline. A maintenance fraction of 5%/4%/3% (conservative/midpoint/optimistic) is adopted, corresponding to the lower end of the simple mechanical processes range.

Key measurements and cross-checks.

- *Throughput:* The base volume of 650 t/h (Whabouchi) is consistent with the Georgia Lake cross-check of 522 t/h for a shallower bench geometry.
- *Powder factor:* The Georgia Lake disclosure reports 50.1 kg explosives per hole and 202 t rock broken per hole, giving a powder factor of $50.1/202 = 0.248$ kg/t. This is consistent with the SME hard rock range of 0.11–0.38 kg/t (converted from 0.3–1.0 kg/m^3 at 2.6–2.8 t/m^3 density). These powder factors are applied at the production step level, not the machine block.
- *Life of Mine unit cost:* The Georgia Lake disclosure reports life-of-mine drilling costs of \$40.9M and blasting costs of \$22.9M for 29.6 Mt of material, giving a combined drilling and blasting unit cost of \$2.15/t. This aggregate value encompasses CAPEX amortization, diesel, labor, maintenance, and consumables, and serves as a high-level validation target for the assembled block once throughput and project parameters are applied.
- *Diesel:* The Sandvik DI550 specification (328 kW, Cat C13B engine) yields 92 L/h at full load; at 75% load factor, 69 L/h for the drill alone. Combined with the explosives truck, the midpoint of 80 L/h is consistent with the operating range for this equipment class.

C.5.3.4 `excavation_loading_generic`

Sources.

Caterpillar [445]; Warren et al. [441]; McCracken et al. [446]; **?**]; Peters and Timmerhaus [183]; Ulrich [185]

Process description.

This block models the combined excavation and loading operation at an open-pit mine face: mechanical fragmentation of blasted or unconsolidated material, loading into haul trucks or conveyors, and associated mine maintenance. The block encompasses one primary hydraulic excavator, one wheel loader (for stockpile management, truck loading from temporary stockpiles, and cleanup), and one track dozer (general other material movement and maintenance). All three machines are diesel-powered; the block consumes no grid electricity.

Equation of merit.

$$\dot{m}_{\text{loose}}\ (\text{t/h}) = \frac{V_{\text{bucket}}\ (\text{m}^3) \times \rho_{\text{loose}}\ (\text{t/m}^3) \times f_{\text{fill}}}{t_{\text{cycle}}\ (\text{min})/60}$$

where V_{bucket} is the excavator bucket volume, ρ_{loose} is the loose (bank-swelled) rock density, f_{fill} is the bucket fill factor, and t_{cycle} is the dig–swing–dump cycle time. For the modeled base case, the cycle time and bucket payload are combined into a single `base_volume` of 10 t/cycle.

Sizing basis.

The step basis is `loose_rock_throughput` in tonnes of loose rock per hour. The block is assigned `process_type = batch` with a base volume of **10 t/cycle**, representing the payload of a single excavator bucket pass. Batch cycle times are 1.5 / 1.0 / 0.8 minutes (conservative / midpoint / optimistic), giving throughputs of 400 / 600 / 750 t/h per block unit; setup time is incorporated into these cycle times (set to 0). The 10 t/cycle payload is consistent with a Cat 395-class mass excavation excavator equipped with a 6.5 m^3 bucket at $\sim$1.8 t/m^3 loose rock density and $\sim$85% fill factor ($6.5 \times 1.8 \times 0.85 \approx 9.9$ t). The conservative scenario of 7 t/cycle accounts for lighter material (e.g., clays at lower bulk density), smaller bucket selection, or reduced fill factor in hard digging conditions.

The scaling exponent is 1.0, reflecting that mobile mining equipment scales as discrete integer units: each additional block unit represents one additional excavator–loader–dozer set. `dedicated_line = FALSE`, as multiple block units can share haul roads and ancillary infrastructure.

Bill of Equipment.

CAPEX.

Table C.23. Bill of equipment for the `excavation_loading_generic` block.

Equipment	**Machine data used**	**Source(s)**
Hydraulic excavator (Cat 390/395 class, ∼90 t)	Excavators	Cat Performance Handbook Ed. 46
Wheel loader (Cat 988 class, ∼50 t)	Loaders	Cat Performance Handbook Ed. 46
Track dozer (Cat D9 class, ∼49 t)	Dozers	Cat Performance Handbook Ed. 46
Auxiliary (service truck, fuel/lube trailer)	Not explicit → inferred	—

Equipment costs are based on delivered mine-specification pricing for new machines. One excavator at \$2.0M purchase → \$2.5M delivered, one wheel loader at \$2.0M purchase → \$2.5M delivered, and one track dozer at \$1.5M delivered. The "delivered" price includes mine-specification outfitting and freight/commissioning, representing a 25% markup over base purchase price for the excavator and loader.

Primary equipment subtotal: \$2.5M + \$2.5M + \$1.5M = \$6.5M per block unit. Auxiliary equipment (service vehicles, fuel/lube trailer, initial spare parts inventory) is added as a percentage of the primary subtotal:

Table C.24. CAPEX by scenario for the `excavation_loading_generic` block.

Scenario	**Auxiliary factor**	**Total installed**
Conservative	25%	≈ \$8.1M
Midpoint	15%	≈ \$7.5M
Optimistic	10%	≈ \$7.2M

Equipment life is 20 years.

Electricity.

Zero; all equipment is diesel-powered and no grid electricity is consumed at the excavation face.

Diesel.

Diesel consumption is derived from the Caterpillar Performance Handbook, Edition 46, which reports hourly fuel consumption by duty category based on VisionLink telematics data

from customer machines worldwide (top and bottom 5% excluded). The duty categories are defined by idle time fraction: Low (∼50% idle), Medium (∼30% idle), and High (∼10% idle).

Equipment fuel consumption data (L/h):

Table C.25. Hourly fuel consumption by duty category.[445]

Machine	**Low (L/h)**	**Medium (L/h)**	**High (L/h)**
Cat 390D (Tier 2, C18)	29.5–49.4	49.4–63.2	63.2–73.2
Cat 988K	23.8–34.1	34.1–44.7	44.7–55.3
Cat D9R	32.0–46.0	46.0–59.5	59.5–73.0

Duty assignments are based on the operational role of each machine. The excavator is the primary production machine, operating in continuous truck-loading mode with minimal idle time: Medium duty at minimum, High duty in hard rock or high-demand scenarios. The wheel loader and track dozer are support machines operating intermittently (stockpile management, bench cleanup, push-up); their duty is assigned as Low when intermittent and Medium when operating under sustained demand. Representative values are taken as the midpoint of each duty band.

Table C.26. Diesel consumption derivation by scenario for the `excavation_loading_generic` block.

Component	**Optimistic**	**Midpoint**	**Conservative**
Excavator duty	Medium–low	Medium	High
Excavator (L/h)	49	56	68
Loader duty	Low	Low	Medium
Loader (L/h)	24	29	39
Dozer duty	Low	Low	Medium
Dozer (L/h)	32	39	53
Fleet total (L/h)	**105**	**124**	**150**

Other utilities.

None. No steam, cooling water, compressed air, or natural gas is consumed.

Consumables — Tires.

The machines all require tires, which we model as consumable tools. Tire cost estimation:

Table C.27. Annual tire cost estimation for the wheel loader.

Parameter	**Optimistic**	**Midpoint**	**Conservative**
Tire unit cost ($/tire)	8,000	12,000	15,000
Tire life (operating hours)	5,000	4,000	3,500
Loader annual hours (h/yr)	4,000	4,400	4,700
Tires replaced per year	3.2	4.4	5.4
Annual tire cost ($/yr)	∼$26,000	∼$53,000	∼$81,000

Loader annual operating hours are estimated as block availability (90%) × 8,760 h/yr × loader utilization factor (50–60%), reflecting the loader's intermittent support role. Tire life of 3,500–5,000 hours spans the range for mining applications on maintained haul roads [445].

Labor.

Labor is 4.5 FTE at base scale with a scaling exponent of 0.5 and `dedicated_labor` = FALSE. The base represents the excavator operator only; loader and dozer operators are assumed to be covered at the facility level through labor pooling, keeping total facility labor within reasonable bounds when multiple mining blocks are instantiated in parallel:

$$\text{FTE} = 1.0 \times 4.5 = 4.5$$

Footprint.

No fixed plant footprint is assigned. Mobile mining equipment operates within the active pit area, which does not contribute to building cost calculations.

Availability.

Availability for this block is set to `proc_avail_factor` = 0.90, with `unplanned_downtime` = 0.05 and `scheduled_maintenance` = 0.05, held constant across all three scenarios. The excavation block does not require a dedicated production line (`dedicated_line = FALSE`); multiple block units can share ancillary infrastructure (fuel bays, maintenance workshops, haul road access).

Maintenance.

Annual maintenance cost is expressed as a fraction of scaled equipment cost. For mobile mining equipment, maintenance includes engine and hydraulic system servicing, replacement,

structural repairs, and scheduled component rebuilds. Tires are excluded (captured in tooling above).

Table C.28. Maintenance cost fractions for the `excavation_loading_generic` block.

Scenario	`maint_override`	**Basis**
Conservative	0.08	Hard rock, heavy undercarriage wear
Midpoint	0.06	Typical mining conditions
Optimistic	0.04	Favorable ground, newer equipment

Published mining equipment maintenance benchmarks (excluding tires) typically fall in the range of 4–10% of replacement cost annually, depending on operating conditions and rebuild cycle timing [433, see Ch. 4.9]. The midpoint of 6% is a central estimate for medium-scale open pit operations.

C.5.3.5 `crushing_cycle`

Sources.

Gao et al. [174]; Khakmardan et al. [193]; Liu et al. [175]; SGS Canada et al. [150]; Samari et al. [155]; Riordan et al. [154]; Darling [433], Chapters 3.3–3.4 (Comminution — Jaw, Impact, Gyratory, and Cone Crushers); AusIMM [184], Chapter 11 (Beneficiation — Comminution); Metso:Outotec [447]; Michaud [448]; Sandvik [449]; [450]; other vendor data.

Process description.

This block models a three-stage crushing circuit reducing run-of-mine ore to a product size of approximately 5–25 mm suitable for downstream classification, attrition scrubbing, or flotation feed preparation. The circuit comprises a primary jaw crusher, two secondary/tertiary cone crushers operating in closed circuit with sizing screens, and associated material handling (ROM receiving, conveyors, transfer chutes, dust collection). Both hard rock (spodumene, lepidolite) and clay pathways use this block, though target product sizes differ: hard rock circuits target ∼15 mm for flotation feed, while clay circuits may accept coarser product for slurry preparation.

Sizing basis.

The step basis is `crushed_ore_throughput` in t/h, with `process_type = continuous` and a base volume of **240 t/h** as a baseline. This represents fresh feed throughput; recirculating loads from closed-circuit screening are internal to the block and do not appear in the step basis. No oversizing factor is assumed (1.0). The scaling exponent is 0.65, consistent with the six-tenths rule for comminution equipment.

The crushing stages are sized as follows (after 911 Metallurgist):

Table C.29. Crushing stage size reduction targets.

Stage	Crusher type	Feed size	Product size
Primary	Jaw (C6X class)	up to 1.5 m	150–300 mm
Secondary	Cone (CH890 class)	150–450 mm	40–130 mm
Tertiary	Cone (CH890 class)	20–150 mm	5–25 mm

Bill of Equipment.

Table C.30. Bill of equipment for the `crushing_cycle` block.

Equipment	**Machine data used**	**Source(s)**
Primary jaw crusher (C6X class, 160 kW)	Jaw crushers	911 Metallurgist; Metso:Outotec; vendor data
Secondary cone crusher (Sandvik CH890 class, 750 kW)	Cone crushers	911 Metallurgist; AusIMM Ch. 11; vendor data
Tertiary cone crusher (Sandvik CH890 class, 750 kW)	Cone crushers	911 Metallurgist; AusIMM Ch. 11; vendor data
ROM receiving (dump pocket, grizzly, feeder)	Not explicit → inferred	—
Sizing screens (scalping, product, recirculation)	Not explicit → inferred	—
Crusher recirculation (return conveyors, splitters, bins)	Not explicit → inferred	—
Conveyors and transfer chutes	Not explicit → inferred	—
Dust collection (baghouses, fans, ducting)	Not explicit → inferred	—

CAPEX.

Crusher capital costs are estimated using a power-based scaling approach anchored to published pricing for a mid-scale primary jaw crusher (C6X125 class, 240 t/h, 160 kW installed motor). The bare equipment cost for this anchor unit is USD 0.45–0.68M, with costs scaling sub-linearly by installed power at exponent $n = 0.65$. Installation multipliers of 1.5–2.0× bare cost are applied for the jaw crusher, increasing to 1.6–2.1× for the cone crushers to reflect additional structural steel, hydraulic systems, and controls scope.

The secondary and tertiary crushers are both Sandvik CH890-class cone crushers at 750 kW rated power. Because the same machine class is used for both stages, they are costed identically at \$3.5M / \$4.5M / \$5.5M installed (optimistic / midpoint / conservative). Auxiliary equipment is added as a percentage of the installed crusher subtotal, itemized in Table C.31.

Table C.31. Auxiliary CAPEX adders for the `crushing_cycle` block (% of crusher subtotal).

Component	**Optimistic**	**Midpoint**	**Conservative**
ROM receiving	20%	30%	45%
Sizing screens	15%	25%	35%
Crusher recirculation	10%	15%	25%
Conveyors and transfer chutes	35%	60%	80%
Dust collection	8%	12%	20%
Auxiliary total	**88%**	**142%**	**205%**

Equipment life is assumed to be 15 years, given the heavy wear and tear these machines undergo.

Electricity.

Total rated crusher motor power is $160 + 750 + 750 = 1,660$ kW. Actual operating power is less than rated; load factors of 65% / 70% / 75% (optimistic / midpoint / conservative) are applied to reflect typical draw during continuous operation. Auxiliary electrical loads (screens, feeders, conveyors, dust collection fans, lubrication systems) are added as a percentage of crusher power.

Table C.32. CAPEX summary for the `crushing_cycle` block.

Item	**Optimistic**	**Midpoint**	**Conservative**
Primary jaw (160 kW), installed	\$0.7M	\$1.0M	\$1.4M
Secondary cone (750 kW), installed	\$3.5M	\$4.5M	\$5.5M
Tertiary cone (750 kW), installed	\$3.5M	\$4.5M	\$5.5M
Crusher subtotal	\$7.7M	\$10.0M	\$12.4M
Auxiliary adder	×1.88	×2.42	×3.05
Total installed CAPEX	**\$14.5M**	**\$24.2M**	**\$37.8M**

Table C.33. Electricity derivation for the `crushing_cycle` block.

Parameter	**Optimistic**	**Midpoint**	**Conservative**
Rated motor power (kW)	1,660	1,660	1,660
Load factor	65%	70%	75%
Crusher draw (kW)	1,079	1,162	1,245
Auxiliary electrical adder	30%	40%	50%
Total electricity (kWh/h)	**1,400**	**1,630**	**1,870**
Specific energy (kWh/t)	5.8	6.8	7.8

Other utilities.

None; no thermal utilities as all equipment is assumed to be electrically driven. Dust suppression water is consumed, but not tracked as a utility stream in this model.

Consumables.

Crusher wear parts (manganese jaw plates, cone mantles and bowl liners) are significant consumables for this block. These are currently not populated (`tooling_cost_base` = 0) and are flagged as a gap requiring project-specific liner consumption data or vendor estimates.

Maintenance.

`maint_override` is not currently populated for this block, and thus inherits a standard 5% value across all scenarios.

Labor.

Labor is 9.0 FTE at base scale with a scaling exponent of 0.5 and `dedicated_labor` = FALSE, summing operator-per-shift factors across the crushing circuit — primary crusher (0.5), secondary/tertiary crushers (0.5), screens and conveyors (0.5), and helper (0.5):

$$\text{FTE} = (0.5 + 0.5 + 0.5 + 0.5) \times 4.5 = 9.0$$

Footprint.

The base plot area is 500 m^2 with a scaling exponent of 0.7, covering the crusher structures, ROM handling, screening, inter-stage conveyors, dust collection, and control room.

Availability.

`proc_avail_factor` = 0.92, with `unplanned_downtime` = 0.03 and `scheduled_maintenance` = 0.05, held constant across scenarios. The 92% availability is representative of continuous crushing circuits with scheduled liner change-outs and preventive maintenance.

C.5.3.6 `attrition_scrubber`

Sources.

Lane et al. [153]; SGS Canada et al. [150]; Riordan et al. [154]; FLSmidth [451] WEMCO Attrition Scrubber product datasheet (No. 120 cell); Ulrich [185]

Process description.

This block models an attrition scrubbing circuit that uses high-intensity mechanical abrasion to liberate clay coatings and deagglomerate particles prior to downstream classification or leaching. Attrition scrubbers operate at high pulp densities (70–80 wt% solids) to maximize inter-particle contact and scrubbing intensity. The block is configured as a single bank of four cells in series, consistent with vendor guidance of 1–2 minutes residence time per cell and 4–6 cells for effective scrubbing. Additional capacity is added as parallel banks (lines).

Sizing basis.

The step basis is `scrubbed_slurry_throughput` in m^3/h of slurry at operating density. The base volume of **158** m^3/h is derived from the WEMCO No. 120 cell maximum capacity of 225 t/h dry solids at 75 wt% solids:

$$\rho_{\text{slurry}} = \frac{1}{w_s/\rho_s + (1-w_s)/\rho_w} = \frac{1}{0.75/2.7 + 0.25/1.0} = 1.895 \text{ t/m}^3$$

$$\dot{m}_{\text{slurry}} = 225/0.75 = 300 \text{ t/h} \implies \dot{V} = 300/1.895 \approx 158 \text{ m}^3/\text{h}$$

where $\rho_s = 2.7$ t/m^3 is a representative hard-rock solids density. The bank residence time is $4 \times 1.2 = 4.8$ minutes at the vendor maximum-capacity point. As such we set the following other process design parameters: `process_type = continuous`, `scaling_exponent` = 1.0 (discrete parallel banks), `oversizing_factor` = 1.15, `dedicated_line = TRUE`.

Bill of Equipment.

Table C.34. Bill of equipment for the `attrition_scrubber` block.

Equipment	Machine data used	Source(s)
Attrition scrubber package (4× WEMCO No. 120 cells)	Attrition scrubbers	WEMCO datasheet
Feed preparation and slurry make-up	Not explicit → inferred	—
Slurry pumping (feed, discharge, transfer)	Not explicit → inferred	—

CAPEX.

The bare equipment cost for a four-cell WEMCO No. 120 package is estimated at approximately $5M. An installation multiplier of 1.6× (structural steel, erection, alignment, electrical terminations, commissioning) yields a base installed cost of $8M. A ±25% uncertainty band defines the scenario range: $10M conservative, $8M midpoint, $6M optimistic.

Auxiliary equipment is added as a percentage of the installed package cost:

Table C.35. Auxiliary CAPEX adders for the `attrition_scrubber` block (% of installed package).

Component	**Optimistic**	**Midpoint**	**Conservative**
Slurry pumps and piping	15%	22.5%	30%
Structural and access steel	10%	15%	20%
Instrumentation and controls	5%	10%	15%
Feed conditioning and density control	10%	17.5%	25%
Auxiliary total	**40%**	**65%**	**90%**

Table C.36. CAPEX summary for the `attrition_scrubber` block.

Item	**Optimistic**	**Midpoint**	**Conservative**
Installed package	$6.0M	$8.0M	$10.0M
Auxiliary adder	×1.40	×1.65	×1.90
Total installed CAPEX	**$8.4M**	**$13.2M**	**$19.0M**

Equipment life is 15 years. `prim_equip_scaling_exponent` = 0.6.

Electricity.

Per-cell motor ratings from the WEMCO datasheet range from 75–150 HP (56–112 kW). Load factors of 65% / 70% / 75% are applied to the four-cell bank to obtain cell demand. Auxiliary electrical loads are itemized:

Pump power is estimated from the slurry hydraulic equation $P = \rho g Q H/\eta$ at $\rho = 1,895$ kg/m^3, $Q = 158$ m^3/h, and pump efficiency $\eta \approx 0.60$, giving approximately 0.82 kW per meter of head. The slurry transfer adder (20–50%) is the dominant variable, driven by inter-area

Table C.37. Auxiliary electrical adders for the `attrition_scrubber` block (% of cell demand).

Component	Optimistic	Midpoint	Conservative
Feed pump(s)	10%	12%	15%
Discharge pump(s)	10%	12%	15%
Discharge screen	5%	5%	5%
Density control	2%	3%	5%
Slurry transfer to downstream	20%	35%	50%
Instrumentation and controls	3%	3%	5%
Auxiliary total	**50%**	**70%**	**95%**

pumping distance and elevation change; the conservative end is consistent with the Thacker Pass disclosure, which reports total "Attrition Scrubbing Area (w/ Slurry Transfer)" demand of 1,204 kW against 600 kW installed cell power.

Table C.38. Electricity summary for the `attrition_scrubber` block.

Parameter	Optimistic	Midpoint	Conservative
Motor rating per cell (HP)	75	125	150
Motor rating per cell (kW)	56	93	112
Rated bank power, 4 cells (kW)	224	373	448
Load factor	0.65	0.70	0.75
Cell demand (kW)	146	261	336
Auxiliary electrical adder	+50%	+70%	+95%
Total electricity (kWh/h)	**219**	**443**	**655**

Other utilities.

None. Process water for slurry make-up is consumed but not tracked as a utility in this block.

Consumables.

None populated; impeller wear parts are a potential consumable but are not currently modeled, which we flag as a gap requiring project-specific liner consumption data or vendor estimates.

Labor.

Labor is 4.5 FTE at base scale with a scaling exponent of 0.5 and `dedicated_labor` = FALSE, assuming one operator per shift for the four-cell bank (cells operate in series as a single train):

$$\text{FTE} = 1.0 \times 4.5 = 4.5$$

Footprint.

The base plot area is 300 m^2 with a scaling exponent of 0.7, covering the four-cell scrubber bank, feed and discharge sumps, slurry pumps, density control equipment, and local motor control.

Availability.

`proc_avail_factor` = 0.92, with `unplanned_downtime` = 0.03 and `scheduled_maintenance` = 0.05, held constant across scenarios. The 92% availability is typical of continuous scrubbing circuits with scheduled inspections and preventive maintenance.

Maintenance.

`maint_override` is not currently populated for this block, and thus inherits a standard 5% value across all scenarios.

C.5.3.7 `classifier_generic`

Sources.

FLSmidth [452]; AusIMM [184]; Metso:Outotec [447]; SGS Canada et al. [150]; Riordan et al. [154]; Lane et al. [153]; **?**]

Process description.

Classification separates particles by size using hydraulic or mechanical means, typically following crushing, attrition, or grinding. In lithium mineral processing flowsheets, classification is most commonly achieved using hydrocyclone clusters, often followed by screening to reject oversize particles. This block models a hydrocyclone cluster with associated feed pumping, piping, and product handling. For consistency with other slurry-based unit operations (attrition, leaching, thickening), the block is parameterized on slurry volumetric flow rate (m^3/h) rather than solids throughput.

Sizing basis.

The step basis is `classified_slurry_throughput` in m^3/h, with `process_type = continuous` and a base volume of **160** m^3/h. The `scaling_exponent` = 0.67, consistent with the six-tenths rule; there is no oversizing (`oversizing_factor` = 1.0), and lines must be dedicated `dedicated_line = TRUE`.

Bill of Equipment.

Table C.39. Bill of equipment for the `classifier_generic` block.

Equipment	Machine data used	Source(s)
Hydrocyclone cluster (FLSmidth Krebs gMAX, 2 operating + 1 standby)	Hydrocyclone classifiers	Krebs gMAX datasheet
Auxiliary (slurry pumps, piping, product handling, instrumentation)	Not explicit → inferred	—

CAPEX.

Installed equipment cost for the hydrocyclone cluster (2 operating + 1 standby units, manifold, distributor, and structural support) is estimated at \$1.5M / \$2.0M / \$2.8M (optimistic / midpoint / conservative). Auxiliary equipment is added as a percentage of the installed cluster cost:

Table C.40. Auxiliary CAPEX adders for the `classifier_generic` block (% of installed cluster).

Component	Optimistic	Midpoint	Conservative
Slurry pumps and piping	20%	25%	35%
Structural and access steel	10%	15%	20%
Instrumentation and controls	8%	12%	18%
Screen package (oversize rejection)	15%	20%	30%
Auxiliary total	**53%**	**72%**	**103%**

Table C.41. CAPEX summary for the `classifier_generic` block.

Item	Optimistic	Midpoint	Conservative
Installed cluster	\$1.5M	\$2.0M	\$2.8M
Auxiliary adder	×1.53	×1.72	×2.03
Total installed CAPEX	**\$2.3M**	**\$3.4M**	**\$5.7M**

The equipment life is estimated at 15 years, based on the heavy expected wear. The primary scaling exponent is `prim_equip_scaling_exponent` = 0.6, consistent with the six-tenths rule.

Electricity.

Hydrocyclones have no moving parts; energy demand is dominated by slurry feed pumping. Pumping power is estimated from the hydraulic equation $P = \rho g Q H / \eta$, expressed as a specific energy intensity:

$$E_{\text{pump}} = \frac{\rho\, g\, H}{\eta \times 3.6 \times 10^{6}} \ \text{kWh/m}^3$$

The range $E_{\text{pump}} = 0.15$–0.40 kWh/m^3 spans slurry densities of 1,100–1,800 kg/m^3, pump heads of 30–100 m, and pump efficiencies of 60–80%. Auxiliary loads (instrumentation, valve actuators, lubrication) add 10–16 kW.

Other utilities.

None. Process water for slurry density adjustment is consumed but not tracked as a utility in this block.

Table C.42. Electricity summary for the `classifier_generic` block.

Parameter	Optimistic	Midpoint	Conservative
Specific energy (kWh/m^3)	0.15	0.25	0.40
Pump power at 160 m^3/h (kW)	24	40	64
Auxiliary (kW)	10	12	16
Total electricity (kWh/h)	**34**	**52**	**80**

Consumables.

Hydrocyclone wear liners (urethane or ceramic) are the principal consumable for this block, but are not estimated for this block (`tooling_cost_base` = 0); liner replacement data from vendor quotations or site operating records are required to populate this field.

Maintenance.

`maint_override` is not currently populated for this block, and thus inherits the facility-level default maintenance fraction.

Labor.

Labor. Labor is 4.5 FTE at base scale with a scaling exponent of 0.5 and `dedicated_labor` = FALSE, assuming one operator per shift for the hydrocyclone cluster and associated pumping:

$$\text{FTE} = 1.0 \times 4.5 = 4.5$$

Footprint.

The base plot area is 50 m^2 with a scaling exponent of 0.7, covering the cyclone cluster structure, feed sump and pump, and discharge launders.

Availability.

`proc_avail_factor` = 0.92, with `unplanned_downtime` = 0.03 and `scheduled_maintenance` = 0.05, held constant across scenarios.

C.5.3.8 `classifier_flotation`

Sources.

Jiao et al. [434]; Garrett [196]; Deng and Feng [180]; Darling [433] Ch. 12 (Flotation); AusIMM [184]; Metso:Outotec [447]; Peters and Timmerhaus [183].

Process description.

This block models the combined classification and froth flotation circuit used to concentrate run-of-mine lepidolite ore prior to high-energy grinding and sulfate roasting. Classification by wet screening or hydrocyclone removes oversize gangue material; froth flotation then selectively recovers lepidolite minerals from the classified fines using amine collectors and phosphate depressants. The combination of these two unit operations into a single block reflects their close physical integration in lepidolite processing plants, where the classifier overflow feeds directly to a conditioning tank and then to the flotation circuit, though future iterations of this flowsheet could separate the two. This block builds on the `classifier_generic` block (C.5.3.7), adding the flotation circuit and associated reagent systems.

The flotation step is the key beneficiation step in the lepidolite pathway. At typical Jiangxi ROM ore grades of less than 0.4% Li_2O (≈1,860 ppm Li), direct roasting of unbeneficiated material would be uneconomical. Froth flotation using dodecylamine collectors and sodium hexametaphosphate depressants selectively recovers lepidolite into a concentrate at approximately 1.6% Li_2O (≈7,430 ppm Li) [196, 434], a 4.0× concentration factor. At a flotation lithium recovery of 60%, the mass reporting to the downstream roasting circuit is reduced by approximately 85% relative to the ROM feed tonnage (i.e., approximately 15 tonnes of concentrate per 100 tonnes of ROM), substantially reducing the energy-intensive roasting throughput, while also generating very significant flotation tailings.

Equation of merit.

$$V_{\text{cells}}\ (\text{m}^3) = \frac{\dot{V}_{\text{slurry}}\ (\text{m}^3/\text{h}) \times \tau_{\text{res}}\ (\text{min})}{60}$$

where $\dot{V}_{\text{slurry}}$ is the volumetric feed rate of classified slurry to the flotation circuit and τ_{res} is the total flotation residence time across rougher, scavenger, and cleaner stages. Total installed cell volume is sized to provide the required residence time at the design throughput; circuit configuration (number of cells, cell volume) then follows from cell-size selection.

Sizing basis.

The step basis is `classified_slurry_throughput` in m^3/h of slurry at approximately 30 wt% solids. The `base_volume` is **160** m^3/h, which at 30 wt% solids and a slurry density of 1.233 t/m^3 (grain density 2.7 t/m^3) corresponds to a total slurry mass flow of $160 \times 1.233 =$

197 t/h and a dry solids throughput of 197 × 0.30 = 59 t/h. An oversizing factor of 1.2 is applied, giving a design throughput of 192 m^3/h to accommodate feed variability and surge. The `scaling_exponent` = 0.67, reflecting modest economies of scale in flotation circuit sizing; lines are not dedicated (`dedicated_line = FALSE`).

Bill of Equipment.

Table C.43. Bill of equipment for the `classifier_flotation` block.

Equipment	**Machine data used**	**Source(s)**
Hydrocyclone cluster (classification)	Hydrocyclone classifiers	Krebs gMAX datasheet
Conditioning tank + agitator	Not explicit → inferred	—
Rougher flotation cells	Mechanical flotation cells	Jiao et al. 2024; Arfania et al. 2017; SME Handbook Ch. 12
Scavenger flotation cells	Mechanical flotation cells	Jiao et al. 2024; Arfania et al. 2017; SME Handbook Ch. 12
Cleaner flotation cells	Mechanical flotation cells	Jiao et al. 2024; Arfania et al. 2017; SME Handbook Ch. 12
Flotation air compressor	Not explicit → inferred	SME Handbook Ch. 12
Feed, froth, and tails pumps	Not explicit → inferred	—
Reagent dosing systems (DDA collector + SHMP depressant)	Dosages from literature	Jiao et al. 2024
Instrumentation and controls	Not explicit → inferred	—

CAPEX.

The block CAPEX is built bottom-up from three components: the classification circuit (costed consistently with the standalone `classifier_generic` block, C.5.3.7), the flotation circuit (sized from required cell volume), and the conditioning and auxiliary equipment.

Flotation cell volume. The required installed cell volume is derived from the equation of merit $V = \dot{V} \times \tau/60$. At 160 m^3/h feed, the rougher and scavenger stages process the full feed stream; the cleaner stages process approximately 20% of the feed volume (the mass pull to rougher concentrate). Residence times for fine-particle lepidolite flotation are longer than typical sulfide circuits due to the fine grain size and the need for multiple cleaner stages to achieve the target 1.6% Li_2O concentrate grade:

Table C.44. Flotation cell volume derivation for the `classifier_flotation` block.

Parameter	**Optimistic**	**Midpoint**	**Conservative**
Rougher + scavenger residence (min)	20	25	35
Cleaner residence (min)	10	12	18
R+S cell volume (m^3)	53	67	93
Cleaner cell volume (m^3)	5	6	10
Total cell volume (m^3)	**59**	**73**	**103**
Number of cells	8	10	14

Flotation cell costing. Individual cell costs are estimated using the power-law cost models of Arfania et al. [453], which regress capital cost against cell volume for standard mechanical flotation cells using InfoMine survey data. Per-cell bare equipment costs (excluding motors) of \$108–162K (2025 USD, escalated from 2013 via CEPCI ratio ≈1.35) are applied, with separate motor costs of \$15–25K per cell. An installation multiplier of 1.5–2.2× is applied to the bare-plus-motor cost to account for structural steel, piping, electrical termination, launders, and commissioning.

Conditioning and auxiliaries. A conditioning tank with high-torque agitator (∼15 m^3, 5–10 min reagent contact time) is estimated at \$250–450K installed. Flotation auxiliaries — air compressor, reagent dosing systems (DDA collector + SHMP depressant), feed/froth/tails pumps, and instrumentation — are estimated at \$430K–1,050K installed, with the range reflecting circuit complexity and instrumentation scope.

The SME Handbook (Ch. 12) cites typical flotation plant CAPEX of approximately \$15–80k per (t/h dry solids) for base-metal sulfide ores, implying \$0.9–4.7M for the flotation portion alone at 59 t/h. The flotation subtotal (\$2.2–7.3M) is within or modestly above this range, with the elevation attributable to the fine-particle mica application requiring longer residence time and more cleaner stages. Equipment life is 15 years, reflecting the rubber-lined and

Table C.45. CAPEX summary for the `classifier_flotation` block.

Component	Optimistic	Midpoint	Conservative
Classifier (per C.5.3.7)	\$2.3M	\$3.4M	\$5.7M
Flotation cells, installed	\$1.5M	\$2.8M	\$5.8M
Conditioning tank, installed	\$0.3M	\$0.4M	\$0.5M
Flotation auxiliaries	\$0.4M	\$0.7M	\$1.0M
Total installed CAPEX	**\$4.5M**	**\$7.2M**	**\$13.0M**
Per t/h dry solids	\$76k	\$122k	\$220k

reagent-wetted components; the `prim_equip_scaling_exponent` = 0.6 per the six-tenths rule.

Electricity.

Electrical demand is derived bottom-up from five components: the classification circuit (per the `classifier_generic` block), flotation cell drives and air blower, conditioning tank agitator, slurry pumps, and minor loads.

Flotation cells and blower. Specific power for mechanical flotation cells is 1.0–2.0 kW per m^3 of active cell volume [453], inclusive of both the impeller drive and the forced-air blower. At the cell volumes derived in the CAPEX section (59 / 73 / 103 m^3), this gives 59 / 110 / 206 kW.

Conditioning tank. A 15 m^3 high-torque agitated tank at 2–5 kW/m^3 (reflecting the high-solids conditioning duty at ∼30 wt% solids) draws 30 / 45 / 75 kW.

Pumps. Feed, froth, and tails pump power is estimated from $P = \rho g Q H/\eta$, with $\rho = 1,233$ kg/m^3, $Q = 160$ m^3/h, pump efficiencies of 55–75%, and total dynamic heads of 15–40 m. The froth pump handles approximately 20% of the feed flow; the tails pump handles approximately 80% at 60% of the feed head.

The SME Handbook reports typical flotation specific energy of 2–8 kWh/t dry solids for fine-particle flotation circuits. All three scenarios fall within this range. The flotation cells are the dominant load at midpoint and conservative (45% and 47% of total respectively); the classifier dominates at the optimistic scenario (23%).

Compressed air.

Flotation cells require a continuous supply of compressed air to generate the bubble stream that carries hydrophobic particles to the froth layer. Industry practice for mechanical

Table C.46. Electricity derivation for the `classifier_flotation` block.

Component	Optimistic	Midpoint	Conservative
Classifier (per C.5.3.7)	34 kW	52 kW	80 kW
Flotation cells + blower	59 kW	110 kW	206 kW
Conditioning tank agitator	30 kW	45 kW	75 kW
Pumps (feed + froth + tails)	18 kW	35 kW	66 kW
Reagent dosing + instrumentation	3 kW	5 kW	8 kW
Total electricity (kWh/h)	**145**	**245**	**435**
Specific energy (kWh/t dry solids)	2.5	4.2	7.4

flotation cells (SME Handbook Ch. 12) cites typical air rates of 0.5–1.5 Nm^3 per m^3 slurry processed, varying with cell design and froth stability requirements. All three scenario values fall within this range.

Table C.47. Compressed air summary for the `classifier_flotation` block.

Parameter	Optimistic	Midpoint	Conservative
Compressed air (Nm^3/h)	80	128	208
Specific air (Nm^3 per m^3 slurry)	0.50	0.80	1.30

The compressor electrical load is already included in the electricity figure above; the compressed air field records volumetric air consumption for mass balance and emissions accounting purposes only.

Other utilities.

None.

Labor.

Labor is 6.75 FTE at base scale with a scaling exponent of 0.5 and `dedicated_labor` = FALSE, summing operator-per-shift factors for the combined circuit — classifier/hydrocyclone (1.0, consistent with the `classifier_generic` block) and automated flotation train

including conditioning, rougher/scavenger/cleaner cells, and reagent dosing (0.5):

$$\text{FTE} = (1.0 + 0.5) \times 4.5 = 6.75$$

Footprint.

The base plot area varies across scenarios — 1,000 / 750 / 370 m^2 (conservative / midpoint / optimistic) — reflecting uncertainty in cell configuration and spacing. The conservative footprint corresponds to a layout with multiple smaller cells and wider inter-cell spacing; the optimistic footprint reflects a compact arrangement using larger modern cells. The scaling exponent is 0.7.

Availability.

`proc_avail_factor` = 0.90, with `unplanned_downtime` = 0.05 and `scheduled_maintenance` = 0.05, held constant across scenarios. Flotation circuits are sensitive to feed quality variability (particle size, reagent balance, slurry density), justifying a slightly lower availability than purely mechanical steps.

Maintenance.

`maint_override` is not currently populated for this block, and thus inherits the facility-level default maintenance fraction.

C.5.3.9 `high_energy_grinding_mill`

Sources.

De Bakker [454]; Brissette [455]; Glencore Technology IsaMill product literature and compendium of papers (Technology [456]); Kumar et al. [457]; Nan et al. [212]; Nan et al. [211]; Zou et al. [213]; Ulrich [185].

Process description.

High-energy grinding (also termed mechanical activation) uses a stirred media mill to reduce particle size and disrupt crystalline structure prior to roasting or leaching. In the lepidolite pathway, this step mechanically activates the flotation concentrate to increase reactive surface area — a fundamentally more energy-intensive duty than conventional regrinding. The dominant equipment class is the horizontal high-speed stirred mill (IsaMill / Netzsch platform) or vertical stirred mill (Metso HIGMill), selected based on target product size and circuit configuration. Stirred mills operate at power intensities of 100–350 kW/m^3 (vs. $\sim$20 kW/m^3 for conventional ball mills), enabling fine and ultrafine grinding in a compact footprint.

Equation of merit.

$$P_{\text{motor}}\ (\text{kW}) = E_{\text{sp}}\ (\text{kWh/t}) \times \dot{m}_{\text{DS}}\ (\text{t/h})$$

where E_{sp} is the specific energy consumption and $\dot{m}_{\text{DS}}$ is the dry solids throughput. Specific energy is the primary sizing variable for stirred mills; it is determined by laboratory or pilot-scale test work and depends on feed size, product size, ore hardness, and media selection.

Sizing basis.

The step basis is `ground_slurry_throughput` in m^3/h of slurry at approximately 50 wt% solids. The `base_volume` is 21 m^3/h, which at 50 wt% solids and a slurry density of 1.46 t/m^3 (grain density 2.7 t/m^3) corresponds to a dry solids throughput of $21 \times 1.46 \times 0.50 = 15.3$ t/h. The `scaling_exponent` = 1.0, reflecting that stirred mills scale as discrete integer units (each additional unit adds one mill); an `oversizing_factor` of 1.2 is applied to accommodate feed variability. The line is not dedicated `dedicated_line = FALSE`.

Bill of Equipment.

CAPEX.

The CAPEX anchor is an IsaMill M3000-class unit (3,000 L grinding chamber, 1,120 kW motor) at an estimated installed cost of \$4.5M (2025 USD), based on vendor literature and AusIMM cost relationships for fine-grinding equipment. Required motor power is derived

Table C.48. Bill of equipment for the `high_energy_grinding_mill` block.

Equipment	**Machine data used**	**Source(s)**
Horizontal high-speed stirred mill (IsaMill M3000 class, 1.12 MW)	IsaMill M3000 as sizing anchor	Glencore Technology
Ceramic grinding media supply system	Not explicit → inferred	—
Slurry feed/discharge handling	Not explicit → inferred	—
Auxiliary (electrical, structural, controls)	Not explicit → inferred	—

from the specific energy and throughput; CAPEX is then scaled from the anchor using the six-tenths rule ($n = 0.65$):

$$\text{CAPEX} = \$4.5\text{M} \times \left(\frac{P_{\text{motor}}}{1{,}120\ \text{kW}} \right)^{0.65}$$

Table C.49. CAPEX summary for the `high_energy_grinding_mill` block.

Parameter	**Optimistic**	**Midpoint**	**Conservative**
Specific energy (kWh/t)	40	70	120
Motor power (kW)	610	1,070	1,840
Scaling ratio $(P/1{,}120)^{0.65}$	0.67	0.97	1.38
Total installed CAPEX	**$3.0M**	**$4.4M**	**$6.2M**

We assume the equipment life is 20 years, with a `prim_equip_scaling_exponent` = 0.65 consistent with the six-tenths rule.

Electricity.

Electricity is typically the dominant operating cost for stirred mills, and is derived directly from the specific energy requirement. The scenario variation reflects the difference between conventional regrind duty (40 kWh/t at P80 ∼40–60 μm) and full mechanical activation ($\geq$120 kWh/t at P80 <15 μm with crystallographic disruption):

Table C.50. Electricity derivation for the `high_energy_grinding_mill` block.

Parameter	Optimistic	Midpoint	Conservative
Target duty	Coarse regrind	Fine regrind	Mech. activation
Specific energy (kWh/t DS)	40	70	120
Dry solids throughput (t/h)	15.3	15.3	15.3
Total electricity (kWh/h)	**610**	**1,070**	**1,840**

The specific energy ranges are anchored to De Bakker [454] and **?**] for conventional regrind (20–60 kWh/t), and to the CN patent literature (CN110983071B, CN118835103A, CN107475537B) for mechanical activation duties (40–120 kWh/t). No auxiliary electrical adder is applied; the motor draw is the total block demand, as feed/discharge handling is gravity-fed or included in the mill package.

Other utilities.

None.

Consumables.

Ceramic grinding media (typically 2–4 mm beads) is the principal consumable and a significant OPEX driver, commonly accounting for 30–50% of total circuit operating cost [454]. Consumption rates of 0.3–1.0 kg/t solids at media costs of \$1–3/kg give an estimated media cost of \$0.30–2.50/t dry solids. `tooling_cost_base` is not currently populated; this is an area for future work.

Labor.

Labor is 2.25 FTE at base scale with a scaling exponent of 0.5 and `dedicated_labor` = FALSE. The stirred mill can be modeled under high automation, requiring only 0.5 operators per shift (with operators shared with adjacent circuit duties):

$$\text{FTE} = 0.5 \times 4.5 = 2.25$$

Footprint.

The base plot area is 150 m^2 with a scaling exponent of 0.6, covering the mill, media handling system, feed/discharge sumps, and controls. Stirred mills have a substantially smaller footprint per installed kW than conventional ball mills due to their high power intensity.

Availability.

`proc_avail_factor` = 0.92, with `unplanned_downtime` = 0.03 and `scheduled_maintenance` = 0.05, held constant across scenarios. This is consistent with IsaMill vendor-reported availability of 90–93%.

Maintenance.

`maint_override` is not currently populated for this block, and thus inherits the facility-level default maintenance fraction.

C.5.3.10 `sulfate_roasting_train_lepidolite`

Sources.

Deng and Feng [180], Yan et al. [205], Gao et al. [174], Park et al. [210], Nan et al. [212], Nan et al. [211], Zou et al. [213], Jiao et al. [434], Garrett [196]; Chinese equipment vendor quotes for mineral sulfation roasting at 50–80 t/h scale.

This block models the complete roasting train for sulfate roasting of dewatered lepidolite concentrate. The train encompasses feed conditioning (blending of lepidolite cake, Na_2SO_4, and lime), a rotary kiln operating at 800–950°C, an afterburner or gas cleaning stage for fluorine management, a heat recovery exchanger, and discharge conveying to the water leach circuit. The roasting step converts lithium from the tightly bound mica lattice into water-soluble lithium sulfate via the central reaction:

$$(Li_2O)(Al_2O_3)(3\,SiO_2)(KF)\,(s) + Na_2SO_4(s) \longrightarrow Li_2SO_4(s) + Al_2O_3(s) + 3\,SiO_2(s) + KF(g/s) \tag{C.39}$$

Fluorine species are captured by lime addition as CaF_2 and AlF_3 precipitates, rather than volatilizing as toxic HF gas. This is the characteristic feature distinguishing sulfate-based from acid- or chlorine-based lepidolite processing routes.

Equation of merit.

$$\dot{Q}_{\text{thermal}}\ (\text{MMBTU/h}) = q_{\text{specific}}\ (\text{MMBTU/t}) \times \dot{m}_{\text{mix}}\ (\text{t/h})$$

where q_{specific} is the specific fuel consumption per tonne of roasting mix and $\dot{m}_{\text{mix}}$ is the roasting mix throughput. This relationship also determines the required kiln size via the thermal load per unit kiln volume.

Sizing basis.

The step basis is `roasting_mix` in tonnes per hour at 80 wt% solids entering the kiln. The base volume is **60 t/h roasting mix**, consistent with the capacity range of industrial rotary kilns used for mineral sulfation (typically 50–80 t/h for analogous applications). The roasting mix includes dry ore solids (0.8 t per tonne wet cake), sodium sulfate addition ($0.5 \times 0.8 = 0.4$ t), lime addition ($0.15 \times 0.8 = 0.12$ t), potassium sulfate addition ($0.1 \times 0.8 = 0.08$ t), and residual moisture (0.2 t), giving a total input-to-process conversion factor (IPCF) of 1.6 per tonne of wet cake feed. At Jianxiawo's approximate annual roasting mix throughput of ∼4.3 Mt/yr, the required number of parallel kiln trains at 60 t/h and 92% availability is approximately 8–9 trains.

The block is assigned `process_type = batch` with `batch_cycle_time = 1 h` and zero setup time, as a modeling convenience for a continuous thermal operation. The oversizing

factor is 1.20, reflecting the standard 20% design margin applied to continuous thermal equipment to accommodate feed variability and maintain campaign availability.

Bill of Equipment.

Table C.51. Bill of equipment for the `sulfate_roasting_train_lepidolite` block.

Equipment	Machine data used	Source(s)
Feed conditioning mixer	Inferred from analogous mineral processing trains	Deng and Feng [180]
Rotary kiln (50–80 t/h class)	Rotary Kiln data	Vendor quotes; Jiao et al. [434]
Afterburner / fluorine gas scrubber	Not explicit → inferred	Deng and Feng [180]; Yan et al. [205]
Heat recovery exchanger	Not explicit → inferred	—
Feed and discharge conveyors	Not explicit → inferred	—
Exhaust gas fans and ducting	Not explicit → inferred	—
Instrumentation and controls	Not explicit → inferred	—

CAPEX.

Capital cost per kiln train is estimated using a bottom-up approach anchored to Chinese equipment vendor pricing, cross-checked against the Jianxiawo project-level capital intensity reported by CRU [206]. The previous values in this block ($67M/$42M/$25M) reflected Western-specification installed plant costs and are not applicable to a Chinese domestic operation.

Bottom-up estimate. Chinese-fabricated metallurgical rotary kilns at 50–80 t/h capacity are quoted at approximately ¥5–15M bare equipment ($0.7–2.1M at 7.1 RMB/USD). A full installed train requires the conditioning mixer, afterburner or gas scrubber, heat recovery exchanger, feed and discharge conveyors, civil foundations, electrical distribution, and instrumentation, together with a fluorine-resistant refractory upcharge for the HF-bearing operating environment. Applying a full-train installation multiplier of 3.0–4.5× the bare kiln cost yields an installed range of approximately $2–10M per train for domestic Chinese specification.

Top-down cross-check. CRU reports Jianxiawo's all-in capital intensity at $5,500/t LCE/yr [206], giving a total project CAPEX of approximately $253M for 46,000 t LCE/yr capacity. With 8–9 parallel roasting trains and the thermal conversion step representing a plausible 25–35% of total processing capital (consistent with Peters and Timmerhaus [183] guidance for calcination-dominated flowsheets), the implied per-train cost is approximately $7–11M at midpoint — consistent with the bottom-up range.

Based on these cross-defined estimates, we estimate the conservative, midpoint, and optimistic CAPEX costs in Table C.52.

Table C.52. CAPEX scenario values for the `sulfate_roasting_train_lepidolite` block (per 60 t/h train, fully installed).

Scenario	`prim_equip_price_base`	**Basis**
Conservative	$18,000,000	Chinese spec, corrosion-resistant refractory upgrade, full balance of plant (BOP), 20% contingency
Midpoint	$10,000,000	Standard Chinese mineral roasting train, full installed BOP; consistent with CRU top-down anchor
Optimistic	$5,000,000	Streamlined domestic fabrication, minimal F-control allowance, low contingency

The CAPEX scaling exponent is 0.60, following the six-tenths rule for process equipment cost. Equipment life is 20 years across all scenarios, with major refractory relining campaigns expected every 5–7 years within that period, though this is currently not modeled. The fixed site infrastructure shared across multiple trains (central control room, switchgear building, shared reagent storage) is not separately modeled at this block level; it is implicitly distributed across the per-train estimates.

Natural gas.

Natural gas is the dominant utility and the primary operating cost driver for this block. The three scenarios span 75–150 MMBTU/h at the 60 t/h base scale, equivalent to 1.25–2.50 MMBTU/t roasting mix (1.32–2.64 GJ/t).

The gross theoretical heat requirement has three components: First, sensible heat to raise the dry solids from ambient to 850°C (∼0.74 GJ/t roasting mix). Second, estimated reaction enthalpy for aluminosilicate decomposition (∼0.50 GJ/t). Third, the drying load to evaporate the residual moisture in the 80 wt% solids input cake: at 0.125 t H_2O per tonne

roasting mix, the drying duty (sensible heating of liquid water from 25°C to 100°C, latent heat of evaporation, and superheating of steam to exhaust temperature) totals approximately $0.125 \times 2.77 \approx 0.35$ GJ/t. The gross theoretical minimum is therefore ∼1.59 GJ/t, or ∼1.50 MMBTU/t.

In practice, kiln exhaust at 850°C carries substantial recoverable heat. The heat recovery exchanger in the train (see Bill of Equipment) preheats incoming feed, partially offsetting the drying load. The effective net minimum therefore lies between the gross minimum (1.50 MMBTU/t, no recovery) and a lower bound approaching the solids-plus-reaction demand alone (1.17 MMBTU/t) as recovery effectiveness increases.

Table C.53. Natural gas scenario derivation for the `sulfate_roasting_train_lepidolite` block.

Scenario	**MMBTU/h**	**MMBTU/t mix**	**Min. recovery req.**	**Basis**
Conservative	150	2.50	none	Older equipment, minimal heat recovery, high F-maintenance downtime
Midpoint	100	1.67	none	Modern kiln, basic heat recovery, Chinese industrial context
Optimistic	75	1.25	>75% of drying load	Well-engineered system with high-effectiveness preheater

The conservative and midpoint scenarios are thermodynamically consistent without heat recovery. The optimistic scenario (1.25 MMBTU/t) lies below the gross theoretical minimum and requires that at least 75% of the drying load be recovered from kiln exhaust, a physically achievable but likely demanding heat integration target. It is retained as the optimistic bound on the assumption of best-practice preheater design, and represents the upper end of what a well-engineered system could plausibly achieve. Natural gas dominates operating costs for this step and is the primary driver of its contribution to the modeled utility costs per tonne of lithium carbonate, as well as a large direct contributor to greenhouse gas emissions.

Electricity.

Electrical demand covers kiln rotation drive, feed and discharge conveying, exhaust gas fans, and instrumentation. It is secondary to natural gas across all scenarios:

Table C.54. Electricity scenario values for the `sulfate_roasting_train_lepidolite` block.

Scenario	`electricity_base_total` (kWh/h)	**Intensity (kWh/t mix)**
Conservative	270	4.50
Midpoint	200	3.33
Optimistic	160	2.67

While electricity plays only a minor role in total operating cost, it contributes meaningfully to Scope 2 emissions given the carbon intensity of the Chinese grid.

Compressed air.

Compressed air consumption is set at 150 Nm^3/h as a round number across all scenarios, held constant across scenarios as it is driven by the control system specification, rather than process throughput quantities. This value assumes wet scrubbing for fluorine management rather than dry bag filtration, which would use substantially more compressed air.

Labor.

Labor is 5.85 FTE per train at base scale with a scaling exponent of 0.2 and `dedicated_labor` = FALSE. The low exponent reflects the reality that one operator shift crew covers multiple kilns once above a minimum staffing threshold; labor grows only weakly with kiln size in this equipment class:

$$\text{FTE} = (0.5 + 0.5 + 0.2 + 0.1) \times 4.5 = 5.85$$

Footprint.

Each train occupies a base plot area of 3,800 m^2 with a scaling exponent of 0.7, covering the kiln shell and drive, feed and discharge buildings, gas handling equipment, heat recovery system, and surrounding access ways.

Availability.

Process availability is 92% with 3% unplanned downtime and 5% scheduled maintenance (totaling 8% total downtime). The availability figure is slightly lower than for purely mechanical steps, reflecting refractory maintenance requirements and the corrosive operating environment characteristic of fluorine-bearing feedstocks.

Other utilities.

None. Diesel, propane, cooling water, and steam are all zero for this block. The dominant energy input is natural gas for kiln firing, described above; electricity covers all remaining

mechanical and control loads.

Notes and flags.

`tooling_cost` and `maint_override` are both zero in the current parameterization. This is a notable gap for a refractory-lined rotary kiln: annual maintenance costs for refractory repair, refractory relining campaigns, and corrosion-related component replacement are material and should be estimated in more detail.

C.5.3.11 `acid_leach_circuit`

Sources.

Wesselkaemper et al. [142], SGS Canada et al. [150], Enterprises [151], Riordan et al. [154], Mousavinezhad et al. [173], Khakmardan et al. [193], Iyer and Kelly [194], Roy et al. [195], Mousavinezhad et al. [356]

The acid leach circuit dissolves lithium and associated metals from classified clay slurry using sulfuric acid, producing a pregnant leach solution (PLS) for downstream processing. Unlike hard-rock pathways, no thermal activation is required; acid attack on the clay mineral lattice proceeds directly at near-ambient temperatures. High-torque agitation is required throughout because the elevated fines content of clay slurries produces viscous, non-Newtonian behavior that standard low-shear impellers cannot adequately handle.

Equation of merit.

$$\tau_{\text{res}}\ (\text{h}) = \frac{V_{\text{tank}}\ (\text{m}^3)}{\dot{V}_{\text{slurry}}\ (\text{m}^3/\text{h})} \times N_{\text{tanks}}$$

The circuit is sized to deliver a total residence time sufficient for lithium extraction. At 1 h residence time per tank and 3 tanks in cascade, the total design residence time is 3 h, consistent with disclosed operating conditions at Tonopah Flats and Thacker Pass.

Bill of Equipment.

Table C.55. Bill of equipment for the `acid_leach_circuit` block.

Equipment	Machine data used	Source(s)
Acid leach tanks (3-tank cascade)	AusIMM leach duty tank estimates	AusIMM Monograph 27
High-torque agitator systems	Mixtec high-torque agitator series	Agitator tank series data
Feed and slurry make-up systems	Not explicit → inferred	—
Acid receiving and storage	Not explicit → inferred	—
Slurry and recycle pumping	Not explicit → inferred	—
Balance-of-plant (structural, piping, electrical)	Not explicit → inferred	—

Sizing basis.

The base volume is 1,100 m^3/h slurry throughput, corresponding to the operating flow rate of a single 3-tank leach cascade. The design tank volume is approximately 1,300 m^3 per tank, giving an operating fill of approximately 1,100 m^3 at a 85% fill level (oversizing factor 1.15, i.e. $1100 \times 1.15 \approx 1265$ $m^3 \approx 1300$ m^3 design). Three tanks operate in series with a 1 h residence time per tank, for a total circuit residence time of 3 h, consistent with Thacker Pass and Tonopah Flats disclosures.

The block is assigned `process_type = continuous` and `scaling_exponent_throughput = 1.0`, as additional throughput capacity is added by operating multiple parallel cascades rather than by scaling individual tank size.

CAPEX.

Installed cost per tank-agitator vessel is estimated from AusIMM leach duty tank benchmarks for heavy-duty, corrosion-resistant steel construction, combined with agitator costs from Mixtec product data. Installed agitator cost is derived by applying an installation factor of 2.0–3.0× to the raw equipment price of approximately $83,000 per agitator unit, yielding $166,000–$249,000 per agitator. Combined with the tank vessel, the fully installed cost per tank-agitator unit spans $1.67M–$3.75M across scenarios.

Per Association for the Advancement of Cost Engineering (AACE) guidance for Class 4–5 estimates [425], primary vessels typically represent 30–50% of total installed circuit cost, with the remainder covering interconnecting piping, structural steel, instrumentation, electrical distribution, civil foundations, and engineering. The midpoint fraction (40%) is used for the midpoint scenario, with the conservative (30%) and optimistic (50%) fractions applied to the respective bounds:

Table C.56. CAPEX derivation for the `acid_leach_circuit` block (3-tank cascade).

Scenario	Per-tank, inst.	3-tank subtotal	Vessel frac.	`prim_equip_price_base`
Conservative	$3.75M	$11.25M	30%	$37.7M
Midpoint	$2.71M	$8.13M	40%	$20.3M
Optimistic	$1.67M	$5.01M	50%	$10.0M

The CAPEX scaling exponent is 0.60 (six-tenths rule). Equipment life is set to 15 years, shorter than the 20-year default used for most mechanical blocks, reflecting the anticipated replacement cycle for rubber-lined steel tanks exposed to continuous sulfuric acid service.

Electricity.

Electrical demand has two components: agitation and pumping. Agitation is fixed at 35 kW per tank (3 tanks = 105 kW total), derived from the Mixtec high-torque agitator specification at operating volume. Pumping demand is estimated using a volumetric energy-intensity approach rather than detailed hydraulic modeling, collapsing pump head, slurry density, and pump-motor efficiency into a single specific energy coefficient.

For slurry transport, a specific energy intensity of 0.15–0.40 kWh/m^3 is assumed, corresponding to moderate transfer heads of approximately 30–60m, a slurry specific gravity of 1.4–1.5, and overall pump-motor efficiencies of approximately 65%. For wash and recycle water, a lower intensity of 0.10–0.25 kWh/m^3 is applied, with water flow estimated at 10–50% of the slurry volumetric flow rate.

Table C.57. Electricity scenario derivation for the `acid_leach_circuit` block.

Scenario	**Agitation (kW)**	**Slurry pumping (kW)**	**Water pumping (kW)**	**Total pumping (kW)**	`electricity_base_total` **kWh/h**
Conservative	105	440	110	550	655
Midpoint	105	302	48	350	455
Optimistic	105	165	11	200	305

Slurry pumping is the dominant electrical consumer, accounting for 65–84% of total demand across scenarios. The agitation contribution (105 kW) is fixed regardless of scenario, as it is determined by tank volume and impeller specification rather than by head or flow uncertainty.

Steam.

Steam is used to maintain the leach circuit at the target operating temperature of 75–90°C. The heating duty has two components: the gross sensible load required to raise the slurry from ambient feed temperature (∼22°C) to the leach temperature, and a substantial offset from the exothermic heat of dilution of concentrated sulfuric acid.

Acid dilution credit. At the disclosed acid addition rate of 0.49 t H_2SO_4 per tonne prepared ore ([195]) and an ore mass flow of approximately 686,000 kg/h at base scale, the total acid addition is roughly 336,000 kg/h. The integral heat of dilution from 98% to process concentration (15%) is approximately 750 kJ/kg H_2SO_4, yielding approximately 70,000 kW of exothermic heat — sufficient to raise 1,595 t/h of slurry (effective $c_p \approx 2.71$ kJ/(kg·°C) at 43% solids) by approximately 58°C from ambient. Acid dilution alone therefore brings the slurry to approximately 80°C under ambient Nevada desert conditions.

Net steam demand. The residual heating duty after the acid credit is small and strongly dependent on the target leach temperature:

Table C.58. Steam demand derivation for the `acid_leach_circuit` block.

Scenario	**Target T (°C)**	**Gross duty (kW)**	**Net duty after acid credit (kW)**	**`steam_base_total` (kg/h)**
Conservative	90	81,700	11,700	∼21,100
Midpoint	84	74,500	4,500	∼8,000
Optimistic	75	63,700	0	∼0

Steam demand is calculated assuming saturated steam at approximately 130°C (3 bar g) with condensate return at 80°C, giving approximately 2,000 kJ/kg useful heat per kilogram of steam. The midpoint value of ∼8,000 kg/h implies a target leach temperature of approximately 84°C, consistent with the 75–90°C operating range.

We use a constant `steam_base_total = 8,000` kg/h across all scenarios, which corresponds to the midpoint derivation above. The conservative and optimistic scenarios differ substantially (21,100 vs. 0 kg/h), and the constant value should be treated as a central estimate only. At the optimistic end, acid dilution heat alone is expected to be sufficient to reach the target temperature and no external steam input is required. This result is sensitive to the assumed acid concentration, ambient temperature, and heat losses; it should be revisited if detailed heat balance data become available.

Labor.

Labor is 4.5 FTE at base scale with a scaling exponent of 0.2 and `dedicated_labor` = FALSE, assuming one operator per shift for the continuous three-tank leach cascade:

$$\text{FTE} = 1.0 \times 4.5 = 4.5$$

Footprint.

The base plot area is 800 m^2 with a scaling exponent of 0.7, covering the three leach tanks, electrical equipment, and additional civil infrastructure.

Availability.

Process availability is 92% with 3% unplanned downtime and 5% scheduled maintenance, giving 8% total downtime. These values are shared with the sulfate roasting block and reflect the corrosive operating environment. The continuous process type means any unplanned downtime directly affects circuit throughput without buffer.

Key measurements and cross-checks.

- Lithium extraction efficiency (fraction of feed Li reporting to PLS); validated against Thacker Pass disclosed recoveries
- Acid consumption per tonne ore: Roy et al. 2024 report 0.49t H_2SO_4 per tonne prepared ore at 3,270 ppm Li; Mousavinezhad et al. 2025 report 15.97 t H_2SO_4 per tonne LCE produced
- Total circuit residence time (3 h target) consistent with Thacker Pass and Tonopah Flats disclosures
- CAPEX cross-check: midpoint of \$20.3M for a 1,100 m^3/h cascade implies approximately \$18.5/(m^3/h), consistent with the Class 4–5 estimate range for stirred-tank leach circuits in the AusIMM cost database

Notes and flags.

The electricity derivation uses a volumetric energy-intensity approach rather than detailed hydraulic modeling. The assumed specific energy range (0.15–0.40 kWh/m^3 for slurry) encompasses significant uncertainty in actual pump head, which depends on circuit elevation differences and pipe run lengths not specified in the available disclosures.

C.5.3.12 `water_leach`

Sources. Gao et al. [174], Deng and Feng [180], Yan et al. [205], Park et al. [210]; Chinese patents (Nan et al. [211, 212], Zou et al. [213]). Tank costs from AusIMM benchmarks (AusIMM [184]); agitator costs from Mixtec product data (Baguley [458]).

Process description.

The water leach circuit dissolves water-soluble lithium sulfate and other sulfate species from sulfate-roasted calcine into aqueous solution, producing a pregnant leach solution (PLS) for downstream impurity removal. Unlike the acid leach circuit used in the clay pathway, no acid is added — the lithium was converted to water-soluble Li_2SO_4 during upstream sulfate roasting, and leaching proceeds by simple dissolution in hot water at approximately 80°C. The circuit also separates the insoluble leach residue (primarily amorphous silica, alumina, and CaF_2 from fluoride capture during roasting) from the PLS for disposal to tailings.

The feed is roasted calcine slurried with process water to approximately 20 wt% solids. At this low solids fraction and with crystalline rather than clay-derived particles, the slurry is Newtonian and substantially less viscous than the 43 wt% clay feed in the acid leach circuit. The leach medium is water with dissolved sulfate salts rather than sulfuric acid, but the hot sulfate solution (∼80°C) may require lined or stainless steel construction depending on dissolved sulfate concentrations.

Equation of merit.

$$\tau_{\text{res}}\ (\text{h}) = \frac{V_{\text{tank}}\ (\text{m}^3)}{\dot{V}_{\text{slurry}}\ (\text{m}^3/\text{h})} \times N_{\text{tanks}}$$

The circuit is sized to deliver a total residence time of 2 hours (1 hour per tank, 2 tanks in cascade), consistent with reported leaching times of 1–2 hours at 60–90°C for sulfate-roasted lepidolite [180, 205].

Bill of Equipment.

Sizing basis.

The base volume is 60 m^3/h of leaching slurry at 20 wt% solids with calcine grain density 2.7 t/m^3, giving a slurry density of 1.144 t/m^3 and a solids throughput of 13.7 t/h. Each tank has a working volume of 60 m^3 (equal to 1 hour of feed at the base flow rate) and a design volume of 72 m^3 after the `oversizing_factor` = 1.20. Two tanks operate in series for a total circuit residence time of 2 hours.

The `scaling_exponent_throughput` = 0.70, `process_type` = continuous, `dedicated_line` = FALSE. Equipment life is assumed to be 15 years, given the caustic nature of the outputs of the sulfuric acid-based roasting process that feed into this water leach.

Table C.59. Bill of equipment for the `water_leach` block.

Equipment	Machine data used	Source(s)
Leach tanks (2-tank cascade, ∼72 m^3 each)	AusIMM tank benchmarks	AusIMM Monograph 27
Standard agitator systems	Mixtec agitator series	Agitator tank series data
Feed and slurry make-up systems	Not explicit → inferred	—
Slurry and recycle pumping	Not explicit → inferred	—
Steam heating system (jacket or coil)	Not explicit → inferred	—
Balance-of-plant (structural, piping, electrical)	Not explicit → inferred	—

CAPEX.

CAPEX is built bottom-up from AusIMM tank volumetric benchmarks and Mixtec agitator costs, with lower installation factors than the acid leach circuit to reflect the smaller tank sizes and absence of acid storage and dosing equipment.

Tanks. Raw tank cost = AusIMM volumetric cost × total design volume (144 m^3). At \$4,500–\$9,000/m^3 (scenario-dependent), the raw tank cost is \$648K–\$1,296K.

Installation. A factor of 2.0–3.0× bare equipment is applied, consistent with the effective range for other agitated tank leach circuits in the model. The scope includes erection, foundations, piping, electrical distribution, and the steam heating circuit (jacket or coil).

Auxiliaries. Slurry pumps, piping, and instrumentation are applied as 20–40% of the installed cost.

Table C.60. CAPEX derivation for the `water_leach` block (2-tank cascade).

Scenario	Raw equip.	Inst. sub.	Aux.	Tot.	`prim_equip_price_base`
Cons.	\$1,456K	\$4,368K	\$1,747K	\$6,115K	**\$6,100,000**
Mid.	\$1,056K	\$2,640K	\$792K	\$3,432K	**\$3,450,000**
Opt.	\$728K	\$1,456K	\$291K	\$1,747K	**\$1,750,000**

Derivation detail (midpoint row). Raw tanks: \$6,500/m^3 × 144 m^3 = \$936K. Raw agitators: 2 × \$60K = \$120K. Total raw: \$1,056K × 2.5 installation = \$2,640K. Auxiliaries at 30%: \$792K. Total: \$3,432K, rounded to \$3,450K.

Electricity.

Electricity demand is dominated by agitation, which accounts for approximately 70% of the midpoint total. The agitation power intensity (0.25–0.65 kW/m^3) is substantially lower than the 0.5–1.5 kW/m^3 used in the acid leach circuit because the 20 wt% calcine slurry is Newtonian and non-viscous. Pumping loads are modest given the low volumetric throughput.

Table C.61. Electricity derivation for the `water_leach` block.

Component	**Opt. (kW)**	**Mid. (kW)**	**Cons. (kW)**
Agitation (2 × 60 m^3)	30	48	78
Slurry and transfer pumps	6	12	20
Heating circuit pump	2	4	6
Instrumentation and auxiliaries	2	3	4
`electricity_base_total`	**40 kWh/h**	**67 kWh/h**	**108 kWh/h**

Steam.

We assume no additional steam `steam_base_total` = 0 across all scenarios. The water leach circuit operates at approximately 80°C, and the zero steam value reflects the assumption that residual sensible heat from the upstream sulfate roasting step (which operates at 800–950°C) is sufficient to bring the calcine–water slurry to the target leach temperature without external steam input. The gross heating duty is approximately 3,700 kW (68.6 t/h of slurry raised by 55°C), which is small relative to the thermal energy available in the hot calcine exiting the roaster.

Other utilities.

Compressed air is consumed at 30 Nm^3/h for tank decontamination and instrumentation.

Consumables.

`tooling_cost_base` = 0. No reagents are consumed in the water leach itself; the leaching medium is process water dosed at the production step level.

Labor.

Labor is 3.15 FTE at base scale with a scaling exponent of 0.2 and `dedicated_labor` = FALSE. The operator count is lower than the acid leach circuit (4.5 FTE) because no acid handling or dosing is required:

$$\text{FTE} = 0.7 \times 4.5 = 3.15$$

Footprint.

The base plot area is 350 m^2 with a scaling exponent of 0.6, covering the two leach tanks, slurry pumps, piping, and access.

Availability.

Process availability is 92% with 3% unplanned downtime and 5% scheduled maintenance. The continuous process type means any unplanned downtime directly affects circuit throughput without buffer.

Maintenance.

Neither `maint_override` nor `opex_fraction_of_capex` is populated; the block inherits the project-level default maintenance rate of 5% of CAPEX. The mild steel construction and absence of corrosive reagents give this block lower maintenance requirements than the acid leach circuit.

Key measurements and cross-checks.

- *CAPEX*: the midpoint of \$3.45M is built from AusIMM tanks at \$6,500/m^3 plus Mixtec agitators, with a 2.5× installation factor and 30% auxiliaries. The installation factor is consistent with other agitated tank leach circuits in the model.
- *Electricity*: Agitation dominates (∼70% of midpoint) at a power intensity of 0.40 kW/m^3 — approximately half the acid leach circuit's intensity, reflecting the less viscous 20 wt% calcine slurry.
- *Steam*: the zero steam demand assumes that residual heat from the upstream sulfate roasting circuit (800–950°C) is sufficient to reach the 80°C leach temperature. The gross heating duty is approximately 3,700 kW, which is small relative to the thermal energy in the hot calcine.
- *Residence time*: 2 hours total (1 h × 2 tanks) is consistent with the 1–2 h range reported in the literature for sulfate-roasted lepidolite water leaching at 60–90°C [180, 205].

C.5.3.13 `solid_liquid_separation`

Sources. Filter press costs from published benchmarks (PORVOO [459]).

Process description.

The solid–liquid separation block performs polishing clarification of a low-solids process solution by direct pressure filtration. It is used in the Jianxiawo (lepidolite) pathway to remove residual precipitates (<1 wt% solids) from neutralized pregnant leach solution after staged neutralization, yielding a clarified lithium sulfate solution suitable for downstream carbonate precipitation.

At the low solids loadings in this application (∼0.5 wt%, or approximately 13 t/day of precipitate fines at base scale), the block assumes a pressure leaf filter or a small plate-and-frame filter press as the primary equipment.

Bill of Equipment.

Table C.62. Bill of equipment for the `solid_liquid_separation` block.

Equipment	Machine data used	Source(s)
Pressure leaf filter or plate-and-frame press (470–630 mm plates)	Porvoo 2025 benchmarks	PORVOO [459]
Feed pump (centrifugal, 100 m^3/h)	Not explicit → inferred	—
Backwash system	Not explicit → inferred	—
Coagulant/flocculant dosing	Not explicit → inferred	—
Instrumentation & controls	Not explicit → inferred	—

No gravity thickener is included. The entire liquid flow passes through the filter; solids are collected on the filter media and periodically discharged. This distinguishes the block from `thickener_generic` and `multi_stage_ccd_thickening`, both of which use gravity settling as the primary separation mechanism.

Sizing basis.

The base volume is 100 m^3/h of feed solution to be clarified. At ∼0.5 wt% solids and a feed density of ∼1.07 t/m^3, the solids throughput is approximately 0.5 t/h (13 t/day). The

filter is sized by liquid throughput and filtration rate rather than by solids settling flux. At a specific filtration rate of 3–8 $m^3/m^2/h$ (typical for polishing-grade pressure filtration), the required filtration area is 12–33 m^2, which is within the capacity of a single 470–630 mm plate-and-frame press or a bank of pressure leaf filter elements.

The `oversizing_factor` = 1.15, `scaling_exponent` = 0.70 (throughput) / 0.60 (equipment), `process_type` = continuous, `dedicated_line` = FALSE. Equipment life is 20 years.

CAPEX.

CAPEX is built bottom-up from the Porvoo filter press benchmark plus auxiliaries. No gravity thickener is included, so the cost is substantially lower than for the `thickener_generic` block.

Polishing filter. Bare equipment costs for 470–630 mm plate-and-frame presses handling 60–150 m^3/h are \$45,000–\$160,000 from published benchmarks [459]. At 100 m^3/h, a single 630 mm press is adequate for the midpoint and optimistic scenarios; the conservative scenario assumes a larger press or parallel units to provide redundancy.

Auxiliaries. The feed pump, backwash system, coagulant dosing, and instrumentation are estimated at \$15,000–\$50,000 depending on the complexity of the backwash and chemical dosing arrangement.

Total plant cost factor. A factor of 2.0–3.0× bare equipment is applied, reflecting the simple installation (no large foundations, no concrete tank, no rake mechanism).

Table C.63. CAPEX derivation for the `solid_liquid_separation` block.

Scenario	**Filter bare**	**Aux bare**	**Tot bare**	**PF**	`prim_equip_price_base`
Cons.	\$160,000	\$50,000	\$210,000	3.0×	**\$630,000**
Mid.	\$100,000	\$30,000	\$130,000	2.5×	**\$325,000**
Opt.	\$55,000	\$15,000	\$70,000	2.0×	**\$140,000**

Derivation detail (midpoint row). A single 630 mm plate-and-frame press at \$100,000 bare handles the full 100 m^3/h liquid throughput at 5 $m^3/m^2/h$ filtration rate (20 m^2 required area). Auxiliaries (feed pump, backwash, coagulant dosing) add \$30,000. The total bare equipment cost of \$130,000 at a 2.5× plant factor gives \$325,000 installed.

Electricity.

Electricity demand is dominated by the feed pump, which must push the full 100 m^3/h liquid flow through the filter media at 2–5 bar pressure drop.

Table C.64. Electricity derivation for the `solid_liquid_separation` block.

Component	**Opt. (kW)**	**Mid. (kW)**	**Cons. (kW)**
Feed pump (100 m^3/h through filter)	5	8	15
Backwash pump (intermittent)	2	3	6
Compressed air (cake discharge)	1	2	5
Filter mechanism (plate shifter if press)	2	2	4
Coagulant/flocculant dosing	1	1	2
Instrumentation & controls	2	2	3
`electricity_base_total`	**13**	**18**	**35**

The feed pump accounts for approximately 45% of the midpoint total, consistent with a configuration where the dominant energy cost is pushing liquid through filter media rather than settling solids by gravity. The scenario spread (13–35 kWh/h) is driven primarily by the assumed filter pressure drop (2 bar optimistic, 3 bar midpoint, 5 bar conservative), which affects the feed pump power via $P = \Delta p \cdot Q/\eta$.

Feed pump cross-check (midpoint). At 100 m^3/h = 0.028 m^3/s, 3 bar = 300 kPa, and $\eta = 0.60$:

$$P = \frac{\Delta p \times Q}{\eta} = \frac{300{,}000 \times 0.028}{0.60} = 14{,}000 \text{ W} = 14 \text{ kW (installed)}$$

The tabulated 8 kW reflects the average draw at partial load (filter media resistance increases during the filtration cycle; the pump operates below design pressure for most of the cycle). An average-to-installed ratio of 0.57 is consistent with centrifugal pump part-load operation.

Consumables.

Filter cloths for the press are a recurring consumable that would be tracked through `tooling_cost_base`, which are currently unmodeled.

Labor.

Labor is 0.9 FTE at base scale with a scaling exponent of 0.2 and `dedicated_labor` = FALSE. The highly automated polishing filter requires only periodic monitoring, backwash initiation, and cloth inspection:

$$\text{FTE} = 0.2 \times 4.5 = 0.9$$

Footprint.

The base plot area is 400 m^2 with a scaling exponent of 0.6. Although the filter press and feed pump alone have a modest footprint, the total includes the coagulant storage and dosing system, backwash water tank, and access.

Availability.

Process availability is 90% with 5% unplanned downtime and 5% scheduled maintenance.

Maintenance.

Neither `maint_override` nor `opex_fraction_of_capex` is populated; the block inherits the project-level default maintenance rate of 5% of CAPEX. The polishing filter has modest maintenance requirements (cloth/element replacement, pump seals).

Key measurements and cross-checks.

- *CAPEX*: the midpoint of \$325K at 100 m^3/h is 19% of the `thickener_generic` midpoint (\$1.70M), consistent with the absence of a gravity thickener and the minimal solids handling equipment. The CAPEX is driven by the filter and feed pump, not by settling infrastructure.

C.5.3.14 `thickener_generic`

Sources.

SGS Canada et al. [150], Peters and Timmerhaus [183], **?**]; vendor data (gravity thickener and filter press cost curves). Thickener bare equipment costs from the CAPCOSTS cost correlation [460]; filter press costs from published filter press price benchmarks [459]. Cross-checked against a published municipal thickening and dewatering project cost estimate of $9.4M–$10.9M installed (ClearGov [461]); rake drive motor power confirmed against manufacturer specification of 5.5 kW for a 24 m center-drive thickener [462].

Process description.

The thickener block combines gravity settling with mechanical dewatering to concentrate a dilute slurry into a dense cake or thickened underflow. A high-rate gravity thickener (with flocculant) settles the feed to an intermediate underflow density, followed by a filter press that mechanically dewaters the underflow to the target output solids concentration. The block is used in three contexts across the model pathways, listed below in decreasing order of solids throughput at the block's base scale.

1. *Thacker Pass clay classification thickening* (15 wt% → 55 wt% solids). Classified smectite clay slurry is thickened to prepare a dense leach feed for acid leaching, with the overflow recycled as process water. This is the heaviest duty and the primary basis for the block's sizing and CAPEX derivation.
2. *Silver Peak brine separation* (∼9 wt% → clarified brine). Precipitated $Mg(OH)_2$ and $CaCO_3$ solids from batch chemical purification are removed by settling and filtration, yielding a clarified lithium brine. The solids throughput is substantially lower than the clay application at comparable volumetric feed rates.
3. *Li_2CO_3 thickening and dewatering* (20 wt% → 70 wt% solids). Precipitated Li_2CO_3 slurry from the soda ash reactor is concentrated for downstream washing and drying. Li_2CO_3 crystals (ρ = 2.11 t/m^3, 20–50 μm) settle rapidly compared to clay, so the thickener is small-diameter at this duty; the filter press handles the final dewatering from ∼35 to 70 wt%.

All three applications share the same two-stage equipment structure (gravity thickener followed by filter press) and differ only in the size of equipment required. The block is parameterized for the heaviest duty (clay thickening at 100 m^3/h), and the model's six-tenths scaling rule reduces the cost for the lighter applications, which involve faster-settling feed materials and lower solids throughputs. Modern high-rate gravity thickeners use polymer flocculant to promote rapid particle aggregation, which can achieve the same throughput at

a fraction of the diameter of a conventional unit. Of note - the Thacker Pass disclosure uses decanter centrifuges rather than filter presses for the dewatering stage; this block uses filter presses as the generic dewatering equipment for broader applicability.

Bill of Equipment.

Table C.65. Bill of equipment for the `thickener_generic` block.

Equipment	Machine data used	Source(s)
High-rate gravity thickener (17–22 m diameter at base scale)	CAPCOSTS Fig. 190	**?**]
Filter press package (630–800 mm plates)	Porvoo 2025 benchmarks	**?**]
Liquids handling (feed, underflow, overflow, recycle pumps)	Not explicit → inferred	—
Output discharge equipment	Not explicit → inferred	—
Auxiliaries (utilities, civil, structural)	Not explicit → inferred	—

Sizing basis.

The base volume is 100 m^3/h of thickener feed slurry at 15 wt% solids with grain density 2.6 t/m^3, giving a slurry density of 1.102 t/m^3 and a solids throughput of 16.5 t/h (396 t/day). Design capacity is 115 m^3/h after the `oversizing_factor` = 1.15. The `scaling_exponent` = 0.60 (six-tenths rule). Lines are not dedicated (`dedicated_line` = FALSE). Equipment life is 15 years.

The gravity thickener is sized from the solids settling flux. The Thacker Pass equipment list specifies a single 55 m diameter classification thickener handling approximately 3,600 tpd of solids, implying a unit settling area of 7.1 ft^2/tpd [150]. This falls within the copper-tailings range of 4–10 ft^2/tpd tabulated in CAPCOSTS [**?** , p. 270], confirming that lithium clay with flocculant (dosed at 260 g/t per the disclosure) settles in the base-metal range rather than the potash or uranium-precipitate range.

At the base volume of 100 m^3/h (396 tpd solids), the required settling area at 6–10 ft^2/tpd is 2,400–3,960 ft^2, corresponding to a thickener diameter of 55–71 ft (17–22 m). The filter press handles the thickener underflow (∼45 m^3/h at 30 wt% solids), pressing it to the 55 wt% target.

CAPEX.

CAPEX is built bottom-up from the CAPCOSTS high-capacity thickener correlation and the Porvoo filter press benchmark.

Thickener. The CAPCOSTS correlation for high-capacity (flocculant-assisted) thickeners is [**?** , Figure 190, p. 271]:

$$\text{Price}_{1998} = 5{,}809 \times D^{0.8786} \quad \text{(USD, } D \text{ in feet, range 16–100 ft, M\&S = 1400)}$$

The price includes the drive, motor, lift, controls, rake mechanism (mild steel), tank shell, and anchor channel. It excludes erection, foundations, freight, and remote controls. Prices are inflated from 1998 to 2025 using the CEPCI ratio of $800/390 \approx 2.05$.

Filter press. Bare equipment costs for 630–800 mm plate filter presses are taken from the Porvoo benchmark at \$75,000–\$200,000 [**?**].

Total plant cost factor. The bare equipment total (thickener + filter press) is multiplied by a total plant cost factor of 2.5–3.5× to account for erection, foundations, piping, electrical, and instrumentation.

Table C.66. CAPEX derivation for the `thickener_generic` block.

Scenario	**SA (ft^2/tpd)**	**D (ft)**	**Thk bare**	**FP bare**	**Tot bare**	**PF**	`prim_equip_price_base`
Cons.	10	71	\$504K	\$200K	\$704K	3.5×	**\$2,450,000**
Mid.	7	59	\$431K	\$130K	\$561K	3.0×	**\$1,700,000**
Opt.	6	55	\$403K	\$75K	\$478K	2.5×	**\$1,200,000**

Derivation detail (midpoint row). The settling area of 7 ft^2/tpd (rounded from the Thacker Pass value of 7.1) at 396 tpd gives a required area of 2,772 ft^2 and a thickener diameter of 59 ft (18 m). The Mular correlation gives a bare thickener cost of \$210,200 in 1998 USD, which inflates to \$431,200 in 2025 USD via CEPCI. Adding a 630–800 mm filter press at \$130,000 bare gives a total bare equipment cost of \$561,200. Applying a total plant cost factor of 3.0× yields the installed cost of \$1,684,000, rounded to \$1,700,000.

Electricity.

Electricity demand is the sum of continuous loads from the gravity thickener (rake drive, underflow pump, flocculant dosing) and cycle-averaged loads from the filter press (hydraulic

power unit, feed pump, plate shifter, cloth wash, and compressed air). The filter press operates in batch cycles while the thickener runs continuously; the cycle-averaged loads for the press reflect intermittent operation at 50–70% load factor.

The rake drive power is anchored to two independent data points: the Thacker Pass classification thickener at 30 kW for a 55 m unit (Table 14-4 of SGS Canada et al. [150]), and a manufacturer specification of 5.5 kW for a 24 m unit [462]. Both data points are consistent with power scaling as D^2:

$$P_{\text{rake}} = 30 \times (18/55)^2 = 3.2 \text{ kW}; \quad P_{\text{rake}} = 5.5 \times (18/24)^2 = 3.1 \text{ kW}$$

The midpoint thickener at 18 m diameter therefore draws approximately 3 kW for the rake mechanism.

Table C.67. Electricity derivation for the `thickener_generic` block.

Component	**Opt. (kW)**	**Mid. (kW)**	**Cons. (kW)**
Thickener rake drive	3	3	3
Underflow pump	5	8	12
Overflow pump + flocculant dosing	4	5	5
Filter press hydraulic unit (cycle-avg.)	8	11	15
Filter press feed pump (cycle-avg.)	8	10	14
Press auxiliaries (shifter, air, cloth wash)	3	5	7
Instrumentation & controls	2	3	3
Total	**33**	**45**	**59**
`electricity_base_total`	**35 kWh/h**	**45 kWh/h**	**60 kWh/h**

The thickener's continuous loads (rake, pumps, flocculant) account for approximately 36% of the midpoint total, with the filter press cycle-averaged loads accounting for the remainder. The Thacker Pass disclosure uses decanter centrifuges for the dewatering stage, which are substantially more power-intensive (355 kW main drive per unit); the filter press assumption in this block gives a lower electricity draw and represents a generic rather than Thacker

Pass-specific configuration.

Consumables.

Polymer flocculant is tracked at the production step level as a secondary input with a dosage of 2.6×10^{-4} kg/m^3 of slurry feed, calibrated to the Thacker Pass disclosed consumption of approximately 260 g/t of solids [150]. Filter cloths for the press are a recurring consumable tracked through `tooling_cost_base`, which are currently unmodeled.

Labor.

Labor is 1.8 FTE at base scale with a scaling exponent of 0.2 and `dedicated_labor` = FALSE, consistent with a highly automated thickener (e.g. automated rake controls and flocculant dosing) combined with a semi-automated filter press requiring periodic supervisory attention for cake discharge and cloth condition:

$$\text{FTE} = 0.4 \times 4.5 = 1.8$$

Footprint.

The base plot area is 600 m^2 with a scaling exponent of 0.7. At the midpoint thickener diameter of 18 m, the plan area of the thickener tank alone is approximately 250 m^2, leaving 350 m^2 for the filter press, piping, and access.

Availability.

Process availability is 92% with 3% unplanned downtime and 5% scheduled maintenance. The high availability reflects the robust, low-complexity nature of gravity settling operations. The filter press component has a somewhat lower individual availability, but the combined block maintains 92% overall because the thickener provides buffer capacity in its underflow inventory.

Maintenance.

Neither `maint_override` nor `opex_fraction_of_capex` is populated; the block inherits the project-level default maintenance rate of 5% of CAPEX. The gravity thickener has modest maintenance requirements (rake bearings, overflow weir adjustment), while the filter press requires regular cloth replacement and hydraulic seal maintenance.

C.5.3.15 `multi_stage_ccd_thickening`

Sources.

SGS Canada et al. [150], Mular and Poulin [460]; filter press costs from published benchmarks (PORVOO [459]).

Process description.

The multi-stage CCD block separates the pregnant leach solution (PLS) from the solid leach residue through eight stages of counter-current decantation, recovering dissolved lithium from entrained solution while minimizing lithium losses to the filter cake. The input is post-neutralization slurry at approximately 34 wt% solids. In each stage, the slurry is fed to a thickener where the solids settle under gravity (aided by flocculant), producing a clarified overflow and a dense underflow. Fresh wash water enters the last stage and flows counter-current to the solids, progressively diluting the entrained mother liquor. The first-stage overflow is the clarified PLS containing the dissolved lithium; the eighth-stage underflow is the washed leach residue routed to final filtration.

The Thacker Pass disclosure describes the CCD thickeners as "Hi-Density" units, achieving a denser underflow (∼33 wt% solids per stage) than a conventional high-rate thickener. The block also includes a final leach-residue filter press to dewater the eighth-stage underflow for disposal or tailings stacking.

Bill of Equipment.

Sizing basis.

The base volume is 100 m^3/h of CCD feed slurry at 34 wt% solids. At grain density 2.6 t/m^3, the slurry density is 1.265 t/m^3 and the solids throughput is 43.0 t/h (1,032 t/day). The `oversizing_factor` = 1.15, `scaling_exponent` = 0.60, `process_type` = continuous, `dedicated_line` = FALSE. Equipment life is assumed to be 15 years.

The settling area per CCD stage is anchored to the Thacker Pass disclosure: eight 36 m CCD thickeners handle an estimated 2,900 tpd of solids (derived as 3,619 tpd of classified ore × 0.70 dissolution residue × 1.15 neutralization precipitates). Each thickener has a settling area of 10,956 ft^2, giving an implied unit settling area of 3.8 ft^2/tpd. This is lower than the classification thickener's 7.1 ft^2/tpd, which is physically consistent: the post-leach residue (predominantly amorphous silica) should settle faster than raw smectite clay. The scenarios use 2.5 / 3.8 / 5.0 ft^2/tpd for optimistic / midpoint / conservative, bracketing the Thacker Pass anchor.

CAPEX.

Table C.68. Bill of equipment for the `multi_stage_ccd_thickening` block.

Equipment	Machine data used	Source(s)
8 Hi-Density CCD thickeners (17–25 m diameter at base scale)	CAPCOSTS Fig. 190	Mular and Poulin [460]
Leach residue filter press (630–800 mm plates)	Porvoo 2025	PORVOO [459]
Underflow and wash-water pumps (per stage)	Not explicit → inferred	—
Flocculant dosing system	Not explicit → inferred	—
Counter-current piping manifolds	Not explicit → inferred	—
Instrumentation & controls	Not explicit → inferred	—

CAPEX is built bottom-up from the CAPCOSTS high-capacity thickener correlation applied to each of the 8 CCD stages, plus auxiliary equipment.

Thickener per stage. The Mular correlation for high-capacity thickeners (16–100 ft range) is $\text{Price}_{1998} = 5,809 \times D^{0.8786}$, inflated to 2025 via CEPCI (×2.05). Each CCD stage requires settling area = SA × 1,032 tpd, giving diameter $D = \sqrt{4A/\pi}$. The eight-stage bare thickener cost is 8× the per-stage cost.

Auxiliaries. The CCD circuit requires underflow pumps (one per stage), wash-water distribution pumps, a shared flocculant dosing system, counter-current piping manifolds, and a leach-residue filter press. Auxiliary costs are estimated at \$350K–\$900K (scenario-dependent) based on the Porvoo filter press benchmark (\$75K–\$200K) plus inferred pump and piping costs.

Plant factor. The total plant cost factor for an 8-stage CCD circuit is lower than the 2.5–3.5× used for a standalone thickener because the eight thickeners share common foundations, flocculant systems, piping manifolds, control systems, and EPCM. A factor of 1.3–1.7× is applied.

Derivation detail (midpoint row). At 3.8 ft^2/tpd and 1,032 tpd, the required area per stage is 3,922 ft^2, giving $D = 70.7$ ft (21.5 m). The Mular correlation gives \$244K per stage in 1998 USD, inflating to \$500K in 2025. Eight stages total \$4,000K bare. Adding \$600K in auxiliaries gives \$4,600K total bare, which at 1.5× plant factor yields \$6,900K installed.

Table C.69. CAPEX derivation for the `multi_stage_ccd_thickening` block.

Scenario	SA	D/stg	8×stg bare	Aux	Tot bare	PF	prim_equip_price_base
Cons.	5.0	81 ft	$4,530K	$900K	$5,430K	1.7×	**$9,200,000**
Mid.	3.8	70 ft	$4,000K	$600K	$4,600K	1.5×	**$6,900,000**
Opt.	2.5	57 ft	$3,340K	$350K	$3,690K	1.3×	**$4,800,000**

Electricity.

Electricity demand is derived from two independently anchored components per CCD stage: a rake drive and an underflow pump, plus shared loads.

Rake drive. The rake motor power is anchored to the Jinhao Mining NZ-24 specification of 5.5 kW for a 24 m center-drive thickener (a pure rake motor rating), and scaled to the per-stage diameter via $P \propto D^2$ [462]. At the midpoint diameter of 21.5 m, the rake power per stage is $5.5 \times (21.5/24)^2 = 4.4$ kW.

Underflow pump. Each stage's underflow at 33 wt% solids (density 1.255 t/m^3) is pumped to the next stage. At 100 m^3/h CCD feed with 43 t/h solids, the underflow volumetric flow per stage is 43.0/0.33/1.255 = 103.8 m^3/h = 0.0288 m^3/s. Pump power is $P = \rho g Q H/\eta = 1255 \times 9.81 \times 0.0288 \times 10/0.60 = 5.9$ kW per stage at 10 m head and 60% efficiency.

Cross-check at Thacker Pass scale. At Phase 1 (36 m thickeners, ∼282 m^3/h CCD feed), the same derivation gives 12.4 kW rake +16.6 kW pump = 29.0 kW per stage, consistent with the disclosed 30 kW per thickener (Table 14-4 of SGS Canada et al. [150]).

Table C.70. Electricity derivation for the `multi_stage_ccd_thickening` block.

Component	Opt. (kW)	Mid. (kW)	Cons. (kW)
Rake drive per stage (D^2 from Jinhao)	2.8	4.2	6.0
Underflow pump per stage ($P = \rho g Q H/\eta$)	4.7	5.9	7.1
× 8 stages	60	81	105
Shared (flocculant, I&C, filter)	8	8	8
`electricity_base_total` (Rounded)	**70 kWh/h**	**90 kWh/h**	**115 kWh/h**

The scenario variation in the underflow pump (4.7–7.1 kW) reflects different assumed pumping heads: 8 m for optimistic (compact installation), 10 m for midpoint, and 12 m for

conservative (taller thickeners with longer transfer lines). The rake and pump components contribute roughly equally to the per-stage total at the midpoint (4.2 vs 5.9 kW), with the pump dominating slightly because of the dense underflow.

Consumables.

Flocculant is tracked at the production step level at 0.04 kg/m^3 of CCD feed, calibrated to the Thacker Pass disclosure's reported polymer dosage for the CCD circuit. Filter cloths for the press are a recurring consumable tracked through `tooling_cost_base`, which are currently unmodeled.

Labor.

Labor is 1.8 FTE at base scale with a scaling exponent of 0.2 and `dedicated_labor` = FALSE. The low operator count reflects the highly automated nature of CCD circuits, where counter-current flow rates, flocculant dosing, and underflow densities could be managed automatically with minimal manual intervention:

$$\text{FTE} = 0.4 \times 4.5 = 1.8$$

Footprint.

The base plot area is 3,500 m^2 with a scaling exponent of 0.7. At the midpoint thickener diameter of 21.5 m per stage, the plan area of eight thickeners arranged in a double row is approximately $8 \times \pi \times 10.75^2 = 2,903$ m^2.

Availability.

Process availability is 92% with 3% unplanned downtime and 5% scheduled maintenance. The CCD circuit has inherent redundancy in that the loss of a single stage degrades washing efficiency but does not halt production; however, pump failures in the counter-current circuit could cascade rapidly.

Maintenance.

Neither `maint_override` nor `opex_fraction_of_capex` is populated; the block inherits the project-level default maintenance rate of 5% of CAPEX. The CCD thickeners themselves have low maintenance requirements (rake bearings, flocculant injection nozzles), but the inter-stage pumps and final filter press have higher wear rates due to the abrasive post-leach slurry.

Key measurements and cross-checks.

- *CAPEX*: the midpoint of \$6.9M represents eight 21 m Hi-Density thickeners plus a filter press and auxiliaries, at a 1.5× total plant cost factor for the series installation. The implied settling area of 3.8 ft^2/tpd per stage is anchored to the Thacker Pass 36 m

CCD thickeners and falls below the Cu-tail range (4–10), consistent with faster-settling post-leach amorphous silica residue.

- *Thacker Pass note*: the actual Thacker Pass Phase 1 CCD uses 8 thickeners at 36 m diameter for an estimated throughput of ∼280 m^3/h. The model would represent this as ∼2.8 machines at base_volume = 100 m^3/h, scaled via the six-tenths rule: CAPEX = \$6.9M × $2.8^{0.6}$ = \$13.0M. This is broadly consistent with the scale of the Thacker Pass process plant capital budget.

C.5.3.16 `mechanical_dewatering`

Sources.

Mular and Poulin [460], PORVOO [459]. Bare equipment costs are cross-checked against both the CAPCOSTS vertical automated plate-and-frame filter press correlation (Figure 182, p. 263) [460] and the Porvoo published filter press price benchmarks [459].

Process description.

The mechanical dewatering block converts thickened or classified slurry into a handleable filter cake by pressure filtration, removing interstitial water through a combination of hydraulic pressure and membrane squeeze. The block is sized for the lepidolite dewatering duty in the Jianxiawo pathway, where ground lepidolite concentrate at 60 wt% solids is dewatered to 80 wt% solids prior to sulfate roasting. The roasting train requires a low-moisture feed; excess water would consume thermal energy in evaporation and could cause handling problems.

The primary equipment is a bank of automated plate-and-frame filter presses operating in parallel batch cycles. At 60 wt% feed solids (already well within the plate-and-frame operating range of 25–60%), no upstream belt filter stage is assumed — the presses receive the dense slurry directly from the grinding circuit.

Bill of Equipment.

Sizing basis.

The base volume is 60 m^3 per batch of slurry being pressed. At 60 wt% solids with grain density 2.7 t/m^3, the slurry density is 1.607 t/m^3. Each batch cycle is 1.2 hours (1.0 hour pressing + 0.2 hours setup and cake discharge), giving an effective throughput of 50 m^3/h. The solids throughput is $50 \times 1.607 \times 0.60 = 48.2$ t/h. The filtrate production is approximately 20 m^3/h (the difference between the water content at 60 wt% and 80 wt% solids).

The product cake at 80 wt% solids has a density of $1/(0.8/2.7 + 0.2/1.0) = 2.015$ t/m^3 and a volume of approximately 36 m^3 per batch. This cake volume determines the number of filter presses required: each 1000 mm press with 60 chambers and 40 mm cake thickness holds approximately 1.9 m^3 per batch, so 19 presses would be needed at this plate size. Using larger 1500 mm presses with 100 chambers (7.8 m^3 per press) reduces this to 5 units.

The `oversizing_factor` = 1.25, `prim_equip_scaling_exponent` = 0.70, `scaling_exponent_throughput` = 1.0 (linear throughput scaling reflects the modular nature of parallel presses). Equipment life is 20 years.

CAPEX.

Table C.71. Bill of equipment for the `mechanical_dewatering` block.

Equipment	Machine data used	Source(s)
Automated plate-and-frame filter presses (5–8 units, 1000–1500 mm plates)	CAPCOSTS Fig. 182; Porvoo 2025	**? ?**]
Feed slurry pumps and piping	Not explicit → inferred	—
Cake discharge conveyors	Not explicit → inferred	—
Filtrate collection and pumping	Not explicit → inferred	—
Cloth washing system	Included in press packages	Vendor standard
Compressed air system (membrane squeeze)	Not explicit → inferred	—
Structural platform and access	Not explicit → inferred	—
Instrumentation and controls	Not explicit → inferred	—

CAPEX is built bottom-up from the per-press bare equipment cost multiplied by the number of presses, plus auxiliaries and a total plant cost factor.

Filter presses. Per-press bare costs are anchored to two sources. The CAPCOSTS automated vertical plate-and-frame correlation at 800 ft^2 (74 m^2) filtration area gives \$146,000 in 1998 USD, inflating to \$300,000 in 2025 via CEPCI [460, Figure 182]. The Porvoo benchmark gives \$150,000–\$350,000 for a 1000 mm press [459]. These are consistent, and a per-press bare cost of \$165,000–\$200,000 is adopted for the midpoint and optimistic scenarios (standard automated presses), with the conservative scenario reflecting larger or more heavily instrumented units.

Auxiliaries. Feed pumps, cake conveyors, filtrate collection, compressed air, structural platform, and instrumentation are estimated at \$50,000–\$150,000 (scenario-dependent), reflecting shared infrastructure across the parallel press bank.

Total plant cost factor. A factor of 2.0–3.0× bare equipment is applied, consistent with the Mular ratio method for skid-mounted mechanical equipment with elevated platforms and chemical-resistant materials.

Table C.72. CAPEX derivation for the `mechanical_dewatering` block.

Scenario	Presses	Press bare	Aux	PF	`prim_equip_price_base`
Cons.	8 × \$165K	\$1,320K	\$150K	3.0×	**\$4,400,000**
Mid.	6 × \$175K	\$1,050K	\$80K	2.5×	**\$2,800,000**
Opt.	5 × \$200K	\$1,000K	\$50K	2.0×	**\$2,100,000**

Derivation detail (midpoint row). Six automated plate-and-frame presses at \$175,000 each give \$1,050,000 in bare press costs. Auxiliaries (shared feed pumps, cake conveyors, filtrate system, compressed air, platform, and instrumentation) add \$80,000. The total bare equipment cost of \$1,130,000 at a 2.5× plant factor yields \$2,825,000, rounded to \$2,800,000.
The scenario variation in press count (5–8 units) reflects uncertainty in the achievable cake thickness and cycle time: more presses are needed if the cake forms more slowly or if thicker cakes are required for handling. The optimistic scenario assumes 1500 mm plates with 7.8 m^3 chamber volume per press (5 units); the conservative scenario assumes smaller or more numerous presses to provide redundancy.

Electricity.

Electricity demand is dominated by the hydraulic power units (HPUs) of the parallel press bank, which provide the clamping force and membrane squeeze pressure. Each press has an installed HPU motor of 15–40 kW (scenario-dependent), operating at a cycle-averaged load factor of 0.55–0.65 because the HPU pressurizes intermittently during squeeze phases and is idle during fill and cake discharge.
HPU derivation (midpoint). Six presses at 30 kW installed HPU × 0.60 load factor = 108 kW cycle-averaged. The load factor reflects the HPU duty cycle: pressurizing during the squeeze phase (∼40% of cycle time at full load), maintaining pressure during hold (∼20% at partial load), and idle during fill and discharge (∼40%).

Other utilities.

Compressed air is consumed at 2 Nm^3/h for membrane squeeze and cake blow-down. Cooling water is consumed at 0.05 m^3/h for hydraulic oil cooling across the press bank.

Consumables.

`tooling_cost_base` = 0. Filter cloths are a recurring consumable, but replacement costs are not yet populated.

Labor.

Table C.73. Electricity derivation for the `mechanical_dewatering` block.

Component	**Opt. (kW)**	**Mid. (kW)**	**Cons. (kW)**
Press HPUs (cycle-avg., $N \times P_{\text{inst}} \times \text{LF}$)	61	108	208
Feed pump	8	12	18
Cake discharge conveyors	3	5	8
Cloth wash system	2	4	6
Compressed air (membrane squeeze)	3	4	6
Instrumentation and controls	2	3	3
Total	**79**	**136**	**249**
`electricity_base_total`	**80 kWh/h**	**135 kWh/h**	**250 kWh/h**

Labor is 0.9 FTE at base scale with a scaling exponent of 0.2 and `dedicated_labor` = FALSE, consistent with a bank of automated presses requiring periodic supervisory attention for cake discharge quality, cloth condition monitoring, and cycle parameter adjustment:

$$\text{FTE} = 0.2 \times 4.5 = 0.9$$

Footprint.

The base plot area is 250 m^2 with a scaling exponent of 0.7, covering a linear bank of 5–8 plate-and-frame presses with auxiliaries.

Availability.

Process availability is 90% with 5% unplanned downtime and 5% scheduled maintenance. The parallel press configuration provides partial redundancy: the loss of a single press reduces throughput but does not halt production.

Maintenance.

Neither `maint_override` nor `opex_fraction_of_capex` is populated; the block inherits the project-level default maintenance rate of 5% of CAPEX. Filter presses are mechanically intensive equipment with regular cloth replacement, hydraulic seal maintenance, and plate inspection requirements.

Key measurements and cross-checks.

- *CAPEX*: the midpoint of $2.8M for 6 automated presses handling 48 tph solids implies $467K per press installed, or approximately 2.7× the Porvoo bare cost of $175K. This is consistent with the Mular ratio method for skid-mounted mechanical equipment.
- *Electricity*: the midpoint of 135 kWh/h is dominated by HPU loads (80% of total), consistent with a bank of parallel presses where the hydraulic squeeze is the primary energy consumer. The previous CSV values of 506–564 kWh/h matched the Thacker Pass centrifuge rating and have been replaced.
- *Press count*: the 5–8 unit range reflects uncertainty in achievable cake thickness and cycle time. The midpoint (6 presses) assumes 1000 mm plates with 60 chambers at 1.9 m^3 cake per press, requiring 19 loads per batch — but with staggered cycling across 6 presses, each press runs approximately 3 cycles per batch, which is feasible within the 1-hour pressing window.

C.5.3.17 `staged_neutralization`

Sources.

SGS Canada et al. [150], Battaglia et al. [360], Liu and Azimi [381], Deng and Feng [180], Yan et al. [205], AusIMM [184], Peters and Timmerhaus [183]; vendor data. Cost methodology scaled from the `liming_system` block (C.5.3.19).

Process description.

The staged neutralization block removes dissolved metal impurities from pregnant leach solution (PLS) through multi-stage pH-controlled precipitation. Target species include Al^{3+}, Fe^{3+}, Mg^{2+}, and Ca^{2+}, with the specific ionic loading depending on the upstream pathway. Typical PLS impurity concentrations entering this block are: Al 50–200 mg/L, Fe 10–50 mg/L, Mg 200–800 mg/L. The key reactions are:

$$Fe^{3+}(aq) + 3\,OH^- \longrightarrow Fe(OH)_3(s)\downarrow \quad (\text{pH } \sim 3.5\text{–}4) \tag{C.40}$$

$$Al^{3+}(aq) + 3\,OH^- \longrightarrow Al(OH)_3(s)\downarrow \quad (\text{pH } \sim 4.5\text{–}5.5) \tag{C.41}$$

$$Mg^{2+}(aq) + Ca(OH)_2 \longrightarrow Mg(OH)_2(s)\downarrow + Ca^{2+}(aq) \quad (\text{pH } \sim 9\text{–}10) \tag{C.42}$$

Each species precipitates at a distinct pH range, necessitating staged addition of neutralizing agents (limestone for bulk acid neutralization, lime for Mg removal, soda ash for residual Ca polishing) with intermediate settling or clarification between stages. This block is modeled as a more complex variant of the `liming_system` block: it handles a more diverse ionic loading, requires multiple pH-controlled stages, and serves a different functional role (impurity removal from PLS while keeping lithium in solution, rather than simple brine softening).

This block is used in both the Thacker Pass and Jianxiawo pathways: for bulk acid neutralization and Fe/Al precipitation following sulfuric acid leaching of clay (Thacker Pass), for Mg/Ca removal from washed PLS (Thacker Pass impurity removal), and for staged neutralization of water leach PLS from lepidolite calcine (Jianxiawo). Reagent dosages and elimination fractions are defined at the production step level and vary by pathway; the block provides the shared equipment-level cost and utility basis.

Equation of merit.

$$V_{\text{working}}\ (\text{m}^3) = \dot{V}_{\text{PLS}}\ (\text{m}^3/\text{h}) \times \tau_{\text{total}}\ (\text{h})$$

The working volume of the multi-stage circuit V_{working} is sized to deliver a PLS throughput $\dot{V}_{\text{PLS}}$ with total residence time τ_{total} across multiple neutralization and conditioning stages. At a base flow of 100 m^3/h and total residence time of ∼1.2 h (consistent with the `liming_`

system and precipitator blocks), the working volume is approximately 120 m^3, distributed across 3–4 reaction stages plus conditioning. The additional stages relative to the liming_system (which uses a single-stage cascade) are the primary driver of the complexity premium applied in the CAPEX derivation.

Bill of Equipment.

Table C.74. Bill of equipment for the staged_neutralization block.

Equipment	Machine data used	Source(s)
Multi-stage neutralization/ conditioning tanks (3–4 stages, ∼120 m^3 total)	Based on liming_ system tank train	Methodology writeup
Lime slaking system (5 m^3 slaker + 25 m^3 storage)	See liming_system	Methodology writeup
Reagent handling (lime, limestone, soda ash)	Not explicit → inferred	—
pH control instrumentation (per-stage probes, dosing loops)	Not explicit → inferred	—
Slurry transfer pumps & piping	Not explicit → inferred	—
Instrumentation & controls	Not explicit → inferred	—

The equipment configuration is more complex than the liming_system: multiple reaction stages operate at different pH setpoints (typically pH ∼3.5 for Fe, pH ∼5 for Al, pH ∼9–10 for Mg), each requiring independent reagent dosing and pH control. Intermediate clarification or settling may occur between stages to remove precipitated solids before the next pH adjustment.

Sizing basis.

The base volume is 100 m^3/h of pregnant leach solution. Design capacity is 115 m^3/h after the oversizing_factor = 1.15. The scaling_exponent = 0.60 (six-tenths rule). Lines are not dedicated (dedicated_line = FALSE). Equipment life is 15 years, matching the liming_system to reflect lime-service wear.

CAPEX.

CAPEX is derived by scaling the liming_system installed cost from 60 m^3/h to 100 m^3/h using the six-tenths rule, then applying a complexity multiplier to account for the

additional stages, instrumentation, and multi-reagent handling:

Step 1: Scale `liming_system` to 100 m^3/h. The scaling factor is $(100/60)^{0.60} = 1.667^{0.60} = 1.359$.

Step 2: Apply complexity multiplier. The multiplier increases from 1.50× (conservative) to 1.75× (optimistic). This inverted pattern reflects a range-compression effect: the staged neutralization requires a minimum equipment configuration (multi-stage pH control, per-stage instrumentation, multi-reagent handling) that establishes a cost floor. This floor prevents the optimistic cost from scaling down as aggressively as the underlying `liming_system` base, compressing the scenario range from 2.07× (`liming_system`) to 1.77×:

Table C.75. CAPEX derivation for the `staged_neutralization` block.

Scenario	**Lime_soft scaled**	**× 1.359**	**Complexity**	`prim_equip_price_base`
Conservative	$1,971K	$2,679K	×1.50	**$4,010,000**
Midpoint	$1,346K	$1,829K	×1.60	**$2,930,000**
Optimistic	$950K	$1,291K	×1.75	**$2,260,000**

Electricity.

Electricity demand is 168 / 120 / 90 kWh/h (conservative / midpoint / optimistic). These values are cross-checked against the `liming_system` block scaled linearly from 50 m^3/h to 100 m^3/h:

Table C.76. Electricity cross-check for the `staged_neutralization` block.

Scenario	`liming_system` **at 100** m^3/h	`electricity_base_total`	**Premium over** `liming_system`
Conservative	150 kWh/h	**172 kWh/h**	∼15%
Midpoint	100 kWh/h	**120 kWh/h**	20%
Optimistic	70 kWh/h	**90 kWh/h**	∼30%

The ∼15–30% premium is consistent with the additional electrical loads of a multi-stage circuit: per-stage pH probes with dosing loops, additional inter-stage transfer pumps, and more intensive mixing at the Fe/Al precipitation stages (which require rapid reagent dispersion to avoid local pH overshoot). The midpoint specific electricity of 1.2 kWh/m^3 is comparable to the `liming_system` (1.0 kWh/m^3 at 50 m^3/h).

Other utilities.

No thermal utilities are consumed. Process water is required for lime slaking if assuming purchasing of quicklime, at an approximately 2:1 water-to-quicklime ratio by weight; this is not tracked as a utility in the block but would need to be further accounted for in the facility water balance.

Consumables.

Limestone ($CaCO_3$), quicklime (CaO), soda ash (Na_2CO_3), and flocculant are tracked at the production step level as secondary inputs with pathway-specific dosages; no tooling is used in this step - `tooling_cost_base` = 0.

Labor.

Labor is 4.5 FTE at base scale with a scaling exponent of 0.2 and `dedicated_labor` = FALSE. The same base FTE as the simpler `liming_system` is used despite the greater operational complexity because the additional stages can be fully automated with controlled pH dosing loops, requiring supervisory rather than manual intervention:

$$\text{FTE} = 1.0 \times 4.5 = 4.5$$

Footprint.

The base plot area is 500 m^2 with a scaling exponent of 0.7, matching the `liming_system`. The additional neutralization stages add modestly to the tank footprint, but the lime slaking and reagent storage systems (which dominate the plot area) are comparable in size.

Availability.

Process availability is 92% with 3% unplanned downtime and 5% scheduled maintenance. The slightly higher availability (vs. 90% for the `precipitator`) reflects the robust operating envelope of neutralization chemistry, which is less sensitive to temperature and stoichiometric control than crystallization circuits.

Maintenance.

Neither `maint_override` nor `opex_fraction_of_capex` is populated; the block inherits the project-level maintenance costs of 5%.

Key measurements and cross-checks.

- *CAPEX*: midpoint of \$2.93M at 100 m^3/h implies \$29.3/(m^3/h), approximately 2.2× the `precipitator`'s \$13/(m^3/h) and 1.6× the `liming_system`'s scaled cost. This premium is consistent with the additional stages and instrumentation.

- *Electricity*: 120 kWh/h midpoint at 100 m^3/h = 1.2 kWh/m^3, comparable to the `liming_system` (1.0) and `batch_treatment` (1.5). The scaling from `liming_system` passes within 20%.
- *Pathway usage*: this block is shared across two pathways (Thacker Pass clay leach and Jianxiawo lepidolite water leach), each with different ionic loadings and reagent combinations. The 100 m^3/h base volume and multi-stage design accommodate both use cases without pathway-specific equipment.

C.5.3.18 `precipitator`

Sources.

Battaglia et al. [360], Liu and Azimi [381], Khakmardan et al. [193], Iyer and Kelly [194], Roy et al. [195], SGS Canada et al. [150], Samari et al. [155], SRK Consulting [178], AusIMM [184], Peters and Timmerhaus [183]; vendor data (e.g. Baguley [458]).

The precipitator block converts dissolved lithium — either as a chloride or sulfate species depending on the upstream pathway, as noted in Appendix C.5.1.6 — into lithium carbonate via reaction with soda ash (Na_2CO_3), yielding Li_2CO_3 as a solid product:

$$2\,\mathrm{LiCl(aq)} + \mathrm{Na_2CO_3(aq)} \longrightarrow \mathrm{Li_2CO_3(s)} + 2\,\mathrm{NaCl(aq)} \tag{C.43}$$

$$\mathrm{Li_2SO_4(aq)} + \mathrm{Na_2CO_3(aq)} \longrightarrow \mathrm{Li_2CO_3(s)} + \mathrm{Na_2SO_4(aq)} \tag{C.44}$$

Precipitation is conducted at 80–95°C to exploit the solubility of Li_2CO_3 and maximize recovery yield. The block encompasses pre-heating of the polished liquor feed, reagent addition, reaction and crystallization, and conditioning/aging for crystal agglomeration prior to downstream solid-liquid separation.

Equation of merit.

$$V_{\text{working}}\ (\mathrm{m^3}) = \dot{V}_{\text{liquor}}\ (\mathrm{m^3}/h) \times \tau_{\text{res}}\ (\mathrm{h})$$

The working volume of the circuit V_{working} is sized to deliver a liquor volume $\dot{V}_{\text{liquor}}$ with total residence time τ_{res}, across a three-tank cascade. [178] For example, with a base liquor flow of 50 $\mathrm{m^3/h}$ and total residence time of 1.2 h, this gives a working volume of 60 $\mathrm{m^3}$.

Bill of Equipment.

Sizing basis.

The base volume is 50,000 L/h (50 $\mathrm{m^3/h}$) of polished brine or purified leach liquor. This corresponds to a reference production rate of approximately 10 t Li_2CO_3/day. The three-tank configuration splits working volume between rapid mixing (∼5 $\mathrm{m^3}$), reaction and crystallization (∼35 $\mathrm{m^3}$), and conditioning/aging (∼10 $\mathrm{m^3}$). Throughput scaling uses an exponent of 0.7, reflecting the sub-linear cost scaling of vessel-dominated circuits with instrumentation.

CAPEX.

Tank costs are scaled from AusIMM rapid mix tank and Mixtec agitator tank benchmarks using the six-tenths rule. The volumetric cost (USD per $\mathrm{m^3}$ of working volume) is applied to the 60 $\mathrm{m^3}$ working volume to obtain the raw tank cost. Subsystem adders cover slurry pumps and piping (20%), soda ash handling and dosing (15%), and instrumentation and controls

Table C.77. Bill of equipment for the `precipitator` block.

Equipment	**Machine data used**	**Source(s)**
Steam preheater	Not explicit → inferred	—
Rapid mix tank (reagent addition, ∼5 m^3)	AusIMM Rapid Mix Tank	Rapid Mix Tanks
Reaction / crystallization tank (∼35 m^3)	Mixtec high-torque agitator series	Agitator Tanks
Conditioning / aging tank (∼10 m^3)	Mixtec high-torque agitator series	Agitator Tanks
Slurry transfer pumps & piping	Not explicit → inferred	—
Reagent handling (Na_2CO_3)	Not explicit → inferred	—
Instrumentation & controls	Not explicit → inferred	—

(10%) applied to the tank cost. An implicit installation factor of ∼1.15× is embedded in the volumetric unit costs, giving an effective total multiplier of approximately 1.67×:

Table C.78. CAPEX derivation for the `precipitator` block.

Scenario	**Volumetric cost**	**Raw tank CAPEX**	**Total mult.**	`prim_equip_price_base`
Conservative	\$9,000/m^3	\$540,000	×1.67	**\$900,000**
Midpoint	\$6,500/m^3	\$390,000	×1.67	**\$650,000**
Optimistic	\$4,800/m^3	\$288,000	×1.67	**\$480,000**

Equipment life is 20 years. The CAPEX scaling exponent is 0.60 (six-tenths rule).

Pre-heating and thermal demand.

Pre-heating the feed liquor to the precipitation temperature is required to exploit the retrograde solubility of Li_2CO_3 and is noted across multiple project disclosures (Thacker Pass, Bonnie Claire, Silver Peak). Precipitation is conducted at 80–95°C. The thermal duty depends on the feed temperature entering the precipitator, which varies significantly between pathways: chloride-pathway feeds (Silver Peak) arrive near ambient (∼22°C) after

batch chemical purification, while sulfate-pathway feeds (Thacker Pass) may retain some heat from upstream processing (∼40–50°C). Given the relatively small amount of material and temperature increase, this heating demand is described here but not yet implemented into the model.

Electricity. Electricity demand for precipitation was estimated as the sum of (i) tank mixing power, (ii) liquor transfer pumping power, and (iii) auxiliary motor loads for reagent make-up and dosing (lime slaking and soda ash dissolution).

Agitation. Mixing power is estimated from the SME Mining Engineering Handbook envelope of 0.2–0.6 kW/m^3, with a midpoint estimate of 0.4 kW/m^3, applied to the 50 m^3/h base flow volume:

Transfer pumping. Estimated from the hydraulic power relation $P = \rho g Q H/\eta$ at $Q = 50/3600 = 0.0139$ m^3/s, $\rho = 1300$ kg/m^3, $\eta = 0.65$, and transfer heads of 20/30/40 m (optimistic/midpoint/conservative), giving 5.5/8.2/10.9kW respectively.

Auxiliary loads (soda ash dissolution, reagent dosing, instrumentation) account for the remaining demand to reach the total:

Table C.79. Electricity scenario derivation for the `precipitator` block.

Scenario	**Mixing (kW)**	**Transfer pump (kW)**	**Auxiliary (kW)**	`electricity_base_total`
Conservative	30	10.9	35	**75 kWh/h**
Midpoint	20	8.2	20	**50 kWh/h**
Optimistic	10	5.5	10	**25 kWh/h**

The auxiliary load (35/20/10 kW) is approximately equal in magnitude to the mixing power across all scenarios, consistent with the additional drive loads for soda ash conveying, dissolution tank agitation, and metering pumps. Total electricity load is then rounded.

Labor.

Labor is 4.5 FTE at base scale with a scaling exponent of 0.2 and `dedicated_labor` = FALSE, assuming one operator per shift for the continuous three-tank precipitation cascade:

$$\text{FTE} = 1.0 \times 4.5 = 4.5$$

Footprint.

The base plot area is 150 m^2 with a scaling exponent of 0.7, covering the three tanks, preheater, reagent handling, and surrounding access.

Availability.

Process availability is 90% with 5% unplanned downtime and 5% scheduled maintenance. The slightly lower availability (relative to the 92% used for leach circuit blocks) reflects the sensitivity of crystallization circuits to upsets in temperature and reagent dosing.

Key measurements and cross-checks.

- Li_2CO_3 precipitation yield (fraction of dissolved Li reporting to product); literature values 85–95% for optimized circuits [360, 381]
- Crystal size distribution and filterability (conditioning/aging tank residence time is the primary lever)
- Soda ash stoichiometry: approximately 2.65 kg Na_2CO_3 per kg Li_2CO_3 produced (theoretical: $M_{Na_2CO_3}/M_{Li_2CO_3} = 106/74 = 1.43$; applied at 85–90% conversion efficiency)
- Midpoint CAPEX of \$650,000 for a 50 $m^3/h/h$ circuit implies \$13 (m^3/h), consistent with Class 4–5 benchmarks for small stirred-tank precipitation at this scale

C.5.3.19 `liming_system`

Sources. Battaglia et al. [360], Liu and Azimi [381], Khakmardan et al. [193], Iyer and Kelly [194], Roy et al. [195], SGS Canada et al. [150], AusIMM [184], Peters and Timmerhaus [183]; vendor data (BHT, Westpro, Sodimate lime slaking systems).

Process description.

The liming system block removes divalent cation impurities—primarily Mg^{2+}, Ca^{2+}, Fe^{2+}, and Mn^{2+}—from brine or pregnant leach solution by reaction with slaked lime ($Ca(OH)_2$). The primary reaction for magnesium removal is:

$$Mg^{2+}(aq) + Ca(OH)_2(s) \longrightarrow Mg(OH)_2(s)\downarrow + Ca^{2+}(aq) \tag{C.45}$$

Precipitated $Mg(OH)_2$ reports to the solids phase, while the calcium introduced by lime addition remains in solution and must be subsequently removed, typically by soda ash addition in a downstream polishing step. Unlike the `precipitator` block, which converts dissolved lithium into a solid product, liming system removes unwanted impurities while keeping lithium in solution. The resulting slurry has a substantially higher solids loading than a lithium carbonate precipitation circuit, requiring more robust agitation and solids-handling equipment. The block encompasses quicklime receiving and slaking, milk-of-lime storage and dosing, rapid reagent mixing, reaction and crystallization in a continuous stirred-tank reactor (CSTR), and conditioning/aging for floc agglomeration prior to downstream solid-liquid separation.

This block is not actively used in the current model but serves as a reference baseline for other tank-based circuits, e.g. `precipitator` C.5.3.18 and `batch_treatment_generic` C.5.3.20.

Equation of merit.

$$V_{\text{circuit}}\ (\text{m}^3) = \dot{V}_{\text{liquor}}\ (\text{m}^3/\text{h}) \times \tau_{\text{circuit}}\ (\text{h})$$

The working volume of the softening circuit V_{circuit} is sized to deliver a liquor flow $\dot{V}_{\text{liquor}}$ with total circuit residence time τ_{circuit} across a three-tank cascade: rapid mix ($\sim$5 m^3, $\tau \approx 6$ min), reaction CSTR ($\sim$35 m^3, $\tau \approx 42$ min), and conditioning ($\sim$20 m^3, $\tau \approx 24$ min), giving $\tau_{\text{circuit}} \approx 1.2$ h and $V_{\text{circuit}} = 60$ m^3 at the 50 m^3/h base flow. An additional 30 m^3 of agitated volume is required for the lime slaking subsystem (5 m^3 detention slaker + 25 m^3 milk-of-lime storage), sized by the required lime production rate rather than the brine throughput. Total agitated volume across the block is 90 m^3.

Bill of Equipment.

Table C.80. Bill of equipment for the `liming_system` block.

Equipment	Machine data used	Source(s)
Detention lime slaker (∼5 m^3)	AusIMM Rapid Mix Tank	Rapid Mix Tanks
Milk-of-lime storage tank (∼25 m^3)	Mixtec Agitator Tank Series	Agitator Tanks
Rapid mix tank — reagent addition (∼5 m^3)	AusIMM Rapid Mix Tank	Rapid Mix Tanks
Reaction / crystallization CSTR (∼35 m^3)	Continuous Stirred Tank Reactor	CSTRs
Conditioning / aging tank (∼20 m^3)	Mixtec Agitator Tank Series	Agitator Tanks
Quicklime feed system (silo, screw feeder)	Not explicit → inferred	—
Grit removal (classifying screen/rake)	Not explicit → inferred	—
Slurry transfer pumps & piping	Not explicit → inferred	—
Lime dosing (metering pumps)	Not explicit → inferred	—
Instrumentation & controls	Not explicit → inferred	—

The lime slaking subsystem is substantially more equipment-intensive than the soda ash handling in the `precipitator` block. A quicklime conditioning system encompasses receiving, slaking, grit removal, slurry storage, and dosing. The solids-handling and mechanical components are expected to dominate the cost of the slaking subsystem.

Sizing basis.

The base volume is 50,000 L/h (50 m^3/h) of brine or leach solution, matching the `precipitator` block throughput for cross-comparability. Design capacity is 60 m^3/h after applying the `oversizing_factor` = 1.2. The `scaling_exponent` = 0.60 (six-tenths rule), consistent with the cost scaling of vessel-dominated circuits. Lines are not dedicated (`dedicated_line` = FALSE). Equipment life is 15 years, shorter than the precipitator's 20-year life to reflect the higher abrasive and scaling demands of lime slurry service.

CAPEX.

CAPEX is built bottom-up from two subsystems: the lime slaking system and the softening circuit proper. Tank costs are scaled from AusIMM rapid mix tank and Mixtec agitator tank volumetric benchmarks, with the same unit costs ($/m^3 of working volume) used in the `precipitator` block. An implicit installation factor of $\sim$1.15$\times$ is embedded in the volumetric unit costs.

Lime slaking system (30 m^3 total tank volume). The raw tank cost covers the 5 m^3 slaker and 25 m^3 storage vessel. A +250% subsystem adder (total 3.5$\times$) is applied to account for the quicklime silo, screw feeder, grit removal, dust scrubber, dosing pumps, and associated piping and controls:

Table C.81. Lime slaking system CAPEX derivation for the `liming_system` block.

Scenario	**Vol. cost**	**Raw tanks**	**$\times$3.5**	**Slaking total**
Conservative	\$9,000/m^3	\$270,000	\$945,000	\$945,000
Midpoint	\$6,500/m^3	\$195,000	\$682,500	\$683,000
Optimistic	\$4,800/m^3	\$144,000	\$504,000	\$504,000

Flag: the +250% subsystem adder is an engineering estimate with no direct vendor or published source. The multiplier is plausible given the number of mechanical components (silo, feeder, grit classifier, dust scrubber) but should be validated against a vendor quote.

Softening circuit (60 m^3 working volume). Subsystem adders are applied to the raw circuit tank cost: slurry pumps and piping (20%), lime handling and dosing (25/40/60% by scenario), and instrumentation and controls (10%). The total is then combined with the lime slaking system:

Table C.82. Softening circuit and total CAPEX derivation for the `liming_system` block.

Scenario	**Raw circuit tanks**	**Adder %**	**Circuit sub.**	**+ Slaking**	`prim_equip_price_base`
Conservative	\$540,000	+90%	\$1,026,000	\$945,000	**\$1,971,000**
Midpoint	\$390,000	+70%	\$663,000	\$683,000	**\$1,346,000**
Optimistic	\$288,000	+55%	\$446,000	\$504,000	**\$950,000**

The adder percentages differ by scenario: the conservative case assumes a larger lime handling and dosing fraction (60%) reflecting corrosion-resistant materials and redundant dosing

pumps, while the optimistic case (25%) reflects a simplified system with standard carbon steel construction.

Electricity.

Electricity demand is rederived from first principles as the sum of (i) tank agitation power across all five agitated vessels, (ii) lime handling system drives, (iii) liquor and slurry transfer pumping, and (iv) auxiliary loads.

Agitation. Mixing power is estimated from the power intensity (kW/m^3) applied to the *working volume* of each agitated vessel, not the throughput flow rate. Intensity ranges are assigned by vessel duty: the lime slaker requires vigorous agitation (1.0–2.0 kW/m^3) to maintain slaking temperature uniformity and prevent drowning; the reaction CSTR uses moderate agitation (0.15–0.40 kW/m^3) for solids suspension, consistent with the 0.2–0.6 kW/m^3 SME envelope applied in the `precipitator` block; and the conditioning tank uses gentle mixing (0.08–0.20 kW/m^3) for floc growth:

Table C.83. Agitation power derivation by vessel for the `liming_system` block.

Vessel	**Vol. (m^3)**	**Optimistic (kW)**	**Midpoint (kW)**	**Conservative (kW)**
Lime slaker	5	5.0 (1.0 kW/m^3)	7.5 (1.5 kW/m^3)	10.0 (2.0 kW/m^3)
Lime storage	25	2.5 (0.10)	3.8 (0.15)	6.3 (0.25)
Rapid mix (circuit)	5	2.5 (0.50)	5.0 (1.0)	7.5 (1.5)
Reaction CSTR	35	5.3 (0.15)	8.8 (0.25)	14.0 (0.40)
Conditioning	20	1.6 (0.08)	2.4 (0.12)	4.0 (0.20)
Agitation subtotal	**90**	**16.9**	**27.4**	**41.8**

Lime handling. The slaking subsystem drives include a screw feeder (3–5 kW), grit removal rake or classifying screen (2–4 kW), and lime dosing metering pump (1–2 kW), totaling 6/8.5/11 kW (optimistic/midpoint/conservative).

Transfer pumping. Estimated from the hydraulic power relation $P = \rho g Q H/\eta$ at $Q = 50/3600 = 0.0139$ m^3/s, $\rho = 1050$ kg/m^3 (dilute brine), $\eta = 0.65$, and transfer heads of 20/30/40 m (optimistic/midpoint/conservative), giving 4.4/6.6/8.8 kW for the main brine pump. An additional 1–2 kW for the lime slurry dosing pump gives a pumping subtotal of 5.4/7.6/10.8 kW.

Auxiliary loads (instrumentation, lighting, sampling systems) account for the remaining 4/6/9 kW.

Table C.84. Electricity scenario derivation for the `liming_system` block.

Scenario	Agitation (kW)	Lime hdlg. (kW)	Pumping (kW)	Aux. (kW)	`elec_base_total`
Conservative	41.8	11	10.8	9	**75 kWh/h**
Midpoint	27.4	8.5	7.6	6	**50 kWh/h**
Optimistic	16.9	6	5.4	4	**35 kWh/h**

Other utilities.

No thermal utilities are consumed. Process water is required for lime slaking if assuming purchasing of quicklime, at an approximately 2:1 water-to-quicklime ratio by weight; this is not tracked as a utility in the block but would need to be further accounted for in the facility water balance.

Consumables.

Equipment-tied consumables (e.g., pump seals, agitator wear parts) are not separately tracked in this model; `tooling_cost_base` = 0.

Labor.

Labor is 4.5 FTE at base scale with a scaling exponent of 0.2 and `dedicated_labor` = FALSE, assuming one operator per shift for the continuous softening circuit. The lime slaking subsystem is assumed to not add incremental operators; modern slaking systems can use automated controls for e.g. lime-to-water ratios, and require only periodic supervisory attention:

$$\text{FTE} = 1.0 \times 4.5 = 4.5$$

Footprint.

The base plot area is 500 m^2 with a scaling exponent of 0.7, covering the softening tanks, lime slaking system (including silo and feed equipment), dosing and pumping skids, and surrounding access. The substantially larger footprint compared to the `precipitator` (150 m^2) reflects the lime receiving and storage infrastructure.

Availability.

Process availability is 92% with 3% unplanned downtime and 5% scheduled maintenance. The slightly higher availability relative to the `precipitator` (90%) reflects the simpler chemistry (no retrograde solubility sensitivity) and more robust operating envelope of lime softening.

Maintenance.

Neither `maint_override` nor `opex_fraction_of_capex` is populated; the block inherits zero maintenance cost. This is a modeling gap: lime slaking systems require above-average maintenance due to grit accumulation, calcium carbonate scaling on piping and valves, and abrasive wear on slurry pumps. A `maint_override` of 3–5% of installed equipment cost would be typical for lime-service equipment and should be considered if this block is activated for production use.

C.5.3.20 `batch_treatment_generic`

Sources.

SRK Consulting [178]; AusIMM [184]; Peters and Timmerhaus [183]; vendor data. Cost methodology adapted from the `precipitator` (C.5.3.18) and `liming_system` (C.5.3.19) blocks.

Process description.

The batch treatment block precipitates dissolved magnesium and calcium impurities from concentrated lithium brine, leaving lithium in solution. It is modeled on the Silver Peak (Nevada) batch chemical purification process, in which discrete batches of concentrated brine are sequentially treated with slaked lime ($Ca(OH)_2$) and soda ash (Na_2CO_3). The primary reactions are:

$$Mg^{2+}(aq) + Ca(OH)_2(s) \longrightarrow Mg(OH)_2(s)\downarrow + Ca^{2+}(aq) \tag{C.46}$$

$$Ca^{2+}(aq) + Na_2CO_3(aq) \longrightarrow CaCO_3(s)\downarrow + 2\,Na^{+}(aq) \tag{C.47}$$

Lime targets magnesium removal; the calcium introduced by the lime is subsequently removed by soda ash addition. Native calcium in the brine is also removed by soda ash. Reagent dosages, stoichiometry, and elimination fractions are described in the production step writeup (C.5.1.3). The block encompasses quicklime slaking and storage, soda ash dissolution and dosing, batch reactor agitation, and conditioning prior to downstream solid-liquid separation. This block is intentionally structured as a composite of the `liming_system` and `precipitator` blocks, with the addition of a soda ash dissolution and dosing subsystem, reflecting the dual-reagent treatment performed in a single batch vessel.

Equation of merit.

$$V_{\text{batch}}\ (\text{m}^3) = \text{batch size (L)} \times 10^{-3}$$

The batch reactor must accommodate the full batch volume V_{batch} with headspace for agitation and reagent addition. At a base volume of 60,000 L/batch = 60 m^3/batch, the design vessel volume is 69 m^3 after the `oversizing_factor` = 1.15 is applied. Each batch cycle takes 7 hours total: 6 hours reaction time (`batch_cycle_time`) plus 1 hour setup (`batch_setup_time`) for filling, draining, and changeover, giving an effective throughput of $60/7 \approx 8.6$ m^3/h. The 60 m^3 batch volume is consistent with the 60 m^3 tank cascade working volume priced in the CAPEX derivation (rapid mix 5 + CSTR 35 + conditioning 20 m^3).

Bill of Equipment.

Table C.85. Bill of equipment for the `batch_treatment_generic` block.

Equipment	Machine data used	Source(s)
Detention lime slaker (∼5 m^3)	AusIMM Rapid Mix Tank	Rapid Mix Tanks
Milk-of-lime storage tank (∼25 m^3)	Mixtec Agitator Tank Series	Agitator Tanks
Rapid mix tank — reagent addition (∼5 m^3)	AusIMM Rapid Mix Tank	Rapid Mix Tanks
Reaction / crystallization tank (∼35 m^3)	Continuous Stirred Tank Reactor	CSTRs
Conditioning / aging tank (∼20 m^3)	Mixtec Agitator Tank Series	Agitator Tanks
Soda ash dissolution tank (∼5 m^3)	Not explicit → inferred	—
Slurry transfer pumps & piping	Not explicit → inferred	—
Reagent handling (CaO + Na_2CO_3)	Not explicit → inferred	—
Instrumentation & controls	Not explicit → inferred	—

The equipment list mirrors the `liming_system` tank cascade (30 m^3 lime slaking + 60 m^3 circuit = 90 m^3 total), with the addition of a soda ash dissolution and dosing subsystem. In batch operation, the three circuit vessels (rapid mix, CSTR, conditioning) collectively hold the 60 m^3 batch; reagent addition, reaction, and conditioning occur sequentially across the cascade rather than in a single vessel.

Sizing basis.

The base volume is 60,000 L/batch (60 m^3) of concentrated lithium brine, with a total cycle time of 7 hours. `process_type` = batch, `dedicated_line` = FALSE. The `scaling_exponent` = 0.60 (six-tenths rule), though this exponent is not applied for batch processes in the model: batch CAPEX scales as `machines_required` × `prim_equip_price_base`, where machines required is determined by line time demand.

CAPEX.

CAPEX is built bottom-up from three subsystems, following the methodology established in the `liming_system` (C.5.3.19) and `precipitator` (C.5.3.18) blocks:

- *Tank train* (60 m^3 working volume): volumetric cost × volume, using the same AusIMM/Mixtec benchmarks as the precipitator. This gives raw tank costs of \$540,000 / \$390,000 / \$288,000 (conservative / midpoint / optimistic).
- *Soda ash dissolution and dosing system*: 15% of raw tank train cost, reflecting a small dissolution tank (∼5 m^3), metering pumps, and dry storage/handling. This gives \$81,000 / \$59,000 / \$43,000.
- *Lime slaking package*: identical to the `liming_system` lime slaking subsystem (30 m^3 at 3.5× multiplier): \$945,000 / \$683,000 / \$504,000.

The pre-auxiliary subtotal is then adjusted with subsystem adders applied to the full package: slurry pumps and piping (20%), additional reagent handling and dosing (15%), and instrumentation and controls (10%), totaling a 45% adder:

Table C.86. CAPEX derivation for the `batch_treatment_generic` block.

Scenario	**Tank train**	**Soda ash**	**Lime slk.**	**Pre-aux sub.**	**×1.45**	`prim_equip_price_base`
Conservative	\$540K	\$81K	\$945K	\$1,566K	\$2,271K	**\$2,270,000**
Midpoint	\$390K	\$59K	\$683K	\$1,132K	\$1,641K	**\$1,640,000**
Optimistic	\$288K	\$43K	\$504K	\$835K	\$1,211K	**\$1,210,000**

Equipment life is 15 years, similar to the operating conditions of the liming system.

Electricity.

Electricity demand is derived for batch operation, where the power draw varies across the 7-hour cycle. The model field `electricity_base_total` represents the cycle-averaged power draw (kWh/h), since the code computes total consumption as `electricity_base_total` × line time required, where line time includes both reaction and setup phases.

Batch tank agitation. The 60 m^3 batch reactor requires time-varying agitation: vigorous mixing during reagent addition (∼2 hr at 0.15–0.45 kW/m^3), gentle agitation during reaction and settling (∼3 hr at 0.03–0.08 kW/m^3), conditioning (∼1 hr at 0.04–0.12 kW/m^3), and no agitation during setup (∼1 hr). The time-weighted average over the 7-hour cycle:

Ancillary loads (cycle-averaged). The lime slaking system operates intermittently (∼1.5 hr per 7-hour cycle, proportional to the 60 m^3 batch reagent demand); the 25 m^3 storage tank runs intermittent keep-alive agitation during the remainder, totaling 2.5–5.0 kW cycle-averaged across slaker, storage, and lime handling drives. Soda ash dissolution (∼0.8 hr at

Table C.87. Batch tank agitation power derivation for the `batch_treatment_generic` block.

Phase	Dur. (hr)	Opt. (kW)	Mid. (kW)	Cons. (kW)
Lime addition + mixing	1.0	12 (0.20 kW/m³)	18 (0.30)	27 (0.45)
Soda ash addition + mixing	1.0	9 (0.15)	12 (0.20)	18 (0.30)
Reaction / settling	3.0	1.8 (0.03)	3 (0.05)	4.8 (0.08)
Conditioning	1.0	2.4 (0.04)	4.2 (0.07)	7.2 (0.12)
Setup (fill / drain)	1.0	0	0	0
Time-wtd. avg. (÷7 hr)		**4.1**	**6.2**	**9.5**

3–5 kW) contributes ∼0.4 kW average. Pumping (fill and drain, ∼1 hr at 4–7 kW) contributes 0.5–1.0 kW average. Instrumentation and controls add 1.5–2.5 kW:

Table C.88. Total electricity derivation for the `batch_treatment_generic` block.

Component	Opt. (kW)	Mid. (kW)	Cons. (kW)
Batch agitation (time-wtd.)	4.1	6.2	9.5
Lime slaking + storage (cycle-avg.)	2.5	3.5	5.0
Soda ash dissolution	0.3	0.4	0.5
Pumping (cycle-avg.)	0.5	0.7	1.0
Auxiliaries (I&C)	1.5	2.0	2.5
Total (cycle-avg.)	**8.9**	**12.8**	**18.5**
`electricity_base_total`	**9 kWh/h**	**13 kWh/h**	**18 kWh/h**

These cycle-averaged values are much lower than a continuous-mode estimate would suggest, because batch operation entails intermittent equipment usage with substantial settling periods where agitation is minimal.

Other utilities.

No thermal utilities are consumed. Process water is required for lime slaking, if quicklime were used as the input, at an approximately 2:1 water-to-quicklime ratio by weight.

Consumables.

Quicklime (CaO) and soda ash (Na_2CO_3) are the primary reagents and are tracked at the production step level as secondary inputs with stoichiometrically derived dosages. `tooling_cost_base` = 0.

Labor.

Labor is 5.4 FTE at base scale with a scaling exponent of 0.2 and `dedicated_labor` = TRUE. The higher base (5.4 vs. 4.5 for the continuous blocks) and dedicated-labor flag reflect the nature of batch operation; operators must be present throughout each 7-hour cycle for reagent charging, pH monitoring, settling observation, and valve sequencing, with additional labor associated with auxiliary equipment and control:

$$\text{FTE} = (1.0 + 0.2) \times 4.5 = 5.4$$

Footprint.

The base plot area is 300 m^2 with a scaling exponent of 0.6, covering the batch reactor, lime slaking system, soda ash storage, and surrounding access.

Availability.

Process availability is 90% with 5% unplanned downtime and 5% scheduled maintenance, matching the `precipitator` block.

Maintenance.

Neither `maint_override` nor `opex_fraction_of_capex` is populated; the block inherits zero maintenance cost. As noted in the `liming_system` block, lime slaking systems require above-average maintenance. A `maint_override` of 3–5% should be considered if this block is refined beyond its current baseline role.

Key measurements and cross-checks.

- *Specific electricity*: midpoint of 13 kWh/h at 8.6 m^3/h effective throughput implies 1.5 kWh/m^3 of brine, compared to ∼1.0 kWh/m^3 for the continuous `precipitator` and `liming_system` blocks. The higher intensity is consistent with the combined lime + soda ash treatment and the overhead of the lime storage keep-alive agitation spread over a smaller batch throughput.
- *CAPEX*: midpoint of $1.64M at ∼8.6 m^3/h effective throughput implies $191/($m^3$/h), approximately 15× the precipitator's $13/($m^3$/h). The premium is driven by the lime

slaking package ($683K, 42% of midpoint total), the dual-reagent system, and the lower effective throughput of batch operation.

C.5.3.21 `wash_purify_combo`

Sources.

SRK Consulting [178], SGS Canada et al. [150], Peters and Timmerhaus [183]; vendor data. Tank costs from AusIMM benchmarks (Rapid Mix Tanks, Agitator Tanks); re-dewatering filter costs from published filter press price benchmarks PORVOO [459].

Process description.

The wash_purify_combo block removes residual mother liquor and soluble impurities (principally Na^+, K^+, SO_4^{2-} or Cl^-) from Li_2CO_3 filter cake by multi-stage counter-current repulp washing. The input is dewatered Li_2CO_3 cake at approximately 70 wt% solids from the upstream `mechanical_dewatering` step. In each wash stage, the cake is re-slurried ("repulped") with clean or partially clean wash water in an agitated tank, dissolving and displacing entrained impurities from the crystal surfaces and interstices. The dilute slurry is then re-dewatered on a modest filter to recover the cake at 70 wt% solids before the next stage. Two to three counter-current stages are required to achieve battery-grade purity (>99.5% Li_2CO_3), with the cleanest water contacting the cleanest cake. The "purification" in the block name refers to this physical displacement of impurities, not to chemical redissolution or recrystallization.

The primary equipment is therefore agitated repulp tanks with a secondary re-dewatering filter — structurally analogous to the tank-based circuits in the `precipitator` and `liming_system` blocks, not to the high-duty hydraulic press in the `mechanical_dewatering` block. The re-dewatering filter handles a lighter duty than the upstream dewatering step: it re-presses dilute slurry from ∼35 wt% back to 70 wt% solids, not from 20 wt% to 70 wt%.

This block is used in both the chloride (Silver Peak) and sulfate (Thacker Pass, Jianxiawo) pathways. Wash liquor handling differs by pathway: in the chloride route, wash water is recycled to the precipitation reactor or evaporation pond; in the sulfate route, it is discharged to wastewater treatment. This latter modeling choice should be revisited under a closer water balance.

Equation of merit.

$$V_{\text{repulp}}\ (\text{m}^3) = V_{\text{cake}}\ (\text{m}^3) + V_{\text{wash}}\ (\text{m}^3)$$

Each repulp tank must hold the cake plus wash water. At a 1:1 wash ratio by volume, 60 m^3 of cake produces approximately 120 m^3 of repulp slurry at ∼35 wt% solids. With three tanks at ∼35 m^3 each (105 m^3 total working volume), the cake is processed in sub-batches of ∼20 m^3 through the counter-current wash circuit within the 1-hour cycle.

Bill of Equipment.

Table C.89. Bill of equipment for the `wash_purify_combo` block.

Equipment	Machine data used	Source(s)
Repulp tanks, SS316 (3× ∼35 m^3)	Mixtec Agitator Tank Series	Agitator Tanks
Re-dewatering filter (630–800 mm plate press or vacuum belt)	Published benchmarks	Porvoo 2025
Wash water pumps & piping	Not explicit → inferred	—
Slurry transfer pumps	Not explicit → inferred	—
Compressed air system (cake blow-down)	Not explicit → inferred	—
Cake discharge conveyor	Not explicit → inferred	—
Instrumentation & controls	Not explicit → inferred	—

Sizing basis.

The base volume is 60 m^3/batch of Li_2CO_3 cake at 70 wt% solids (approximately 95 tonnes wet cake containing 66.5 tonnes Li_2CO_3). While additional water is added and thus the equipment physically is sized to contain more than 120 m^3 of the process basis slurry, the batched nature (`process_type` = batch) means that sizing to the contained cake volume is appropriate as long as units are consistent. Each batch cycle is 1.2 hours (1 hour wash + 0.2 hours setup), giving an effective throughput of 50 m^3/h. Equipment life is assumed to be 20 years.

CAPEX.

CAPEX is built bottom-up from two components: repulp tanks (costed from AusIMM volumetric benchmarks, as in the `precipitator` block) and a re-dewatering filter (costed from published filter press price benchmarks). A stainless steel premium of 1.3–1.5× is applied to all wetted equipment to meet battery-grade Li_2CO_3 purity requirements.

Repulp tanks (105 m^3 total working volume, 3 tanks × 35 m^3): raw tank cost = volumetric cost × volume, then × SS premium.

Re-dewatering filter: a 630–800 mm plate filter press sized for intermittent re-pressing of repulped slurry from ∼35 wt% to 70 wt% solids. Bare cost anchored to the published

benchmark of $75,000–$220,000 for 630–800 mm presses, with a stainless-steel premium and installation applied.

Subsystem adders: wash water pumps and piping (15%), instrumentation and controls (10%), totaling 25% on the combined tank + filter cost.

Table C.90. CAPEX derivation for the `wash_purify_combo` block.

Scenario	**Tanks (stainless)**	**Filter (installed)**	**Wash water sys.**	`prim_equip_price_base`
Conservative	$1,418K	$330K	$150K	**$2,370,000**
Midpoint	$956K	$200K	$100K	**$1,570,000**
Optimistic	$655K	$130K	$60K	**$1,060,000**

Derivation details (midpoint row). The raw tank cost is $6,500/m^3 × 105 m^3 = $683K, which becomes $956K after applying a 1.4× stainless steel premium. The filter cost starts from a 630 mm press at $108K bare, multiplied by 1.4× for installation and 1.3× for stainless steel construction, giving $197K ≈ $200K installed. The wash water system (holding tank, heater stub, metering pumps) is estimated at $100K. The subtotal of $1,256K is then multiplied by 1.25× for piping and instrumentation adders, giving a total of $1,570K.

Electricity.

Electricity demand is derived from the major loads during the 1.2-hour batch cycle. Each of the 2–3 wash stages involves repulping (agitation), pumping, and re-dewatering on the filter. Equipment operates intermittently; `electricity_base_total` represents the cycle-averaged power draw.

Agitation. Repulp tanks at 0.20–0.40 kW/m^3 on the working slurry volume (∼35 m^3 per tank), with 1–2 tanks active, running ∼30–40 min of the 72-min cycle.

Re-dewatering filter. The filter re-presses repulped slurry 2–3 times per batch, running ∼30–45 min total. Installed motor power for a 630–800 mm filter press with hydraulic closure is 15–55 kW; cycle-averaged draw is lower due to intermittent HPU operation.

Pumping. Wash water and slurry transfer pumps from $P = \rho g Q H/\eta$, running during fill and drain phases.

The re-dewatering filter remains the dominant load (∼40% of midpoint total), consistent with the repeated pressing of repulped slurry.

Other utilities.

Cooling water is consumed at 0.2 m^3/h for hydraulic oil cooling on the re-dewatering filter. Compressed air is consumed at 1 Nm^3/h for cake blow-down between wash stages.

Table C.91. Electricity derivation for the `wash_purify_combo` block.

Component	Opt. (kW)	Mid. (kW)	Cons. (kW)
Repulp agitation (cycle-avg.)	8	15	22
Re-dewatering filter (cycle-avg.)	18	38	55
Wash water pumps (cycle-avg.)	8	13	18
Slurry transfer pumps	7	12	16
Compressed air (blow-down)	3	5	7
Auxiliaries (I&C, conveyor)	5	7	9
`electricity_base_total`	**∼50 kWh/h**	**∼90 kWh/h**	**∼130 kWh/h**

No thermal utilities are tracked; wash water temperature conditioning (if used to exploit retrograde solubility of Li_2CO_3) would require steam or process heat integration, but is not currently modeled.

Consumables.

`tooling_cost_base` = 0. Filter cloths for the re-dewatering filter are a recurring consumable (replacement every 500–2,000 press cycles) but are not yet populated. This gap is less significant than for the `mechanical_dewatering` block given the lighter filter duty.

Labor.

Labor is 3.15 FTE at base scale with a scaling exponent of 0.5 and `dedicated_labor` = FALSE, consistent with a semi-automated wash operation sharing operators with adjacent dewatering and drying equipment. The higher scaling exponent (0.5 vs. 0.2 for tank-only circuits) reflects the additional operator involvement in managing the re-dewatering filter:

$$\text{FTE} = (0.5 + 0.2) \times 4.5 = 3.15$$

Footprint.

The base plot area is 100 m^2 with a scaling exponent of 0.7, covering the repulp tanks, re-dewatering filter, wash water system, and cake discharge area.

Availability.

Process availability is 90% with 5% unplanned downtime and 5% scheduled maintenance.

Maintenance.

Neither `maint_override` nor `opex_fraction_of_capex` is populated; the block inherits zero maintenance cost. A `maint_override` of 2–4% should be considered, driven by the re-dewatering filter's cloth and seal replacement requirements.

Key measurements and cross-checks.

- *CAPEX*: bottom-up midpoint of \$1.57M at 50 m^3/h effective throughput implies \$31/(m^3/h), approximately 2.4× the `precipitator`'s \$13/(m^3/h). The premium is consistent with a stainless steel construction requirement and the re-dewatering filter.
- *Electricity*: midpoint of 90 kWh/h (1.5 kWh/m^3 of cake) is dominated by the re-dewatering filter (∼42%).
- *Wash efficiency*: 2–3 counter-current stages at ∼1:1 wash ratio total 120–180 m^3 of wash water per 60 m^3 batch. This demand is tracked at the production step level.

C.5.3.22 `drying_packaging` (Generic and Propane options)

Sources.

SGS Canada et al. [150], Lane et al. [153], Samari et al. [155], SRK Consulting [178], Riordan et al. [186], Dai et al. [415]

Process description.

Following precipitation and solid-liquid separation, lithium carbonate is produced as a wet cake at approximately 70 wt% solids and 30 wt% moisture. This model block dries the cake to a product specification of ≤0.4 wt% moisture and packages the dry product for shipment. The block is modeled as a compact finishing circuit centered on an indirect gas-fired dryer with baghouse dust capture and a bulk bagging line. Two fuel variants share identical equipment configurations and all non-fuel parameters: `drying_packaging_generic` uses natural gas (`natural_gas_base_total`) and is used for the Thacker Pass and Jianxiawo pathways; `drying_packaging_propane` substitutes propane (`propane_base_total`) and is used for Silver Peak, where no natural gas supply is available at the remote Nevada site [178].

Equation of merit.

$$\dot{m}_{\mathrm{H_2O}}\ (\mathrm{t/h}) \approx \dot{m}_{\mathrm{cake}}\ (\mathrm{t/h}) \times 0.297$$

where the factor 0.297 is the water evaporated per tonne of 70 wt% solids wet cake, derived from:

$$w_{\mathrm{evap}} = w_{\mathrm{water}} - \frac{w_{\mathrm{solids}} \times w_{\mathrm{product}}}{1 - w_{\mathrm{product}}} = 0.30 - \frac{0.70 \times 0.004}{0.996} = 0.2972\ \mathrm{t/t\ cake}$$

Thermal duty scales with the water evaporation rate; electricity and CAPEX scale with wet cake throughput.

Sizing basis.

The step basis is `cake_dry_throughput` and `base_volume` = 2.5, representing 2.5 t/h of wet cake feed to the dryer. This value originates from the EverBatt battery materials cost model [415], where a capacity adjustment of $1/0.4 = 2.5$ is applied to a baseline dryer parameterized at $x = 0.4$. At the 2.5 t/h wet cake base:

- Water evaporated: $2.5 \times 0.297 = \mathbf{0.743}$ t/h
- Dry Li_2CO_3 product: $2.5 \times 0.7028 = 1.757$ t/h at ≤0.4 wt% moisture

Bill of Equipment.

Table C.92. Bill of equipment for the `drying_packaging` blocks.

Equipment	Machine data used	Source(s)
Indirect gas-fired dryer	Indirect dryer	Feasibility disclosures, [415]
Baghouse + exhaust fan	Inferred from dryer package	—
Bulk bagging line	Standard Li_2CO_3 packaging	Feasibility disclosures
Feed handling & conveying	Not explicit → inferred	—
Product bins, silo & loadout	Not explicit → inferred	—
Utilities, controls & instrumentation	Not explicit → inferred	—

CAPEX.

The EverBatt battery materials cost model (Argonne National Laboratory) parameterizes the bare dryer purchase cost as $C = (591{,}236 \times x^{0.6}) \times 1.483$ (2019$), giving $506,014 at $x = 0.4$. Inflated to 2025$ yields $666,000. A ±30% range on this anchor reflects Class 4–5 uncertainty in equipment specification, giving bare dryer costs of $466,000 / $666,000 / $866,000 (optimistic / midpoint / conservative).

Auxiliary bare equipment is added as fractions of the bare dryer cost following Peters & Timmerhaus Table 6-11: baghouse and exhaust fan at 25% / 32% / 40% and bulk bagging line at 18% / 22% / 30% (optimistic / midpoint / conservative). Balance-of-plant items (feed handling and conveying, product silo and loadout, utilities and controls) are added at 30% / 40% / 52% of the primary equipment subtotal per Peters & Timmerhaus Table 6-17 for fluid-solid processing. A Lang-type installation factor of 2.3× / 2.8× / 3.3× (optimistic / midpoint / conservative) is then applied to total bare equipment, covering direct installation labor, piping, instrumentation, electrical, structural, civil, engineering, and contingency: These values apply to both variants. The CAPEX scaling exponent is 0.60, following the six-tenths rule for process equipment cost (which is also used in EverBatt), and equipment life is 20 years.

Electricity.

Electrical demand covers the dryer drive and exhaust fan / baghouse system, with all minor motor loads (conveyor drives, bagging line, instrumentation) treated as auxiliary adders.

Table C.93. Total installed circuit cost for the `drying_packaging` blocks.

Scenario	Bare dryer	Total bare	Install factor	`prim_equip_price_base`
Conservative	\$866,000	\$2,238,000	×3.3	**\$7,400,000**
Midpoint	\$666,000	\$1,437,000	×2.8	**\$4,000,000**
Optimistic	\$466,000	\$867,000	×2.3	**\$2,000,000**

Electricity is identical across the natural gas and propane variants; only the fuel supply differs.

Natural gas (`drying_packaging_generic`).

Thermal duty is derived from first principles. The drying load per tonne of water evaporated comprises sensible heat of liquid water from 25°C to 100°C (≈419 kJ/kg), latent heat of evaporation at 100°C (≈2,257 kJ/kg), and a 10% solids heating penalty. An additional air-overheating penalty of 10% / 20% / 30% (optimistic / midpoint / conservative) accounts for energy required to superheat the exhaust stream. Dividing by 80% combustion efficiency gives the fuel input per tonne of water evaporated:

Table C.94. Natural gas derivation for `drying_packaging_generic` at 2.5 t/h wet cake.

Scenario	**Net** (MMBTU/t H_2O)	**Fuel** (MMBTU/t H_2O)	`natural_gas_base_total` (At 0.743 t/h H_2O)
Conservative	3.63	4.54	**3.37**
Midpoint	3.35	4.18	**3.11**
Optimistic	3.07	3.84	**2.85**

Propane (`drying_packaging_propane`). Propane consumption is converted directly from the equivalent natural gas thermal demand using the propane higher heating value of 24,174 Btu/L (liquid propane at standard conditions), giving 41.37 L propane per MMBTU. `natural_gas_base_total` is set to `False` for this variant.

Other utilities.

- `cooling_water_base_total`: 0.1 m^3/h (generic variant only) — minor seal and bearing cooling for the dryer drive
- `compressed_air_base_total`: 0.5 Nm^3/h (both variants) — instrumentation air only; no bag filter pulse-jet system is modeled at this scale

Table C.95. Propane derivation for `drying_packaging_propane` at 2.5 t/h wet cake.

Scenario	NatGas equiv. (MMBTU/h)	Conversion (L/MMBTU)	`propane_base_total`
Conservative	3.37	41.37	**139 L/h**
Midpoint	3.11	41.37	**129 L/h**
Optimistic	2.85	41.37	**118 L/h**

- Steam, diesel: zero for both variants

Labor. Labor is 4.05 FTE at base scale with a scaling exponent of 0.5 and `dedicated_labor` = FALSE, summing operator-per-shift factors across the three principal equipment groups in the circuit — indirect dryer (0.2), separator/baghouse/conveying (0.2), and packaging line (0.5):

$$\text{FTE} = (0.2 + 0.2 + 0.5) \times 4.5 = 4.05$$

Footprint. The base plot area is 400 m^2 with a scaling exponent of 0.7, covering the dryer shell, baghouse, bagging line, product silo, and surrounding access, and is assumed to be identical across both fuel variants.

Availability. Process availability is 90% with 5% unplanned and 5% scheduled maintenance, consistent with a straightforward mechanical finishing circuit. Identical across both variants.

Key measurements and cross-checks.

- Product moisture at $\leq$0.4 wt% — the primary dryer performance specification
- Total energy: 401 kWh/t wet cake (midpoint, gas + electric combined) is consistent with the EverBatt electric-only benchmark once normalised to the same throughput basis and energy carrier
- CAPEX: midpoint \$4.0M implies the bare dryer (\$666k from EverBatt) represents 17% of total installed circuit cost, with installation and auxiliaries accounting for the remainder — consistent with Peters & Timmerhaus guidance for small solid-fluid processing circuits